

Bergey's Manual of
**Systematic
Bacteriology**

SECOND EDITION

Volume Two

The Proteobacteria

Part C The *Alpha*-, *Beta*-, *Delta*- and *Epsilon*proteobacteria

BERGEY'S MANUAL® OF
**Systematic
Bacteriology**
Second Edition

Volume Two
The *Proteobacteria*

Part B
The *Gammaproteobacteria*

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TO COME

*This volume is dedicated to our colleagues,
David R. Boone, Don J. Brenner,
Richard W. Castenholz, and Noel R. Krieg, who
retired from the Board of Trustees of Bergey's Manual
Trust as this edition was in preparation. We deeply
appreciate their efforts as editors and authors; they
have devoted their time and many years in helping
the Trust meet its objectives.*

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Preface to Volume Two of the Second Edition of *Bergey's Manual*[®] of Systematic Bacteriology

There is a long-standing tradition for the Editors of *Bergey's Manual* to open their respective editions with the observation that the new edition is a departure from the earlier ones. As this volume goes to press, however, we recognize a need to deviate from this practice, by offering a separate preface to each volume within this edition. In part, this departure is necessary because the size and complexity of this edition far exceeded our expectations, as has the amount of time that has elapsed between publication of the first volume of this edition and this volume.

Earlier, we noted that systematic procaryotic biology is a dynamic field, driven by constant theoretical and methodological advances that will ultimately lead to a more perfect and useful classification scheme. Clearly, the pace has been accelerating as evidenced in the super-linear rate at which new taxa are being described. Much of the increase can be attributed to rapid advances in sequencing technology, which has brought about a major shift in how we view the relationships among *Bacteria* and *Archaea*. While the possibility of a universally applicable natural classification was evident as the First Edition was in preparation, it is only recently that the sequence databases became large enough, and the taxonomic coverage broad enough to make such an arrangement feasible. We have relied heavily upon these data in organizing the contents of this edition of *Bergey's Manual of Systematic Bacteriology*, which will follow a phylogenetic framework based on analysis of the nucleotide sequence of the small ribosomal subunit RNA, rather than a phenotypic structure. This departs from the First Edition, as well as the Eighth and Ninth Editions of the *Determinative Manual*. While the rationale for presenting the content of this edition in such a manner should be evident to most readers, they should bear in mind that this edition, as in all preceding ones represents a progress report, rather than a final classification of procaryotes.

The Editors remind the readers that the *Systematics Manual* is a peer-reviewed collection of chapters, contributed by authors who were invited by the Trust to share their knowledge and expertise of specific taxa. Citation should refer to the author, the chapter title, and inclusive pages rather than to the Editors. The Trust is indebted to all of the contributors and reviewers, without whom this work would not be possible. The Editors are grateful for the time and effort that each expended on behalf of the entire scientific community. We also thank the authors for their good grace in accepting comments, criticisms, and editing of

their manuscripts. We would also like to thank Drs. Hans Trüper, Brian Tindall, and Jean Euzéby for their assistance on matters of nomenclature and etymology.

We would like to express our thanks to the Department of Microbiology and Molecular Genetics at Michigan State University for housing our headquarters and editorial office and for providing a congenial and supportive environment for microbial systematics. We would also like to thank Connie Williams not only for her expert secretarial assistance, but also for unflagging dedication to the mission of Bergey's Manual Trust and Drs. Julia Bell and Denise Searles for their expert editorial assistance and diligence in verifying countless pieces of critical information and to Dr. Timothy G. Lilburn for constructing many of the phylogenetic trees used in this volume. We also extend our thanks to Alissa Wesche, Matt Chval and Kristen Johnson for their assistance in compilation of the bibliography.

A project such as the *Systematics Manual* also requires the strong and continued support of a dedicated publisher, and we have been most fortunate in this regard. We would also like to express our gratitude to Springer-Verlag for supporting our efforts and for the development of the Bergey's Document Type Definition (DTD). We would especially like to thank our Executive Editor, Dr. William Curtis for his courage, patience, understanding, and support; Catherine Lyons for her expertise in designing and developing our DTD, and Jeri Lambert and Leslie Grossberg of Impressions Book and Journal Services for their efforts during the pre-production and production phases. We would also like to acknowledge the support of ArborText, Inc., for providing us with state-of-the-art SGML development and editing tools at reduced cost. Lastly, I would like to express my personal thanks to my fellow trustees for providing me with the opportunity to participate in this effort, to Drs. Don Brenner, Noel Krieg, and James Staley for their enormous efforts as volume editors and to my wife, Nancy, and daughter, Jane, for their continued patience, tolerance and support.

Comments on this edition are welcomed and should be directed to Bergey's Manual Trust, Department of Microbiology and Molecular Genetics, 6162 Biomedical and Physical Sciences Building, Michigan State University, East Lansing, MI, USA 48824-4320. Email: garrity@msu.edu

George M. Garrity

Preface to the First Edition of *Bergey's Manual*[®] of *Systematic Bacteriology*

Many microbiologists advised the Trust that a new edition of the *Manual* was urgently needed. Of great concern to us was the steadily increasing time interval between editions; this interval reached a maximum of 17 years between the seventh and eighth editions. To be useful the *Manual* must reflect relatively recent information; a new edition is soon dated or obsolete in parts because of the nearly exponential rate at which new information accumulates. A new approach to publication was needed, and from this conviction came our plan to publish the *Manual* as a sequence of four subvolumes concerned with systematic bacteriology as it applies to taxonomy. The four subvolumes are divided roughly as follows: (a) the Gram-negatives of general, medical or industrial importance; (b) the Gram-positives other than actinomycetes; (c) the archaeobacteria, cyanobacteria and remaining Gram-negatives; and (d) the actinomycetes. The Trust believed that more attention and care could be given to preparation of the various descriptions within each subvolume, and also that each subvolume could be prepared, published, and revised as the area demanded, more rapidly than could be the case if the *Manual* were to remain as a single, comprehensive volume as in the past. Moreover, microbiologists would have the option of purchasing only that particular subvolume containing the organisms in which they were interested.

The Trust also believed that the scope of the *Manual* needed to be expanded to include more information of importance for systematic bacteriology and bring together information dealing with ecology, enrichment and isolation, descriptions of species and their determinative characters, maintenance and preservation, all focused on the illumination of bacterial taxonomy. To reflect this change in scope, the title of the *Manual* was changed and the primary publication becomes *Bergey's Manual of Systematic Bacteriology*. This contains not only determinative material such as diagnostic keys and tables useful for identification, but also all of the detailed descriptive information and taxonomic comments. Upon completion of each subvolume, the purely determinative information will be assembled for eventual incorporation into a much smaller publication which will continue the original name of the *Manual*, *Bergey's Manual of Determinative Bacteriology*, which will be a similar but improved version of the present *Shorter Bergey's Manual*. So, in the end there will be two publications, one systematic and one determinative in character.

An important task of the Trust was to decide which genera should be covered in the first and subsequent subvolumes. We were assisted in this decision by the recommendations of our Advisory Committees, composed of prominent taxonomic au-

thorities to whom we are most grateful. Authors were chosen on the basis of constant surveillance of the literature of bacterial systematics and by recommendations from our Advisory Committees.

The activation of the 1976 Code had introduced some novel problems. We decided to include not only those genera that had been published in the Approved Lists of Bacterial Names in January 1980 or that had been subsequently validly published, but also certain genera whose names had no current standing in nomenclature. We also decided to include descriptions of certain organisms which had no formal taxonomic nomenclature, such as the endosymbionts of insects. Our goal was to omit no important group of cultivated bacteria and also to stimulate taxonomic research on "neglected" groups and on some groups of undoubted bacteria that have not yet been cultivated and subjected to conventional studies.

The invited authors were provided with instructions and exemplary chapters in June 1980 and, although the intended deadline for receipt of manuscripts was March 1981, all contributions were assembled in January 1982 for the final preparations. The *Manual* was forwarded to the publisher in June 1982.

Some readers will note the consistent use of the stem -var instead of -type in words such as biovar, serovar and pathovar. This is in keeping with the recommendations of the Bacteriological Code and was done against the wishes of some of the authors.

We have deleted much of the synonymy of scientific names which was contained in past editions. The adoption of the new starting date of January 1, 1980 and publication of the Approved Lists of Bacterial Names has made mention of past synonymy obsolete. We have included synonyms of a name only if they have been published since the new starting date, or if they were also on the Approved Lists and, in rare cases with certain pathogens, if the mention of an old name would help readers associate the organism with a clinical problem. If the reader is interested in tracing the history of a name we suggest he or she consult past editions of the *Manual* or the *Index Bergeyana* and its *Supplement*. In citations of names we have used the abbreviation AL to denote the inclusion of the name on the Approved Lists of Bacterial Names and VP to show the name has been validly published.

In the matter of citation of the *Manual* in the scientific literature we again stress the fact that the *Manual* is a collection of authored chapters and the citation should refer to the author, the chapter title and its inclusive pages, not the Editor.

To all contributors, the sincere thanks of the Trust is due; the Editor is especially grateful for the good grace with which the authors accepted comments, criticisms and editing of their manuscripts. It is only because of the voluntary and dedicated efforts of these authors that the *Manual* can continue to serve the science of bacteriology on an international basis.

A number of institutions and individuals deserve special acknowledgment from the Trust for their help in bringing about the publication of this volume. We are grateful to the Department of Biology of the Virginia Polytechnic Institute and State University for providing space, facilities and, above all, tolerance for the diverted time taken by the Editor during the preparation of the book. The Department of Microbiology at Iowa State University of Science and Technology continues to provide a welcome home for the main editorial offices and archives of the Trust and we acknowledge their continued support. A grant (LM-03707) from the National Library of Medicine, National

Institutes of Health to assist in the preparation of this and the next volume of the *Manual* is gratefully acknowledged.

A number of individuals deserve special mention and thanks for their help. Professor Thomas O. McAdoo of the Department of Foreign Languages and Literatures at the Virginia Polytechnic Institute and State University has given invaluable advice on the etymology and correctness of scientific names. Those assisting the Editor in the Blacksburg office were R. Martin Roop II, Don D. Lee, Eileen C. Falk and Michael W. Friedman and their help is sincerely appreciated. In the Ames office we were ably assisted by Gretchen Colletti and Diane Triggs during the early period of preparation and by Cynthia Pease during the major portion of the editing process. Mrs. Pease has been responsible for the construction of the List of References and her willingness to handle the cumbersome details of text editing on a big computer is gratefully acknowledged.

John G. Holt

Preface to the First Edition of *Bergey's Manual® of Determinative Bacteriology*

The elaborate system of classification of the bacteria into families, tribes and genera by a Committee on Characterization and Classification of the Society of American Bacteriologists (1911, 1920) has made it very desirable to be able to place in the hands of students a more detailed key for the identification of species than any that is available at present. The valuable book on "Determinative Bacteriology" by Professor F. D. Chester, published in 1901, is now of very little assistance to the student, and all previous classifications are of still less value, especially as earlier systems of classification were based entirely on morphologic characters.

It is hoped that this manual will serve to stimulate efforts to perfect the classification of bacteria, especially by emphasizing the valuable features as well as the weaker points in the new system which the Committee of the Society of American Bacteriologists has promulgated. The Committee does not regard the classification of species offered here as in any sense final, but merely a progress report leading to more satisfactory classification in the future.

The Committee desires to express its appreciation and thanks to those members of the society who gave valuable aid in the compilation of material and the classification of certain species. . . .

The assistance of all bacteriologists is earnestly solicited in the correction of possible errors in the text; in the collection of descriptions of all bacteria that may have been omitted from the text; in supplying more detailed descriptions of such organisms as are described incompletely; and in furnishing complete descriptions of new organisms that may be discovered, or in directing the attention of the Committee to publications of such newly described bacteria.

David H. Bergey, *Chairman*
Francis C. Harrison
Robert S. Breed
Bernard W. Hammer
Frank M. Huntoon
Committee on Manual.
August, 1923.

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Phylum XIV. Proteobacteria *phyl. nov.*

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Pro.te.o.bac.te'ri.a. Gr. n. *Proteus* ocean god able to change shape; Gr. n. *bakterion* a small rod; M.L. fem. pl. n. *Proteobacteria* the phylum of bacteria having 16S rRNA gene sequences related to those of the members of the order *Pseudomonadales*.

The phylum *Proteobacteria* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rRNA gene sequences. The phylum contains Gram-negative bacteria in the classes *Alpha-*

proteobacteria, *Betaproteobacteria*, *Gammaproteobacteria*, *Deltaproteobacteria*, and *Epsilonproteobacteria*.

Type order: **Pseudomonadales** Orla-Jensen 1921, 270.

Class III. Gammaproteobacteria *class. nov.*

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Gam.ma.pro.te.o.bac.te'ri.a. Gr. n. *gamma* name of third letter of Greek alphabet; Gr. n. *Proteus* ocean god able to change shape; Gr. n. *bakterion* a small rod; M.L. fem. pl. n. *Gammaproteobacteria* class of bacteria having 16S rRNA gene sequences related to those of the members of the order *Pseudomonadales*.

The class *Gammaproteobacteria* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rRNA sequences; the class contains the orders *Acidithiobacillales*, *Aeromonadales*, *Alteromonadales*, *Cardiobacteriales*, *Chromatiales*, *Enterobacteriales*, *Legion-*

ellales, *Methylococcales*, *Oceanospirillales*, *Pasteurellales*, *Pseudomonadales*, *Thiotrichales*, *Vibrionales*, and *Xanthomonadales*.

Type order: **Pseudomonadales** Orla-Jensen 1921, 270.

Order I. Chromatiales *ord. nov.*

JOHANNES F. IMHOFF

Chro.ma.ti.a'les. M.L. neut. n. *Chromatium* type genus of the order; *-ales* ending to denote an order; M.L. fem. pl. n. *Chromatiaceae* the *Chromatium* order.

The order *Chromatiales* consists of unicellular Gram-negative bacteria belonging to the *Gammaproteobacteria*. Cells are spherical, vibrioid, spiral, or rod shaped. Multiply by binary fission, motile by flagella or nonmotile; some contain gas vesicles.

The order *Chromatiales* contains anoxygenic phototrophic bacteria, known as phototrophic purple sulfur bacteria, able to perform photosynthesis under anoxic conditions without oxygen production. In addition, the order contains nonphototrophic, purely chemotrophic relatives. Color of cell suspensions of the phototrophic species ranges from purple-violet to purple-red, red, orange-brown, yellowish brown, brownish red, brown, and green. Common to all phototrophic species is the presence of bacteriochlorophyll *a* or *b* (see Table BXII.γ.1), and various carotenoids of the spirilloxanthin, okenone, or rhodopinal groups (Table BXII.γ.2). Photosynthetic pigments are located in the cy-

toplasmic membrane, and in internal membrane systems of different fine structure which originate from, and are continuous with, the cytoplasmic membrane.

Phototrophic species are able to grow by a photolithoautotrophic or photoorganoheterotrophic metabolism under anoxic conditions. The photosynthetic metabolism differs from that of the cyanobacteria, algae, and green plants in that water cannot serve as electron donor substrate. Photosynthetic CO₂ assimilation depends on the utilization of external electron donors such

TABLE BXII.γ.1. Bacteriochlorophylls of phototrophic purple bacteria

Pigment	Absorption maxima (nm) in living cells
Bacteriochlorophyll <i>a</i>	375–378, 590, 796–805, 820–898
Bacteriochlorophyll <i>b</i>	400, 605, 835–850, 980–1030

TABLE BXII.γ.2. Carotenoid groups of phototrophic purple bacteria^a

Carotenoid group	Major components
Normal spirilloxanthin group	spirilloxanthin, lycopene, rhodopin
Alternative spirilloxanthin group	spheroidene, spheroidenone, chloroxanthin, (spirilloxanthin)
Okenone group	okenone
Rhodopinal group	rhodopinal, rhodopin, lycopenal, lycopene, lycopanol, spirilloxanthin

^aData from Schmidt (1978).

as reduced sulfur compounds, reduced iron, molecular hydrogen, or organic compounds. In the presence of sulfide and light, cells of phototrophic members of the *Chromatiales* (purple sulfur bacteria) form highly refractile globules of elemental sulfur either inside or outside their cells. CO₂ is photoassimilated through the reductive pentose phosphate cycle. Fixation of N₂ has been demonstrated in many species. All species contain cytochromes, quinones, and non-heme iron proteins of the ferredoxin type. In addition to their phototrophic capabilities, many species of this order can grow chemotrophically under microoxic to oxic conditions in the dark, but some species, such as *Arhodomonas aquaeolei* (see family *Ectothiorhodospiraceae*), may be purely chemotrophic and unable to grow phototrophically.

Type genus: Chromatium Perty 1852, 174 emend. Imhoff, Sül-
ing and Petri 1998b, 1138.

FURTHER DESCRIPTIVE INFORMATION

The available 16S rDNA sequence information for *Ectothiorhodospiraceae* (Imhoff and Sül-
ing, 1996) and *Chromatiaceae* species (DeWeerd et al., 1990; Wahlund et al., 1991; Caumette et al., 1997; Guyoneaud et al., 1997, 1998b; Imhoff et al., 1998b) supports the classification of these bacteria in two different families (Imhoff, 1984b). At the same time, it indicates the existence of different genera within the family *Ectothiorhodospiraceae* (Imhoff and Sül-
ing, 1996), and the inconsistency of the previous taxonomic classification of the family *Chromatiaceae* (see Pfennig and Trüper, 1989) with its phylogeny (Imhoff et al., 1998b). In addition, the phylogenetic relatedness of both families and their distance to other bacteria of the *Gammaproteobacteria* support their treatment within one order, for which the name *Chromatiales* is proposed. The order *Chromatiales* includes the phototrophic purple sulfur bacteria of the *Ectothiorhodospiraceae* and *Chromatiaceae*

(Imhoff, 1984b, 1989; Pfennig and Trüper, 1989) and their purely chemotrophic relatives (see Figs. BXII.γ.3 and BXII.γ.4 of the chapter describing the family *Chromatiaceae*). The purely chemotrophic bacteria known so far are specifically related to *Ectothiorhodospiraceae*. The 16S rDNA sequence of *Arhodomonas aquaeolei* (Adkins et al., 1993; Imhoff and Sül-
ing, 1996) is similar to those of *Halorhodospira* species, but clearly distinct at the genus level. In addition, sequences of *Nitrococcus mobilis* (Teske et al., 1994) and *Nitrosococcus oceanii* (Head et al., 1993) place these bacteria in phylogenetic proximity to the *Ectothiorhodospiraceae*, though on a more distant level than the genus *Arhodomonas*.

Differentiation of Chromatiaceae and Ectothiorhodospiraceae

Traditionally, the most important and easily recognized distinguishing property within the phototrophic members of the order *Chromatiales* is the deposition of S⁰ during growth on sulfide. Sulfur globules appear exclusively inside the cells (*Chromatiaceae*) or outside the cells (*Ectothiorhodospiraceae*). A clear distinction between the two families is also possible based on differences in quinone, lipid, and fatty acid composition (Imhoff and Bias-Imhoff, 1995). Several glucolipids are present in *Chromatiaceae* species, but absent from members of *Ectothiorhodospiraceae* (Imhoff et al., 1982a). In addition, the lipopolysaccharides show significant differences between members of the two families (Weckesser et al., 1979, 1995). The lipid A of all investigated *Chromatiaceae* species (*Allochromatium vinosum*, *Thermochromatium tepidum*, *Thiocystis violacea*, *Thiocapsa roseopersicina*, *Thiococcus pfennigii*) is characterized by a phosphate-free backbone with D-glucosamine as the only amino sugar, having terminally attached D-mannose and amide-bound C_{14:0} 3OH. In the lipid A of all tested *Ectothiorhodospiraceae* species (*Ectothiorhodospira vacuolata*, *E. shaposhnikovii*, *E. haloalkaliphila*, and *Halorhodospira halophila*), phosphate is present, 2,3-diamino-2,3-dideoxy-D-glucose is the major amino sugar (D-glucosamine is also present), D-mannose is lacking (D-galacturonic acid, and D-glucuronic acid are present instead) and, quite remarkably, C_{10:0} 3OH is present as an amide-bound fatty acid (Zahr et al., 1992; Weckesser et al., 1995; Imhoff and Sül-
ing, 1996). These distinctive properties of the lipid A appear to be characteristic features of the two families. In addition, sequences of 16S rDNA are significantly different in representatives of the two families, as demonstrated by a number of characteristic signatures (Table BXII.γ.3) and by their overall dissimilarity (Figs. BXII.γ.3 and BXII.γ.4 of the chapter describing the family *Chromatiaceae*).

Key to the phototrophic members of the families of the Chromatiales

- I. *Gammaproteobacteria* that contain bacteriochlorophyll *a* or *b* and different carotenoid pigments, have internal photosynthetic membrane structures and are able to grow phototrophically under anoxic conditions in the light.
 - A. In the presence of both sulfide and light, globules of S⁰ are formed inside the cells and further oxidized to sulfate.

Chromatiaceae
 - B. In the presence of both sulfide and light, globules of S⁰ are formed outside the cells (only exceptionally also in the peripheral part of the cell) and may be further oxidized to sulfate. Growth is dependent on saline and alkaline growth conditions.

Ectothiorhodospiraceae

TABLE BXII.γ.3. Differential characteristics of the phototrophic members of the families *Chromatiaceae* and *Ectothiorhodospiraceae*

Characteristic	<i>Chromatiaceae</i>	<i>Ectothiorhodospiraceae</i>
Sulfur deposition	inside cells	outside cells
Characteristics of polar lipids ^a	several glycolipids	glycolipids absent
Major fatty quinones ^a	Q8/MK8	Q7 or Q8/MK7 or MK8
Major fatty acids ^a :		
C _{16:0}	20–35%	11–25%
C _{16:1}	25–37%	<10%
C _{18:1}	38–45%	50–75%
Properties of lipopolysaccharides ^b :		
Major aminosugar	glucosamine	diamino-dideoxy-glucose
Phosphate	present	absent
D-Mannose	+	—
D-Galacturonic acid	—	+
D-Glucuronic acid	—	+
Amide-bound fatty acid	C _{14:0} 3OH	C _{12:0} 3OH
Signature 16S rDNA sequence ^c :		
Position 217	T	G/A ^d
Position 234–235	CA	TG/CG ^d
Position 821–822	TC	AG
Position 876–877	GA	CT
Position 985	A	T/G ^d

^aData from Imhoff and Bias-Imhoff (1995).^bData from Weckesser et al. (1995).^cNumbering based on *E. coli* 16S rDNA sequence.^dBases above the line are found in the genera *Ectothiorhodospira* and *Thiorhodospira*; bases below the line are found in the genera *Arhodomonas* and *Halorhodospira*.**Family I. Chromatiaceae** Bavendamm 1924, 125^{AL} emend. Imhoff 1984b, 339

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Chro.ma.ti.a'ce.ae. M.L. neut. n. *Chromatium* type genus of the family; *-aceae* ending to denote a family; M.L. fem. pl. n. *Chromatiaceae* the *Chromatium* family.

Cells are spherical, ovoid, vibrioid, spiral, or rod shaped, multiply by binary fission, may contain gas vesicles, may be motile by flagella or nonmotile. Motile forms have either monotrichous or multitrichous polar flagella. **Gram negative, belong to the Gammaproteobacteria, and contain internal photosynthetic membranes**, which are continuous with the cytoplasmic membrane and of vesicular type in most species (Fig. BXII.γ.1), though some species have tubular internal membranes (Fig. BXII.γ.2) or lamellae. **As photosynthetic pigments, bacteriochlorophyll *a* or *b* and carotenoids of the spirilloxanthin, okenone, or rhodopinal groups are present** in most species, but other carotenoids such as tetrahydrospirilloxanthin, may occur. In general, the color of cultures of strains with carotenoids of the spirilloxanthin group is orange-brown to brownish red or pink, of those with okenone, purple-red, and of those with carotenoids of the rhodopinal group, purple-violet.

Under anoxic conditions in the light, all species are capable of photolithoautotrophic growth with sulfide or S⁰ as electron donor. S⁰ globules accumulate as an intermediate oxidation product inside the cells. Many species are able to use H₂ as an electron donor under reducing culture conditions, and some use reduced iron ions. All species are potentially mixotrophic and photoassimilate a number of simple organic compounds, of which acetate and pyruvate are the most widely used. Some species are able to photoassimilate organic substances in the absence of sulfide or sulfur, or grow photoorganoheterotrophically. Many species are strictly anaerobic and obligately phototrophic, while others are capable of chemolithoautotrophic or chemoorganohet-

erotrophic growth under microoxic to oxic conditions in the dark.

The pathway of autotrophic CO₂ assimilation was shown to be the reductive pentose phosphate cycle in all species tested so far. The fixation of dinitrogen has been demonstrated in several species. Storage materials are polysaccharides, poly-β-hydroxybutyrate, elemental sulfur, and polyphosphate. Vitamin B₁₂ is required by several species.

In nature, members of the *Chromatiaceae* occur in the illuminated, anoxic, and sulfide-containing parts of a large variety of aquatic environments and sediments from moist and muddy soils, ditches, ponds, lakes, rivers, sulfur springs, estuaries, marine habitats, and even salt and soda lakes.

The mol% G + C of the DNA is: 45.5–70.4.

Type genus: **Chromatium** Perty 1852, 174 emend. Imhoff, Sülting and Petri 1998b, 1138.

TAXONOMIC COMMENTS

In the eighth edition of the *Determinative Manual*, the family *Chromatiaceae* was described to include the genus *Ectothiorhodospira*, purple sulfur bacteria that form sulfur globules outside the cells (Pfennig and Trüper, 1974). With the establishment of the new family *Ectothiorhodospiraceae* and an emended description of the family *Chromatiaceae* Bavendamm 1924 (Imhoff, 1984b), the family *Chromatiaceae* exclusively comprises phototrophic sulfur bacteria capable of depositing globules of S⁰ inside the cells. This description agrees with Molisch's (1907) definition of the "*Thiorhodaceae*" and is in line with their phylogenetic separation from



FIGURE BXII.γ.1. Ultrathin section of a cell of *Thiocystis gelatinosa* DSM 215 showing internal photosynthetic membranes as numerous small vesicles, which are evenly distributed throughout the cell. Bar = 1 μ m (Courtesy of H.G. Trüper and J.B. Waterbury.)

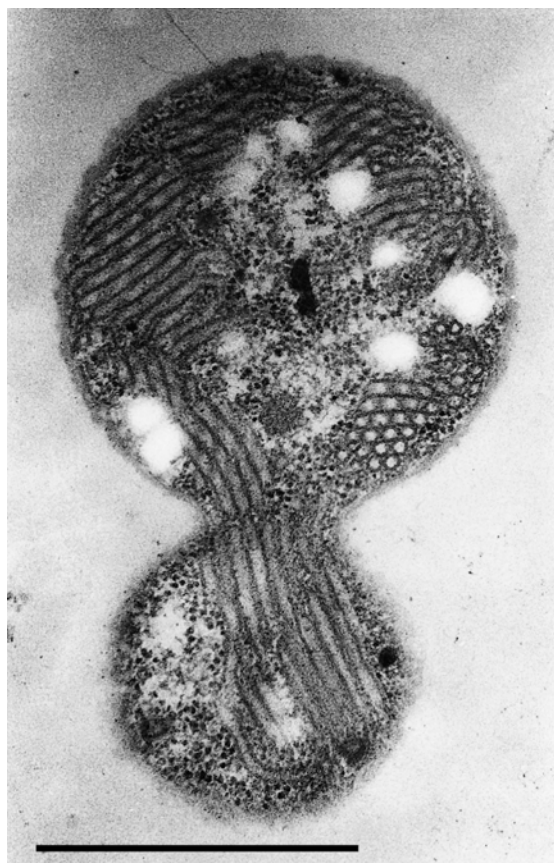


FIGURE BXII.γ.2. Ultrathin section of a cell of *Thiococcus pfennigii* strain 9111 showing tubular internal photosynthetic membranes in longitudinal and cross section. Bar = 0.5 μ m (Courtesy of H.G. Trüper and J.B. Waterbury.)

other phototrophic purple sulfur bacteria (Fowler et al., 1984; Imhoff and Söling, 1996; Imhoff et al., 1998b).

In the past, the taxonomy of the phototrophic purple bacteria, including the family *Chromatiaceae*, was entirely based on simple phenotypic characteristics (see the previous editions of this *Manual*, Pfennig and Trüper, 1974, 1989). Until recently, little molecular and sequence-based information, such as rRNA oligonucleotide data for selected species (Fowler et al., 1984) and a few 16S rDNA sequences (DeWeerd et al., 1990; Wahlund et al., 1991), was available for *Chromatiaceae* species. During the past few years, complete 16S rDNA sequences for most of the *Chromatiaceae* species have been determined (see Caumette et al., 1997; Guyoneaud et al., 1998b; Imhoff et al., 1998b). At the time of writing, 16S rDNA sequences of quite a few type strains of purple sulfur bacteria have not yet been determined or are not available: *Chromatium weissei*, *Thiospirillum jenense*, *Thiodictyon elegans*, *Thiodictyon bacillosum*, *Lamprobacter modestohalophilus*, and *Thiopedia rosea*.

The results of 16S rDNA sequence analysis, shown in Figs. BXII.γ.3 and BXII.γ.4, revealed that separate major phylogenetic branches of the family *Chromatiaceae* contain i) truly marine and halophilic species; ii) species that are motile by polar flagella, do not contain gas vesicles, and are primarily freshwater species; and iii) species with ovoid to spherical cells that are primarily freshwater species (some also occur in brackish and marine habitats), the majority of which are nonmotile (except *Lamprocystis roseopersicina*), and most of which contain gas vesicles. Because the phylogenetic relationships were not in congruence with tra-

ditional classification, a reclassification based on 16S rDNA sequence similarity, supported by selected phenotypic properties, was proposed (Guyoneaud et al., 1998b; Imhoff et al., 1998b). Positive correlation between 16S rDNA sequence similarity values and particular phenotypic properties of the species were taken as an indication of the importance of these properties in achieving a phylogenetically oriented taxonomy that includes both genetic and phenotypic information (Imhoff et al., 1998b).

Ecological source and adaptation to specific environmental conditions are of importance for phylogeny and taxonomy. Evolution occurs in particular natural environments, and separate phylogenetic lineages are expected to result from bacterial evolution in distinct habitats that, due to their particular properties (e.g., extremes of temperature, salinity, and pH), allow only the development of specifically adapted bacteria. In principle, major phylogenetic branches of the family *Chromatiaceae* can be distinguished on the basis of their salt requirement, which implies a separate evolutionary development in the marine and in the freshwater environment. This suggests that the salt response is an important taxonomic criterion in a phylogenetically oriented taxonomic system of these bacteria. The salt requirement has been considered in the taxonomy of purple nonsulfur bacteria and *Ectothiorhodospiraceae*. It proved to be of relevance for distinguishing the genera *Halorhodospira* and *Ectothiorhodospira* (Imhoff and Söling, 1996), *Rhodovulum*, *Rhodobacter* (Hiraishi and Ueda, 1994), and genera of the spiral, purple nonsulfur *Alphaproteobacteria*, the former *Rhodospirillum* species (Imhoff et al., 1998b).

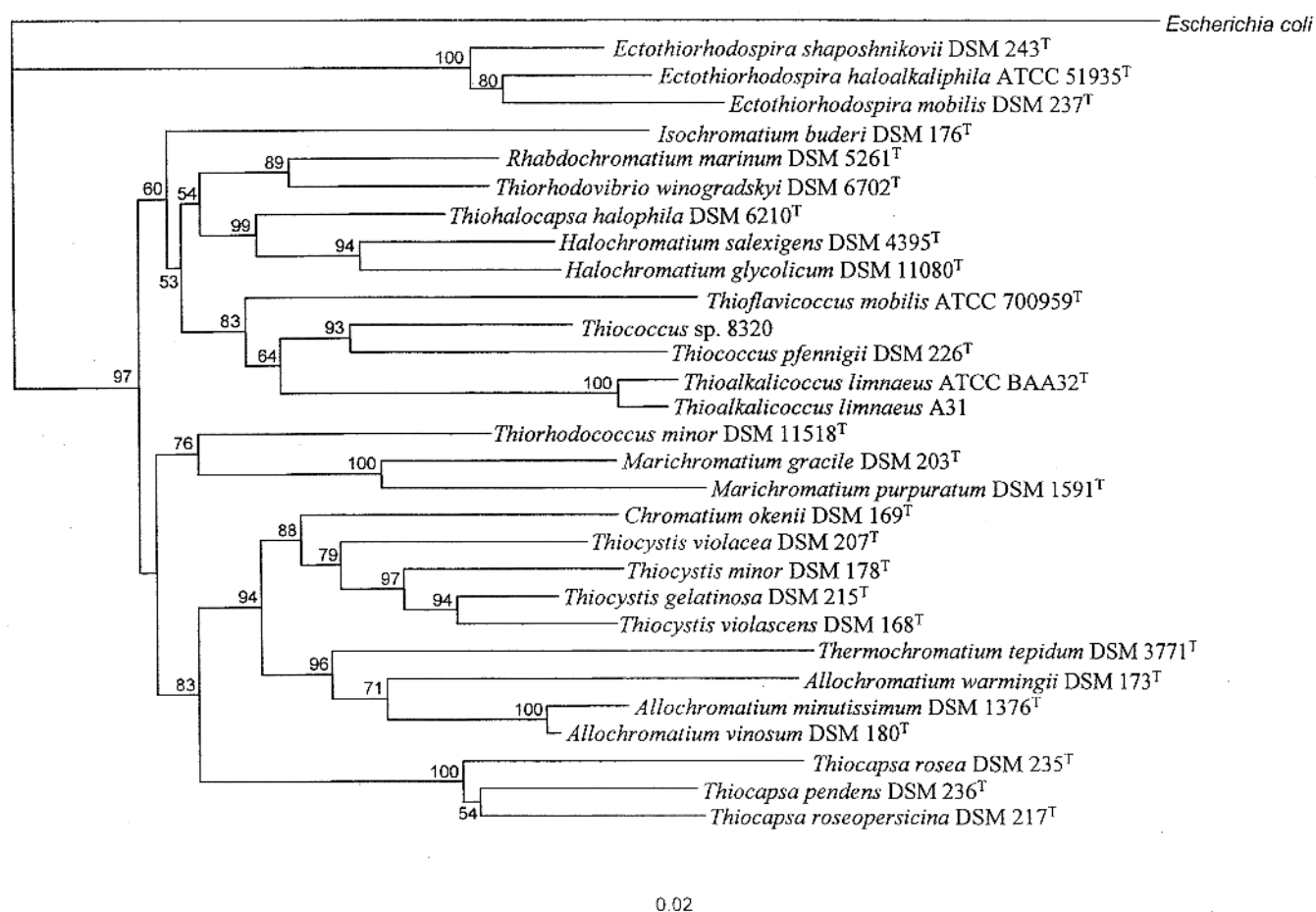


FIGURE BXII.γ.3. 16S rDNA-derived phylogenetic tree of the family Chromatiaceae including species of *Chromatium*, *Thiocystis*, *Allochromatium*, *Thermochromatium*, and the halophilic species *Isochromatium*, *Rhabdochromatium*, *Thiorhodovibrio*, *Thiohalocapsa*, *Thiococcus*, *Halochromatium*, *Thiorhodococcus* and *Marichromatium*. Representatives of *Thiocapsa* and *Thiolamprovum* and *Ectothiorhodospira* were included to complete the view on purple sulfur bacteria. The phylogeny of these genera is shown in more detail in Fig. BXII.γ.4 and in Fig. BXII.γ.19 in the chapter on *Ectothiorhodospiraceae*. Sequence data used were published previously (Imhoff et al., 1998b). Sequence alignments, calculation of distance data and tree construction were performed as described earlier (Petri and Imhoff 2000).

Within the family Chromatiaceae, both the phylogenetic relationship and the salt responses enable us to distinguish the halophilic *Halochromatium* species from the marine *Marichromatium* and the freshwater *Allochromatium* species, all of which were previously known as species of the genus *Chromatium* (Imhoff et al., 1998b). A differentiation of groups of species is also possible based on compatible solute production. Different compatible solutes are accumulated as osmotica by Chromatiaceae species compared to *Ectothiorhodospiraceae* species (Severin et al., 1992; Imhoff, 1993; Welsh and Herbert, 1993). Ectoine, trehalose, and glycine betaine are common among *Ectothiorhodospira* and *Halorhodospira* species, but not found in Chromatiaceae. Instead of trehalose, the less compatible sugar sucrose was found as the major component in freshwater species including *Allochromatium vinosum*, *Thiocystis violacea*, *Thiocystis minor*, *Thiocapsa roseopersicina*, and *Thiocapsa rosea* (Welsh and Herbert, 1993). Among the halophilic and marine species (*Halochromatium salexigens*, *Thiohalocapsa halophila*, and *Marichromatium purpuratum*), glycine betaine appears to occur as the major component, along with sucrose and N-acetyl-glutaminylglutamine amide (Severin et al., 1992; Imhoff, 1993).

Although morphological properties of the cells appear to be of lower phylogenetic significance, it is apparent that one group of freshwater Chromatiaceae species contains motile single cells

without gas vesicles (e.g., *Chromatium* and *Allochromatium* species, see Fig. BXII.γ.3), and the possession of gas vesicles is restricted to a second group with predominantly spherical to ovoid cells that are nonmotile (e.g., *Thiocapsa* and *Thiolamprovum* species, with the exception of *Lamprocystis roseopersicina*, see Fig. BXII.γ.4).

The DNA base ratio within the family Chromatiaceae, expressed as the mol% G + C content, has been found to span a large scale. Thus, in the previous edition of the *Manual* the genus *Chromatium* (Pfennig, 1989b) had values ranging from 48.0 to 70.4. Because the mol% G + C content is crudely indicative of the genetic relatedness of bacteria, these values already suggested excessive diversity within the genus *Chromatium*. A phylogenetic system would be expected to group bacteria that have much less variation in their mol% G + C content together. Indeed, after rearrangement of the species of the family Chromatiaceae according to their genetic relatedness on the basis of 16S rDNA sequence analysis (Guyoneaud et al., 1998b; Imhoff et al., 1998b), a quite narrow range of mol% G + C content is found within most of the newly established genera. The ranges are ~63–66 in the cluster including species of *Thiocapsa*, *Lamprocystis*, and *Thiolamprovum*, 69–70 in *Marichromatium* species, 64–66 in *Halochromatium* species, and 48–50 in the true *Chromatium* species. This correlation appears of such significance, that any case of high

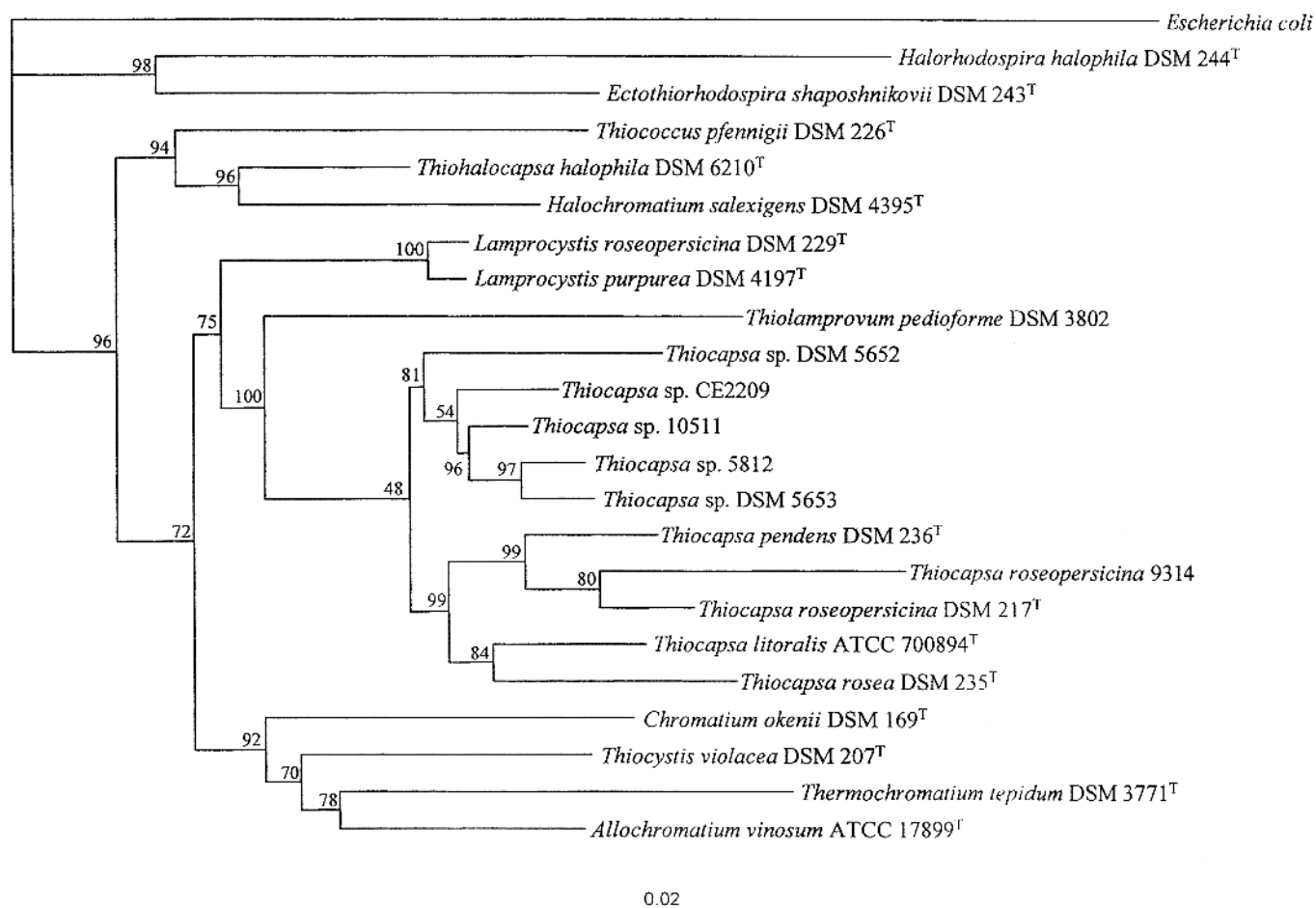


FIGURE BXII.4. 16S rDNA-derived phylogenetic tree of the family Chromatiaceae including species of *Thiocapsa*, *Lamprocystis* and *Thiolamprovum*. Representatives of *Ectothiorhodospiraceae* and other *Chromatiaceae* genera were included to complete the view on purple sulfur bacteria. The phylogeny of these genera is shown in more detail in Fig. BXII.γ.3 and in Figure BXII.γ.19 in the chapter on *Ectothiorhodospiraceae*. Sequence data used were published previously (Guyoneaud et al., 1998b; Imhoff et al., 1998b). Sequence alignments, calculation of distance data and tree construction were performed as described earlier (Petri and Imhoff 2000).

intrageneric and intraspecies variation of the mol% G + C content necessitates experimental examination. For example, the high intraspecies variation that has been noted for *Allochromatium vinosum* (61.3–66.3), *Allochromatium warmingii* (55.1–60.2), and *Thiocystis violacea* (62.8–67.9) suggests misclassification of some strains. Further studies are needed to clarify these problems.

Although fatty acid and quinone composition is of great significance for classification of purple nonsulfur bacteria of the *Alphaproteobacteria* and *Betaproteobacteria*, and these compounds are well-accepted tools for taxonomic differentiation of phototrophic bacteria (Imhoff, 1991, 1984a; Imhoff and Bias-Imhoff, 1995; Thiemann and Imhoff, 1996), both show only minor variation and therefore are of minor significance for differentiation within the family *Chromatiaceae* (Imhoff and Bias-Imhoff, 1995). Polar lipids have been analyzed so far in only a limited number of *Chromatiaceae* species. First results, obtained with several freshwater species, showed significant correlations with the new classification, i.e., identical polar lipid composition was found i) in *Allochromatium vinosum* and *Allochromatium warmingii*, ii) in all four *Thiocystis* species, and iii) in the two *Chromatium* species (Imhoff, unpublished results). Thus, polar lipid composition may well turn out to be a relevant property for distinguishing between genera or groups of genera of the family *Chromatiaceae*. This implies that the polar lipid composition reflects phylogenetic relationships and is a rather stable property in evolutionary

terms. More analytical data on members of the family *Chromatiaceae* are required to support this concept.

Further Comments Physiological properties have always been important characteristics in taxonomy, although their taxonomic significance is ambiguous. On the one hand, members of the family *Chromatiaceae* are quite conservative in their use of sulfur sources and their restricted use of organic carbon sources: all species are able to grow photolithoautotrophically under anoxic conditions in the light using sulfide or S^0 as an electron donor. On the other hand, we can distinguish two major physiological groups, versatile and specialized species. (i) The specialized species depend on strictly anoxic conditions and are obligately phototrophic. Sulfide is required; thiosulfate and hydrogen are not used as electron donors. Only acetate and pyruvate (or propionate) are photoassimilated in the presence of sulfide and CO_2 . These bacteria do not grow with organic electron donors, chemotrophic growth is not possible, and sulfate is not assimilated as a sulfur source. Among these species are *Chromatium okenii*, *Chromatium weissei*, *Allochromatium warmingii*, *Isochromatium buderi*, *Thiospirillum jenense*, and *Thiococcus pfennigii*. (ii) The versatile species photoassimilate a larger variety of organic substrates. Most of them are able to grow in the absence of reduced sulfur sources with organic substrates as electron donors for photosynthesis, and to assimilate sulfate as the sole sulfur

source. Some species even grow chemoautotrophically or chemoheterotrophically (Gorlenko, 1974; Kondrat'eva et al., 1976; Kämpf and Pfennig, 1980, 1986). Among these species are *Allochrochromatium vinosum*, *Thiocystis violacea*, *Thiocapsa roseopersicina*, *Thiocapsa rosea*, *Thiocapsa pendens*, and *Lamprobacter modestohalophilus*.

Depending on the environmental conditions in nature or on the culture conditions, all species are able to develop either in the form of single cells or in immobile cell aggregates of variable size and shape embedded in slime. The different modes of growth described by Winogradsky (1888) may be obtained experimentally by varying sulfide concentration, light intensity, pH, salinity, temperature, and oxygen tension. At high sulfide concentration (2–4 mM) and high light intensity (2000–4000 lux), all flagellated species become immobile as the result of slime formation that embeds groups of cells. For a given light intensity, the various species differ from each other with respect to the sulfide concentration at which the cells develop functional flagella and become motile. For a given sulfide concentration, the species differ with respect to the light intensity and mode of

illumination (diurnal light and dark periods) which allow the cells to become motile. The lower the light intensity (100–200 lux), the higher the sulfide concentration at which a given strain can persist in the motile stage. These characteristics must be considered for reliable identification of pure cultures.

Differentiation of the genera of the Chromatiaceae Three major genetically distinct groups of the family *Chromatiaceae* are known: i) a group of truly halophilic and salt-requiring species of the genera, including *Thiorhodovibrio*, *Marichromatium*, *Isochromatium*, *Rhabdochromatium*, *Thiohalocapsa*, *Halochromatium*, *Thiorhodococcus*, *Thiococcus*, *Thioflavicoccus*, and *Thioalkalicoccus*; ii) a group that is dominated by spherical or rod-shaped nonmotile species, many of which produce gas vesicles and that are primarily freshwater bacteria (including species of the genera *Thiocapsa*, *Lamprocystis*, and *Thiolamprovum*); and iii) a group of motile species without gas vesicles that are primarily freshwater bacteria and include the genera *Chromatium*, *Allochrochromatium*, *Thermochromatium*, and *Thiocystis*. In both of the freshwater clusters, strains or species that may behave indifferently to low salt concentra-

TABLE BXII.γ.4. Differential characteristics of the motile *Chromatiaceae* without gas vesicles typically found in freshwater habitats^a

Characteristic	<i>Chromatium okenii</i>	<i>Chromatium weissii</i>	<i>Allochrochromatium vinosum</i>	<i>Allochrochromatium minutissimum</i>	<i>Allochrochromatium warmingii</i>	<i>Thermochromatium tepidum</i>	<i>Thiocystis violacea</i>	<i>Thiocystis gelatinosa</i>	<i>Thiocystis minor</i>	<i>Thiocystis violascens</i>	<i>Thiospirillum jenense</i>
Cell shape	rod	rod	rod	rod	rod	rod	sphere	sphere	rod	rod	spiral
Cell size (μm)	4.5–6.0 × 8.0–16.0	3.5–4.5 × 7.0–14.0	2.0 × 2.5–6.0	1.0–1.2 × 2	3.5–4.0 × 5.0–11.0	1.0–2.0 × 2.8–3.2	2.5–3.0	3.0	2.0 × 2.5–6.0	2.0 × 2.5–6.0	2.5–4.5 × 30–40
Motile by flagella	+	+	+	+	+	+	+	+	+	+	+
Color of cell suspensions ^b	Pur	Pur	Br	Br	Puv	R	Puv	Pur	Pur	Puv	Ob
Carotenoid group	ok	ok	sp	sp	ra	sp	ra	ok	ok	ra	ra
Mol% G + C of DNA	48.0	48.0	61.3–66.3	63.7	55.1–60.2	61.5	62.8–67.9	61.3	62.2	61.8–64.3	45.5
B ₁₂ requirement	+	+	–	–	+	–	–	–	–	–	+
Sulfate assimilation	–	–	+	+	–	nd	±	–	–	+	nd
Chemotrophic growth	–	–	+	+	–	–	+	+	+	+	–
pH optimum	7.0	7.0	7.0–7.3	7.0–7.3	7.0	7.0	7.0–7.3	7.0–7.3	7.0–7.3	7.0–7.3	7.0
Temperature optimum, °C	20–35	20–35	25–35	25–35	25–30	48–50	25–35	30	30	25–35	20–25
Salt requirement	none	none	none ^c	none	none	none	0–2%	0–1%	none	none	none
<i>Substrates used:</i>											
Hydrogen	–	–	+	+	–	–	+	nd	nd	+	nd
Sulfide	+	+	+	+	+	+	+	+	+	+	nd
Thiosulfate	–	–	+	+	–	–	+	–	+	+	nd
Sulfur	+	+	+	+	+	+	+	+	+	+	nd
Sulfite	nd	nd	+	+	nd	–	±	–	nd	+	nd
Formate	–	–	+	+	–	–	nd	nd	–	+	nd
Acetate	+	+	+	+	+	+	+	+	+	+	+
Pyruvate	+	+	+	+	+	+	+	+	+	+	nd
Propionate	–	–	+	+	nd	–	±	–	–	+	nd
Butyrate	–	–	±	±	–	–	nd	nd	–	±	nd
Lactate	–	–	nd	nd	–	nd	–	–	nd	nd	nd
Fumarate	–	–	+	+	–	–	+	–	–	–	nd
Succinate	–	–	+	+	–	–	±	–	–	–	nd
Malate	–	–	+	+	–	–	nd	nd	–	–	nd
Fructose	–	–	–	–	–	–	±	–	–	–	nd
Glucose	–	–	–	–	–	–	±	–	+	–	nd
Ethanol	–	–	–	–	–	nd	–	–	–	–	nd
Propanol	–	–	–	–	–	nd	±	–	–	–	nd
Glycerol	–	–	–	–	–	nd	±	–	–	–	nd

^aSymbols: +, positive in most strains; –, negative in most strains; ±, variable; nd, not determined; ok, okenone; sp, spirilloxanthin; ra, rhodopinal.

^bBr, brownish red; Ob, orange-brown; Pur, purple red; Puv, purple violet; R, red.

^cMarine strains may tolerate low concentrations of NaCl.

tions, or require only small amounts of salts occur. Major differentiating characteristics of species and genera of the family *Chromatiaceae* are presented in Tables BXII.γ.4, BXII.γ.5, and BXII.γ.6. Their genetic relationship is shown in Figs. BXII.γ.3 and BXII.γ.4.

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TABLE BXII.γ.5. Differential characteristics of the truly marine and halophilic *Chromatiaceae*^a

Characteristic	<i>Halochromatium salexigens</i>	<i>Halochromatium glycolicum</i>	<i>Isochromatium buderi</i>	<i>Marichromatium gracile</i>	<i>Marichromatium purpuratum</i>	<i>Rhalochromatium marinum</i>	<i>Thioalkalicoccus limnaeus</i>	<i>Thiococcus pfennigi</i>	<i>Thioflavococcus mobilis</i>	<i>Thiohalocapsa halophila</i>	<i>Thiorhodococcus minor</i>	<i>Thiorhodovibrio winogradskyi</i>
Cell shape	rod	rod	rod	rod	rod	rod	sphere	sphere	sphere	sphere	sphere	vibrio
Cell size (μm)	2.0–2.5	0.8–1.0	3.5–4.5	1.0–1.3	1.2–1.7	1.5–1.7	1.3–1.8	1.2–1.5	0.8–1.0	1.5–2.5	1.0–2.0	1.2–1.6
	×	×	×	×	×	×						×
	4.0–7.5	2.0–4.0	4.5–9.0	2.0–6.0	3.0–4.0	16–32						2.6–4.0
Motility	+	+	+	+	+	+	–	–	+	–	+	+
Color of cell suspensions	P, Rr	P, Pr	P, Puv	Br	Pur	B, Ob	Yb, Ob	Ob	Yb, Ob	Pur	Bo	P, Pr
Carotenoid group	sp	sp	ra	sp	ok	lycopene	ths	ths	ths	ok	sp	sp
Mol% G + C of DNA	64.6	66.1–66.5	62.2–62.8	68.9–70.4	68.4–68.9	60.4	63.6–64.8	69.4–69.9	66.5	65.9–66.6	66.9	61.0
B ₁₂ requirement	+	–	+	–	–	–	–	–	–	+	–	–
Sulfate assimilation	–	–	–	+	nd	nd	nd	–	ndo	–	–	nd
Chemolithotrophic growth	+	+	–	+	–	–	–	–	–	+	+	+
pH optimum	7.4–7.6	7.2–7.4	7.2–7.4	7.2–7.4	7.2–7.6	7.2–7.3	8.8–9.5	7.2–7.4	7.2–7.4	7.0	7.0–7.2	7.0–7.4
Temperature optimum (°C)	20–30	25–35	25–35	20–35	25–30	30	20–25	20–35	25–30	20–30	30–35	33
Salinity range (%)	4–20	2–20	1–5	0.5–8	2–7	1–6.5	0.5–7	0–3	1–3	3–20	0.5–9	up to 7.2
NaCl optimum (%)	8–11	4–6	2–3	2–3	5	1.5–5	1–6	0.5–2	2	4–8	2	2.2–3.2
<i>Substrates used:</i>												
Hydrogen	+	+	–	+	nd	+	nd	nd	nd	+	+	nd
Sulfide	+	+	+	+	+	+	+	+	+	+	+	+
Thiosulfate	+	+	–	+	+	+	–	–	–	+	+	–
Sulfur	+	+	+	+	+	+	+	+	+	+	+	+
Sulfite	+	+	–	+	nd	nd	nd	nd	nd	+	–	nd
Formate	–	(+)	–	+	–	–	–	–	–	–	–	–
Acetate	+	(+)	+	+	+	+	+	+	+	+	+	+
Pyruvate	+	(+)	+	+	+	+	+	+	+	+	+	+
Propionate	–	–	–	+	+	+	+	+	–	–	+	+
Butyrate	–	–	–	+	+	–	–	–	–	–	–	–
Lactate	–	–	–	+	+	–	–	+	–	+	+	–
Fumarate	–	+	–	+	+	–	(+)	+	–	–	+	–
Succinate	–	+	–	+	+	–	+	+	–	–	+	–
Malate	–	–	–	+	+	–	+	+	–	–	(+)	–
Fructose	–	nd	–	–	(+)	–	–	+	–	+	+	–
Glucose	–	–	–	–	–	–	–	–	–	(+)	–	–
Ethanol	–	–	–	–	–	–	–	–	–	–	+	–
Propanol	–	nd	–	nd	–	–	nd	–	–	–	+	–
Glycerol	–	+	–	–	–	–	–	–	–	(+)	–	–
Glycolate	–	+	nd	nd	nd	–	–	–	–	–	+	–
Crotonate	–	–	–	+	nd	–	nd	–	–	–	–	–
Valerate	–	nd	–	nd	+	–	–	–	–	–	–	–
Casamino acids	–	(+)	–	+	(+)	–	–	(+)	–	–	–	–

^aSymbols: +, positive in most strains; –, negative in most strains; ±, variable; (+), weak growth; nd, not determined; P, pink; Pr, pinkish red; Rr, rose red; Puv, purple-violet; Br, brownish red; Pur, purple-red; B, beige; Ob, orange-brown; Yb, yellowish brown; Bo, brown-orange; ok, okenone; sp, spirilloxanthin; ra, rhodopinal; ths, tetrahydrospirilloxanthin.

TABLE BXII.γ.6. Differential characteristics of the *Chromatiaceae* group characterised by nonmotile cells with gas vesicles typically found in freshwater habitats, including the motile *Lamprocystis roseopersicina* and *Lamprobacter modestohalophilus*^a

Characteristic	<i>Lamprobacter modestohalophilus</i>	<i>Lamprocystis roseopersicina</i>	<i>Lamprocystis purpurea</i>	<i>Thiocapsa roseopersicina</i>	<i>Thiocapsa rosea</i>	<i>Thiocapsa pendens</i>	<i>Thiodictyon elegans</i>	<i>Thiodictyon bacillosum</i>	<i>Thiolamprovum pediforme</i>	<i>Thiopetia rosea</i>
Cell shape	ovoid to rod	sphere	sphere	sphere	sphere	sphere	rod	rod	sphere	sphere
Motility	+	+	—	—	—	—	—	—	—	—
Cell diameter (μm)	2.0–2.5	2.0–3.5	1.9–2.3	1.2–3.0	2.0–3.0	1.5–2.0	1.5–2.0	1.5–2.0	2.0	2.0–2.5
Gas vesicles	+	+	+	—	+	+	+	+	+	+
<i>Aggregate formation:</i>										
Clumps of cells			+							
Irregular aggregates		+		+	+	+		+		
Irregular clumps										
Irregular net-ishlike aggregates							+			
Platelets									+	+
Tetrads				+						
Color of cell suspensions ^b	Pur ok	Pv ra	Pur ok	P, Rr sp	P, Rr sp	P, Rr sp	Puv ra	Puv ra	P, Rr sp	Pur ok
Carotenoid group	60–64	63.8	63.5–63.6	63.3–66.3	64.3	65.3	65.3–66.3	66.3	65.5	62.5–63.5
Mol% G + C of DNA	+	nd	nd	—	+	+	nd	nd	—	—
B ₁₂ requirement	—	nd	—	+	—	—	nd	nd	—	—
Sulfate assimilation	+	—	+	+	+	—	—	—	+	—
Chemolithotrophic growth	7.4–7.6	7.0–7.3	7.0–7.3	7.3	6.7–7.5	6.7–7.5	6.7–7.3	6.7–7.3	7.4–7.6	7.3–7.5
pH optimum	23–27	20–30	23–25	20–35	20–35	20–35	20–25	20–30	37	20
Temperature optimum (°C)	1–4	none	none	none ^c	none	none	none	none	none	none
<i>Substrates used:</i>										
Hydrogen	+	nd	—	+	—	—	nd	nd	—	—
Sulfide	+	+	+	+	+	+	+	+	+	+
Thiosulfate	+	nd	+	+	+	+	nd	nd	+	—
Sulfur	+	+	+	+	+	+	+	+	+	+
Formate	—	nd	+	—	—	—	nd	nd	—	nd
Acetate	+	+	+	+	+	+	+	+	+	+
Pyruvate	+	+	+	+	+	+	+	+	+	+
Propionate	—	nd	+	±	+	+	nd	nd	—	—
Butyrate	nd	nd	—	—	—	—	nd	nd	—	+
Lactate	+	nd	+	±	+	+	nd	nd	+	—
Fumarate	—	nd	—	+	—	—	nd	nd	—	±
Succinate	—	nd	—	+	—	—	nd	nd	—	±
Malate	—	nd	—	+	±	+	nd	nd	—	±
Fructose	—	nd	—	+	+	—	nd	nd	—	±
Glucose	nd	nd	+	—	—	+	nd	nd	—	+
Ethanol	+	nd	—	—	—	nd	nd	nd	—	—
Propanol	+	nd	—	—	—	nd	nd	nd	—	—
Glycerol	+	nd	—	+	—	—	nd	nd	—	—
Glycolate	nd	nd	—	—	nd	nd	nd	nd	—	—
Crotonate	nd	nd	—	nd	nd	nd	nd	nd	—	—
Valerate	nd	nd	—	nd	—	—	nd	nd	—	+
Casamino acids	nd	nd	—	nd	nd	nd	nd	nd	—	—

^aSymbols: +, positive in most strains; —, negative in most strains; ±, variable; nd, not determined; ok, okenone; sp, spirilloxanthin; ra, rhodopinal.^bP, pinkish; Pur, purple red, Puv, purple violet, Pv, pinkish violet; Rr, rose-red.^cMarine strains may tolerate low concentrations of NaCl.

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Genus I. Chromatium Perty 1852, 174^{AL} emend. Imhoff, Sling and Petri 1998b, 1138

JOHANNES F. IMHOFF

Chro.ma'ti.um. Gr. n. *chroma* color; M.L. neut. n. *Chromatium* one which is colored.

Cells are **straight or slightly curved rods**, more than 3 µm wide, single or in pairs, multiply by binary fission. **Motile** by a polar tuft of multitrichous flagella that can be seen in the light microscope, Gram negative, belong to the *Gamma*proteobacteria, and contain **internal photosynthetic membranes of vesicular type** in which the photosynthetic pigments bacteriochlorophyll *a* and carotenoids are located.

Strictly anaerobic, obligately phototrophic, and require sulfide-reduced media. **Photolithoautotrophic growth occurs under anoxic conditions in the light with sulfide and S⁰ as electron donors**. Thiosulfate and hydrogen are not used. During oxidation of sulfide, S⁰ is transiently stored inside the cells in the form of highly refractile globules. Sulfate is the final oxidation product. In the presence of sulfide or sulfur and bicarbonate, a few simple organic substrates are photoassimilated. Assimilatory sulfate reduction is lacking. Ammonia is used as nitrogen source, N₂ may be used, nitrate is not reduced. Storage materials: polysaccharides, poly-β-hydroxybutyrate, S⁰, and polyphosphates. Vitamin B₁₂ required.

Mesophilic freshwater bacteria with optimum growth temperatures from 20 to 35°C and without a distinct requirement for salt.

Habitat: water and sediment surface layers of stagnant freshwater habitats such as ditches, ponds, and lakes, containing hydrogen sulfide and exposed to light.

The mol% G + C of the DNA is: 48.0–50.0.

Type species: Chromatium okenii (Ehrenberg 1838) Perty 1852, 174 (*Monas okenii* Ehrenberg 1838, 15.)

FURTHER DESCRIPTIVE INFORMATION

Under unfavorable culture conditions, cells of *Chromatium* species may not show active motility when viewed under the microscope. Diurnal light and dark phases, low concentrations of sulfide, and low light intensities may be required to sustain active flagellar motility. Optimal growth conditions are clearly recognized by the bioconvection patterns that arise because of the swarming activity of the cells and that can be seen with the naked eye (Pfennig, 1962).

ENRICHMENT AND ISOLATION PROCEDURES

Chromatium okenii and *C. weissei* have so far been found only in sulfide-containing freshwater habitats, which is of primary importance for the design of enrichment experiments. Because of the selectivity of any enrichment condition, even species and strains that were not detected in the original sample from the natural habitat may become dominant in liquid enrichment cultures. Therefore, enrichment cultures are only indicated when a certain species should be enriched and isolated from a natural sample, and selective conditions for this bacterium are known. When information on the diversity of culturable species in a natural sample is desired, enrichment cultures should be avoided and dilution series in agar deeps should be prepared directly from the natural sample. Agar cultures should be incubated under conditions similar to those used for liquid cultures of the desired bacteria. The defined medium¹ proved to be relatively nonspecific and may be used for the cultivation of not only *Chromatium* species, but also most of the freshwater and, with the addition of appropriate concentrations of NaCl, marine *Chro-*

matia species. A somewhat simpler, but less widely applicable, culture medium was described by Pfennig and Trper (1981). In agar shake dilution cultures, growth may be enhanced by the addition of 3 mM acetate to the defined medium.

Chromatium species have a selective advantage when cultures with relatively low sulfide concentrations (1–2 mM) are incubated at about 20°C, at low light intensities of 50–300 lux with diurnal light and dark phases (e.g., 16 h light, 8 h dark). They keep swarming throughout the whole bottle and can be further enriched by using cell suspensions from the upper part of the enrichment culture as inoculum for subsequent enrichments (Pfennig, 1962).

The sulfide concentration initially provided in fresh culture medium does not support much growth. To achieve reasonably high population densities in enrichments or pure cultures, repeated additions of neutralized sodium sulfide solution are necessary. Additions are made when the previously added sulfide is consumed and the transiently stored S⁰ is nearly oxidized. The sulfide solution for feeding of cultures can be prepared by neutralizing a stirred sodium sulfide solution (60 mM) with a certain amount of sterile 2 M sulfuric acid to a pH of ~7.5. This neutralized solution has to be applied immediately. Preferably, a special device for preparation of sterile sulfide solution, neutralized and stored under CO₂-pressure, would be used (Siefert and Pfennig, 1984).

Pure cultures are obtained by repeated application of the agar shake dilution method. Water agar is prepared with 1.8% (w/v) of agar. Before the solution is prepared, the agar should be washed several times in distilled water. Depending on the salinity of the sample from nature or the enrichment culture medium, appropriate amounts of NaCl and MgCl₂·6H₂O are added. The agar solution is dispensed in 3-ml amounts into test tubes, which are stoppered with cotton plugs and autoclaved. The agar tubes are kept molten in a water bath at 50°C. Ready-prepared, defined

1. Defined medium for *Chromatium* and other freshwater and marine *Chromatiaceae* species after Pfennig (Pfennig, 1965, 1989b; Pfennig and Trper, 1981, 1992): The medium is prepared in an Erlenmeyer flask with an outlet near the bottom at one side. Connected to the outlet is a silicon rubber tube with a pinchcock, and a bell for aseptic distribution of the medium into bottles or tubes. The flask is closed by a silicon rubber stopper with an inlet and an outlet for gas, and a screw-capped glass tube through which additions can be made or samplings taken.

The defined basal medium has the following composition (per liter of distilled water): KH₂PO₄, 0.34 g; NH₄Cl, 0.34 g; KCl, 0.34 g; CaCl₂·2H₂O, 0.25 g; NaCl (only for seawater medium), 20.0 g; MgSO₄·7H₂O (3.0 g for seawater medium), 0.5 g; and trace element solution, 1 ml. After the medium has been autoclaved and cooled under an atmosphere of N₂, the following components are added aseptically per liter of medium from sterile stock solutions, while access of air is prevented by continuous flushing with N₂: 15 ml of a 10% (w/v) solution of NaHCO₃ (saturated with CO₂ and autoclaved under a CO₂ atmosphere), 4 ml of a 10% (w/v) solution of Na₂S·9H₂O (autoclaved under an N₂ atmosphere), and 1 ml of a vitamin B₁₂ solution containing 2 mg vitamin B₁₂ in 100 ml of distilled water. The pH of the medium is adjusted to pH 7.2 by stirring under an atmosphere of CO₂ (0.5 bar pressure) for approx. 40 min. The medium is then dispensed aseptically under pressure of N₂ into sterile 100-ml bottles with metal screw caps containing autoclavable rubber seals. A small pea-sized air bubble is left in each bottle to meet possible pressure changes. Trace element solution (SL12 of Pfennig and Trper, 1992) contains (per liter of distilled water): ethylenediaminetetraacetate-Na₂, 3.0 g; FeSO₄·7H₂O, 1.1 g; CoCl₂·6H₂O, 190 mg; MnCl₂·4H₂O, 40 mg; ZnCl₂, 42 mg; Na₂MoO₄·2H₂O, 18 mg; NiCl₂·6H₂O, 24 mg; H₃BO₃, 300 mg; and CuCl₂·2H₂O, 2 mg. The EDTA-Na₂ is dissolved first, followed by the other components. More information on preparation of media and on isolation procedures is described in Pfennig and Trper (1992).

culture medium is prewarmed to 50°C, and 6-ml amounts are added to the tubes of liquefied agar. Exposure to air is minimized by dipping the tip of the pipette into the agar medium. Starting with a few drops of sample from nature or an enrichment culture as inoculum and using 6–8 tubes, serial dilutions are made. All tubes are then hardened in cold water and immediately sealed with a sterile overlay consisting of 1 part paraffin wax and 3 parts paraffin oil. The overlay should be 2 cm thick. Alternatively, the tubes are finally flushed with N₂/CO₂ (90:10) and sealed with butyl rubber stoppers. The tubes are kept in the dark for 6–12 h and subsequently incubated at the desired light intensity and temperature. During the first 2 days of incubation, the paraffin overlay is gently reheated to achieve a complete sealing effect. Well-separated pink, purple-red, purple-violet, or orange-brown colonies that develop in the higher dilutions can be removed for microscopic examination and further purification with sterile Pasteur pipettes, without breaking the tube. The cells are suspended in 0.5–1.0 ml of anoxic medium and used as the inoculum for subsequent agar shake cultures. The process is repeated until a pure culture is achieved.

When pure agar cultures are obtained, individual colonies are isolated and inoculated into liquid medium. It is advisable to start with small-sized bottles or screw-capped tubes (10 or 25 ml), and then to scale up to the regularly used sizes. Purity is checked both microscopically and by use of AC medium (Difco), adjusted to the appropriate salinity.

MAINTENANCE PROCEDURES

Pure cultures of *Chromatium* species can be maintained in the defined mineral medium, preferably in 100-ml screw-capped bottles. The freshly grown cultures are supplemented with neutralized sulfide solution to a final concentration of 1.5 mM, and kept in the light for about 2 h until the cells have formed intracellular sulfur globules. At this stage, the stock cultures can be stored at 4°C for 2–3 months. The cultures keep well if they are put back under dim light at room temperature and supplemented again with neutralized sulfide solution. After formation of sulfur globules, the cultures are put back into the refrigerator.

Long-term preservation of *Chromatium* species is successfully performed by storage in liquid nitrogen. For this purpose, dense cell suspensions of liquid cultures are supplemented with DMSO as a protective agent to a final concentration of 5%, using an autoclaved 50% DMSO stock solution. Such suspensions are dispensed into 2-ml plastic ampules, which are then sealed and freeze-stored. *Chromatium* species do not survive lyophilization and cannot be preserved in this way.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *CHROMATIUM*

The most significant known difference between *C. okenii* and *C. weissei* is the cell width (Table BXII.γ.4 of the chapter describing the family *Chromatiaceae*).

DIFFERENTIATION OF THE GENUS *CHROMATIUM* FROM OTHER GENERA

According to 16S rDNA sequence analysis (Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*), the genus *Chromatium* belongs to the branch of the family *Chromatiaceae* that characteristically contains freshwater species with flagellar motility. Species of this genus have very low mol% G + C values (48–50), large-sized cells (more than 3 µm wide), require vitamin B₁₂, and lack the capability for chemotrophic growth. Characteristics for differentiation of *Chromatium* species from other phototrophic members of the *Chromatiaceae* are shown in Table BXII.γ.4 of the chapter describing the family *Chromatiaceae*. The genetic relationship between *Chromatium* species and other members of the *Chromatiaceae*, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

In the past, all rod-shaped, motile, purple sulfur bacteria without gas vesicles that store S⁰ inside their cells were classified as *Chromatium* species (Pfennig and Trüper, 1989). These species were not only phenotypically quite diverse, but also distantly related on a genetic basis, as indicated by the large variation in DNA base ratio within these bacteria (48.0–70.4 mol% G + C). 16S rDNA sequence analysis of recognized species of the family *Chromatiaceae* gave support for a new classification, in which the type species of the genus, *Chromatium okenii*, appeared distant and separate from most of the other species previously assigned to this genus. The only other species showing sufficient phenotypic similarity to *C. okenii* to allow assignment to this genus is *C. weissei* (Imhoff et al., 1998b), although 16S rDNA sequence data for this species are currently not available.

FURTHER READING

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List of species of the genus *Chromatium*

1. ***Chromatium okenii*** (Ehrenberg 1838) Perty 1852, 174^{AL} (*Monas okenii* Ehrenberg 1838, 15.)
o.ken'i.i. M.L. gen. n. *okenii* of Oken, named for L. Oken, a German naturalist.

Cells straight or slightly curved rods, 4.5–6.0 × 8–16 µm, occasionally longer; motile by a polar tuft of flagella which is 1.5–2 times the cell length and visible by brightfield (Fig. BXII.γ.5) or phase-contrast microscopy. In the pres-

ence of sulfide and light, globules of elemental sulfur appear evenly distributed within the cell. Color of individual cells and of cell suspensions is purple-red. Photosynthetic pigments are bacteriochlorophyll *a* and the carotenoid okenone.

Strictly anaerobic and obligately phototrophic. Sulfide-reduced media required. Sulfide and S⁰ used as photosynthetic electron donors. In the presence of sulfide and bi-

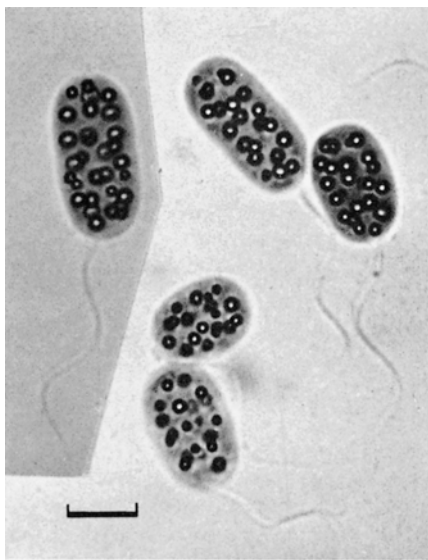


FIGURE BXII.γ.5. *Chromatium okenii* DSM 169 cultured photoautotrophically with sulfide. Note the evenly distributed intracellular globules of S^0 and the spiral tufts of polar flagella. Brightfield micrograph. Bar = 5 μ m.

carbonate, acetate and pyruvate are photoassimilated. Not utilized: thiosulfate, molecular hydrogen, sugars, sugar alcohols, alcohols, higher fatty acids, amino acids, benzoate, formate, and most intermediates of the tricarboxylic acid cycle. Assimilatory sulfate reduction is absent. Nitrogen sources: ammonium salts, and urea. Vitamin B₁₂ required.

Mesophilic freshwater bacterium with optimum growth at 20–35°C and pH 7.0 (range 6.5–7.3).

Habitat: water and sediment surface layers of stagnant freshwater habitats such as ditches, ponds, and lakes, containing hydrogen sulfide and exposed to light.

The mol% G + C of the DNA is: 48.0 (Bd).

Type strain: 1111, BN 6010, DSM 169.

GenBank accession number (16S rRNA): AJ223234, Y12376.

2. *Chromatium weissei* Perty 1852, 174^{AL}

weiss' se. i. M.L. gen. n. *weissei* of Weisse, named for J.F. Weisse, a German zoologist.

Cells are rod shaped, 3.5–4.5 × 7–14 μ m. Nitrogenase activity is present. Other characteristics are as for *C. okenii*.

The mol% G + C of the DNA is: 48.0 (Bd).

Type strain: DSM 171.

Genus II. *Allochro-matium* Imhoff, Sling and Petri 1998b, 1140^{VP}

JOHANNES F. IMHOFF

Al'lo.chro.ma'ti.um. Gr. adj. *allos* the other; *Chromatium* a genus name; M.L. neut. n. *Allochro-matium* the other *Chromatium*.

Cells straight to slightly curved rods, single or in pairs, multiply by binary fission, motile by polar flagella, Gram negative, belong to the *Gammaproteobacteria*, and contain internal photosynthetic membranes of vesicular type in which the photosynthetic pigments bacteriochlorophyll *a* and carotenoids are located.

Photolithoautotrophic growth occurs under anoxic conditions in the light with sulfide and S^0 as electron donors. During oxidation of sulfide, S^0 is transiently stored inside the cells in the form of highly refractile globules. Sulfate is the final oxidation product. Molecular hydrogen and thiosulfate may be used as electron donors. In the presence of sulfide and bicarbonate, organic substrates can be photoassimilated, potentially mixotrophic. Some species may grow chemoautotrophically or chemorganotrophically under microoxic to oxic conditions in the dark. Storage materials: polysaccharides, poly- β -hydroxybutyrate, S^0 , and polyphosphates. Vitamin B₁₂ is required.

Mesophilic bacteria with growth at 25–35°C and pH 6.5–7.6. No distinct requirement for salt, except for strains that originate from the marine and brackish water environment and may tolerate or require low concentrations of salt.

Habitat: ditches, ponds, and lakes with stagnant freshwater containing hydrogen sulfide and exposed to the light; also sewage lagoons, estuaries, and salt marshes.

The mol% G + C of the DNA is: 55.1–66.3.

Type species: *Allochro-matium vinosum* (Ehrenberg 1838) Imhoff, Sling and Petri 1998b, 1141 (*Chromatium vinosum* (Ehrenberg 1838) Winogradsky 1888, 99; *Monas vinosa* Ehrenberg 1838, 11.)

ENRICHMENT AND ISOLATION PROCEDURES

Media and culture conditions are the same as described for *Chromatium* species. *Allochro-matium vinosum* is one of the most com-

mon purple sulfur bacteria in nature. Since it is a nonfastidious purple sulfur bacterium, it can be readily isolated. *Allochro-matium vinosum* is enriched at high sulfide concentrations (3–4 mM), at high light intensities of more than 1000 lux and at incubation temperatures of 20–30°C. This species is among the dominant members of the *Chromatiaceae* in enrichment cultures under strictly autotrophic conditions, but also in the presence of organic substrates and complex nutrients, if present in the sample. *Allochro-matium* species can also be isolated directly by agar shake dilution cultures from samples collected in nature, as described for *Chromatium* species.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *ALLOCHROMATIUM* FROM OTHER GENERA

According to 16S rDNA sequence analysis, the genus *Allochro-matium* belongs to the branch of the family *Chromatiaceae* containing motile freshwater species and characterized by rod-shaped bacteria. Characteristics for differentiation of *Allochro-matium* species from other phototrophic members of the *Chromatiaceae* are shown in Table BXII.γ.4 of the chapter describing the family *Chromatiaceae*. The phylogenetic relationships of *Allochro-matium* species to other members of the family *Chromatiaceae*, based on 16S rDNA sequence comparison, are shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

Although salt requirements are obviously important properties for the taxonomic characterization of *Chromatiaceae* species, the

congruence between phylogenetic position and salt response requires further attention in a few species. Species such as *Allochro-matium vinosum* have been reported to occur in freshwater, brackish water, and marine habitats. Large differences in the mol% G + C content of the DNA of strains assigned to this species (61.3–66.3) may indicate genetic heterogeneity. So far, clear proof of identity of various isolates from marine habitats tentatively assigned to *Allochro-matium vinosum* is lacking, because of improper physiological and genetic (DNA–DNA hybridization, 16S rDNA sequences) characterization. Further studies are needed, to show whether differences in base composition are related to the marine or freshwater nature of these bacteria.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *ALLOCHROMATIUM*

The genus *Allochro-matium* contains two easily distinguished groups of species. *A. vinosum* and *A. minutissimum* are physiologically versatile species that do not require growth factors, grow chemoautotrophically, have rather high tolerance towards sulfide, and use thiosulfate, hydrogen, and organic compounds as photosynthetic electron donors. *A. warmingii*, on the other hand, is a specialized species that requires vitamin B₁₂ as a growth factor,

The strain HPC, which was tentatively identified as a strain of “*Chromatium vinosum*” (Bauld et al., 1987), was certainly misidentified and does not belong to this species. This strain has an optimum salt concentration for growth of 2.5–4.5% NaCl, which may indicate membership to the genus *Halochromatium*. Further studies should allow its taxonomic position to be determined.

FURTHER READING

Imhoff, J.F., Süling, J. and Petri, R. 1998. Phylogenetic relationships among the *Chromatiaceae*, their taxonomic reclassification and description of the new genera *Allochro-matium*, *Halochromatium*, *Isochro-matium*, *Marichromatium*, *Thiococcus*, *Thiohalocapsa*, and *Thermochro-matium*. Int. J. Syst. Bacteriol. 48: 1129–1143.

tolerates only rather low concentrations of sulfide, and is a strictly anaerobic and phototrophic bacterium. Differential characteristics of *Allochro-matium* species are shown in Table BXII.γ.4 of the chapter describing the family *Chromatiaceae*. Presently known *Allochro-matium* species can easily be distinguished on the basis of their cell diameter: Cells 2.0 μm wide, *A. vinosum*; cells 1–1.2 μm wide, *A. minutissimum*; cells 3.5–4.0 μm wide, *A. warmingii*.

List of species of the genus *Allochro-matium*

1. ***Allochro-matium vinosum*** (Ehrenberg 1838) Imhoff, Süling and Petri 1998b, 1141^{VP} (*Chromatium vinosum* (Ehrenberg 1838) Winogradsky 1888, 99; *Monas vinosa* Ehrenberg 1838, 11.)

vi.no'sum. L. neut. adj. *vinosum* full of wine.

Cells are rod shaped, 2 × 2.5–6 μm, occasionally longer. Globules of S⁰ evenly distributed within the cell (Fig. BXII.γ.6). Single cells are colorless, color of growing cultures first yellowish to orange-brown, later brownish red. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the normal spirilloxanthin group.

Grow phototrophically under anoxic conditions in the light or chemotrophically under microoxic to semioxic conditions in the dark (Kämpf and Pfennig, 1980). Both chemotrophic and phototrophic growth is possible under autotrophic and heterotrophic conditions. Photosynthetic electron donors: sulfide, sulfur, thiosulfate, sulfite, molecular

hydrogen, formate, acetate, propionate, pyruvate, fumarate, malate, and succinate. Some strains utilize butyrate. Not utilized: sugars, sugar alcohols, alcohols, benzoate, citrate, and amino acids. Assimilatory sulfate reduction occurs (Thiele, 1968). Nitrogen sources: ammonium salts and N₂.

Mesophilic freshwater bacterium with optimum growth at 25–35°C and pH 7.0–7.3 (range 6.5–7.6). Marine isolates may tolerate or require low concentrations of NaCl.

Habitat: ponds and lakes with stagnant freshwater, sewage lagoons, brackish waters, estuaries, salt marshes, and marine habitats containing hydrogen sulfide and exposed to light. Among the most widely occurring species of the family *Chromatiaceae*.

The mol% G + C of the DNA is: 61.3–66.3 (Bd); type strain 64.3 (Bd).

Type strain: D, BN 5110, ATCC 17899, DSM 180.

GenBank accession number (16S rRNA): M26629.

2. ***Allochro-matium minutissimum*** (Winogradsky 1888) Imhoff, Süling and Petri 1998b, 1141^{VP} (*Chromatium minutissimum* Winogradsky 1888, 100.)

mi.nu.tis'si.mum. L. neut. sup. adj. *minutissimum* very small, smallest.

Cells are rod shaped, 1–1.2 × 2 μm. Other characteristics are the same as for *A. vinosum*.

The mol% G + C of the DNA is: 63.7 (Bd).

Type strain: MSV, BN 5310, DSM 1376.

GenBank accession number (16S rRNA): Y12369.

3. ***Allochro-matium warmingii*** (Cohn 1875) Imhoff, Süling and Petri 1998b, 1141^{VP} (*Chromatium warmingii* (Cohn 1875) Migula 1900, 1048; *Monas warmingii* Cohn 1875, 167.)

war.min'gi.i. M.L. gen. n. *warmingii* of Warming; named for E. Warming, a Danish botanist.

Cells ovoid to rod shaped, 3.5–4.0 × 5–11 μm, sometimes longer, motile by a flagellar tuft which is usually 1.5–2 times the cell length and visible by brightfield or phase-contrast microscopy. In the presence of sulfide and light, globules of S⁰ are predominantly located at the two poles

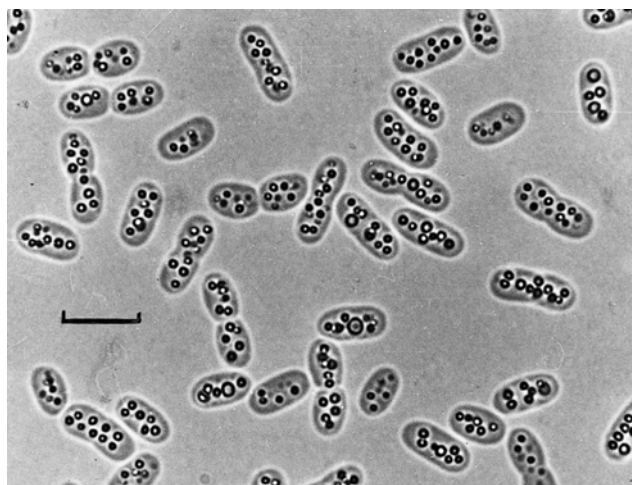


FIGURE BXII.γ.6. *Allochro-matium vinosum* ATCC 17899 cultured photoautotrophically with sulfide. The cells contain sulfur globules. Bright-field micrograph. Bar = 5 μm.

of the cell. Dividing cells form additional sulfur globules near the central division plane (Fig. BXII.γ.7). Color of individual cells is grayish to slightly pink, color of cell suspensions pinkish to purple-violet. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the rhodospinal group.

Strictly anaerobic and obligately phototrophic. Sulfide-reduced media required. Sulfide and S^0 used as photosynthetic electron donors. In the presence of sulfide and bicarbonate, acetate and pyruvate are photoassimilated. Not utilized: thiosulfate, sugars, alcohols, higher fatty acids, amino acids, benzoate, formate, and most intermediates of the tricarboxylic acid cycle. Assimilatory sulfate reduction is absent. Nitrogen sources: ammonium salts, urea, and N_2 . Vitamin B_{12} required.

Freshwater bacterium with optimum growth at 25–30°C and pH 7.0 (pH range 6.5–7.3).

Habitat: ditches, ponds, and lakes with stagnant freshwater containing hydrogen sulfide and exposed to light.

The mol% *G* + *C* of the DNA is: 55.1–60.2 (Bd); type strain 55.1 (Bd).

Type strain: 6512, ATCC 14959, BN 5810, DSM 173.

GenBank accession number (16S rRNA): Y12365.

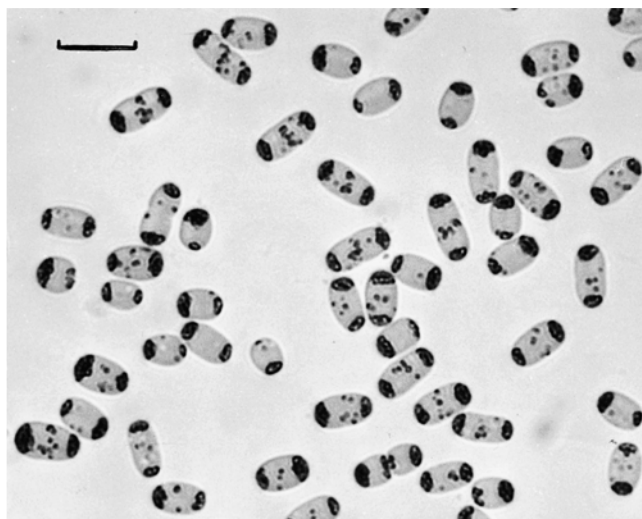


FIGURE BXII.γ.7. *Allochromatium warmingii* DSM 173 cultured photoautotrophically with sulfide. Note the polarly located globules of S^0 . Dividing cells also deposited sulfur globules at the newly formed cross-walls. Bright-field micrograph. Bar = 5 μ m.

Genus III. *Halochromatium* Imhoff, Sling and Petri 1998b, 1139^{VP}

JOHANNES F. IMHOFF AND PIERRE CAUMETTE

Ha'lo.chro.ma' ti.um. Gr. gen. n. *halos* of the salt; *Chromatium* a genus name; M.L. neut. n. *Halochromatium* the *Chromatium* of the salt.

Cells straight to slightly curved rods, single or in pairs, multiply by binary fission. **Motile by polar flagella**, Gram negative, belong to the *Gammaproteobacteria*, and contain internal **photosynthetic membranes of vesicular type** with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Photolithoautotrophic growth under anoxic conditions in the light with sulfide and S^0 as electron donor. During oxidation of sulfide, S^0 is formed as an intermediate and stored in the form of highly refractile globules inside the cells. Sulfate is the final oxidation product. Hydrogen may be used as electron donor. In the presence of sulfide and bicarbonate, organic substrates can be photoassimilated. **Chemolithotrophic and chemoorgano-trophic growth under microoxic conditions in the dark is possible**.

Mesophilic bacteria with optimum growth at 20–35°C, and pH 7.2–7.6. **Elevated salt concentrations are required for growth**.

Habitat: Reduced sediments in salinas, microbial mats in hypersaline environments, and coastal lagoons with elevated salt concentrations, which are exposed to the light.

The mol% *G* + *C* of the DNA is: 64.6–66.3.

Type species: *Halochromatium salexigens* (Caumette, Baulaigue and Matheron 1988) Imhoff, Sling and Petri 1998b, 1140 (*Chromatium salexigens* Caumette, Baulaigue and Matheron 1988, 291.)

ENRICHMENT AND ISOLATION PROCEDURES

Methods for enrichment and isolation are the same as those described for the genus *Chromatium*. *Halochromatium* species are generally found in microbial mats at the sediment surface of hypersaline environments, under salt (gypsum) deposits of salt-erns, in shallow salt lakes, and in evaporative lagoons with salinities of approx. 10–20%. From samples of purple layers in mats, these bacteria can be enriched or isolated by direct inoculation

of deep agar dilution series in synthetic media containing 7–10% NaCl, as described by Caumette et al. (1988).

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *HALOCHROMATIUM* FROM OTHER GENERA

Halochromatium species are characterized by their requirement for elevated salt concentrations and by having motile rod-shaped cells. Characteristics for differentiation of the genus *Halochromatium* from other members of the *Chromatiaceae* are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*. The phylogenetic relationships of *Halochromatium* species to other members of the family *Chromatiaceae*, based on 16S rDNA sequence comparison, are shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

The two species presently comprising the genus *Halochromatium* were originally described as species of the genus *Chromatium*, because they have rod-shaped, motile cells without gas vesicles, features that previously defined the species of the genus *Chromatium* (Pfennig and Trper, 1974, 1989). As a consequence of 16S rDNA sequence analyses, which showed their phylogenetic distance from true *Chromatium* species and other members of the family *Chromatiaceae*, they were transferred to the new genus *Halochromatium* (Imhoff et al., 1998b). *Halochromatium* species are among the most halophilic species of the family *Chromatiaceae* and belong to the branch of truly marine and halophilic *Chromatiaceae* species.

FURTHER READING

- Caumette, P., Baulaigue, R. and Matheron, R. 1988. Characterization of *Chromatium salexigens* sp. nov., a halophilic *Chromatiaceae* isolated from Mediterranean salinas. *System. Appl. Microbiol.* 10: 284–292.
- Caumette, P., Imhoff, J.F., Söling, J. and Matheron, R. 1997. *Chromatium glycolicum* sp. nov., a moderately halophilic purple sulfur bacterium that uses glycolate as a substrate. *Arch. Microbiol.* 167: 1129–1143.
- Imhoff, J.F., Söling, J. and Petri, R. 1998. Phylogenetic relationships among the *Chromatiaceae*, their taxonomic reclassification and description of the new genera *Allochromatium*, *Halochromatium*, *Isochromatium*, *Marichromatium*, *Thiococcus*, *Thiohalocapsa*, and *Thermochromatium*. *Int. J. Syst. Bacteriol.* 48: 1129–1143.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *HALOCHROMATIUM*

Characteristic properties for distinguishing *Halochromatium* species are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*.

List of species of the genus *Halochromatium*

1. ***Halochromatium salexigens*** (Caumette, Baulaigue and Matheron 1988) Imhoff, Söling and Petri 1998b, 1140^{VP} (*Chromatium salexigens* Caumette, Baulaigue and Matheron 1988, 291.)

sal.ex' i.gens. L. n. *sal* salt; L. v. *exigo* demanding, to demand; M.L. part. adj. *salexigens* salt-demanding.

Cells straight or slightly curved rods, 2.0–2.5 × 4.0–7.5 μm, motile by means of a polar or subpolar tuft of flagella. Color of cell suspension pinkish red to slightly violet. Photosynthetic pigments are bacteriochlorophyll *a* and the carotenoids spirilloxanthin, anhydrorhodovibrin, rhodopin, lycopene, and rhodovibrin.

Grow photolithoautotrophically under anoxic conditions in the light with sulfide, S⁰, sulfite, thiosulfate, and H₂ as electron donors. In the presence of sulfide and carbonate, acetate and pyruvate are photoassimilated. Chemolithotrophic growth under microoxic conditions in the dark is possible with sulfide and thiosulfate, chemoorganotrophic growth with acetate and pyruvate. Assimilatory sulfate reduction is absent. Vitamin B₁₂ required.

Mesophilic and halophilic bacterium with optimum growth at 20–30°C and pH 7.4–7.6 (pH range 7.0–8.0). Sodium chloride required for growth, optimum salinity for growth at 8–11% NaCl, salinity range for growth from 4 to 20% NaCl.

Habitat: Reduced sediments in coastal salinas with salt deposits exposed to light.

The mol% G + C of the DNA is: 64.6 (Bd).

Type strain: SG3201, BN 6310, DSM 4395.

GenBank accession number (16S rRNA): X98597.

2. ***Halochromatium glycolicum*** (Caumette, Imhoff, Söling and Matheron 1997) Imhoff, Söling and Petri 1998b, 1140^{VP}

(*Chromatium glycolicum* Caumette, Imhoff, Söling and Matheron 1997, 17.)

gly.co' li.cum. M.L. neut. adj. *glycolicum* related to the ability to use glycolate as substrate.

Cells straight to slightly curved rods, motile by polar flagella. Autotrophically grown cells 0.8–1.0 × 2–4 μm; in the presence of organic substrates 1.0–3.0 × 2–11 μm. Color of cell suspension pinkish red. Photosynthetic pigments are bacteriochlorophyll *a* and the major carotenoids spirilloxanthin and anhydrorhodovibrin, together with small amounts of rhodopin, lycopene, and rhodovibrin.

Grow photolithoautotrophically under anoxic conditions in the light with hydrogen, sulfide, S⁰, sulfite, and thiosulfate as electron donors. In the presence of sulfide and hydrogen carbonate, glycolate, glycerol, fumarate, and succinate are photoassimilated. Formate, acetate, pyruvate, and casamino acids slightly increase the growth yield. Chemolithotrophic growth under microoxic conditions in the dark is possible with sulfide and thiosulfate, chemoorganotrophic growth with glycolate and glycerol. Assimilatory sulfate reduction is absent.

Mesophilic and halophilic bacterium with optimum growth at 25–35°C (no growth above 40°C) and pH 7.2–7.4 (pH range 6.2–9.0). Sodium chloride required for growth, optimum salinity at 4–6% NaCl, salinity range from 2–20% NaCl.

Habitat: Anoxic, sulfidic, and light-exposed habitats in hypersaline environments such as coastal lagoons subject to evaporative water losses, salt lakes, and salinas at elevated salt concentrations.

The mol% G + C of the DNA is: 66.3 (HPLC).

Type strain: SL3201, ATCC 700202, DSM 11080.

GenBank accession number (16S rRNA): X93472.

Genus IV. *Isochromatium* Imhoff, Söling and Petri 1998b, 1141^{VP}

JOHANNES F. IMHOFF

I'so.chro.ma' ti.um. Gr. adj. *isos* equal, similar; *Chromatium* a genus name; M.L. neut. n. *Isochromatium* the similar *Chromatium*, the bacterium similar to *Chromatium*.

Cells straight to slightly curved rods, multiply by binary fission, **motile by a polar tuft of flagella**. Gram negative, belong to the *Gamma*proteobacteria, and contain internal **photosynthetic membranes of vesicular type** with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth under anoxic conditions in the light with sulfide

and S⁰ as electron donors. During oxidation of sulfide, S⁰ stored inside the cells in the form of highly refractile globules. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates are photoassimilated.

Mesophilic bacteria with optimum growth at 25–35°C and a pH range from 6.5 to 7.6. **Salt is required for growth** at concentrations typical for marine bacteria.

Habitat: estuarine salt flats and salt marshes.

The mol% G + C of the DNA is: 62.2–62.8.

Type species: **Isochromatium buder**i (Trüper and Jannasch 1968) Imhoff, Süling and Petri 1998b, 1141 (*Chromatium buder*i Trüper and Jannasch 1968, 364.)

ENRICHMENT AND ISOLATION PROCEDURES

The media described for *Chromatium* species can also be used for *Isochromatium* and other marine members of the family *Chromatiaceae*, if supplemented with 2–3% NaCl as required. The culture conditions and isolation procedures are the same as described for the genus *Chromatium*.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

List of species of the genus *Isochromatium*

1. **Isochromatium buder**i (Trüper and Jannasch 1968) Imhoff, Süling and Petri 1998b, 1141^{VP} (*Chromatium buder*i Trüper and Jannasch 1968, 364.)
bu' deri. M.L. gen. n. *buder*i of Buder; named for J. Buder, a German plant physiologist.

Cells ovoid to rod shaped, 3.5–4.5 × 4.5–9 µm during exponential growth, about 3–4 µm in the stationary phase, motile by a polar tuft of flagella which is usually 1.5–2 times the cell length and may be visible in the light microscope. Globules of S⁰ appear evenly distributed within the cell. Color of individual cells grayish, color of cell suspensions pinkish violet to purple-violet. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the rhodospinal group.

Obligately phototrophic and strictly anaerobic. Sulfide and S⁰ used as electron donors for photolithoautotrophic

DIFFERENTIATION OF THE GENUS *ISOCHROMATIUM* FROM OTHER GENERA

Characteristics for differentiation of the genus *Isochromatium* from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*. The genetic relationship between *Isochromatium buder*i and other *Chromatiaceae* species, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

*Isochromatium buder*i was previously known as *Chromatium buder*i (Trüper and Jannasch, 1968). It was transferred to the new genus *Isochromatium* based on significant genetic and physiological differences from *Chromatium* species and other members of the *Chromatiaceae* (Imhoff et al., 1998b). *Isochromatium buder*i belongs to the branch of truly marine and halophilic *Chromatiaceae* species.

growth. In the presence of sulfide and bicarbonate, acetate and pyruvate are photoassimilated. Not utilized: thiosulfate, molecular hydrogen, sugars, alcohols, higher fatty acids, amino acids, benzoate, formate, and most intermediates of the tricarboxylic acid cycle. Assimilatory sulfate reduction is absent. Nitrogen sources: ammonium salts. Vitamin B₁₂ required.

Mesophilic marine bacterium requiring NaCl for growth. Optimum growth at 1–3% (w/v) NaCl, otherwise extremely pleomorphic, pH optimum for growth 7.2–7.4 (range 6.5–7.6), and temperature growth optimum 25–35°C.

Habitat: estuarine salt flats and salt marshes.

The mol% G + C of the DNA is: 62.2–62.8 (Bd), type strain: 62.2 (Bd).

Type strain: 8111, ATCC 25588, DSM 176.

GenBank accession number (16S rRNA): AJ224430.

Genus V. **Lamprobacter** Gorlenko, Krasil'nikova, Kikina and Tatarinova 1988, 220^{VP} (Effective publication: Gorlenko, Krasil'nikova, Kikina and Tatarinova 1979, 765)

VLADIMIR M. GORLENKO AND JOHANNES F. IMHOFF

Lam'pro.bac'ter. Gr. adj. *lampros* bright, brilliant; M.L. masc. n. *bacter* rod; M.L. masc. n. *Lamprobacter* brilliant rod.

Cells are rod shaped or ovoid, 2.0–2.5 × 4–5 µm, multiply by binary fission, do not form aggregates. **Gas vesicles formed** in cell periphery or in the entire cytoplasm. **Motile by flagella** during certain stages of development. Motile cells normally devoid of gas vesicles. Gram negative, belong to the *Gammaproteobacteria*, and contain internal photosynthetic membranes of vesicular type with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Photolithoautotrophic growth under anoxic conditions in the light with reduced sulfur compounds as electron donors. During oxidation of sulfide or thiosulfate, S⁰ transiently stored inside the cells in the form of highly refractile globules. In the presence of sulfide and bicarbonate, simple organic substrates are photoassimilated. Chemotrophic growth under microoxic conditions in the dark.

The mol% G + C of the DNA is: 60–64.

Type species: **Lamprobacter modestohalophilus** Gorlenko,

Krasil'nikova, Kikina and Tatarinova 1988, 220 (Effective publication: Gorlenko, Krasil'nikova, Kikina and Tatarinova 1979, 765.)

FURTHER DESCRIPTIVE INFORMATION

L. modestohalophilus exhibits considerable polymorphism under various growth conditions (Gorlenko et al., 1979). On the mineral medium of Pfennig (1965) with sulfide or thiosulfate, rod-shaped or ovoid cells of different strains may be 0.5–1.5 × 2.5–5 µm. These cells display dimorphism during normal development of cultures on bicarbonate-containing media with acetate, glycerol, or other organic compounds. In freshly transferred cultures, cells are rod shaped, motile by means of flagella, generally devoid of gas vesicles, and contain globules of S⁰. Late in the exponential growth phase, cells lose their motility, gas vesicles appear, and slime capsules are formed. In stationary phase, slime capsules consolidate and cells are transformed into cysts. Cells

with gas vesicles tend to concentrate in the upper part of the bottle at room temperature. Gas vesicles, short point-ended cylinders of 60×80 nm to 120×140 nm, are grouped near the cell periphery, while sulfur is deposited in the central vesicle-free part of the cells. Multiple passages on mineral salts medium result in a temporary loss of the ability to produce motile cells, and bacteria that are surrounded by loose or dense slime capsules and contain gas vesicles during all stages of growth. Cells are large and irregularly shaped at high NaCl concentrations and an acid pH of 5.9–6.6, but are thin, elongated, and slightly bent at a pH of 8.0 or higher.

The cell wall is of the Gram-negative type and, characteristically, the outer membrane is overlaid by an external layer (S-layer) consisting of hexagonal subunits which are 20 nm long. Vegetative cells possess a loose fibrous slime capsule. Slime capsules of the cyst-like cells are dense, with a clear margin. The photosynthetic membrane system occupies the major part of the cytoplasm in bacteria growing anaerobically in the light. Storage materials are abundantly produced during growth with glycerol, both under chemotrophic (aerobically in the dark) as well as under phototrophic (anaerobically in the light) growth conditions.

Suspensions of bacteria grown anaerobically in the light appear purple-pink. *In vivo* absorption maxima at 370, 655, 804, and 827 nm and a shoulder at 880 nm indicate the presence of bacteriochlorophyll *a*, while a major peak at 518 nm and shoulders at 486 and 549 nm reveal the presence of carotenoids of the okenone group. By chemical analysis, the major components were identified as okenone (64.2%), the compound thiothece-OH-484 (18.5%), also present in *Thiocystis gelatinosa* (Schmidt, 1978), and spirilloxanthin (10.5%) (Sidorova et al., 1998).

Among the organic compounds that are photoassimilated, glycerol, acetate, pyruvate, lactate, maltose, and lactose strongly increase the yield. Glycerol metabolism involves the functioning of glycerol kinase and α -glycerophosphate dehydrogenase independent of pyridine nucleotides (Krasil'nikova et al., 1979). Organic acids are assimilated through the incomplete tricarboxylic acid cycle. The glyoxylate cycle is absent (Krasil'nikova, 1985; Krasil'nikova and Kondrat'eva, 1979). Sugars are utilized through the Embden–Meyerhof pathway.

Optimum conditions for chemotrophic growth in the dark are ~10% O₂ in the gas phase. Phototrophic growth in the light

is suppressed at O₂ concentrations above 2%. In a stab culture in 0.8% agar medium in the dark, growth occurs 1 cm below the surface. Thiosulfate or S⁰ are required as electron donor and sulfur source for assimilatory purposes (Krasil'nikova, 1981). Optimum concentration of thiosulfate is 0.2%. At a concentration of 0.5% Na₂S₂O₃·5H₂O, growth of aerobic cultures is strongly inhibited. In the dark, thiosulfate is oxidized to S⁰ and sulfate. S⁰ is deposited inside the cells as an intermediate product, accumulation of S⁰ and generation of sulfate occur in parallel. Vitamin B₁₂ is required for chemotrophic aerobic growth (Kondrat'eva et al., 1981). Best growth in the dark occurs with glycerol as carbon source, while lactate, pyruvate, and propionate also stimulate growth.

ENRICHMENT AND ISOLATION PROCEDURES

L. modestohalophilus is widely distributed in microbial mats of shallow, warm, and saline water bodies (0.5–12% NaCl) with high sulfide content of the southern Crimea, the Donetsk region (Ukraine), and Turkmenia (former U.S.S.R.). *Lamprobacter* was not found in the White Sea littoral zone.

When water and mud samples are inoculated into a medium containing 0.5–4% NaCl, 0.05% Na₂S·9H₂O, 0.1% Na₂S₂O₃·5H₂O, and 0.1% sodium acetate, bacteria containing gas vesicles are concentrated in the upper part of the culture vessel. This fraction is removed by using a sterile capillary and is transferred into agar dilution series, as described for the genus *Lamprocystis*.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*. For intervals of 1–3 months, cultures can be maintained in liquid medium stored in a refrigerator at low light intensity.

DIFFERENTIATION OF THE GENUS *LAMPROBACTER* FROM OTHER GENERA

The genus *Lamprobacter* is characterized and distinguished from other members of the family *Chromatiaceae* by rod-shaped motile cells that develop gas vesicles under certain growth conditions. Characteristics used for differentiation of the genus *Lamprobacter* from other members of the *Chromatiaceae* are shown in Table BXII.7.6 of the chapter describing the family *Chromatiaceae*.

List of species of the genus *Lamprobacter*

1. ***Lamprobacter modestohalophilus*** Gorlenko, Krasil'nikova, Kikina and Tatarinova 1988, 220^{VP} (Effective publication: Gorlenko, Krasil'nikova, Kikina and Tatarinova 1979, 765.) *mo.des'to.ha.lo'phi.hus*. L. n. *modestus* moderate; Gr. n. *hals* salt; Gr. adj. *philos* loving; M.L. masc. adj. *modestohalophilus* moderate salt-loving.

Cells are rod shaped or ovoid, $2.0\text{--}2.5 \times 4\text{--}5$ μm , polymorphic under changing conditions of incubation (Fig. BXII.7.8). Life cycle with two alternating morphological forms: nonmotile cells with gas vesicles and slime capsules, and cells motile by means of flagella and devoid of gas vesicles. Gas vesicles frequently located in the cell periphery, globules of S⁰ in center of cells. Photosynthetic membranes of the vesicular type. Vesicles in cells grown in mineral medium measure about 70 nm in diameter. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the okenone group.

Growth under anoxic conditions in the light and under

microoxic conditions in the dark. Under anoxic conditions in the light, cell suspensions appear purple to pink. Under aerobic conditions in the dark, synthesis of pigments is reduced and cultures become colorless. Photosynthetic electron donors used are sulfide, thiosulfate, S⁰, and hydrogen, but not sulfite. Electron donors in the dark are thiosulfate and sulfur. Sulfide and thiosulfate are oxidized to S⁰ and sulfate. Assimilatory sulfate reduction is absent and reduced sulfur compounds are required.

CO₂ assimilated via the Calvin cycle. In the presence of sulfide (or thiosulfate) and bicarbonate, alcohols (glycerol, *n*-propanol, *n*-butanol, ethanol, isobutanol, and isopropanol), and organic acids (lactate, pyruvate, and acetate) are photoassimilated. Methanol, isomyl alcohol, sorbitol, dulcitol, mannitol, succinate, fumarate, malate, lactose, maltose, xylose, galactose, and fructose are not utilized. Growth inhibited by *n*-amyl alcohol, allyl alcohol, formate, propionate, benzoate, arabinose, rhamnose, and raffinose. Ni-

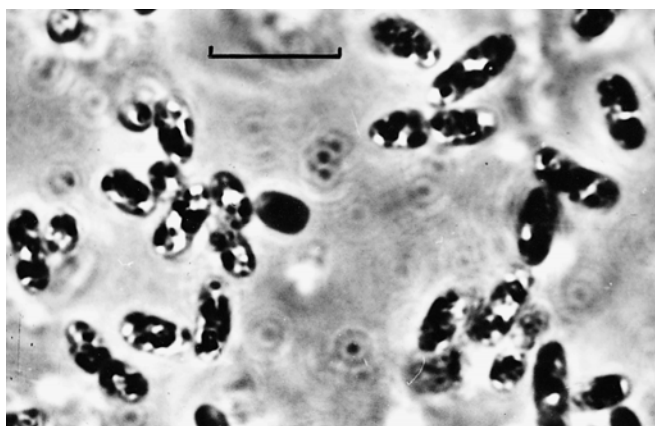


FIGURE BXII.γ.8. *Lamprobacter modestohalophilus* strain RO-1. Phase-contrast micrograph. Bar = 10 μm. (Reproduced with permission from V.M. Gorlenko et al., *Izvestiya Akademii Nauk S.S.S.R. Seriya Biologicheskaya* (Moskva) 5: 765, 1979© MAIK Nauka/Interperiodic Publishing, Moscow.)

trogen sources used: ammonium salts, urea, casein hydrolysate, glutamic acid, and N₂. Hydrogenase and catalase activity occur. Storage materials: S⁰, poly-β-hydroxybutyrate, polyphosphate, and polysaccharide. Vitamin B₁₂ required.

Optimum growth is at 23–27°C, pH 7.4–7.6, 1–4% NaCl (up to 9% NaCl). Sodium chloride is required for growth. Favorable light intensity is 3000 lux.

Habitat: hydrogen sulfide-containing mud and water of saline water bodies with 1–11% salinity.

The mol% G + C of the DNA is: 60–64.

Type strain: Type strain INMI RO-1 is lost; no neotype strain designated.

Additional Remarks: Available candidate for neotype: “Syvash” (Institute of Microbiology, Russia Academy of Sciences, Moscow, Russia), isolated from a lagoon of salt lake Syvash (Crimea) with 6.7% salinity, pH range for growth 6.9–7.6, temperature range for growth from 20 to 30°C.

Genus VI. *Lamprocystis* Schroeter 1886, 151^{AL}

JOHANNES F. IMHOFF

Lam'pro.cys'tis. Gr. adj. *lampros* bright, brilliant; Gr. n. *cystis* the bladder, a bag; M.L. fem. n. *Lamprocystis* brilliant bag.

Cells spherical to ovoid, 1.9–3.8 μm in diameter, diplococcus-shaped before cell division and may occur in irregular aggregates, multiply by binary fission, **nonmotile or motile by means of a single flagellum**, **form gas vesicles** in central part. Gram negative and belong to the *Gammaproteobacteria*. At high sulfide concentration (4–6 mM) and light intensity (1000–2000 lux), cells may grow as long, branching cell aggregates embedded in slime. Under favorable growth conditions, cell aggregates may break up into smaller clusters and spheroidal microcolonies, which become motile by flagella; finally, individual motile cells are liberated. Internal **photosynthetic membranes are of vesicular type** and contain the photosynthetic pigments bacteriochlorophyll *a* and carotenoids.

Photolithoautotrophic growth under anoxic conditions in the light with reduced sulfur compounds such as sulfide and S⁰ as electron donors. During oxidation of sulfide, S⁰ is transiently stored in the peripheral part of the cells that is free of gas vesicles. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates are photoassimilated. Obligately phototrophic or facultatively chemotrophic under microoxic to oxic conditions in the dark.

The mol% G + C of the DNA is: 63.4–64.1.

Type species: *Lamprocystis roseopersicina* (Kützing 1849) Schroeter 1886, 151 (*Protococcus roseopersicinus* Kützing 1849, 196.)

ENRICHMENT AND ISOLATION PROCEDURES

Media and culture conditions are the same as described for *Chromatium* species. *Lamprocystis* species, like other purple sulfur bacteria containing gas vesicles, may be selectively enriched by making use of the buoyancy of their cells at lower temperatures of 4–10°C (Pfennig et al., 1968). For enrichment cultures, an inoculum from a natural habitat, in which the presence of *Lamprocystis* cells and cell aggregates was established microscopically, should be used. The culture medium and incubation conditions are the same as described for the genus *Chromatium*. Low sulfide

concentrations (1–2 mM) and low light intensity (100–300 lux) with diurnal light and dark periods (e.g., 18 h light, 6 h dark) are employed at a room temperature of about 20°C. The bottles are incubated in a lying position to avoid accumulation of cells in the nonilluminated area under the screw cap. After two or three supplementations of the enrichment culture with neutralized sulfide solution, the well-developed culture is stored in a refrigerator in an upright position. This is not done until the various types of purple sulfur bacteria have almost completely oxidized the sulfur globules inside the cells. After 1–2 weeks of storage at a temperature between 4 and 10°C, purple sulfur bacteria with gas vesicles, including *Lamprocystis* species, accumulate at the surface of the liquid medium under the screw cap. A small amount of the floating material is removed with an inoculation loop and inspected microscopically under brightfield illumination. For further enrichment, the floating cell mass is carefully pipetted from the surface and transferred to fresh medium. Alternatively, or in addition, the enriched material is used to inoculate one or two series of agar shake cultures as described for the isolation of *Chromatium* species.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *LAMPROCYSTIS* FROM OTHER GENERA

Characteristics used for differentiation of the genus *Lamprocystis* from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.6 of the chapter describing the family *Chromatiaceae*. The phylogenetic relationship of *Lamprocystis* species to other members of the *Chromatiaceae*, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.4 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

Only a few strains of *Lamprocystis roseopersicina* have thus far been studied in pure culture. From observations in natural samples, it is apparent that cells showing the characteristics of *Lamprocystis* are quite frequent and that cells of different diameter occur. Further studies using a greater variety of pure cultures of these bacteria are needed, to show whether clusters of strains of different cell diameter represent different species. According to 16S rRNA oligonucleotide pattern analysis, *Lamprocystis roseopersicina* is distantly related to *Thiocapsa* and *Thiodictyon* species (Fowler et al., 1984). Sequence analyses of 16S rDNA place this species into the group of spherical to ovoid, nonmotile freshwater *Chromatiaceae* species, many of which contain gas vesicles (Fig. BXII.γ.4 of the chapter describing the family *Chromatiaceae*), and in close proximity to the species originally described as *Amoebobacter purpureus* (Eichler and Pfennig, 1988). In accordance with the rules of the International Code of Nomenclature of Bacteria (Lapage et al., 1992), the transfer of this species to the genus *Lamprocystis* as the new combination *Lamprocystis purpurea* comb. nov. was proposed (Imhoff, 2001b).

The genus *Amoebobacter* comprised spherical, nonmotile species containing gas vesicles (Winogradsky, 1888; Pfennig, 1989a). Two species, including the type species, were transferred to the genus *Thiocapsa* and a third species to the new genus *Thiolamprovum* (Guyoneaud et al., 1998b; see genera *Thiocapsa* and *Thiolamprovum*). Because the type species had been removed from the genus *Amoebobacter*, the names of other species of this genus had to change, according to rule 37a, and it was illegitimate to retain the name *Amoebobacter purpureus* (Guyoneaud et al., 1998b). The transfer of this species to the new genus *Pfennigia* has been proposed (Tindall, 1999). However, the high degree of similarity between 16S rDNA sequences from this bacterium and *Lamprocystis roseopersicina* demonstrated the close relationship of the two bacteria, and precluded their separation into different genera (see Fig. BXII.γ.4 of the chapter describing the family

Chromatiaceae). Because of the priority of the genus and species name of *Lamprocystis roseopersicina*, *Pfennigia purpurea* Tindall 1999 (*Amoebobacter purpureus* Eichler and Pfennig, 1988) was transferred to *Lamprocystis purpurea* comb. nov. (Imhoff, 2001b).

Lamprocystis purpurea (*Amoebobacter purpureus*) has been described based on six isolates, which were classified into two groups according to cell size and substrate utilization (Eichler and Pfennig, 1988). The described differences between the groups may be sufficient to allow the recognition of two different species. The description of *Lamprocystis purpurea* given below includes properties of the type strain and an additional strain (ThSchm4). Strains of the second group differ from *Pfennigia purpurea* in the following characteristics: cells are $3.3\text{--}3.8 \times 3.5\text{--}4.5\ \mu\text{m}$, do not utilize formate, and have a mol% G + C of 63.4–64.1 (Eichler and Pfennig, 1988).

FURTHER READING

- Pfennig, N., Markham, M.C. and Liaaen-Jensen, S. 1968. Carotenoids of *Thiorhodaceae*. 8. Isolation and characterization of a *Thiothece*, *Lamprocystis* and *Thiodictyon* strain and their carotenoid pigments. *Arch. Mikrobiol.* 62: 178–191.
- Eichler, B. and Pfennig, N. 1988. A new purple sulfur bacterium from stratified freshwater lakes, *Amoebobacter purpureus* sp. nov. *Arch. Microbiol.* 149: 395–400.
- Guyoneaud, R., Süling, J., Petri, R., Matheron, R., Caumette, P., Pfennig, N. and Imhoff, J.F. 1998. Taxonomic rearrangements of the genera *Thiocapsa* and *Amoebobacter* on the basis of 16S rRNA sequence analysis and description of *Thiolamprovum* gen. nov. *Int. J. Syst. Bacteriol.* 48: 957–964.
- Tindall, B.J. 1999. Transfer of *Amoebobacter purpureus* to the genus *Pfennigia* gen. nov., as *Pfennigia purpurea* comb. nov., on the basis of the illegitimate proposal to make *Amoebobacter purpureus* the type species of the genus *Amoebobacter*. *Int. J. Syst. Bacteriol.* 49: 1307–1308.
- Imhoff, J.F. 2001. Transfer of *Pfennigia purpurea* Tindall 1999 (*Amoebobacter purpureus* Eichler and Pfennig 1988) to the genus *Lamprocystis* as *Lamprocystis purpurea* comb. nov. *Int. J. Syst. Evol. Microbiol.* 51: 1699–1701.

List of species of the genus *Lamprocystis*

1. ***Lamprocystis roseopersicina*** (Kützing 1849) Schroeter 1886, 151^{AL} (*Protococcus roseopersicinus* Kützing 1849, 196.) *ro' se. o. per. si. ci' na*. L. adj. *roseus* rosy; Gr. n. *persicus* the peach; M.L. fem. adj. *roseopersicina* rosy peach-colored.

Cells spherical to ovoid, 2.0–3.5 μm in diameter (Fig. BXII.γ.9), motile by flagella, contain gas vesicles. Color of cell suspensions pinkish violet to purple-violet. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the rhodopinal group, with lycopene as main component (Pfennig et al., 1968).

Obligate phototrophic and strictly anaerobic. Photoautotrophic growth with sulfide and S^0 as electron donors. In the presence of sulfide and bicarbonate, acetate and pyruvate are photoassimilated. Nitrogen sources: ammonium salts.

Mesophilic freshwater bacterium with optimum growth at 20–30°C and pH 7.0–7.3.

Habitat: mud and stagnant water of ponds and lakes containing hydrogen sulfide; common planktonic purple sulfur bacterium in the sulfide-containing hypolimnion of freshwater lakes.

The mol% G + C of the DNA is: 63.8 (Bd).

Type strain: 3012, BN 4510, DSM 229.

GenBank accession number (16S rRNA): AJ006063.

2. ***Lamprocystis purpurea*** comb. nov. (Eichler and Pfennig

1988) Imhoff 2001b, 1700^{VP} (*Pfennigia purpurea* (Eichler and Pfennig 1988) Tindall 1999, 1308; *Amoebobacter purpureus* Eichler and Pfennig 1988, 399.)

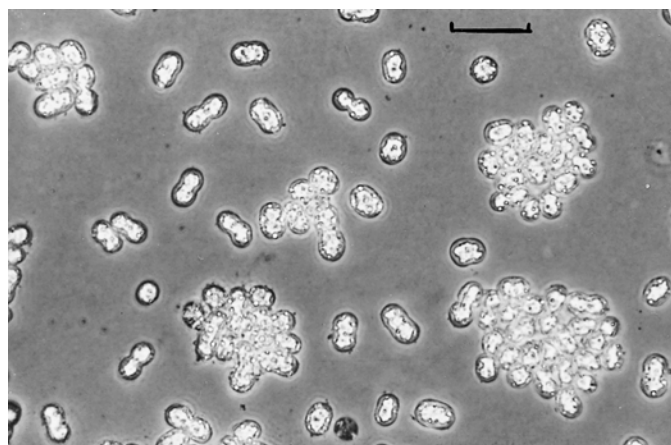


FIGURE BXII.γ.9. *Lamprocystis roseopersicina* DSM 229 cultured photoautotrophically with sulfide. Single cells and cell colonies are motile in liquid medium. The irregular whitish areas within the cells are the gas vacuoles. The small spherical bodies in the cells are the sulfur globules. Phase-contrast micrograph. Bar = 10 μm .

pur.pur' e.a. L. fem. adj. *purpurea* purple or purple-red.

Cells nearly spherical to oval, $1.9\text{--}2.3 \times 2.0\text{--}3.2 \mu\text{m}$, nonmotile, and contain gas vesicles in central parts of cells. Cells single or in irregular aggregates of up to 40 cells. Color of dense cell suspensions purple-red. Photosynthetic pigments are bacteriochlorophyll *a* and the carotenoid okenone is a major component.

Photolithoautotrophic growth under anoxic conditions in the light with sulfide, thiosulfate, and S^0 as electron donors. In the presence of sulfide and bicarbonate, acetate, propionate, pyruvate, lactate, glucose, and formate are photoassimilated. Chemolithoautotrophic growth possible with sulfide and thiosulfate as electron donors under microoxic

conditions in the dark. Assimilatory sulfate reduction is absent.

Mesophilic freshwater bacterium with optimum growth at $23\text{--}25^\circ\text{C}$ and pH 7.0–7.3.

Habitat: Anoxic sulfide-containing water of stratified freshwater lakes, mud, and stagnant water of ponds and lakes, may occur as dominant bloom-forming bacterium, together with *Thiocapsa roseopersicina*, in wastewater lagoons containing degradable organic substances.

The mol% G + C of the DNA is: 63.5–63.6 (Bd).

Type strain: ThSch12, DSM 4197.

GenBank accession number (16S rRNA): Y12366, AJ223235.

Genus VII. *Marichromatium* Imhoff, Siling and Petri 1998b, 1140^{VP}

JOHANNES F. IMHOFF

Ma'ri.chro.ma'ti.um. L. gen. n. *maris* of the sea; *Chromatium* a genus name; M.L. neut. n. *Marichromatium* the *Chromatium* of the sea, the truly marine *Chromatium*.

Cells straight to slightly curved rods, motile by polar flagella and multiply by binary fission. Cells single or in pairs, and may stick together and form clumps. Gram negative, belong to the *Gammaproteobacteria*, and contain internal **photosynthetic membranes of vesicular type** with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Grow photolithoautotrophically under anoxic conditions in the light with sulfide and S^0 as electron donors. During oxidation of sulfide, S^0 is formed as an intermediate and stored in the form of highly refractile globules inside the cells. Final oxidation product is sulfate. Molecular hydrogen may be used as electron donor. In the presence of sulfide and bicarbonate, organic substrates can be photoassimilated. **Chemolithotrophic and chemoorganotrophic growth may be possible under microoxic conditions in the dark.**

Mesophilic bacteria with optimum growth at $25\text{--}35^\circ\text{C}$, pH 6.5–7.6, and at salt concentrations typical of marine bacteria.

Habitat: anoxic marine sediments and stagnant waters containing hydrogen sulfide and exposed to the light, marine sponges and other marine invertebrates.

The mol% G + C of the DNA is: 68.9–70.4.

Type species: *Marichromatium gracile* (Strzeszewski 1913) Imhoff, Siling and Petri 1998b, 1140 (*Chromatium gracile* Strzeszewski 1913, 321.)

ENRICHMENT AND ISOLATION PROCEDURES

The media described for *Chromatium* species can also be used for *Marichromatium* species and other marine members of the family *Chromatiaceae*, if supplemented with appropriate amounts

of salt as required. The culture conditions and isolation procedures are the same as described for the genus *Chromatium*.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *MARICHRMATIUM* FROM OTHER GENERA

Marichromatium species are distinct from other members of the family *Chromatiaceae* by their specific salt requirement and the very high mol% G + C content of their DNA. Characteristics for differentiation of the genus *Marichromatium* from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

Species of this genus have previously been classified as belonging to the genus *Chromatium* (Pfennig and Truper, 1974; Pfennig, 1989b). However, because of significant phenotypic and genetic differences to true *Chromatium* species and other members of the family *Chromatiaceae*, they have been transferred to the new genus *Marichromatium* (Imhoff et al., 1998b).

FURTHER READING

Imhoff, J.F., Siling, J. and Petri, R. 1998. Phylogenetic relationships among the *Chromatiaceae*, their taxonomic reclassification and description of the new genera *Allochromatium*, *Halochromatium*, *Isochromatium*, *Marichromatium*, *Thiococcus*, *Thiohalocapsa*, and *Thermochromatium*. Int. J. Syst. Bacteriol. 48: 1129–1143.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *MARICHRMATIUM*

Marichromatium species can be distinguished on the basis of cell suspension color and pigment content, differences in substrate utilization, as well as slight differences in cell size and salt response. Characteristic properties for distinguishing *Marichromatium* species are shown in Table BXII.γ.5 of the chapter describ-

ing the family *Chromatiaceae*. The phylogenetic relationships of *Marichromatium* species to other members of the family *Chromatiaceae* is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

List of species of the genus *Marichromatium*

1. *Marichromatium gracile* (Strzeszewski 1913) Imhoff, Siling and Petri 1998b, 1140^{VP} (*Chromatium gracile* Strzeszewski 1913, 321.)

gra'ci.le. L. neut. adj. *gracile* thin, slender.

Cells are rod shaped, $1.0\text{--}1.3 \times 2\text{--}6 \mu\text{m}$. Globules of S^0 evenly distributed within cells. Single cells colorless. Color

of growing cultures orange-brown to brownish red. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the normal spirilloxanthin group.

Photolithoautotrophic growth under anoxic conditions in the light with sulfide, sulfur, thiosulfate, sulfite, and molecular hydrogen as electron donors. Organic substrates formate, acetate, propionate, pyruvate, fumarate, malate, succinate, lactate, propionate, butyrate, crotonate, and Casamino acids are used. Not utilized: sugars, sugar alcohols, alcohols, benzoate, citrate, and amino acids. Chemoautotrophic and mixotrophic growth under microoxic to semi-oxic conditions possible. Assimilatory sulfate reduction occurs (Thiele, 1968). Nitrogen sources: ammonium salts, N₂. Vitamins not required.

Habitat: brackish waters, estuaries, salt marshes, and marine habitats containing hydrogen sulfide and exposed to the light.

Mesophilic, marine bacterium with optimum growth at 2–3% NaCl, pH 7.2–7.4 (range 6.8–7.6), and a temperature range of 20–35°C.

The mol% *G* + *C* of the DNA is: 68.9–70.4 (Bd); type strain 69.9 (Bd).

Type strain: 8611, BN 5210, DSM 203.

GenBank accession number (16S rRNA): X93473.

2. **Marichromatium purpuratum** (Imhoff and Trüper 1980) Imhoff, Siling and Petri 1998b, 1140^{VP} (*Chromatium purpuratum* Imhoff and Trüper 1980, 69.)

pur.pur.a' tum. L. neut. adj. *purpuratum* dressed in purple.

Cells are rod shaped, 1.2–1.7 × 3–4 µm, when grown under autotrophic conditions, motile by a single polar flagellum. Color of cell suspensions purple-red. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the okenone group.

Obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth with sulfide, sulfur, and thiosulfate as electron donors; photoorganoheterotrophic growth with acetate, propionate, butyrate, valerate, lactate, pyruvate, fumarate, malate, and succinate in the presence of bicarbonate. Fructose and casamino acids are poor substrates. Not utilized: sugars, sugar-alcohols, alcohols, higher fatty acids, benzoate, and amino acids. Nitrogen sources: ammonium salts. Vitamins not required.

Mesophilic marine bacterium that requires NaCl for growth, optimum salinity 5% NaCl, salinity range 2–7% NaCl, pH range 7.2–7.6, temperature range 25–30°C.

Habitat: anoxic and sulfidic marine sediments and waters exposed to the light, marine sponges, and other marine invertebrates, such as didemnids and copepods.

The mol% *G* + *C* of the DNA is: 68.9 (*T_m*).

Type strain: BN 5500, DSM 1591.

GenBank accession number (16S rRNA): AF001580, AJ224439.

Genus VIII. **Nitrosococcus** Winogradsky 1892, 127^{AL} (Nom. Cons. Opin. 23 Jud. Comm. 1958b, 169)

HANS-PETER KOOPS AND ANDREAS POMMERENING-RÖSER

Ni.tro.so.coc' cus. M.L. adj. *nitrosus* nitrous; *coccus* a grain, berry; M.L. masc. n. *Nitrosococcus* nitrous sphere.

Cells are spherical to ellipsoidal (Fig. BXII.γ.10A). Typical Gram-negative cell wall, but **some strains have additional layers composed of repeating subunits arranged in a molecular array** (Watson and Remsen, 1969, 1970). **Extensive intracytoplasmic membrane systems arranged centrally in the protoplasm** (Fig. BXII.γ.10B). Carboxysomes not observed. **Obligate salt requirement.** Urea can be used as ammonia source by one of the two described species. Motile cells possess **a tuft of flagella or a single flagellum**. Species are **distributed in oceans and in salt lakes**.

Type species: **Nitrosococcus nitrosus** (Migula 1900) Buchanan 1925, 402 (Nom. Cons. Opin. 23 Jud. Comm. 1958b, 169) (*Micrococcus nitrosus* Migula 1900, 194.)

TAXONOMIC COMMENTS

Isolates of coccoid ammonia-oxidizers were first described by Winogradsky (1892, 1904). One of these strains has been named *Nitrosococcus nitrosus* by Migula (1900). However, this type species of the genus *Nitrosococcus* was not preserved in culture. Only data

on the cell shape and size are available from the original description (Winogradsky, 1892). Since this species cannot be identified with certainty, it is a candidate for placement on the list of rejected names.

Another coccoid ammonia-oxidizer, originally placed in *Nitrosococcus* as *N. mobilis* (Koops et al., 1976), has been proved via phylogenetic analyses to be a member of the genus *Nitrosomonas* (Woese et al., 1984b; Head et al., 1993; Teske et al., 1994).

The remaining two described species of the genus *Nitrosococcus*, *N. oceani* and "*N. halophilus*", are both members of the *Gammaproteobacteria*. They are closely related to each other, as demonstrated by DNA–DNA hybridizations. Using the S₁ nuclease technique, a DNA similarity of 9–10% was estimated (Pommerening-Röser, 1993).

In the future, the taxonomically revised genus *Nitrosococcus* should be reserved for representatives of ammonia-oxidizing bacteria belonging to the *Gammaproteobacteria* and being closely related to the grouping of *N. oceani* and "*N. halophilus*".

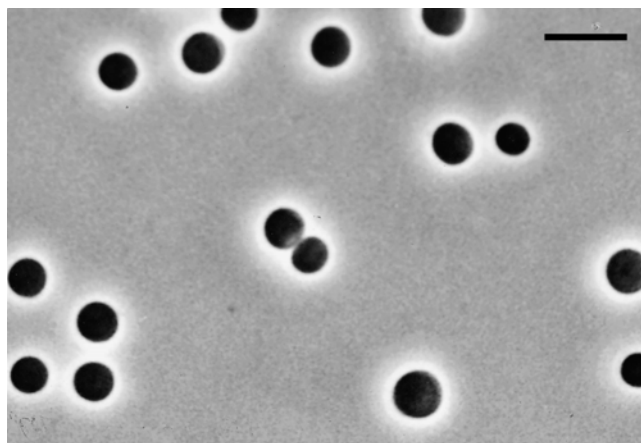
List of species of the genus *Nitrosococcus*

1. **Nitrosococcus nitrosus** (Migula 1900) Buchanan 1925, 402^{AL} (Nom. Cons. Opin. 23 Jud. Comm. 1958b, 169) (*Micrococcus nitrosus* Migula 1900, 194.)
ni.tro' sus. M.L. adj. *nitrosus* nitrous.

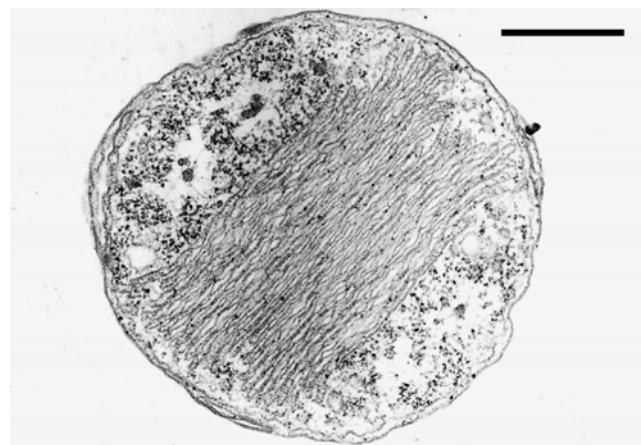
Cells are spheres 1.5–1.7 µm in diameter. No further phenotypic characteristics are available.

The mol% *G* + *C* of the DNA is: unavailable.

Type strain: no longer in culture.



A



B

FIGURE BXII.γ.10. (A) Phase contrast micrograph of a thin section of cells of *Nitrosococcus oceanii*. Bar = 5 μm. (B) Electron micrograph of a thin section of cells of *N. oceanii*. Bar = 0.5 μm.

2. ***Nitrosococcus oceanii*** (Watson 1965) Watson 1971, 267^{AL} (*"Nitrosocystis oceanus"* Watson 1965, R279.)
o.ce.a' nus. M.L. gen. n. *oceanii* of the ocean.

The cells are spherical to ellipsoidal, 1.8–2.2 μm in diameter. Cells generally occur singly or in pairs, but sometimes aggregates are formed via production of exopolymeric substances. Motile cells possess 1–20 flagella. Cells have a typical Gram-negative envelope, with two additional cell wall layers composed of subunits arranged in rectilinear and hexagonal arrays. Cells possess an extensive intracytoplasmic membrane system, arranged as a centrally located stack of parallel, flattened vesicles. Cells have an obligate salt requirement, with optimum NaCl concentrations for growth of 400–500 mM. The maximum concentration of ammonium salts is around 1000 mM at pH 7.8. Urease positive. All strains have been isolated from marine environments.

The mol% G + C of the DNA is: 50.5 (T_m , Bd).

Type strain: C-107, ATCC 19707.

GenBank accession number (16S rRNA): M96395.

3. ***"Nitrosococcus halophilus"*** Koops, Böttcher, Möller, Pommerening-Röser and Stehr 1990, 247.
ha.lo' phi.lus. Gr. n. *halos* salt; Gr. adj. *philos* loving; M.L. adj. *halophilus* salt loving.

Cells are spherical to ellipsoidal, 1.8–2.5 μm in diameter. Gram-negative cell wall. Additional outer layers, as demonstrated for *N. oceanii*, may exist, but have not been reported. Motile cells have a tuft of flagella. Intracytoplasmic membranes are arranged as a stack of centrally located flattened vesicles. Cells are obligately halophilic, optimum NaCl concentrations for growth are 600–800 mM, up to 1600 mM is tolerated. Ammonium compounds are tolerated up to concentrations around 600 mM. Urease negative. Isolates originate from salt lagoons and salt lakes.

The mol% G + C of the DNA is: 50.5 (T_m).

Deposited strain: Nc 4.

Genus IX. *Pfennigia* Tindall 1999, 1308^{VP*}

BRIAN J. TINDALL

Pfen.nig'ia. M.L. *ia* ending; M.L. fem. n. *Pfennigia* named after Norbert Pfennig, in recognition of his contribution to the biology and taxonomy of anoxygenic phototrophic bacteria.

Oval to spherical cells, 1.9–3.8 × 2.0–4.5 μm. May form large aggregates (up to 40 cells). Gas vacuoles occur in the central part of the cytoplasm. The intracytoplasmic membrane system is of the vesicular type. **Nonmotile.** Gram negative. **Photolithotrophic under anaerobic conditions in the light;** chemoautotrophic under microaerobic conditions in the dark. **Hydrogen sulfide, thiosulfate, and S⁰ serve as electron donors. Globules of S⁰ appear in cells during oxidation of sulfide or thiosulfate to sulfate. Major photosynthetic pigments are bacteriochlorophyll *a* and the carotenoid okenone.** Isolated from the chemocline of freshwater lakes.

The mol% G + C of the DNA is: 63.4–64.1.

Type species: ***Pfennigia purpurea*** (Eichler and Pfennig 1988)

Tindall 1999, 1308 (*Amoebobacter purpureus* Eichler and Pfennig 1988, 399.)

TAXONOMIC COMMENTS

The trend towards the use of 16S rDNA sequences in the taxonomy of anoxygenic phototrophic bacteria has indicated that many genera, such as *Chromatium*, *Thiocapsa*, and *Amoebobacter*, which are phenotypically relatively heterogeneous are also phylogenetically heterogeneous. However, this has also led to a tendency to rely heavily on the 16S rDNA sequences and to reduce many phenotypic parameters to being "unreliable". 16S rDNA sequence analysis of the type strain of *Pfennigia purpurea* (DSM 4197) indicates that this strain has a high degree of sequence similarity to *Lamprocystis roseopersicina* DSM 229. However, two 16S rDNA sequences have been deposited as AJ223235 and Y12366, both being from the type strain of *Pfennigia purpurea* DSM 4197, although they are sufficiently different to infer that they come from different species. Thus, while *Pfennigia purpurea* (DSM 4197;

*Editorial Note: The type species of *Pfennigia*, *P. purpurea*, has been transferred to the genus *Lamprocystis* as *L. purpurea* (Eichler and Pfennig 1988) Imhoff 2001b comb. nov.

GenBank accession no. AJ223235) shows less than 1% 16S rDNA sequence divergence from *Lamprocystis roseopersicina* (DSM 229; GenBank accession no. AJ006063), the degree of sequence divergence between the latter and *Pfennigia purpurea* (DSM 4197; GenBank accession no. Y12366) may be more than 3%. In the first case, it would be appropriate to examine the possibility that *Pfennigia purpurea* (DSM 4197; Y12366) and *Lamprocystis roseopersicina* (DSM 229; AJ006063) are members of the same species, whereas in the latter case their membership in the same genus is questionable. These two possibilities cannot be decided based

on currently available sequence data. It should be pointed out that it is implicit in the evaluations of Guyoneaud et al. (1998b), Imhoff et al. (1998b), or Puchkova et al. (2000) that species sharing 5% or less 16S rDNA sequence divergence belong in the same genus. However, this is usually without adequate supporting data or in the light of other phenotypic data being "unreliable". In other genera, sequence similarities may be higher than 95% and rely on supporting phenotypic data (see, for example, Lincoln et al., 1999).

List of species of the genus *Pfennigia*

1. ***Pfennigia purpurea*** (Eichler and Pfennig 1988) Tindall 1999, 1308^{VP} (*Amoebobacter purpureus* Eichler and Pfennig 1988, 399.)
pur.pur'ea. L. fem. adj. *purpurea* purple or purple-red.

The characteristics are as given for the genus.

The mol% G + C of the DNA is: 63.4–64.1

Type strain: ThSch12, DSM 4197.

GenBank accession number (16S rRNA): AJ223235, Y12366.

Genus X. *Rhabdochromatium* Dilling, Liesack and Pfennig 1996, 362^{VP} (Effective publication: Dilling, Liesack and Pfennig 1995, 130)

JOHANNES F. IMHOFF

Rhab'do.chro.ma'tium. Gr. n. *rhabdos* a rod; Gr. n. *chromatium* color, paint; M.L. neut. n. *Rhabdochromatium* colored rod.

Cells straight or slightly bent rods, motile by a polar tuft of flagella, multiply by binary fission. Gram negative, belong to the *Gamma*proteobacteria, and contain internal **photosynthetic membranes of vesicular type** in which the photosynthetic pigments bacteriochlorophyll *a* and carotenoids are located.

Metabolism obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth with hydrogen, sulfide, S⁰, and thiosulfate as electron donors. During oxidation of sulfide, globules of sulfur are stored inside the cells as intermediary product. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates are photoassimilated.

Mesophilic marine bacteria that require NaCl for growth, optimum growth at neutral pH and 20–35°C.

The mol% G + C of the DNA is: 60.4.

Type species: ***Rhabdochromatium marinum*** Dilling, Liesack and Pfennig 1996, 362 (Effective publication: Dilling, Liesack and Pfennig 1995, 130.)

ENRICHMENT AND ISOLATION PROCEDURES

The media described for *Chromatium* species can also be used for *Rhabdochromatium* and other marine members of the family *Chromatiaceae*, if supplemented with 2–3% NaCl as required. The culture conditions and isolation procedures are the same as described for the genus *Chromatium*.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *RHABDOCHROMATIUM* FROM OTHER GENERA

Characteristics for differentiation of the genus *Rhabdochromatium* from other phototrophic members of the family *Chromatiaceae*

are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*. The genetic relationship of *Rhabdochromatium marinum* to other members of the family *Chromatiaceae*, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

The genus *Rhabdochromatium* was established by Winogradsky (1888) for long, rod- or spindle-shaped, red sulfur bacteria observed in samples from natural habitats. From microscopic observations of reddish cells containing sulfur globules, he described three species that were distinguished by cell size: *R. roseum*, *R. minus*, and *R. fusiforme*. Two more species were described later (see Bavendamm, 1924). Since none of the described species has so far been isolated in pure culture or observed again in samples from nature and enrichment cultures, the genus was not included in the Approved Lists of Bacterial Names. Indeed, this genus was included in the genus *Chromatium* (Pfennig and Trüper, 1974), because long and irregularly swollen cells have been observed in pure cultures of *Chromatium okenii*, *Chromatium weissei*, *Isochromatium buderii*, and *Allochromatium warmingii*, when the mineral salts concentrations of the culture media were inappropriate. The name *Rhabdochromatium* was revived, according to Rule 33c of the International Code of Nomenclature of Bacteria (Lapage et al., 1992), for a long, rod-shaped, red sulfur bacterium, the morphology of which has proven to be stable in pure culture for several years. Only one species is currently known. *Rhabdochromatium marinum* belongs to the cluster of truly marine species of the family *Chromatiaceae*.

FURTHER READING

Dilling, W., Liesack, W. and Pfennig, N. 1995. *Rhabdochromatium marinum* gen. nom. rev., sp. nov., a purple sulfur bacterium from a salt marsh microbial mat. Arch. Microbiol. 164: 125–131.

List of species of the genus Rhabdochromatium

1. **Rhabdochromatium marinum** Dilling, Liesack and Pfennig 1996, 362^{VP} (Effective publication: Dilling, Liesack and Pfennig 1995, 130.)

ma.ri'num. L. neut. adj. *marinum* marine.

Cells straight rods with slightly conical ends, sometimes slightly bent, $1.5\text{--}1.7 \times 16\text{--}32 \mu\text{m}$, motile by a polar tuft of flagella of about 10–20 bipolar fibrils, may be flagellated at both ends. Color of cell suspension beige to orange-brown. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids with lycopene as major component.

Obligately phototrophic and strictly anaerobic. Hydrogen, sulfide, S^0 , and thiosulfate used as electron donors for

photolithoautotrophic growth. In the presence of sulfide and bicarbonate, acetate, propionate, and pyruvate are photoassimilated. Growth factors not required.

Mesophilic marine bacterium requiring NaCl for growth: optimum growth at 1.5–5% NaCl, no growth below 1% and above 6.5%, optimum pH 7.2–7.3, optimum temperature 30°C, no growth below 8°C and above 35°C.

Habitat: laminated microbial mat at Great Sippewissett Salt Marsh, Cape Cod, USA.

The mol% G + C of the DNA is: 60.4 (Bd).

Type strain: 8315, DSM 5261.

GenBank accession number (16S rRNA): X84316.

Genus XI. Thermochromatium Imhoff, Siling and Petri 1998b, 1140^{VP}

JOHANNES F. IMHOFF AND MICHAEL T. MADIGAN

Ther'mo.chro.ma'ti.um. Gr. adj. *thermos* hot; *Chromatium* a genus name; M.L. neut. n. *Thermochromatium* the hot *Chromatium*.

Cells straight to slightly curved rods, multiply by binary fission, **motile**, most likely by polar flagella. Gram negative, belong to the *Gammaproteobacteria*, and contain internal **photosynthetic membranes of vesicular type** in which the photosynthetic pigments bacteriochlorophyll *a* and carotenoids are located.

Strictly anaerobic, obligately phototrophic, and require sulfide-reduced media. **Photolithoautotrophic growth under anoxic conditions in the light with sulfide and S^0 as electron donors**. During oxidation of sulfide, S^0 is transiently stored inside the cells in the form of highly refractile globules. Sulfate is the final oxidation product. In the presence of sulfide and bicarbonate, organic substrates can be photoassimilated.

Thermophilic freshwater bacteria with optimum growth at temperatures higher than 40°C, at neutral pH 7.0, and without a requirement for salt.

Habitat: sulfide-containing hot springs of neutral to alkaline pH having temperatures between 35 and 60°C.

The mol% G + C of the DNA is: 60–63.

Type species: **Thermochromatium tepidum** (Madigan 1986) Imhoff, Siling and Petri 1998b, 1140 (*Chromatium tepidum* Madigan 1986, 226.)

ENRICHMENT AND ISOLATION PROCEDURES

Media and culture conditions are the same as described for *Chromatium* species. Enrichment cultures with conditions suitable for *Chromatiaceae* species in general, in the presence of sulfide and acetate, and incubated at 45–55°C and neutral pH, give rise to the development of *Thermochromatium tepidum*, if present in the inoculum.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS THERMOCHROMATIUM FROM OTHER GENERA

According to the results of 16S rDNA sequence analyses, the genus *Thermochromatium* belongs to the branch of the family *Chro-*

matiaceae containing motile freshwater bacteria, characterized by high temperature optima, above 40°C (*T. tepidum* at 48–50°C) and lack of capability for chemotrophic growth. The type strain of *T. tepidum* has an unusual and quite characteristic absorption spectrum. Intact cells and photosynthetic membrane preparations show a near-infrared absorption peak at 920 nm (Garcia et al., 1986). This absorption is due to a novel light-harvesting complex, and is the longest wavelength-absorbing band observed for a photosynthetic light-harvesting complex containing bacteriochlorophyll *a*. Whether this unusual complex is related to the thermophilic lifestyle of *T. tepidum* is unknown.

Morphological and physiological characteristics for the differentiation of *Thermochromatium* species from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.4 of the chapter describing the family *Chromatiaceae*. The phylogenetic relationship of *Thermochromatium tepidum* to other members of the family *Chromatiaceae*, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

The temperature range for growth of *Thermochromatium tepidum* restricts development of this bacterium to hot springs and other thermal environments with sulfide and light (Madigan, 1984; Castenholz and Pierson, 1995). The successful adaptation to growth at high temperatures was regarded as a taxonomically relevant property of *Thermochromatium tepidum*, which is also phylogenetically distinct from mesophilic freshwater *Chromatiaceae* species. Therefore, *Chromatium tepidum* was transferred to the new genus *Thermochromatium* (Imhoff et al., 1998b).

FURTHER READING

Imhoff, J.F., Siling, J. and Petri, R. 1998. Phylogenetic relationships among the *Chromatiaceae*, their taxonomic reclassification and description of the new genera *Allochromatium*, *Halochromatium*, *Isochromatium*, *Marichromatium*, *Thiococcus*, *Thiohalocapsa*, and *Thermochromatium*. Int. J. Syst. Bacteriol. 48: 1129–1143.

Madigan, M.T. 1986. *Chromatium tepidum* sp. nov., a thermophilic phototrophic bacterium of the family *Chromatiaceae*. Int. J. Syst. Bacteriol. 36: 222–227.

List of species of the genus Thermochromatium

1. **Thermochromatium tepidum** (Madigan 1986) Imhoff, Sül-
ing and Petri 1998b, 1140^{VP} (*Chromatium tepidum* Madigan
1986, 226.)

te' pi.dum. L. neut. adj. *tepidum* lukewarm.

Cells are rod shaped, $1-2 \times 2.8-3.2 \mu\text{m}$, occasionally motile by flagella. Photosynthetic pigments are bacteriochlorophyll *a*, esterified with phytol, and carotenoids of the normal spirilloxanthin group, with rhodovibrin and spirilloxanthin as predominant components.

Strictly anaerobic and obligately phototrophic. Growth occurs photolithoautotrophically in mineral media supplemented with sulfide as electron donor. Hydrogen and thiosulfate not utilized as electron donors. Acetate and pyruvate photoassimilated in the presence of sulfide. Citric

acid cycle intermediates, other than acetate, fatty acids, and sugars not utilized. Ammonia, urea, and glutamine serve as nitrogen sources. No growth factors required. Poly- β -hydroxybutyrate is a storage material.

Thermophilic freshwater bacterium with optimum growth at 48–50°C, no growth below 34°C or above 57°C. Optimum pH is 7.0. NaCl is not required for growth and is inhibitory at concentrations above 1% (w/v).

Habitat: Sulfide-containing hot springs of neutral to alkaline pH at temperatures from 35 to 60°C.

The mol% G + C of the DNA is: 61.5 (T_m).

Type strain: MC, ATCC 43061, DSM 3771.

GenBank accession number (16S rRNA): M59150.

Genus XII. Thioalkalicoccus Bryantseva, Gorlenko, Kompantseva and Imhoff 2000b, 2161^{VP}

JOHANNES F. IMHOFF

Thi.o.al' ka.li.coc' cus. Gr. n. *thios* sulfur; Arab n. *al kali* potash, soda; L. masc. n. *coccus* sphere; M.L. masc. n. *Thioalkalicoccus* sulfur sphere from soda.

Cells are spherical or oval, typically form diplococcus-shaped cells during cell division, multiply by binary fission, and are Gram negative. Internal membranes are of the tubular type. **Photosynthetic pigments are bacteriochlorophyll *b* and carotenoids with absorption properties similar to tetrahydrospirilloxanthin.** The metabolism is strictly anaerobic and obligately phototrophic. During photolithoautotrophic growth with sulfide as electron donor, globules of S^0 are accumulated inside the cytoplasm. The final oxidation product is sulfate. In the presence of sulfide and bicarbonate, organic substrates are photoassimilated. **Mesophilic, obligate alkaliphilic bacterium** with optimum growth at 20–25°C. Optimal development is dependent on sodium salts in low concentrations and on alkaline conditions.

Habitat: Surface of sediments rich in organic matter, and microbial mats of soda lakes containing hydrogen sulfide and exposed to the light.

The mol% G + C of the DNA is: 63.6–64.8.

Type species: Thioalkalicoccus limnaeus Bryantseva, Gorlenko, Kompantseva and Imhoff 2000b, 2162.

DIFFERENTIATION OF THE GENUS *THIOALKALICOCCUS* FROM OTHER GENERA

According to the results of 16S rDNA sequence analyses (Fig. BXII.γ.3) of the chapter describing the family *Chromatiaceae*, the genus *Thioalkalicoccus* belongs to the branch of the family *Chromatiaceae* that characteristically contains marine and halophilic species. Characteristics for differentiation from other phototrophic *Chromatiaceae* species are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*. *Thioalkalicoccus limnaeus* is one of the very few species (in addition to *Thiococcus pfennigii* and *Thioflaviccoccus mobilis*) that contain tubular internal photosynthetic membranes. The phylogenetic relationships of *Thioalkalicoccus* species to other members of the family *Chromatiaceae*, based on 16S rDNA sequence comparison, are shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

List of species of the genus Thioalkalicoccus

1. **Thioalkalicoccus limnaeus** Bryantseva, Gorlenko, Kompantseva and Imhoff 2000b, 2162^{VP}
lim.nae' us. Gr. fem. n. *limne* lake, pond, swamp; Gr. adj. *limnaios* pertaining to, living in lakes, swamps; M.L. masc. adj. *limnaeus* living in lakes and swamps.

Cells are spherical or oval in shape, multiply by binary fission, and are Gram negative. Internal membranes are of the tubular type. Cells are cocci of 1.3–1.8 μm diameter, usually nonmotile, and surrounded by a thin capsule. Occasionally cells with a single flagellum are observed. Internal photosynthetic membranes of tubular type are formed by invagination of the cell membrane and fill most of the cytoplasm. Color of cell suspensions is yellowish to orange-brown. The absorption spectrum of intact cells exhibits maxima at 410, 462, 492, 530, and 1030 nm, with shoulders at 602 and 835 nm. Photosynthetic pigments are bacteriochlorophyll *b* and carotenoids with absorption properties similar to tetrahydrospirilloxanthin.

The metabolism is strictly anaerobic. Photolithoautotrophic growth occurs in the light with hydrogen sulfide and S^0 as electron donors. Thiosulfate was not used for phototrophic growth. During growth with sulfide as electron donor, globules of S^0 are accumulated inside the cells. The final oxidation product is sulfate. In the presence of sulfide and sodium bicarbonate, acetate, yeast extract, malate, propionate, pyruvate, succinate, and fumarate were used as organic substrates for phototrophic growth. Growth factors were not required.

Mesophilic, obligate alkaliphilic bacterium with optimum growth at pH 8.8–9.5 (range from 8–10) and 20–25°C. Development is dependent on sodium salts in low concentrations, and good growth occurs over a broad range of salt concentrations without exhibiting a strong optimum, up to 6% NaCl (in the presence of 0.5% sodium carbonates) and up to 8.5% sodium carbonates (in the presence of 0.05% NaCl).

Habitat: Surface of sediments rich in organic matter, and microbial mats of soda lakes containing hydrogen sulfide and exposed to the light.

The mol% G + C of the DNA is: 64.0–64.5 (T_m).

Type strain: A26, ATCC BAA32.

GenBank accession number (16S rRNA): AJ277023.

Genus XIII. *Thiocapsa* Winogradsky 1888, 84^{AL} emend Guyoneaud, Süling, Petri, Matheron, Caumette, Pfennig and Imhoff 1998b, 962

JOHANNES F. IMHOFF AND PIERRE CAUMETTE

Thi.o.cap'sa. Gr. n. *thios* sulfur; L. n. *capsa* box; M.L. fem. n. *Thiocapsa* sulfur box.

Cells spherical, 1.0–3.0 μ m in diameter, diplococci before multiplication by binary fission, **nonmotile**. Tetrads may be formed after consecutive division in two perpendicular planes. Individual cells surrounded by a strong slime capsule and may contain gas vesicles. Gram negative, belong to the *Gammaproteobacteria*, and have internal **photosynthetic membranes of vesicular type** containing the photosynthetic pigments bacteriochlorophyll *a* and carotenoids.

Photolithotrophic growth under anoxic conditions in the light possible with sulfide, thiosulfate, and sulfur as electron donors. Hydrogen may also be used. During oxidation of sulfide, S^0 is transiently stored inside the cells in the form of highly refractile globules. Sulfate is the final oxidation product. In the presence of sulfide and bicarbonate, simple organic substrates can be photoassimilated. **Some species may grow chemoautotrophically or chemoheterotrophically under microoxic to oxic conditions in the dark.** Storage materials: polysaccharides, poly- β -hydroxybutyrate, S^0 , polyphosphates. Vitamin B₁₂ may be required.

The mol% G + C of the DNA is: 63.3–66.3.

Type species: *Thiocapsa roseopersicina* Winogradsky 1888, 84.

FURTHER DESCRIPTIVE INFORMATION

Under unfavorable growth conditions, *Thiocapsa* species may form aggregates of cells embedded in slime, similar to species of *Thiocystis*, *Allochromatium*, and *Marichromatium*. Depending on culture conditions, the cell diameter of *Thiocapsa* species may vary considerably. Two of the presently known species, *Thiocapsa rosea* and *Thiocapsa pendens*, contain gas vesicles; *Thiocapsa roseopersicina* does not.

Thiocapsa roseopersicina has been found in freshwater, brackish water, and marine habitats, and a characteristic of this species may be the rather indifferent response to low salt concentrations, allowing development and successful competition in freshwater, brackish water, and marine environments. However, comparative sequence analyses indicate that a number of strains isolated from the marine environment and tentatively assigned to this species, could possibly be distinguished at the species level (Guyoneaud et al., 1998b). These bacteria behave indifferently to low salt concentrations and may be regarded as halotolerant, but not halophilic.

ENRICHMENT AND ISOLATION PROCEDURES

The methods are the same as those described for the genus *Chromatium*. *Thiocapsa roseopersicina* is one of the most common purple sulfur bacteria in nature. It is particularly abundant in ponds, pools, or lagoons receiving sewage or wastewater rich in organic matter. Since this species is the least fastidious purple sulfur bacterium known, its isolation presents no problems. The other species of *Thiocapsa*, *Thiocapsa pendens* and *Thiocapsa rosea*, can be isolated directly from samples collected in nature by agar shake dilution cultures, as described for the *Chromatium* species.

The color of colonies in agar is characteristically pinkish. Because of their gas vesicle content, colonies of *Thiocapsa pendens* and *Thiocapsa rosea* maintain a chalky appearance, even if the cells are free of S^0 .

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *THIOCAPSA* FROM OTHER GENERA

Characteristics for differentiation of the genus *Thiocapsa* from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.6 of the chapter describing the family *Chromatiaceae*. *Thiocapsa* cells may form tetrads, and therefore can be mistaken for *Thiopedia rosea* or *Thiolumprobum pediforme*. The cells of these latter species are, however, readily distinguishable from *Thiocapsa* by the formation of platelets, which usually consist of 8, 16, or more regularly arranged cells. Other differential phenotypic traits of these genera are listed in Table BXII.γ.6 of the chapter describing the family *Chromatiaceae*. The phylogenetic relationship of *Thiocapsa* species to other members of the family *Chromatiaceae*, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.4 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

Previously, nonmotile, spherical, purple sulfur bacteria without gas vesicles have been united within the genus *Thiocapsa*. However, the results of genetic analyses indicate that two of the three species placed in this genus (the former *Thiocapsa halophila* and *Thiocapsa pfennigii*) were phylogenetically quite distant from the type species, *Thiocapsa roseopersicina*. Both species belong to the cluster containing truly marine *Chromatiaceae* species Figs. BXII.γ.3 and BXII.γ.4 of the chapter describing the family *Chromatiaceae*. Consequently, they were removed from the genus *Thiocapsa* and transferred to *Thiococcus* and *Thiohalocapsa* as *Thiococcus pfennigii* and *Thiohalocapsa halophila*, respectively (Imhoff et al., 1998b).

Furthermore, *Thiocapsa rosea* and *Thiocapsa pendens*, previously known as *Amoebobacter roseus* and *Amoebobacter pendens* (Pfennig and Trüper, 1971b; Pfennig, 1989a), are closely related genetically to *Thiocapsa roseopersicina* and were included in the genus *Thiocapsa* (Guyoneaud et al., 1998b).

The type species of the genera *Amoebobacter* and *Thiocapsa* were united and therefore a decision concerning the genus name had to be made. Because both genera and their type strains were included in the Approved List and had been legitimately published at the same date (Winogradsky, 1888), none of the genera and species had priority according to the rules of Nomenclature

(rules 23a and 24b). *Thiocapsa* is the better-known genus and the more appropriate name, and *Thiocapsa roseopersicina* one of the best-studied species of purple sulfur bacteria. Therefore, according to rule 42 and recommendation 42 of the International Code of Nomenclature of Bacteria (Lapage et al., 1992), the two bacteria were united within the genus *Thiocapsa*, and *Thiocapsa roseopersicina* retained as the type species (Guyoneaud et al., 1998b). Because the nomenclatural type had been removed from the genus *Amoebobacter*, the names of other *Amoebobacter* species were changed, according to rule 37a (see also genera *Thiolamprovum* and *Lamprocystis*).

DIFFERENTIATION OF THE SPECIES OF THE GENUS *THIOCAPSA*

I. Cells without gas vesicles, capable of assimilatory sulfate reduction, able to use hydrogen as electron donor.

Thiocapsa roseopersicina

II. Cells with gas vesicles, not capable of assimilatory sulfate reduction, not able to use hydrogen as electron donor.

A. Able to grow under microoxic conditions in the dark, not able to use glucose.

Thiocapsa rosea

B. Not able to grow under microoxic conditions, strict anaerobe, able to use glucose.

Thiocapsa pendens

List of species of the genus *Thiocapsa*

1. ***Thiocapsa roseopersicina*** Winogradsky 1888, 84^{AL}
ro.se.o.per.si.ci'na. L. adj. *roseus* rosy; Gr. n. *persicus* the peach; M.L. fem. adj. *roseopersicina* rosy peach-colored.

Cells spherical, 1.2–3.0 µm, usually 1.5 µm in diameter (Fig. BXII.γ.11), and surrounded by a strong slime capsule. Aggregates of two, four, or more cells common; irregular clumps of cells usually surrounded by slime. Gas vesicles absent. Individual cells colorless, color of cell suspensions pink to rose-red. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the normal spirilloxanthin group (Schmidt et al., 1965).

Photolithoautotrophic growth under anoxic conditions in the light with hydrogen, sulfide, thiosulfate, and sulfur as electron donors. Acetate, pyruvate, fumarate, malate, succinate, glycerol, and fructose are photoassimilated. Not utilized: lactate, propionate, butyrate, tartrate, α-oxoglutarate, citrate, benzoate, and alcohols. Chemoautotrophic or chemoheterotrophic growth under microoxic to oxic conditions in the dark is possible. Most strains capable of assim-

ilatory sulfate reduction. Nitrogen sources: ammonium salts. Nitrogenase activity present. Growth factors not required.

Mesophilic bacterium with optimum growth at 20–35°C and pH 7.3 (pH range 6.5–7.5).

Habitat: stagnant water and mud of ponds, pools, or wastewater lagoons containing degradable organic substances and hydrogen sulfide. Also common in estuaries and salt marshes.

The mol % G + C of the DNA is: 63.3–66.3 (Bd); type strain: 65.3 (Bd).

Type strain: 1711, BN 4210, DSM 217.

GenBank accession number (16S rRNA): Y12364.

2. ***Thiocapsa litoralis*** Puchkova, Imhoff and Gorlenko 2000, 1446^{VP}

li.to.ra'lis. L. adj. *litoralis* from the shore.

Cells are spherical and nonmotile, 1.5–2.5 µm in diameter and organized in regular platelets of 4, 8, 16, or more cells. Photosynthetic membranes are of vesicular type. Color of cell suspensions is pink to rose-red. Bacteriochlorophyll *a* and, most probably, pigments of the spirilloxanthin series are present as photosynthetic pigments.

Phototrophic growth occurs under anoxic conditions in the light and chemoautotrophic growth is possible under microoxic conditions in the dark. Electron donors used for photoautotrophic growth are hydrogen sulfide, elemental sulfur, thiosulfate, and sulfite. Globules of elemental sulfur are stored inside the cells as an intermediary product. In the presence of sulfide and bicarbonate, acetate, propionate, butyrate, valerate, pyruvate, lactate, malate, succinate, fumarate, fructose, glucose, Casamino acids, and yeast extract are photoassimilated. Caprylate, tartrate, formate, ascorbate, asparaginate, benzoate, malonate, citrate, maltose, galactose, mannitol, sorbitol, glycerol, methanol, and ethanol not utilized. Assimilatory sulfate reduction occurs. Chemolithoautotrophic growth occurs with hydrogen sulfide and thiosulfate as electron donors. Vitamin B₁₂ is required as growth factor. Storage materials are poly-β-hydroxybutyrate and polyphosphates. Mesophilic bacterium with optimum growth at pH 6.5, 30°C, and 1% NaCl.

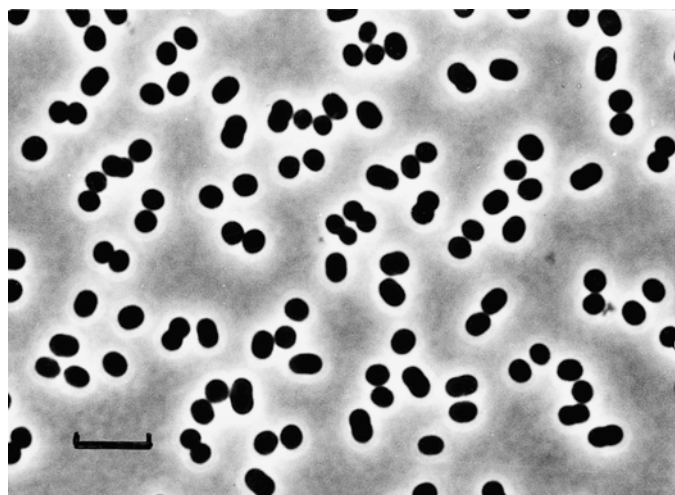


FIGURE BXII.γ.11. *Thiocapsa roseopersicina* DSM 219 cultured photoautotrophically with sulfide. Phase-contrast micrograph. Bar = 5 µm.

Habitat: Microbial mat communities of the White Sea littoral shore.

The mol% G + C of the DNA is: 64.0 (T_m).

Type strain: BM5, ATCC 700894.

GenBank accession number (16S rRNA): AJ24772.

3. **Thiocapsa pendens** (Molisch 1906) Guyoneaud, Söling, Petri, Matheron, Caumette, Pfennig and Imhoff 1998b, 962^{VP} (*Amoebobacter pendens* (Molisch 1906) Pfennig and Trüper 1971b, 13; *Rhodotherce pendens* Molisch 1906, 230 .) *pen' dens*. L. part. adj. *pendens* hanging.

Cells spherical, 1.5–2.0 µm in diameter, sometimes up to 2.5 µm in diameter. Individual cells surrounded by slime capsules, which give rise to irregular cell aggregates of different size. Because of copious slime formation, the culture medium becomes viscous. Gas vesicles present. Individual cells colorless, color of cell suspensions pink to rose-red. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the spirilloxanthin group, with spirilloxanthin as the dominant component.

Strictly anaerobic and obligately phototrophic. No growth under microoxic to oxic conditions in the dark. Photolithoautotrophic growth under anoxic conditions in the light occurs with sulfide, thiosulfate, sulfite, and sulfur as electron donors. Acetate, pyruvate, propionate, lactate, malate, and glucose are the only organic substrates photoassimilated. Assimilatory sulfate reduction is absent. Nitrogen sources: ammonium salts. Vitamin B₁₂ required for growth.

Mesophilic freshwater bacterium with optimum growth at 20–35°C, pH range 6.7–7.5.

Habitat: stagnant freshwater and mud containing sulfide and exposed to the light.

The mol% G + C of the DNA is: 65.3 (Bd).

Type strain: DSM 236 (neotype strain).

GenBank accession number (16S rRNA): AJ002797.

4. **Thiocapsa rosea** (Winogradsky 1888) Guyoneaud, Söling, Petri, Matheron, Caumette, Pfennig and Imhoff 1998b, 962^{VP} (*Amoebobacter roseus* Winogradsky 1888, 77.) *ro' se.a*. L. fem. adj. *rosea* rosy, rose-colored, pink.

Cells spherical, 2.0–3.0 µm in diameter. Individual cells and irregular cell aggregates surrounded by slime. Gas vesicles present. Individual cells colorless, color of cell suspensions pink to rose-red. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the spirilloxanthin group, with spirilloxanthin as the major component.

Photolithoautotrophic growth under anoxic conditions in the light occurs with sulfide, thiosulfate, and sulfur as electron donors. Acetate, pyruvate, propionate, lactate, malate, fructose, and Casamino acids are photoassimilated. Not utilized: other sugars, alcohols, other fatty acids, succinate, tartrate, benzoate, or citrate. Chemoautotrophic or mixotrophic growth under microoxic to oxic conditions in the dark is possible. Assimilatory sulfate reduction does not occur. Nitrogen source, ammonia. Vitamin B₁₂ required.

Mesophilic freshwater bacterium with optimum growth at 20–35°C, pH range 6.7–7.5.

The mol% G + C of the DNA is: 64.3 (Bd).

Type strain: DSM 235.

GenBank accession number (16S rRNA): AJ002798, AJ006062.

Genus XIV. **Thiococcus** Imhoff, Söling and Petri 1998b, 1139^{VP}

JOHANNES F. IMHOFF

Thi.o.coc' cus. Gr. n. *thios* sulfur; L. masc. n. *coccus* sphere; M.L. masc. n. *Thiococcus* sphere with sulfur.

Cells spherical, with diplococcus-shaped division stages, **non-motile**, multiply by binary fission. Gram negative, belong to the *Gammaproteobacteria*, and contain internal **photosynthetic membranes consisting of bundles of ribbon-like branched tubes** continuous with the cytoplasmic membrane. Individual cells colorless, color of cell suspensions is yellowish to orange-brown. Photosynthetic pigments are bacteriochlorophyll *b* and carotenoids.

Obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth under anoxic conditions in the light with sulfide and S⁰ as electron donors. During oxidation of sulfide, S⁰ transiently stored inside the cells in the form of highly refractile globules. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates photoassimilated.

Mesophilic bacteria with optimum growth at 20–35°C and neutral pH (pH range 6.5–7.5). May require salt for optimum growth.

The mol% G + C of the DNA is: 69.4–69.9.

Type species: **Thiococcus pfennigii** (Eimhjellen 1970) Imhoff, Söling and Petri 1998b, 1139 (*Thiocapsa pfennigii* Eimhjellen 1970, 193.)

ENRICHMENT AND ISOLATION PROCEDURES

Thiococcus pfennigii is the only species of the family *Chromatiaceae* that contains bacteriochlorophyll *b* described so far. Owing to the presence of this bacteriochlorophyll, the bacterium has an

in vivo absorption maximum in the infrared region of the spectrum, between 1020 nm and 1030 nm. By use of selective filters, this specific absorption has been successfully applied for the selective enrichment culture of *Thiococcus* species (Eimhjellen et al., 1967; Eimhjellen, 1970). Otherwise, the culture conditions and isolation procedures are the same as described for the genus *Chromatium*. The media described for *Chromatium* species can also be used for *Thiococcus* species, if supplemented with 1–3% NaCl as required.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *THIOCOCCUS* FROM OTHER GENERA

Characteristics for differentiation of the genus *Thiococcus* from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*. The phylogenetic relatedness of *Thiococcus pfennigii* to other members of the family *Chromatiaceae*, based on 16S rDNA analysis, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

The genus name *Thiococcus* was proposed for this bacterium by Eimhjellen et al. (1967). However, this name was never validly published and later the bacterium was validly named *Thiocapsa pfennigii* (Eimhjellen, 1970; Approved List of Bacterial Names, Skerman et al., 1980). Because of the striking genetic and phenotypic difference between *Thiococcus pfennigii* and members of the genus *Thiocapsa*, it was transferred to a new genus and the name *Thiococcus* was revived (Imhoff et al., 1998b). A possible candidate for the neotype strain of *Thiococcus pfennigii* (strain DSM 226, Trüper 8013, Imhoff 4250) belongs to the cluster of truly marine species of the family *Chromatiaceae* (Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*) and requires at least 1% NaCl for growth. The salt dependence and natural distribution of this species are poorly understood. The original type strain was isolated from a freshwater source (river water), but

unfortunately was lost. Additional strains were isolated later from freshwater and marine sources. However, chemotaxonomic and genetic analyses have not been performed on these strains. In addition, salt dependence has not yet been determined for all of the isolates.

FURTHER READING

- Eimhjellen, K.E. 1970. *Thiocapsa pfennigii* sp. nov., a new species of phototrophic sulfur bacteria. Arch. Microbiol. 73: 193–194.
- Eimhjellen, K.E., Steensland, H. and Traetteberg, H. 1967. A *Thiococcus* sp. nov. gen., its pigments and internal membrane system. Arch. Microbiol. 59: 82–92.
- Imhoff, J.F., Söling, J. and Petri, R. 1998. Phylogenetic relationships among the *Chromatiaceae*, their taxonomic reclassification and description of the new genera *Allochro-matium*, *Halochromatium*, *Isochromatium*, *Marichromatium*, *Thiococcus*, *Thiohalocapsa*, and *Thermochromatium*. Int. J. Syst. Bacteriol. 48: 1129–1143.

List of species of the genus *Thiococcus*

1. ***Thiococcus pfennigii*** (Eimhjellen 1970) Imhoff, Söling and Petri 1998b, 1139^{VP} (*Thiocapsa pfennigii* Eimhjellen 1970, 193.)

pfennigii n. sp. M.L. gen. n. *pfennigii* of Pfennig, named after N. Pfennig, a German microbiologist.

Cells spherical, 1.2–1.5 µm in diameter; diplococcus-shaped division stages about 2.5 µm long; stationary phase cells 0.8–1.0 µm in diameter. Internal photosynthetic membranes are bundles of ribbon-like branched tubes continuous with the cytoplasmic membrane. Individual cells colorless, color of cell suspensions yellowish to orange-brown. Photosynthetic pigments are bacteriochlorophyll *b* and the carotenoid 3,4,3',4'-tetrahydrospirilloxanthin (Eimhjellen et al., 1967).

Obligately phototrophic and strictly anaerobic. Photo-

lithoautotrophic growth with sulfide and S⁰ as electron donors. In the presence of sulfide and bicarbonate, acetate, pyruvate, propionate, lactate, fumarate, succinate, malate, and fructose are photoassimilated. Not utilized: thiosulfate and butyrate. Nitrogen sources: ammonium salts and N₂.

Mesophilic bacterium with optimum growth at 0.5–2% NaCl, 20–35°C, and pH 7.0 (range 6.5–7.5).

Habitat: freshwater, brackish, and marine waters and sediments with hydrogen sulfide and exposed to the light.

The mol% G + C of the DNA is: 69.4–69.9 (Bd).

Type strain: DSM 1375 (RG3, Nidelven) is lost; no neotype strain is designated.

GenBank accession number (16S rRNA): Y12373 (strain DSM 226).

Additional Remarks: The GenBank accession number for strain BN 4254 (Pfennig 8320) is AJ010125.

Genus XV. *Thiocystis* Winogradsky 1888, 60^{AL} emend. Imhoff, Söling and Petri 1998b, 1138

JOHANNES F. IMHOFF

Thi.o.cys'tis. Gr. n. *thios* sulfur; Gr. n. *cystis* the bladder, a bag; M.L. fem. n. *Thiocystis* sulfur bag.

Cells are rod shaped or ovoid to spherical, before cell division often diplococcus-shaped, motile by flagella, occurring singly or in pairs; may grow in irregular aggregates surrounded by slime, multiply by binary fission. Gram negative, belong to the *Gammaproteobacteria*, and contain internal photosynthetic membranes of vesicular type in which the photosynthetic pigments bacteriochlorophyll *a* and carotenoids are located.

Photolithotrophic growth occurs under anoxic conditions in the light with sulfide and S⁰ as electron donors. Thiosulfate and hydrogen may also be used. During oxidation of sulfide, S⁰ is transiently stored inside the cells in the form of highly refractile globules. Sulfate is the final oxidation product. In the presence of sulfide and bicarbonate, simple organic substrates can be photoassimilated. Some species grow chemoautotrophically or chemoorganotrophically under microoxic to oxic conditions in the dark. Storage materials: polysaccharides, poly-β-hydroxybutyrate, S⁰, and polyphosphates.

Ubiquitous, mesophilic bacteria with optimum growth at 25–35°C and pH 6.5–7.6, that can be isolated from freshwater, brackish water, and marine environments.

Habitat: anoxic sulfide-containing water and mud of fresh-

water ponds and lakes, as well as brackish water and marine environments exposed to light.

The mol% G + C of the DNA is: 61.3–67.9.

Type species: *Thiocystis violacea* Winogradsky 1888, 65.

FURTHER DESCRIPTIVE INFORMATION

Species of the genus *Thiocystis* are frequently found in the same natural habitats as *Allochro-matium* species. Bacteria of both genera share many physiological properties. This is true not only for the phototrophic metabolism of reduced sulfur compounds and organic substrates, but also for facultatively chemoautotrophic or chemoorganotrophic metabolism under microoxic to oxic conditions in the dark (Kämpf and Pfennig, 1980), which occurs in some species of this genus.

ENRICHMENT AND ISOLATION PROCEDURES

Media and culture conditions are the same as described for *Chromatium* species.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *THIOCYSTIS* FROM OTHER GENERA

According to the results of 16S rDNA sequence analyses, the genus *Thiocystis* belongs to the branch of the family *Chromatiaceae* characterized by motile freshwater species. Characteristics for differentiation of *Thiocystis* species from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.4 of the chapter describing the family *Chromatiaceae*. The phylogenetic relationship of *Thiocystis* species to other *Chromatiaceae* species, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

Previously, the genus *Thiocystis* was distinguished from other genera of the family *Chromatiaceae* based on its spherical morphology, flagellar motility, and lack of gas vesicles (Pfennig and Trüper, 1989). Winogradsky (1888) distinguished *Thiothece* from *Thiocystis*

as a second genus of spherical, motile, purple sulfur bacteria, based on the location of the intracellular sulfur globules. In the monotypic genus *Thiothece*, the sulfur globules were located at the inner periphery of the cells, whereas in *Thiocystis* the globules were distributed at random in the cell (Winogradsky, 1888). Pfennig and Trüper (1971b) included *Thiothece gelatinosa* Winogradsky 1888 in the genus *Thiocystis* because they considered the maintenance of *Thiothece*, in addition to *Thiocystis*, unjustified.

Similarity studies on the 16S rRNA oligonucleotide catalogs of *Chromatiaceae* species (Fowler et al., 1984) and complete 16S rDNA sequence analyses (Imhoff et al., 1998b) not only substantiated the close relationship of *Thiocystis gelatinosa* and *Thiocystis violacea* (see Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*), but also revealed a high degree of genetic relatedness between these species and *Thiocystis minor* (formerly *Chromatium minus*) and *Thiocystis violascens* (formerly *Chromatium violascens*) (Imhoff et al., 1998b).

DIFFERENTIATION OF THE SPECIES OF THE GENUS *THIOCYSTIS*

Characteristics for diagnosis of *Thiocystis* species are shown in Table BXII.γ.4 of the chapter describing the family *Chromatiaceae*.

I. Cells spherical.

- A. Sulfur globules appear randomly distributed within the cell, cells contain carotenoids of the rhodopinal group, and cultures are purple-violet.

Thiocystis violacea

- B. Sulfur globules occur only at the inner periphery of the

cell, cells contain carotenoids of the okenone group, and cultures are purple-red.

Thiocystis gelatinosa

II. Cells rod shaped.

- A. Carotenoids of the okenone group are present; only acetate, pyruvate, and glucose are utilized.

Thiocystis minor

- B. Carotenoids of the rhodopinal group are present; hydrogen, but no sugars are utilized.

Thiocystis violascens

List of species of the genus *Thiocystis*

1. ***Thiocystis violacea*** Winogradsky 1888, 65^{AL}
vi.o.la'ce.a. L. fem. adj. *violacea* violet-colored.

Cells spherical and about 2.5–3 μm in diameter. Depending on the culture conditions, larger individual cells may occur. Irregular cell aggregates surrounded by slime are formed under unfavorable conditions. Sulfur globules appear randomly distributed within the cell (Fig. BXII.γ.12). Color of individual cells grayish, color of cell suspensions purple-violet. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the rhodopinal group (Schmidt et al., 1965).

Photoautotrophic growth occurs under anoxic conditions in the light with sulfide, sulfur, thiosulfate, sulfite (used by some strains), and molecular hydrogen as electron donors. In the presence of sulfide and bicarbonate, acetate, pyruvate, and fumarate are photoassimilated. In addition, some strains use propionate, succinate, oxoglutarate, glucose, fructose, propanol, or glycerol. Chemoautotrophic or mixotrophic growth possible under microoxic to oxic conditions in the dark. Some strains capable of assimilatory sulfate reduction. Nitrogen sources: ammonium salts, urea, and N₂.

Mesophilic bacterium with optimum growth at 25–35°C and pH 7.0–7.3 (range 6.5–7.6). Marine isolates may tolerate or require low concentrations of NaCl.

Habitat: ponds and lakes with stagnant freshwater, brackish water, or seawater containing hydrogen sulfide; sewage lagoons, estuaries, salt marshes, and sulfur springs.

The mol% G + C of the DNA is: 62.8–67.9 (Bd); type strain 63.1 (Bd).

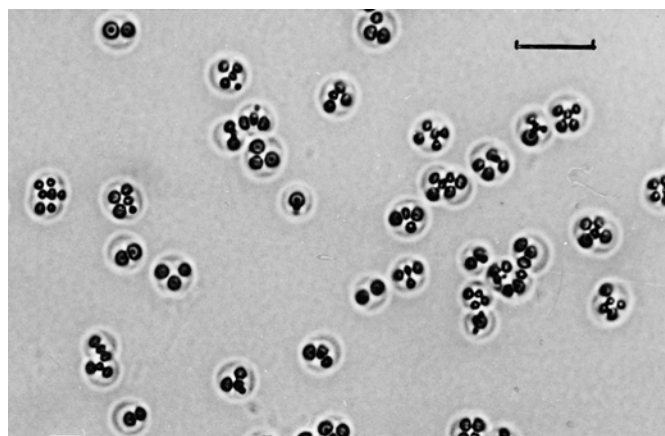


FIGURE BXII.γ.12. *Thiocystis violacea* DSM 207 cultured photoautotrophically with sulfide. The cells contain sulfur globules. Brightfield micrograph. Bar = 5 μm.

Type strain: 2711, BN 4340, DSM 207.

GenBank accession number (16S rRNA): Y11315.

2. ***Thiocystis gelatinosa*** (Winogradsky 1888) Pfennig and Trüper 1971b, 11^{AL} (*Thiothece gelatinosa* Winogradsky 1888, 82.) *ge.la.ti.no'sa.* L. part. adj. *gelatus* frozen, stiffened; M.L. n. *gelatinum* that which stiffens; M.L. fem. adj. *gelatinosa* gelatinous.

Cells spherical and about 3 μm in diameter. Under unfavorable conditions, elongated ovoid cells may occur. At

high sulfide concentration (4–6 mM) and light intensity (>1000 lux), cells become immobile, growing in irregular aggregates surrounded by slime. Under optimum growth conditions, globules of S^0 occur only in the peripheral part of the cytoplasm (Fig. BXII.γ.13). Color of individual cells slightly pink, color of cell suspensions purple-red. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the okenone group (Pfennig et al., 1968).

Photolithoautotrophic growth under anoxic conditions in the light with sulfide and sulfur as electron donors. In the presence of sulfide and bicarbonate, acetate and pyruvate are photoassimilated. Chemoautotrophic or mixotrophic growth is possible under microoxic to semioxic conditions. Nitrogen sources: ammonium salts.

Mesophilic bacterium with optimum growth at 30°C and pH 7.0–7.3 (range 6.5–7.6). The type strain requires 1% NaCl.

Habitat: stagnant water containing hydrogen sulfide, hypolimnion of meromictic lakes.

The mol% G + C of the DNA is: 61.3 (Bd).

Type strain: 2611, BN 4310, DSM 215.

GenBank accession number (16S rRNA): Y11317.

3. *Thiocystis minor* (Winogradsky 1888) Imhoff, Söling and Petri 1998b, 1139^{VP} (*Chromatium minus* Winogradsky 1888, 99.)

mi' nor. L. fem. comp. adj. *minor* less, smaller.

Cells are rod shaped, ~2 × 2.5–6 μm, with polar flagella.

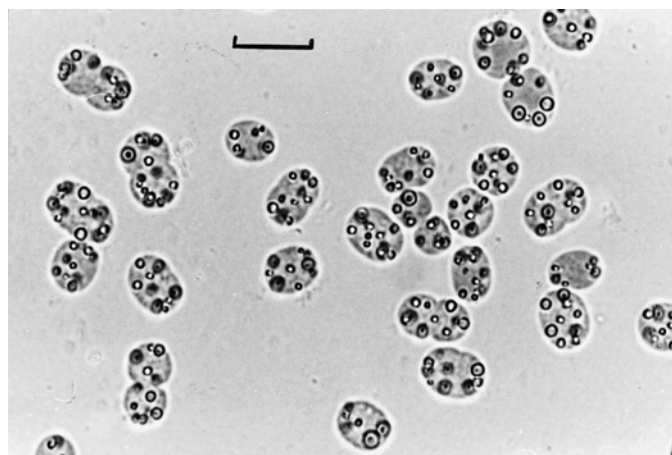


FIGURE BXII.γ.13. *Thiocystis gelatinosa* DSM 215 cultured photoautotrophically with sulfide. The peripheral deposition of the sulfur globules is clearly visible in some of the cells. Brightfield micrograph. Bar = 5 μm.

Globules of S^0 appear evenly distributed within the cell. Single cells colorless to pinkish, color of cell suspensions purple-red. Photosynthetic pigments are bacteriochlorophyll *a* and the carotenoid okenone.

Photolithoautotrophic growth under anoxic conditions in the light with sulfide, thiosulfate, and sulfur as electron donors. In the presence of sulfide and bicarbonate, acetate, pyruvate, and glucose are photoassimilated. Not utilized: sugar-alcohols, alcohols, higher fatty acids, amino acids, benzoate, formate, and most intermediates of the tricarboxylic acid cycle (Thiele, 1968). Chemoautotrophic or mixotrophic growth possible under microoxic to semioxic conditions (Kämpf and Pfennig, 1980). Assimilatory sulfate reduction is absent. Nitrogen sources: ammonium salts and N_2 .

Mesophilic freshwater bacterium with optimum growth at 30°C and pH 7.0–7.3 (pH range 6.5–7.6).

Habitat: freshwater ponds and lakes containing hydrogen sulfide and exposed to light.

The mol% G + C of the DNA is: 62.2 (Bd).

Type strain: 1711, BN 5710, DSM 178.

GenBank accession number (16S rRNA): Y12372.

4. *Thiocystis violascens* (Perty 1852) Imhoff, Söling and Petri 1998b, 1139^{VP} (*Chromatium violascens* Perty 1852, 174.) *vi.o.la' scens*. L. part. *violascens* becoming violet.

Cells are rod shaped, ~2 × 2.5–6.0 μm, occasionally longer. Color of cell suspensions purple-violet. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the rhodopinal group.

Phototrophic growth under anoxic conditions in the light with sulfide, sulfur, thiosulfate, sulfite, H_2 , formate, acetate, propionate, and pyruvate as electron donors. Some strains utilize butyrate. Not utilized: sugars, sugar alcohols, alcohols, benzoate, citrate, and amino acids. Chemoautotrophic or mixotrophic growth is possible under microoxic to semioxic conditions (Kämpf and Pfennig, 1980). Assimilatory sulfate reduction present (Thiele, 1968). Nitrogen sources: ammonium salts and N_2 .

Mesophilic bacterium with optimum growth at 25–35°C and pH 7.0–7.3 (range 6.5–7.6). Marine isolates may require up to 2% NaCl.

Habitat: ponds and lakes with stagnant freshwater, sewage lagoons, brackish waters, estuaries, salt marshes, and marine habitats containing hydrogen sulfide and exposed to the light.

The mol% G + C of the DNA is: 61.8–64.3 (Bd); type strain: 62.2 (Bd).

Type strain: 6111, BN 5410, ATCC 17096, DSM 198.

GenBank accession number (16S rRNA): AJ224438.

Genus XVI. *Thiodictyon* Winogradsky 1888, 80^{AL}

JOHANNES F. IMHOFF

Thi'o.dic'tyon. Gr. n. *thios* sulfur; Gr. n. *dictyon* a net; M.L. neut. n. *Thiodictyon* sulfur net.

Cells are rod shaped with rounded ends, sometimes appearing spindle-shaped, multiply by binary fission, **nonmotile under all conditions**, may form aggregates in which the cells are arranged end-to-end in an irregular netlike structure, the shape of which is variable. May also form clumps that are more compact or break up into individual cells. Contain large, **irregularly shaped gas**

vesicles in the central part of the cell. Gram negative, belong to the *Gamma*proteobacteria, and contain internal **photosynthetic membranes of vesicular type** with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth under anoxic conditions in the light with sulfide

and S^0 as electron donors. During oxidation of sulfide, S^0 is transiently stored in the peripheral part of the cell that is free of gas vesicles. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates are photoassimilated.

Habitat: mud and stagnant water of ponds and lakes containing hydrogen sulfide; common as planktonic bacterium in the sulfide-containing hypolimnion of freshwater lakes.

The mol% G + C of the DNA is: 65.3–66.3.

Type species: **Thiodictyon elegans** Winogradsky 1888, 82.

ENRICHMENT AND ISOLATION PROCEDURES

Media and culture conditions are the same as described for *Chromatium* species. Prerequisite for the successful enrichment and isolation of *Thiodictyon* from anoxic water or mud samples is the microscopic detection of the bacterium in the particular sample. Since in its natural habitats *Thiodictyon elegans* usually grows in the form of readily recognizable netlike cell aggregates, recognition of this bacterium is relatively easy.

For the isolation of pure cultures, it is advisable to prepare both liquid enrichment cultures and agar shake dilution cultures that are directly inoculated from the sample of the natural habitat. In liquid enrichment cultures, *Thiodictyon* may be outgrown by other purple sulfur bacteria containing gas vesicles. The culture medium and incubation conditions are the same as those described for *Lamprocystis* species.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *THIODICTYON*

- I. Cells able to grow in form of typical netlike aggregates under certain culture conditions.

Thiodictyon elegans

- II. Cells that do not form netlike aggregates, usually growing in the form of free single cells.

Thiodictyon bacillosum

List of species of the genus Thiodictyon

1. **Thiodictyon elegans** Winogradsky 1888, 82^{AL}.
e' le.gans. L. adj. *elegans* choice, elegant.

Cells are rod shaped, $1.5\text{--}2.0 \times 3\text{--}8 \mu\text{m}$, grow in the form of irregular netlike aggregates, dependent on the sulfide concentration and light intensity (Fig. BXII.γ.14). In cultures of higher population densities, cell chains break apart and the cultures contain predominantly single cells (Fig. BXII.γ.15). Color of single cells grayish, color of cell suspensions light violet to purple-violet. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the rhodopinal group, with rhodopinal and rhodopin as major components (Pfennig et al., 1968).

Obligately phototrophic and strictly anaerobic. Photo-lithoautotrophic growth under anoxic conditions in the light with sulfide and S^0 as electron donors. During oxidation of sulfide, S^0 is transiently stored in the peripheral part of the cell that is free of gas vesicles. In the presence of sulfide and bicarbonate, acetate and pyruvate are photoassimilated. Nitrogen sources: ammonium salts.

Mesophilic freshwater bacterium with optimum growth at 20–25°C, pH range 6.7–7.3.

The mol% G + C of the DNA is: 65.3–66.3 (Bd); type strain 65.4 (Bd).

Type strain: 3011, DSM 232.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *THIODICTYON* FROM OTHER GENERA

Characteristics used for differentiation of the genus *Thiodictyon* from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.6 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

The genus *Thiodictyon* presently comprises two species: *T. elegans*, the type species of the genus, and *T. bacillosum*. The latter species was originally described by Winogradsky (1888) as *Amoebobacter bacillosus*. Pfennig and Trüper (1971b) transferred *Amoebobacter bacillosus* to the genus *Thiodictyon*, because the rod-shaped cells are very similar to those of *Thiodictyon elegans*. In addition, according to the original description of the type species of the genus *Amoebobacter* (*Amoebobacter roseus* Winogradsky 1888), this genus should comprise only spherical, nonmotile species with gas vesicles. According to 16S rRNA oligonucleotide pattern analysis, *Thiodictyon elegans* is related to *Thiocapsa* species, including *Thiocapsa pendens* and *Thiocapsa rosea* (Fowler et al., 1984). Studies on 16S rDNA sequence similarities of *Thiodictyon* species are expected to confirm this relationship. Because of the lack of genetic sequence information, *Thiodictyon* species have not been included in a recent reclassification of *Chromatiaceae* species (Guyonaeud et al., 1998b; Imhoff et al., 1998b).

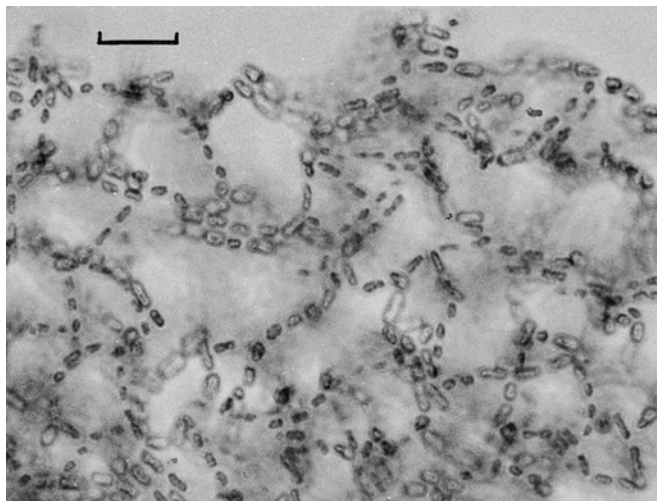


FIGURE BXII.γ.14. *Thiodictyon elegans* DSM 232 showing the typical, somewhat irregular netlike arrangement of the cells. Brightfield micrograph. Bar = 12 μm .

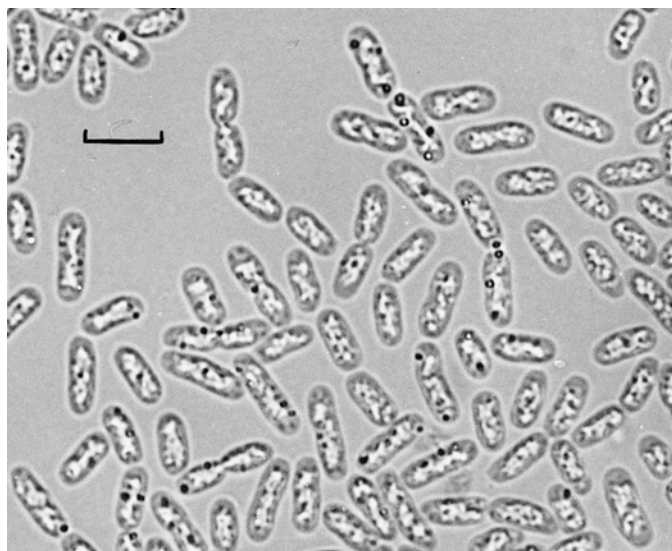


FIGURE BXII.γ.15. *Thiodictyon elegans* DSM 232 cultured photoautotrophically with sulfide. The irregular whitish areas within the cells are the gas vacuoles. The cells also contain small, blackish-appearing globules of S⁰. Brightfield micrograph. Bar = 5 μm.

2. **Thiodictyon bacillosum** (Winogradsky 1888) Pfennig and Trüper 1971b, 12^{AL} (*Amoebobacter bacillosum* Winogradsky 1888, 78.)

ba.cil.lo'sum. L. dim. n. *bacillus* a small rod; M.L. neut. adj. *bacillosum* full of or made up of small rods.

Cells are rod shaped, 1.5–2.0 × 3–6 μm, may grow in irregular clumps surrounded by slime. Netlike aggregates not formed. Color of cell suspensions light violet to purple-violet. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the rhodopinal group, with rhodopinal as major component (Schmidt et al., 1965).

In the presence of sulfide and bicarbonate, acetate and pyruvate are photoassimilated. Nitrogen sources: ammonium salts.

Mesophilic freshwater bacterium with optimum growth at 20–30°C, pH range 6.7–7.3.

The mol% G + C of the DNA is: 66.3 (Bd).

Type strain: 1814, DSM 234.

Genus XVII. **Thioflaviccoccus** Imhoff and Pfennig 2001, 109^{VP}

JOHANNES F. IMHOFF

Th.io fla'vi.coc'cus. Gr. n. *thios* sulfur; L. masc. adj. *flavus* golden-yellow, beige; L. masc. n. *coccus* sphere; M.L. masc. n. *Thioflaviccoccus* beige-yellow coccus with sulfur.

Cells are spherical with diplococcus-shaped division stages, motile by flagella, multiply by binary fission. Gram negative; belong to the *Gammaproteobacteria*. Contain internal photosynthetic membranes of tubular shape. Photosynthetic pigments are bacteriochlorophyll *b* and carotenoids.

Obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth under anoxic conditions in the light with sulfide and elemental sulfur as electron donors. During oxidation of sulfide, elemental sulfur transiently stored inside the cells in the form of highly refractile globules. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates photoassimilated.

Mesophilic bacteria growing well at 20–30°C and neutral pH (pH range 6.5–7.5) and **requiring sodium chloride for optimum growth**.

The mol% G + C of the DNA is: 66.5.

Type species: **Thioflaviccoccus mobilis** Imhoff and Pfennig 2001, 109.

DIFFERENTIATION OF THE GENUS *THIOFLAVICOCCUS* FROM OTHER GENERA

According to the results of 16S rDNA sequence analyses (Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*), the genus *Thioflaviccoccus* belongs to the branch of the family *Chromatiaceae* that characteristically contains marine and halophilic species. Characteristics for differentiation from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*. *Thioflaviccoccus mobilis* is one of the very few species (in addition to *Thiococcus pfennigii* and *Thioalkalicoccus limmaeus*) that contain tubular internal photosynthetic membranes. The phylogenetic relationship of the *Thioflaviccoccus* species to other *Chromatiaceae* species, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

List of species of the genus *Thioflaviccoccus*

1. **Thioflaviccoccus mobilis** Imhoff and Pfennig 2001, 109^{VP}
mo'bi.lis. L. adj. *mobilis* mobile.

Cells are coccoid, short, rod-shaped to diplococcus-shaped before cell division. Cocci 0.8–1.0 μm in diameter. Motile by monopolar monotrichous flagella. Internal photosynthetic membrane system of tubular type. Color of cell suspensions is yellowish beige to orange-brown. The absorption spectrum of living cell suspensions exhibits maxima at 410, 462, 492, 530, and 1025 nm, with shoulders at 602 and 835 nm. Photosynthetic pigments are bacterio-

chlorophyll *b* and 3,4,3',4'-tetrahydro-spirilloxanthin as the main carotenoid. The metabolism is obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth occurs in the light with hydrogen sulfide as electron donor. Globules of elemental sulfur are accumulated inside the cells. The final oxidation product is sulfate. Thiosulfate is not used. In the presence of sulfide and bicarbonate, acetate, pyruvate, and ascorbate are used as organic substrates. Growth factors are not required. Sodium chloride is required for growth. Good growth occurs at 25–30°C, pH

7.2–7.4, with 1–3% NaCl (optimum is at 2%), and at 500 lux light intensity from a tungsten lamp. Habitat: laminated microbial mats of salt marshes.

The mol% G + C of the DNA is: 66.5 (Bd).

Type strain: 8321, ATCC 700959.

GenBank accession number (16S rRNA): AJ010126.

Genus XVIII. *Thiohalocapsa* Imhoff, Süling and Petri 1998b, 1139^{VP}

JOHANNES F. IMHOFF AND PIERRE CAUMETTE

Thi'o.ha.lo.cap'sa. Gr. n. *thios* sulfur; Gr. gen. n. *halos* of the salt; L. fem. n. *capsa* capsule; M.L. fem. n.

Thiohalocapsa the sulfur capsule of the salt.

Cells spherical, nonmotile, multiply by binary fission. Individual cells surrounded by a strong slime capsule and cell aggregates formed. Gram negative, belong to the *Gammaproteobacteria*, and contain internal **photosynthetic membranes of vesicular type** with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Photolithoautotrophic growth under anoxic conditions in the light with sulfide, S⁰, and possibly other electron donors. During oxidation of sulfide, S⁰ transiently stored inside the cells in the form of highly refractile globules. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates are photoassimilated. **Chemolithotrophic or chemoorganotrophic growth is possible under microoxic to oxic conditions in the dark.**

Mesophilic bacteria with optimum growth at 20–30°C and pH 6.5–7.5. Elevated concentrations of sodium chloride required for growth.

Habitat: Sulfide-containing waters and sediments of saltern and other coastal habitats with elevated salt concentrations and exposed to the light.

The mol% G + C of the DNA is: 65.9–66.6.

Type species: *Thiohalocapsa halophila* (Caumette, Baulaigue and Matheron 1991) Imhoff, Süling and Petri 1998b, 1139 (*Thiocapsa halophila* Caumette, Baulaigue and Matheron 1991, 175.)

ENRICHMENT AND ISOLATION PROCEDURES

Methods for enrichment and isolation are the same as those described for the genus *Chromatium*. *Thiohalocapsa* species often form dense purple layers in microbial mats of hypersaline environments with shallow waters at salinities from 10 to 20‰, such as salterns, shallow salt lakes, and evaporitic lagoons. They are also found at the surface of sediments below top layers of salt deposits. *Thiohalocapsa* species can be enriched from samples of the natural environment using media composed of natural brines, as described by Caumette et al. (1991). They can be purified by repeated deep-agar dilution series. After achievement of pure cultures, a single colony is transferred into synthetic media containing 7–10% NaCl (Caumette et al., 1991).

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*. Maintenance is easier in media composed of natural brine than in synthetic media.

DIFFERENTIATION OF THE GENUS *THIOHALOCAPSA* FROM OTHER GENERA

Characteristics for differentiation of the genus *Thiohalocapsa* from other members of the family *Chromatiaceae* are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*. The genetic relationship of *Thiohalocapsa halophila* to other *Chromatiaceae* species, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

Thiohalocapsa halophila was originally described as a *Thiocapsa* species (Caumette et al., 1991), because its phenotypic properties were in agreement with this genus classification, according to Pfennig and Trüper (1989). Cells are spherical and nonmotile purple sulfur bacteria. Because phylogenetic relationships and salt response clearly distinguish *Thiohalocapsa halophila* from species of the genus *Thiocapsa* (Guyoneaud et al., 1998b; Imhoff et al., 1998b), this bacterium was transferred to the new genus *Thiohalocapsa* (Imhoff et al., 1998b).

FURTHER READING

Guyoneaud, R., Süling, J., Petri, R., Matheron, R., Caumette, P., Pfennig, N. and Imhoff, J.F. 1998. Taxonomic rearrangements of the genera *Thiocapsa* and *Amoebobacter* on the basis of 16S rDNA sequence analyses and description of *Thiolamprovum* gen. nov. Int. J. Syst. Bacteriol. 48: 957–964.

Caumette, P., Baulaigue, R. and Matheron, R. 1991. *Thiocapsa halophila*, sp. nov., a new halophilic phototrophic purple sulfur bacterium. Arch. Microbiol. 155: 170–176.

Imhoff, J.F., Süling, J. and Petri, R. 1998. Phylogenetic relationships among the *Chromatiaceae*, their taxonomic reclassification and description of the new genera *Allochromatium*, *Halochromatium*, *Isochromatium*, *Marichromatium*, *Thiococcus*, *Thiohalocapsa* and *Thermochromatium*. Int. J. Syst. Bacteriol. 48: 1129–1143.

List of species of the genus *Thiohalocapsa*

1. *Thiohalocapsa halophila* (Caumette, Baulaigue and Matheron 1991) Imhoff, Süling and Petri 1998b, 1139^{VP} (*Thiocapsa halophila* Caumette, Baulaigue and Matheron 1991, 175.)

ha.lo'phi.la. Gr. n. *hals* salt; Gr. adj. *philos* loving; M.L. fem. adj. *halophila* salt-loving.

Cells spherical, 1.5–2.5 µm in diameter, form aggregates of 2, 4, or more cells, nonmotile. Color of cell suspensions purple-red. Photosynthetic pigments are bacteriochlorophyll *a* and okenone.

Photolithoautotrophic growth occurs with hydrogen, sulfide, S⁰, sulfite, and thiosulfate as electron donors. In presence of sulfide and bicarbonate, acetate, pyruvate, lactate, and fructose are photoassimilated. Glycerol and glucose poorly used. Chemolithotrophic or chemoorganotrophic growth is possible under microoxic conditions in the dark. Assimilatory sulfate reduction is absent. Vitamin B₁₂ required.

Mesophilic and halophilic bacterium with optimum

growth at 20–30°C, pH 7.0 (pH range 6.0–8.0), and at 4–8% NaCl (salinity range 3–20% NaCl).

Habitat: Reduced sediments in coastal saltern with salt deposits exposed to light.

The mol% G + C of the DNA is: 65.9–66.6 (T_m).

Type strain: SG3202, ATCC 49740, DSM 6210.

GenBank accession number (16S rRNA): AJ002796.

Genus XIX. *Thiolamprovum* Guyoneaud, Süling, Petri, Matheron, Caumette, Pfennig and Imhoff 1998b, 963^{VP}

PIERRE CAUMETTE, RÉMI GUYONEAUD AND JOHANNES F. IMHOFF

Thi'o.lam.pro'vum. Gr. n. *thios* sulfur; Gr. n. *lampros* bright, brilliant; L. n. *ovum* egg; M.L. neut. n.

Thiolamprovum bright egg with sulfur.

Cells nearly spherical to oval, $2 \times 2\text{--}3\ \mu\text{m}$, may occur in regular platelets of 4–16 cells, **nonmotile**, and multiply by binary fission. Gram negative, belong to the *Gamma*proteobacteria, and have internal photosynthetic membranes of vesicular type containing the photosynthetic pigments bacteriochlorophyll *a* and carotenoids.

Photolithoautotrophic growth under anoxic conditions in the light occurs with sulfide, thiosulfate, and sulfur as electron donors. During oxidation of sulfide, S^0 transiently stored inside the cells in the form of highly refractile globules. Sulfate is the final oxidation product. In the presence of sulfide and bicarbonate, simple organic substrates can be photoassimilated. **Chemoautotrophic or mixotrophic growth under microoxic to oxic conditions in the dark is possible.**

The mol% G + C of the DNA is: 64.5–66.5 (Bd).

Type species: ***Thiolamprovum pedioforme*** (Eichler and Pfennig 1986) Guyoneaud, Süling, Petri, Matheron, Caumette, Pfennig and Imhoff 1998b, 963 (*Amoebobacter pedioformis* Eichler and Pfennig 1986, 300.)

ENRICHMENT AND ISOLATION PROCEDURES

Media and culture conditions are the same as described for *Chromatium* species. No specific enrichment or isolation procedures are applicable for *Thiolamprovum* species. Inoculation of agar shake dilution cultures directly from samples collected in nature is recommended as described for the *Chromatium* species.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *THIOLAMPROVUM* FROM OTHER GENERA

Characteristics for differentiation of *Thiolamprovum* species from other phototrophic members of the family *Chromatiaceae* are shown in Table BXII.γ.6 of the chapter describing the family *Chromatiaceae*. *Thiolamprovum* is characterized by the formation of platelets of 8–16 or more regularly arranged cells, similar to *Thiopedia rosea*. These bacteria can be distinguished by their different carotenoid content and color of cultures and colonies. *Thiolamprovum* cells contain carotenoids of the spirilloxanthin group with a pinkish color of the colonies, whereas *Thiopedia rosea* cells contain okenone with a purple color of the colonies. The phylogenetic relationship of *Thiolamprovum pedioforme* to other *Chromatiaceae* species, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.4 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

At present, *Thiolamprovum pedioforme* is the only representative of this genus. It was previously known as *Amoebobacter pedioformis* (Eichler and Pfennig, 1986). Sequence analyses of 16S rRNA genes have shown that *Thiolamprovum pedioforme* is genetically distant from *Lamprocystis* and *Thiocapsa* species (Guyoneaud et al., 1998b). Therefore, it has been transferred to *Thiolamprovum pedioforme* (Guyoneaud et al., 1998b).

FURTHER READING

Eichler, B. and Pfennig, N. 1986. Characterization of a new platelet-forming purple sulfur bacterium, *Amoebobacter pedioformis* sp. nov. Arch. Microbiol. 146: 295–300.

Guyoneaud, R., Süling, J., Petri, R., Matheron, R., Caumette, P., Pfennig, N. and Imhoff, J.F. 1998. Taxonomic rearrangements of the genera *Thiocapsa* and *Amoebobacter* on the basis of 16S rDNA sequence analyses and description of *Thiolamprovum* gen. nov. Int. J. Syst. Bacteriol. 48: 957–964.

List of species of the genus *Thiolamprovum*

1. ***Thiolamprovum pedioforme*** (Eichler and Pfennig 1986) Guyoneaud, Süling, Petri, Matheron, Caumette, Pfennig and Imhoff 1998b, 963^{VP} (*Amoebobacter pedioformis* Eichler and Pfennig 1986, 300.)
pe' di.o.for' me. Gr. n. *pedion* a plain, a flat area; L. n. *forma* shape; M.L. neut. adj. *pedioforme* flat shaped.

Spherical to oval cells occur singly or in regular platelets of 4–16 cells. Individual cells $2 \times 2\text{--}3\ \mu\text{m}$, nonmotile, and contain gas vesicles in peripheral parts of the cell. Color of cell suspensions pinkish red. Photosynthetic pigments are bacteriochlorophyll *a* and the carotenoids spirilloxanthin, anhydrorhodovibrin, rhodopin, and OH-spirilloxanthin.

Photolithoautotrophic growth with sulfide, thiosulfate,

and S^0 as electron donors. In the presence of sulfide and bicarbonate, acetate, lactate, and pyruvate photoassimilated. Chemolithoautotrophic growth under microoxic conditions in the dark is possible with sulfide and thiosulfate as electron donors. Assimilatory sulfate reduction does not occur.

Mesophilic freshwater bacterium with optimum growth at 37°C and pH 7.4–7.6.

Habitat. freshwater and waste water ponds, type strain originates from a wastewater pond of a sugar factory.

The mol% G + C of the DNA is: 65.5 (Bd).

Type strain: Strain CML2, DSM 3802.

GenBank accession number (16S rRNA): Y12297.

Genus XX. Thiopedia Winogradsky 1888, 85^{AL}

JOHANNES F. IMHOFF

Thi.o.pe'di.a. G. n. *thios* sulfur; Gr. n. *pedium* a plain, a flat area; M.L. fem. n. *Thiopedia* sulfur plain.

Cells spherical to ovoid, nonmotile, and multiply by binary fission. Rectangular platelets formed with 4, 8, 16, 32, or more regularly arranged cells (up to 128 or 256 cells may stick together). Gram negative, **contain irregularly shaped gas vesicles in the central part of the cell and internal photosynthetic membranes of vesicular type** with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Photolithoautotrophic growth under anoxic conditions in the light with sulfide and S⁰ as electron donors. During oxidation of sulfide, S⁰ transiently stored in the peripheral part of the cell in the form of highly refractile globules. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates are photoassimilated. Reduced sulfur compounds required as sulfur source.

Habitat: mud and stagnant water of ponds and lakes containing hydrogen sulfide, common planktonic bacterium in the sulfide-containing hypolimnion of freshwater lakes.

The mol% G + C of the DNA is: 62.5–63.5.

Type species: ***Thiopedia rosea*** Winogradsky 1888, 85.

ENRICHMENT AND ISOLATION PROCEDURES

A prerequisite for the successful enrichment and isolation of *Thiopedia* species is the use of an inoculum from anoxic water or mud samples in which the characteristic *Thiopedia* platelets were microscopically detected. For the isolation of pure cultures, it is advisable to prepare enrichment cultures in liquid media, as well as agar shake dilution cultures that are directly inoculated from the sample of the natural habitat. In liquid media, *Thiopedia* may be outgrown by other purple sulfur bacteria containing gas vesicles.

The culture medium and incubation conditions are similar to those described for *Chromatium* species. Not more than 1.5 mM sulfide and 3 mM acetate should be used for growth of *Thiopedia*, and the trace element solution must not contain a chelating agent (e.g., EDTA). Growth is enhanced by the addition of 100 µM dithionite to the medium. Incubation is carried out at a temperature of 20–25°C and a light intensity of 100–200 lux. *Thiopedia* species grow in the form of individual cells, or platelets with 4–128 cells. Colonies and cell suspensions of *Thiopedia rosea* exhibit a unique bright purple-red color and can, therefore, be differentiated readily from other purple sulfur bacteria.

MAINTENANCE PROCEDURES

Stock cultures may be maintained in liquid medium in tightly closed 100-ml screw-capped bottles at 4–8°C. Such cultures should be transferred every 2 months. For long-term preservation, vials of cell suspensions in anoxic culture medium containing 5% (v/v) DMSO are stored in liquid nitrogen.

DIFFERENTIATION OF THE GENUS *THIOPEDIA* FROM OTHER GENERA

Characteristics that differentiate *Thiopedia* species from other phototrophic members of the family *Chromatiaceae* are: i) the arrangement of the cells in rectangular platelets, ii) possession of gas vesicles, and iii) the purple-red color of cultures and colonies, which is due to the carotenoid okenone. Platelets similar to those of *Thiopedia* species are formed by *Thiolamprovirus pediforme*.

oforme. Characteristics for differentiation of *Thiopedia* species from other spherical and nonmotile purple sulfur bacteria and from other phototrophic *Chromatiaceae* species are shown in Table BXII.γ.6 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

Purple sulfur bacteria having the morphology first described by Winogradsky for *Thiopedia rosea* are widely distributed in sulfide-containing ponds and lakes (Utermöhl, 1925; Eichler and Pfennig, 1991). Fine-structure studies of *Thiopedia*-like cell material collected from a purple bacteria bloom in a small forest pond (Michigan, USA) revealed the presence of at least two cell types with different internal membrane systems (Hirsch, 1973). One type contained a vesicular membrane system, while the other type contained a membrane system that consisted of stacks of disk-shaped lamellae. Thin section micrographs of the type strain of *Thiopedia rosea*, showing a vesicular membrane system, were published by Remsen (1978).

The description of *T. rosea* given here is based on information from five strains described by Eichler and Pfennig (1991). Little information on this species was previously available. It should be noted, however, that the properties of the five isolates differ in some aspects from earlier descriptions. Winogradsky (1888) indicated a cell diameter of 1–2 µm and Pfennig (1989c) reported cells being 1.2–1.6 × 1.5–2.0 µm and requiring vitamin B₁₂. The latter property was not substantiated by Eichler and Pfennig (1991) and cells were reported to be 2.0–2.5 × 2.5–3.0 µm by these authors.

Unfortunately, the ability of *Thiopedia* to multiply in rectangular platelets, as characteristically observed in many natural habitats, is gradually lost over periods of months or years in pure culture. Therefore, it is not a taxonomically reliable character. Because the formation of rectangular platelets was considered a genus-specific property, distinguishing *Thiopedia* from other *Chromatiaceae* species (Pfennig and Trüper, 1989), recognition of the taxonomic unreliability of this property raises questions about the existence of the genus *Thiopedia*. Further studies are needed to show whether *Thiopedia rosea* is a close relative of *Thiolamprovirus pediforme*, which also forms rectangular platelets (Eichler and Pfennig, 1986), or *Thiocapsa* species, that do not form large platelets, but may form tetrads of cells.

FURTHER READING

- Eichler, B. and Pfennig, N. 1986. Characterization of a new platelet-forming purple sulfur bacterium, *Amoebobacter pediformis* sp. nov. Arch. Microbiol. 146: 295–300.
- Eichler, B. and Pfennig, N. 1991. Isolation and characteristics of *Thiopedia rosea* (neotype). Arch. Microbiol. 155: 210–216.
- Hirsch, P. 1973. Fine structure of *Thiopedia* spp. In Drews (Editor), Abstracts of the Symposium on Prokaryotic Photosynthetic Organisms, Freiburg. p. 184–185.
- Pfennig, N. 1973. Culture and ecology of *Thiopedia rosea*. In Drews (Editor), Abstracts of the Symposium on Prokaryotic Photosynthetic Organisms, Freiburg. p. 75–76.
- Remsen, C.C. 1978. Comparative subcellular architecture of photosynthetic bacteria. In Clayton and Sistrom (Editors), The Photosynthetic Bacteria, Plenum Press, New York. p. 31–60.
- Utermöhl, H. 1925. Limnologische Phytoplanktonstudien. Arch. Hydrobiol. Suppl. 5: 251–277.

List of species of the genus *Thiopedia*

1. ***Thiopedia rosea*** Winogradsky 1888, 85^{AL}
ro'se.a. L. adj. *rosea* rose-colored, pink.

Cells spherical to ovoid, 2.0–2.5 × 2.5–3.0 µm, with regular arrangements in rectangular platelets of up to 64 or more cells (Fig. BXII.γ.16). Nonmotile and contain gas vesicles in the central part of the cells. Individual cells colorless, color of cell suspensions bright purple-red. Internal photosynthetic membranes of vesicular type. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids with okenone as major component.

Obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth with sulfide and S⁰ as electron donors. Sulfide concentrations above 0.6 mM are growth inhibitory. In the presence of sulfide and bicarbonate, the following substrates are photoassimilated: acetate, pyruvate, butyrate, valerate (not more than 0.03% (w/v) of each to be added), and glucose. Some strains also utilize succinate, fumarate, malate, and fructose (0.05%). Ammonium salts serve as nitrogen source. Reduced sulfur compounds required as sulfur source. Assimilatory sulfate reduction does not occur. Growth enhanced by the addition of 100 µM dithionite to growth media.

Mesophilic planktonic freshwater bacterium with optimum growth at 23°C and pH 7.3–7.5.

Habitat: anoxic and sulfidic water of stratified freshwater lakes.

The mol% G + C of the DNA is: 62.5–63.5 (Bd).

Type strain: 4711, DSM 1236 (neotype strain; no type strain available).

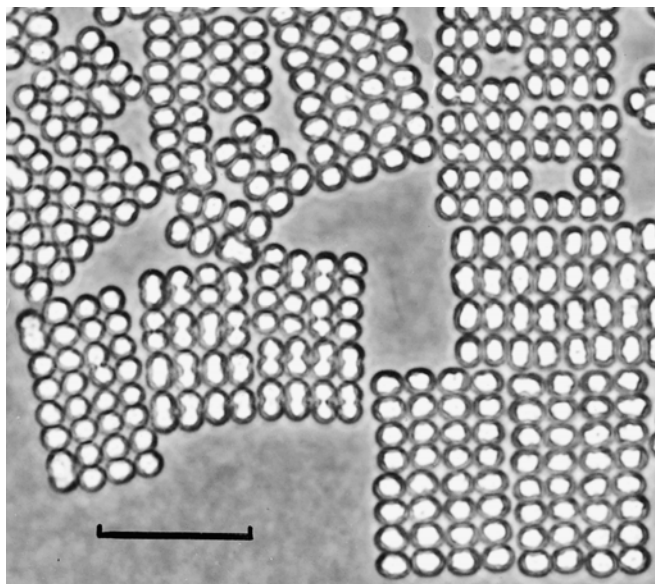


FIGURE BXII.γ.16. *Thiopedia rosea* strain 4211 cultured photoautotrophically with sulfide and acetate. The *light areas* inside the cells are the gas vacuoles. In some of the platelets, synchronously dividing cells can be recognized. Phase-contrast micrograph. Bar = 10 µm.

Genus XXI. *Thiorhodococcus* Guyoneaud, Matheron, Liesack, Imhoff and Caumette 1998a, 328^{VP} (Effective publication: Guyoneaud, Matheron, Liesack, Imhoff and Caumette 1997, 22)

RÉMI GUYONEAUD, PIERRE CAUMETTE AND JOHANNES F. IMHOFF

Thi'o.rho.do.co'ccus. Gr. n. *thios* sulfur; Gr. n. *rhodon* the rose; L. n. *coccus* sphere; M.L. masc. n. *Thiorhodococcus* rose sphere with sulfur.

Cells spherical to slightly ovoid, diplococcus-shaped before cell division, motile by flagella. Cells single or in pairs but may form irregular aggregates, multiply by binary fission. Gram negative, belong to the *Gammaproteobacteria*, and contain **internal photosynthetic membranes of vesicular type** with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Photolithoautotrophic growth under anoxic conditions in the light with sulfide and S⁰ as electron donors. During oxidation of sulfide, S⁰ transiently stored inside the cells in the form of highly refractile globules. Final oxidation product is sulfate. In the presence of sulfide and bicarbonate, simple organic substrates are photoassimilated. Chemotrophic growth is possible under microoxic conditions in the dark.

Habitat: water and sediment of marine and brackish environments containing hydrogen sulfide.

The mol% G + C of the DNA is: 66.8–67.0.

Type species: ***Thiorhodococcus minor*** Guyoneaud, Matheron, Liesack, Imhoff and Caumette 1998a, 328 (Effective publication: Guyoneaud, Matheron, Liesack, Imhoff and Caumette 1997, 22.)

ENRICHMENT AND ISOLATION PROCEDURES

The media described for *Chromatium* species can also be used for *Thiorhodococcus* and other marine *Chromatiaceae* species, if supplemented with 2–3% NaCl as required. The culture conditions and isolation procedures are the same as described for the genus *Chromatium*.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*. Long term preservation by freezing at –80°C in 40% glycerol is also possible.

DIFFERENTIATION OF THE GENUS *THIORHODOCOCCUS* FROM OTHER GENERA

The genus *Thiorhodococcus* is distinguished from other members of the family *Chromatiaceae* by phenotypic and genetic traits. The spherical, motile, purple sulfur bacteria of the genus *Thiorhodo-*

coccus are morphologically similar to *Thiocystis* species, but differ from them by their salt requirement and 16S rDNA sequence composition. *Thiorhodococcus minor* forms a separate lineage not specifically affiliated with other *Chromatiaceae* species (Imhoff et al., 1998b; see Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*). Characteristics useful for the differentiation of the genus *Thiorhodococcus* from other members of the family *Chromatiaceae* are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*.

List of species of the genus *Thiorhodococcus*

1. ***Thiorhodococcus minor*** Guyoneaud, Matheron, Liesack, Imhoff and Caumette 1998a, 328^{VP} (Effective publication: Guyoneaud, Matheron, Liesack, Imhoff and Caumette 1997, 22.)

mi' nor. L. masc. adj. *minor* small.

Cells spherical to slightly ovoid, 1.0–2.0 μm in diameter, highly motile by flagella, divide by binary fission, and form diplococcus-shaped division stages (Fig. BXII.γ.17). Color of individual cells grayish, color of cell suspensions brown-orange. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the normal spirilloxanthin group, with rhodopin as major pigment.

Photolithoautotrophic growth under anoxic conditions in the light with hydrogen, sulfide, thiosulfate, and S⁰ as electron donors. In the presence of sulfide and hydrogen carbonate, acetate, propionate, lactate, glycolate, pyruvate, malate, fumarate, succinate, fructose, sucrose, ethanol, and propanol are photoassimilated. Chemolithotrophic growth with sulfide and thiosulfate, and chemoorganotrophic growth with acetate and fructose are possible under microoxic conditions in the dark. Some strains (but not the type strain) assimilate sulfate. Nitrogen sources: ammonium salts and N₂.

Mesophilic marine bacterium requiring NaCl for growth. Optimal growth at 2% NaCl (salinity range 0.5–9% NaCl), pH 7.0–7.2 (pH range 6.0–8.0), and 30–35°C.

Habitat: anoxic sediments from coastal lagoons and marine environments rich in sulfide.

FURTHER READING

Guyoneaud, R., Matheron, R., Liesack, W., Imhoff, J.F. and Caumette, P. 1997. *Thiorhodococcus minus*, gen. nov., sp. nov., a new purple sulfur bacterium isolated from coastal lagoon sediments. *Arch. Microbiol.* 168: 16–23.

Guyoneaud, R., Matheron, R., Baulaigue, R., Podeur, K., Hirschler, A. and Caumette, P. 1996. Anoxygenic phototrophic bacteria in eutrophic coastal lagoons of the French Mediterranean and Atlantic Coast (Prévoist Lagoon, Arcachon Bay, Certes Fishponds). *Hydrobiologia* 329: 33–43.

The mol% G + C of the DNA is: 66.9 (HPLC).

Type strain: CE2203, ATCC 700259, DSM 11518.

GenBank accession number (16S rRNA): Y11316.

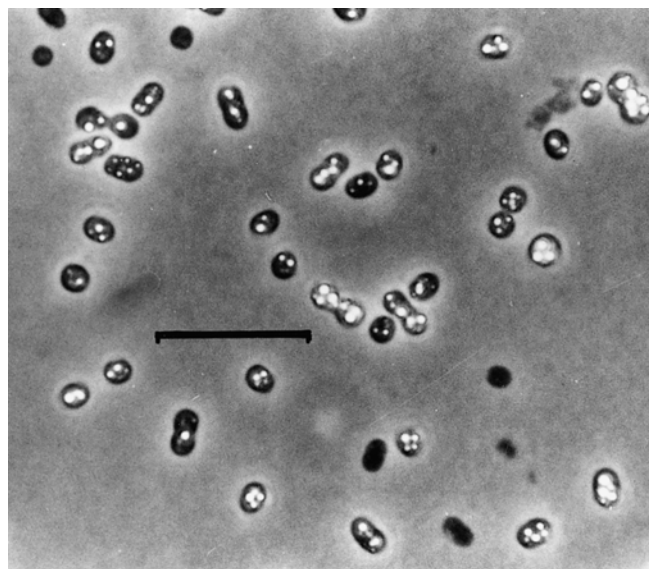


FIGURE BXII.γ.17. *Thiorhodococcus minor* DSM 11518^T grown photolithoautotrophically with sulfide. Phase-contrast microphotograph. Bar = 10 μm.

Genus XXII. ***Thiorhodovibrio*** Overmann, Fischer and Pfennig 1993, 188^{VP} (Effective publication: Overmann, Fischer and Pfennig 1992, 334)

JOHANNES F. IMHOFF AND JÖRG OVERMANN

Thi' o.rho.do.vi' brio. Gr. n. *thios* sulfur; Gr. n. *rhodon* the rose; L. v. *vibrio* vibrates; M.L. masc. n. *vibrio* that which vibrates; *Thiorhodovibrio* rose vibrio with sulfur.

Individual cells vibrioid to spirilloid, motile by a single polar flagellum, multiply by binary fission. Gram negative, belong to the *Gammaproteobacteria*, and contain internal membranes of vesicular type with bacteriochlorophyll *a* and carotenoids as photosynthetic pigments.

Photolithoautotrophic growth under anoxic conditions in the light with sulfide and S⁰ as electron donors. During oxidation of sulfide, S⁰ stored inside the cells in the form of highly refractile globules. Final oxidation product is sulfate. In the presence of

sulfide and bicarbonate, simple organic substrates are photoassimilated. Chemotrophic growth at reduced oxygen partial pressure.

Habitat: surface layers of marine and salt lake sediments and microbial mats containing hydrogen sulfide.

The mol% G + C of the DNA is: 61–63.

Type species: ***Thiorhodovibrio winogradskyi*** Overmann, Fischer and Pfennig 1993, 188 (Effective publication: Overmann, Fischer and Pfennig 1992, 334.)

ENRICHMENT AND ISOLATION PROCEDURES

The media described for *Chromatium* species can also be used for *Thiorhodovibrio* and other marine members of the family *Chromatiaceae*, if supplemented with 2–3% NaCl as required. The culture conditions and isolation procedures are the same as described for the genus *Chromatium*.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for the genus *Chromatium*.

DIFFERENTIATION OF THE GENUS *THIORHODOVIBRIO* FROM OTHER GENERA

Characteristics for differentiation of the genus *Thiorhodovibrio* from other phototrophic *Chromatiaceae* species are shown in Table BXII.γ.5 of the chapter describing the family *Chromatiaceae*. The genetic relationship of *Thiorhodovibrio winogradskyi* to other members of the family *Chromatiaceae*, based on 16S rDNA sequence comparison, is shown in Fig. BXII.γ.3 of the chapter describing the family *Chromatiaceae*.

TAXONOMIC COMMENTS

Thiorhodovibrio winogradskyi was described based on two isolates that show significant phenotypic differences (Overmann et al., 1992). Both strains may be considered different species, and the description of *Thiorhodovibrio winogradskyi* given here is that of the type strain (DSM 6702, SSP1). The following properties distinguish strain 06511 from the type strain of *Thiorhodovibrio winogradskyi*: Cells are spirilloid, $0.7\text{--}0.9 \times 2.5\text{--}3.9\text{ }\mu\text{m}$. Thiosulfate is used as photosynthetic electron donor and the mol% G + C content of the DNA is 62.4. Propionate does not support growth of strain 06511. However, in addition to the substrates used by the type strain, succinate, fumarate, malate, tartrate, malonate, glycerol, and peptone stimulate growth slightly (Overmann et al., 1992) for strain O6511.

FURTHER READING

Overmann, J., Fischer, U. and Pfennig, N. 1992. A new purple sulfur bacterium from saline littoral sediments, *Thiorhodovibrio winogradskyi* gen. nov. and sp. nov. Arch. Microbiol. 157: 329–335.

List of species of the genus *Thiorhodovibrio*

1. ***Thiorhodovibrio winogradskyi*** Overmann, Fischer and Pfennig 1993, 188^{VP} (Effective publication: Overmann, Fischer and Pfennig 1992, 334.) *wi.no.grad'skyi*. M.L. gen. n. *winogradskyi* of Winogradsky, named for S.N. Winogradsky, a Russian microbiologist, who did the first comprehensive studies on the purple sulfur bacteria.

Cells vibrioid, $1.2\text{--}1.6 \times 2.6\text{--}4.0\text{ }\mu\text{m}$, motile by a single polar flagellum. Gram negative and belong to the *Gammaproteobacteria*. Anaerobically grown cultures pink to light pinkish purple. Living cell suspensions have absorption maxima at 370–372, 483–484, 510, 590–591, 794–795, and 867 nm. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the spirilloxanthin group, with rhodopin as the major component, and spirilloxanthin, lycopen, anhydrohodovibrin, and rhodovibrin.

Photolithoautotrophic growth with sulfide and S⁰ as pho-

tosynthetic electron donors. When growing on sulfide, up to ten small sulfur globules are formed inside the cell, often in a row along its long axis. High growth rates are obtained with S⁰ as electron donor. In the presence of sulfide and hydrogen carbonate, acetate, pyruvate, and propionate are photoassimilated. Chemotrophic growth possible under mixotrophic conditions at reduced oxygen partial pressure (~1% O₂). No vitamins required.

Mesophilic marine bacterium requiring NaCl, with optimum growth at 2.2–3.2% NaCl (range 0.2–7.2% NaCl), pH 7.0–7.4 and 33°C (temperature range 14–37°C).

Habitat: sulfide-containing surface layers of salt lake sediments and marine microbial mats.

The mol% G + C of the DNA is: 61.0 (HPLC).

Type strain: SSP1, DSM 6702.

GenBank accession number (16S rRNA): AB016986, AJ006214, Y12368.

Genus XXIII. *Thiospirillum* Winogradsky 1888, 104^{AL}

JOHANNES F. IMHOFF

Thi'o.spi.ril'lum. Gr. n. *thios* sulfur; M.L. dim. neut. n. *Spirillum* a bacterial genus; M.L. neut. n. *Thiospirillum* sulfur *Spirillum*.

Cells **curved rods or spirals** (sigmoid), dividing by binary fission, **motile by (multitrichous) polar flagella**. Cells may be immobile and grow in irregular aggregates surrounded by slime. Gram negative, belong to the *Gammaproteobacteria*, and contain **internal photosynthetic membranes of vesicular type** in which the photosynthetic pigments bacteriochlorophyll *a* and carotenoids are located.

Obligately phototrophic and strictly anaerobic. Photolithoautotrophic growth under anoxic conditions in the light with sulfide and S⁰ as electron donors. During oxidation of sulfide, S⁰ transiently stored inside the cells in the form of highly refractile globules. Sulfate is the final oxidation product. Storage materials are polysaccharides, poly-β-hydroxybutyrate, S⁰, and polyphosphate.

The mol% G + C of the DNA is: 45.5.

Type species: **Thiospirillum jenense** (Ehrenberg 1838) Migula 1900, 1050 (*Ophidomonas jenensis* Ehrenberg 1838, 44.)

FURTHER DESCRIPTIVE INFORMATION

Thiospirillum jenense is the only species of this genus studied in pure culture so far. *T. jenense* is a typical freshwater bacterium. Information obtained from rRNA oligonucleotide patterns (Fowler et al., 1984) indicates that *T. jenense* belongs to the family Chromatiaceae and is distantly related to *Chromatium* and *Allochro-matium* species. Complete sequences of the 16S rDNA of this bacterium are not yet available.

ENRICHMENT AND ISOLATION PROCEDURES

Conditions for enrichment and cultivation of *Thiospirillum jenense* are essentially the same as those described for *Chromatium* species. The large cells of *T. jenense* (generally 30–40 µm in length) grow only in agar shake cultures, when the final concentration of thoroughly washed agar is below 0.6–0.8% (w/v). In such agar cultures, colonies of *Thiospirillum* appear irregular and wisp-shaped. Individual cells can be recognized by using a low power, binocular microscope. Cultures containing not more than 1–2 mM sulfide should be incubated at low light intensity (200–500 lux). Colonies from agar shake cultures grow well in liquid medium, provided culture vessels of small volume are used first, e.g., screw-capped test tubes of 10–15-ml volume.

MAINTENANCE PROCEDURES

Maintenance of strains in liquid culture medium and long-term preservation in liquid nitrogen are carried out as described for

the genus *Chromatium*. *Thiospirillum* species cannot be preserved by lyophilization.

DIFFERENTIATION OF THE GENUS THIOSPIRILLUM FROM OTHER GENERA

Characteristics used for the differentiation of *Thiospirillum* species from other phototrophic members of the family Chromatiaceae are shown in Table BXII.γ.4 of the chapter describing the family Chromatiaceae.

TAXONOMIC COMMENTS

In addition to *Thiospirillum jenense*, two other species of *Thiospi-rillum*, "*Thiospirillum sanguineum*" and "*Thiospirillum rosenbergii*", have been described from observations of water and mud samples collected in nature. The former species was described as having the same size and morphology as *Thiospirillum jenense*, with which it could have been confused. More recent reports of the existence of "*Thiospirillum rosenbergii*" are lacking. Winogradsky (1888) originally designated *Thiospirillum sanguineum* as the type species of this genus. However, because this species, in contrast to *Thiospi-rillum jenense*, has never been obtained in pure culture, *Thiospi-rillum jenense* was later recognized as the type species (Pfennig, 1989d). Further studies of strains obtained in pure culture and 16S rDNA sequence analyses are required to determine the relatedness of *Thiospirillum* species with each other and with other members of the family Chromatiaceae. According to 16S rRNA oligonucleotide catalogs, *Thiospirillum jenense* is not similar to *Chromatium* and *Allochro-matium* species (Fowler et al., 1984), though it may be distantly related to these bacteria and belong to the corresponding group of motile freshwater Chromatiaceae species.

List of species of the genus Thiospirillum

1. **Thiospirillum jenense** (Ehrenberg 1838) Migula 1900, 1050^{AL} (*Ophidomonas jenensis* Ehrenberg 1838, 44.)
je.nen' se. M.L. neut. adj. *jenense* jenense pertaining to Jena, Germany, the city where Ehrenberg discovered this organism.

Cells curved rods, sigmoid or spiral-shaped, 2.5–4.0 µm wide. Sigmoid cells usually 30–40 µm long, spiral cells up to 100 µm long. Complete turns may measure 15–40 µm and have a coil depth of 3–7 µm. Motile by a polar tuft of flagella, which is usually 10–12 µm long and visible by brightfield or phase-contrast microscopy (Fig. BXII.γ.18). Cells rarely tufted at both ends. Color of individual cells is pale yellow, color of cell suspensions yellowish to orange-brown. Photosynthetic pigments are bacteriochlorophyll *a* and the carotenoids lycopene and rhodopin (Schmidt et al., 1965).

Obligately phototrophic and strictly anaerobic. Photo-lithoautotrophic growth with sulfide and S⁰ as electron donors. Thiosulfate not utilized. In the presence of sulfide and bicarbonate, acetate is photoassimilated. Nitrogen sources: ammonium salts. Vitamin B₁₂ required for growth.

Mesophilic freshwater bacterium with optimum growth at 20–25°C and pH 7.0 (range 6.5–7.5).

Habitat: mud and stagnant water of ditches and fresh-water ponds containing hydrogen sulfide.

The mol% G + C of the DNA is: 45.5 (Bd).

Type strain: 1112, DSM 216.

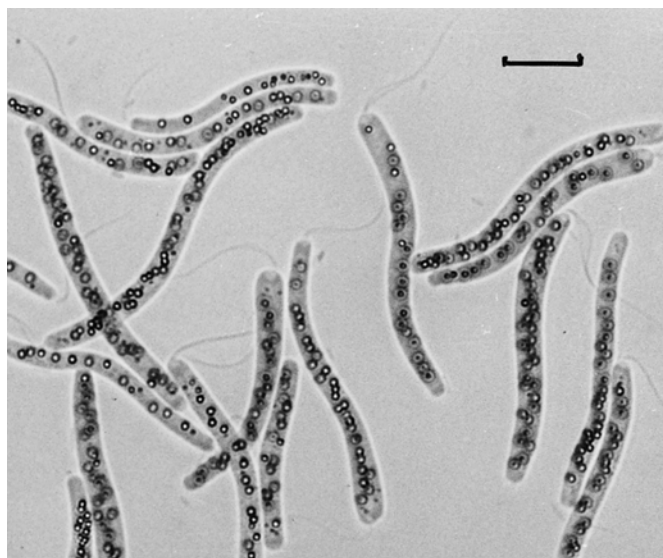


FIGURE BXII.γ.18. *Thiospirillum jenense* DSM 216 cultured photoauto-trophically with sulfide. Note the intracellular sulfur globules and the short, slightly sigmoidal tufts of polar flagella. Brightfield micrograph. Bar = 10 µm.

Family II. **Ectothiorhodospiraceae** Imhoff 1984b, 339^{VP}

JOHANNES F. IMHOFF

Ec'to.thi'o.rho'do.spi.ra'ce.ae. M.L. fem. n. *Ectothiorhodospira* type genus of the family; *-aceae* ending to denote a family; M.L. fem. pl. n. *Ectothiorhodospiraceae* the *Ectothiorhodospira* family.

This family contains **phototrophic purple sulfur bacteria** that perform **anoxygenic photosynthesis with bacteriochlorophylls and carotenoids as photosynthetic pigments and strictly chemotrophic bacteria**. Cells spiral, vibrioid, or rod-shaped, motile by means of polar flagella, divide by binary fission, with or without gas vesicles. **Gram negative and belong to the Gammaproteobacteria**. Phototrophic representatives preferably grow anaerobically in the light using reduced sulfur compounds as electron donors, **oxidize sulfide to S⁰, which is deposited outside the cells, and eventually in the peripheral periplasmic space of the cell body**, forming sulfate as the final oxidation product. **Phototrophic Ectothiorhodospiraceae are found in marine to extremely saline environments containing sulfide and having an alkaline to extremely alkaline pH**. Species of this family are the most halophilic eubacteria.

The mol% G + C of the DNA is: 50.5–69.7.

Type genus: **Ectothiorhodospira** Pelsh 1936, 120.

TAXONOMIC COMMENTS

The taxonomic position of bacteria now classified as *Ectothiorhodospiraceae* has been disputed since their discovery by Pelsh (1936). He distinguished these bacteria, which he called “*Ectothiorhodaceae*”, from those phototrophic purple bacteria with elemental sulfur inside their cells, which he called “*Endothiorhodaceae*” (Pelsh, 1937). Pelsh’s isolates were poorly characterized and were lost soon after their isolation. Trüper (1968) reisolated *Ectothiorhodospira mobilis*, and Pfennig and Trüper (1971a) included *Ectothiorhodospira* as an exceptional genus into the *Chromatiaceae* because of its ability to perform an oxidative dissimilatory sulfur metabolism similar to that of other phototrophic purple sulfur bacteria. Because bacteria known at that time as *Ectothiorhodospira* species were significantly different from all other genera of the *Chromatiaceae*, and because representatives of the purple nonsulfur bacteria had been found to oxidize sulfide to extracellular elemental sulfur, Pfennig (1977) suggested the removal of *Ectothiorhodospira* from the *Chromatiaceae* and its placement in the purple nonsulfur bacteria. However, on the basis of significant differences in their polar lipid composition, three major groups of phototrophic purple bacteria were distinguished by Imhoff et al. (1982a): *Chromatiaceae*, the bacteria known at that time as *Ectothiorhodospira*, and the purple nonsulfur bacteria (see also Imhoff and Bias-Imhoff, 1995). A clear differentiation of *Ectothiorhodospira* species from *Chromatiaceae* was later demonstrated by the oligonucleotide patterns of 16S rRNA molecules (Stackebrandt et al., 1984). With the support of this information, it was proposed that the *Ectothiorhodospiraceae* be separated as a family distinct from the *Chromatiaceae* (Imhoff, 1984b). The separation of two groups of species within the family had

been recognized not only based on physiological properties and lipid composition (Tindall, 1980; Imhoff et al., 1982a; Imhoff, 1984a), but also by similarities of rRNA oligonucleotide patterns (Stackebrandt et al., 1984). However, a formal separation into genera was not proposed until later. The complete sequence analysis of the 16S rDNA from a large number of strains, including all available type strains of *Ectothiorhodospiraceae*, not only supported their separation from the *Chromatiaceae*, but also the divergence of the two groups within this family (Imhoff and Söling, 1996). Because of the large phylogenetic distance between these two groups, which is also reflected in a number of distinctive phenotypic properties, the extremely halophilic species were transferred to the new genus *Halorhodospira* (Imhoff and Söling, 1996). With this taxonomic reevaluation of the *Ectothiorhodospiraceae*, their taxonomy now follows their phylogenetic relationships, based on 16S rDNA sequences.

Two new isolates have expanded our knowledge of the *Ectothiorhodospiraceae*. A new phototrophic sulfur bacterium isolated from a Siberian soda lake, *Thiorhodospira sibirica*, has been shown to be genetically related to the genus *Ectothiorhodospira*, and many of its properties resemble those of *Ectothiorhodospira* species (Bryantseva et al., 1999b). This impressively large bacterium is unique in that it forms S⁰ globules outside the cells as well as inside. Sulfur globules are located in the cells only in the very peripheral part, which is represented by an apparently large periplasmic space; they are not located in the cell interior. Therefore, the mechanism of S⁰ formation and the location of the involved enzymes appear to be the same as in other *Ectothiorhodospiraceae*. However, particular care is needed to distinguish between internal and external deposition of S⁰.

Another new halophilic and strictly chemotrophic bacterium has been found to be genetically related to the *Ectothiorhodospiraceae*, in particular to *Halorhodospira* (Adkins et al., 1993; Imhoff and Söling, 1996). This bacterium, *Arhodomonas aquaeolei*, is a close, purely chemotrophic relative to the phototrophic *Ectothiorhodospiraceae*. It uses a restricted spectrum of carbon sources for growth. It is considered as a member of the *Ectothiorhodospiraceae* primarily because of its phylogenetic relationship by 16S rDNA sequence.

Differentiation of the genera of *Ectothiorhodospiraceae* Characteristics for differentiation of the genera *Ectothiorhodospira*, *Halorhodospira*, *Thiorhodospira*, and *Arhodomonas* are summarized in Table BXII.γ.7. Characteristic fatty acid compositions of *Halorhodospira* and *Ectothiorhodospira* species are shown in Table BXII.γ.8. The genetic relationships of *Ectothiorhodospiraceae* based on 16S rDNA sequences are presented in Fig. BXII.γ.19.

Key to the genera of the Ectothiorhodospiraceae

- I. Halophilic *Gammaproteobacteria* that require >10% total salts for optimum growth.
 - A. Cells contain photosynthetic pigments typical of phototrophic purple bacteria, perform a strictly phototrophic metabolism, and require alkaline pH for optimum growth.
Genus *Halorhodospira*
 - B. Cells without photosynthetic pigments and with a strictly chemotrophic metabolism.
Genus *Arhodomonas*

- II. Halophilic phototrophic purple sulfur bacteria that require alkaline pH and salt concentrations below 10% total salts for optimum growth.
- A. S⁰ globules deposited exclusively outside the cells.
Genus *Ectothiorhodospira*
- B. S⁰ globules deposited both outside and inside the cells.
Genus *Thiorhodospira*
- III. Halophilic chemolithotrophic bacteria that require a pH range of 6.8–8.0 and carry out aerobic oxidation of nitrate to nitrite.
Genus *Nitrococcus*

ACKNOWLEDGMENTS

Sequence alignments and the construction of the phylogenetic tree by Dr. J. Süling (IFM Kiel, Germany) are gratefully acknowledged.

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TABLE BXII.γ.7. Differential characteristics of the genera of the *Ectothiorhodospiraceae*

Characteristic	<i>Ectothiorhodospira</i>	<i>Arhodomonas</i> ^a	<i>Halorhodospira</i>	<i>Thiorhodospira</i> ^a
Gram stain	negative	negative	negative	negative
Phylogenetic group	<i>Gammaproteobacteria</i>	<i>Gammaproteobacteria</i>	<i>Gammaproteobacteria</i>	<i>Gammaproteobacteria</i>
Flagellation	polar tuft of flagella	polar flagellum	bipolar flagella	polar tuft
Internal membranes	lamellar stacks	nd ^b	lamellar stacks	lamellar stacks
Cell form	vibrioid to spirilloid	straight rods	vibrioid to spirilloid	vibrioid to spirilloid
Cell width (μm)	0.7–1.5	0.8–1.0	0.5–1.2	3–4
Growth medium ^c	EM 5% or mMPM	complex medium	EM 15–30%	see genus <i>Thiorhodospira</i>
Optimum salinity	1–7%	15%	15–25%	0.5–1.0%
Compatible solutes	glycine betaine, sucrose	nd	glycine betaine, ectoine, trehalose	nd
Fatty acid cluster ^d	V–VII	nd	I–IV	nd.
Major quinone components ^e	MK7 and Q7 or Q8	–	MK8, Q8, and MK4/5	–
Mol% G + C of DNA	61.4–68.4	67	50.5–69.7	56.0–57.4
<i>16S rDNA signature sequences:</i>				
<i>E. coli</i> position 92–96	CTGAC	CTGGC	GCGGC	CTGAC
<i>E. coli</i> position 135–137	TGG	CTT	CTT	TGG
<i>E. coli</i> position 149–151	CAA	TAG	TAG	TAA
<i>E. coli</i> position 221–225	TATCA	CGAAG	CGAAG	TATCA
<i>E. coli</i> position 667–669	AGA	GAG	GTA	AGG
<i>E. coli</i> position 761	T	T	C	T
<i>E. coli</i> position 1010–1011	CA	CA	TG	CA

^aData on *Arhodomonas* and *Thiorhodospira* are from Adkins et al. (1993) and Bryantseva et al. (1999b), respectively.

^bnd, not determined.

^cEM: standard medium for haloalkaliphilic *Ectothiorhodospiraceae* species according to Imhoff (1988a) with varying salt concentrations; mMPM: modified marine Pfennig's medium according to Trüper (1970).

^dFatty acid clusters according to Thiemann and Imhoff (1996).

^eMajor quinone components according to Imhoff (1984a) and Ventura et al. (1993).

TABLE BXII.γ.8. Fatty acid clusters of *Ectothiorhodospira* and *Halorhodospira* species^{a,b}

Characteristic	<i>E. mobilis</i>	<i>E. haloalkaliphila</i>	<i>E. marina</i>	<i>E. marismortui</i>	<i>E. shaposhnikovii</i>	<i>E. vacuolata</i>	<i>H. halophila</i>	<i>H. halophila</i>	<i>H. abdelmalekii</i>	<i>H. halochloris</i>
Fatty acid cluster	V	VII	VII	V	VI	VI	I	II	III	IV
C _{12:0}	0.7–0.8	0.9–1.2	0.9	1.1	1.3–1.7	nd	2.4–3.4	2.8–5.0	2.4	1.1–2.9
C _{16:0}	35.1–36.4	29.0	24.8	33.2	24.4–26.4	21.6	9.2–11.1	10.8–14.5	21.4	21.0–24.0
C _{16:1 ω7c}	2.7–3.5	4.8–5.1	5.9	3.8	4.1–4.6	3.6	0.4–0.6	0.6–0.8	1.4	1.2–1.6
C _{18:0}	5.7–7.6	6.8–7.1	9.2	5.0	5.6–6.8	5.5	14.2–17.7	11.3–13.2	6.0	6.3–11.6
C _{18:1 ω7c}	12.1–16.2	38.2–38.6	29.2	18.5	57.1–59.0	65.3	62.8–65.5	58.6–62.8	62.8	57.7–57.9
C _{19:0 cyc}	33.5–37.7	15.9–16.3	25.8	34.7	nd	nd	2.5–3.4	3.0–5.8	0.4	1.3–2.0

^and, not determined.

^bData from Thiemann and Imhoff (1996).

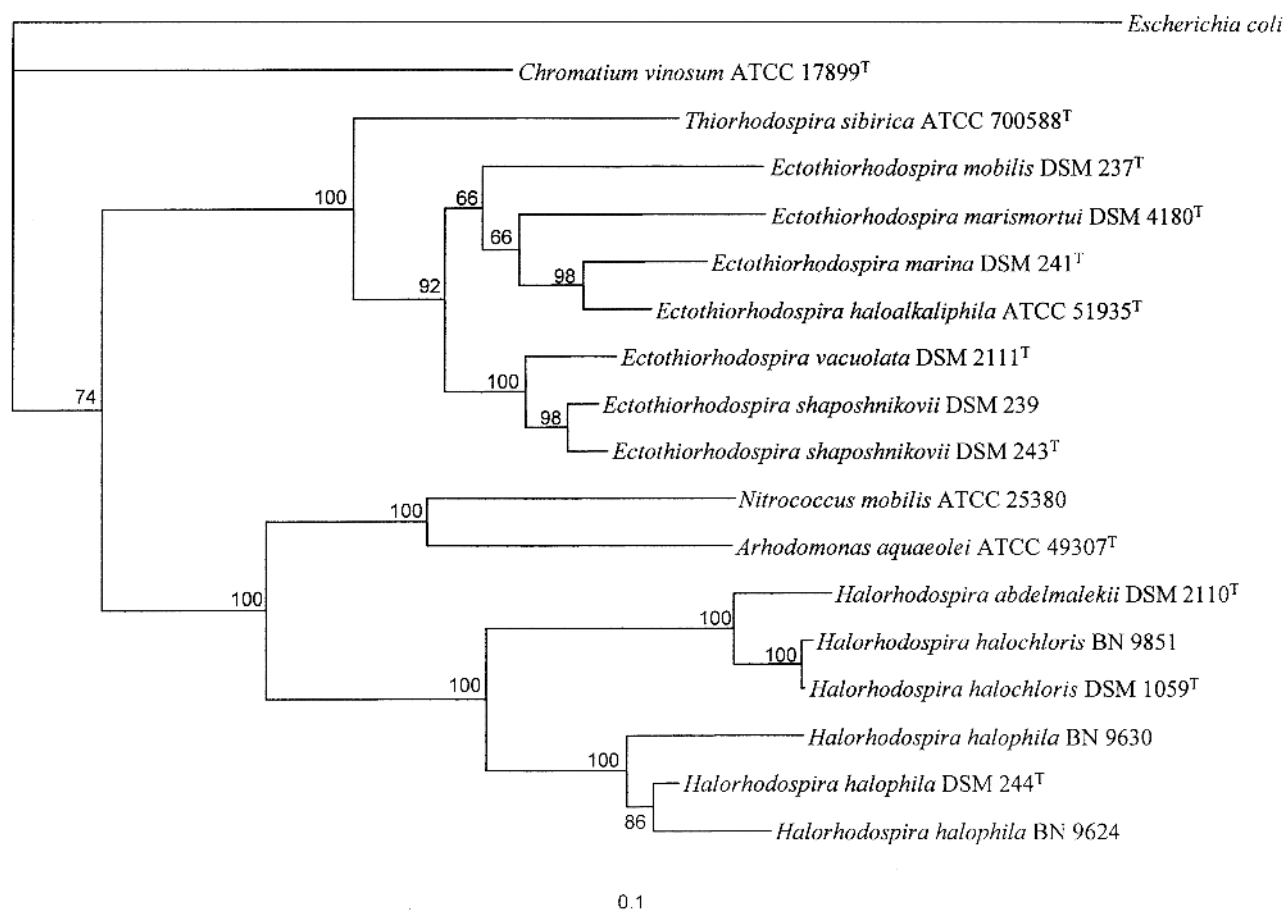


FIGURE BXII.γ.19. 16S rDNA-derived phylogenetic tree of the family Ectothiorhodospiraceae including species of *Ectothiorhodospira*, *Halorhodospira*, *Thiorhodospira* and *Arhodomonas*. Sequence data used were published previously (Imhoff and Süling, 1996). In addition, 16S rDNA sequences of *Arhodomonas aquaeolei* (M26631) and *Thiorhodospira sibirica* (AJ006530) were used. Sequence alignments, calculation of distance data and tree construction were performed as described earlier (Petri and Imhoff, 2000).

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Genus I. *Ectothiorhodospira* Pelsh 1936, 120^{AL}

JOHANNES F. IMHOFF

Ec'to.thi'o.rho'do.spi'ra. Gr. prep. *ectos* outside; Gr. n. *thios* sulfur; Gr. n. *rhodon* the rose; Gr. n. *spira* the spiral; M.L. fem. n. *Ectothiorhodospira* spiral rose with sulfur outside.

Cells rod-shaped or vibrioid, also appearing as true spirals, 0.7–1.5 µm in diameter, **motile by a polar tuft of flagella**, multiply by binary fission, and may contain gas vesicles. **Gram-negative and belong to the Gammaproteobacteria**. **Internal photosynthetic membranes are lamellar stacks** that are continuous with the cytoplasmic membrane. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids.

Grow photoautotrophically under anoxic conditions with reduced sulfur compounds or hydrogen as electron donors, or photoheterotrophically with a limited number of organic compounds. **Sulfide oxidized to sulfate, with S⁰, which is deposited outside the cells, as an intermediary product**. Some species grow microaerobically to aerobically in the dark. **Sodium chloride is**

required for growth, which is dependent on saline and alkaline conditions. Compatible solutes include glycine betaine, sucrose, and *N*-carbamoyl-L-glutamine amide. Growth factors are not required, but vitamin B₁₂ enhances growth of some strains. Storage products are polysaccharides, poly-β-hydroxybutyrate, and polyphosphate.

Most species live in marine and saline environments that contain sulfide, have slightly to extremely alkaline pH, and are exposed to light, such as estuaries, salt flats, salt lakes, and soda lakes. Occasionally found in soil.

The mol% G + C of the DNA is: 61.4–68.4.

Type species: *Ectothiorhodospira mobilis* Pelsh 1936, 120.

FURTHER DESCRIPTIVE INFORMATION

Cell morphology of the *Ectothiorhodospira* species varies with the growth conditions, in particular with pH and salinity. Cells of *Ectothiorhodospira mobilis* are spirals under optimum growth conditions, but tend to form vibrioid cells or bent rods. *Ectothiorhodospira shaposhnikovii* and *Ectothiorhodospira vacuolata* form straight or slightly bent rods under suitable growth conditions. Carotenoids are of the normal spirilloxanthin series with spirilloxanthin as the predominant component (Schmidt and Trüper, 1971). The *in vivo* absorption spectra of *Ectothiorhodospira* species are characterized by long wavelength absorption bands at 798, 822–827, 851–855, and 892–895 nm. The band at ~850 nm is the absolute maximum; the band at ~890 nm is mostly recognized as a shoulder, sometimes as a maximum of similar strength to that at 850 nm; and the one at ~820–830 nm can be seen only in second derivative spectra (Imhoff, unpublished results).

All species grow well under anoxic conditions in the light with reduced sulfur compounds as photosynthetic electron donors and in the presence of organic carbon sources and inorganic carbonate. *Ectothiorhodospira mobilis*, *Ectothiorhodospira shaposhnikovii*, and *Ectothiorhodospira haloalkaliphila* also grow microaerobically in the dark if sulfide is present. During phototrophic growth with sulfide as electron donor, the oxidation of sulfide and S^0 strictly follow each other, as has been shown for *E. mobilis* (see Trüper, 1978). Under the alkaline growth conditions, which are optimal for *Ectothiorhodospira* species, polysulfides are stable intermediates in sulfide oxidation. As a result, polysulfides are the first measurable oxidation products, and the medium becomes translucent yellow at this stage. After sulfide depletion, S^0 droplets are formed rather rapidly, and the medium becomes opaque and whitish. Upon further growth, cultures become pinkish and finally red, if S^0 disappears. The knowledge of enzymes involved in the oxidation of reduced sulfur compounds by *Ectothiorhodospira* species has been summarized by Trüper and Fischer (1982).

The fixation of carbon dioxide via the Calvin cycle is apparently the major route of carbon assimilation in *Ectothiorhodospira* species under autotrophic growth conditions. High activities of ribulosebiphosphate carboxylase have been found in *E. shaposhnikovii* (Firsov et al., 1974) and *E. mobilis* (Sahl and Trüper, 1977). However, the assimilation of several organic carbon sources, such as acetate and propionate, depends on the presence of carbon dioxide and proceeds via several carboxylation reactions (Firsov and Ivanovskii, 1974, 1975). Under these conditions, a considerable proportion of the cellular carbon is therefore derived from carbon dioxide, which is not assimilated via the ribulosebiphosphate pathway. Phosphoenolpyruvate carboxylase, ferredoxin-dependent pyruvate synthase and oxoglutarate synthase have been found in *E. shaposhnikovii* (Firsov et al., 1974); phosphoenolpyruvate carboxylase, phosphoenolpyruvate carboxykinase, and pyruvate carboxylase have been found in *E. mobilis* (Sahl and Trüper, 1977). All enzymes of the glycolytic pathway and the tricarboxylic acid cycle, with the exception of oxoglutarate dehydrogenase, are present in *E. shaposhnikovii* (Krasil'nikova, 1975). Cells grown on acetate demonstrate increased activities of isocitrate lyase, which indicates activity of the glyoxylic acid pathway. The major reserve material formed from acetate and butyrate in the absence of carbon dioxide is poly- β -hydroxybutyric acid. In the presence of carbon dioxide and from other organic carbon sources, carbohydrates are formed instead (Novikova, 1971).

Ammonia and glutamine are suitable nitrogen sources for all

species. Dinitrogen fixation has been demonstrated for some species and is probably a property inherent to the genus as a whole. Glutamate dehydrogenase (NADH-dependent) and glutamine synthetase/glutamate synthase (NADH-dependent) have been found in *E. mobilis* (Bast, 1977). Nitrate can be used as a nitrogen source by *E. shaposhnikovii*, and its reduction is catalyzed by a ferredoxin-dependent nitrate reductase, which is associated with or bound to the membranes and induced during growth with nitrate (Malofeeva et al., 1975).

In regard to their salt responses and the salt concentrations required for optimum growth, *Ectothiorhodospira* species are slightly halophilic or marine. Based on the salt concentration at which optimum growth occurs, species are grouped as nonhalophilic bacteria (<0.1 M NaCl), brackish water bacteria (~0.1–0.35 M NaCl), marine or slightly halophilic bacteria (~0.35–1.2 M NaCl), moderately halophilic bacteria (~1.2–2.5 M NaCl), and extremely halophilic bacteria (>2.5 M NaCl) (see Imhoff, 1993, 2001d). Bacteria of each of these groups may show up with different degrees of salt tolerance. Even nonhalophilic bacteria may exhibit extreme salt tolerances and grow at very high concentration, though at a suboptimal level. In this respect, *Ectothiorhodospira* species have salt optima at the upper border of marine bacteria, and some, in particular *E. haloalkaliphila*, are moderately halotolerant. Because of its extended salt tolerance, this species may successfully compete with *Halorhodospira* species in moderately saline soda lakes. To cope with the salt and osmotic stress, *Ectothiorhodospira* species (*E. haloalkaliphila*, *E. marismortui*) accumulate glycine betaine, sucrose, and a third component (α -N-carbamoyl-L-glutamine amide) as osmotica (Oren et al., 1991; Severin et al., 1992; Imhoff, 1993; Imhoff and Riedel, unpublished results), but not ectoine and trehalose, which along with glycine betaine have been found as major components in *Halorhodospira* species (Galinski et al., 1985; Severin et al., 1992).

ENRICHMENT AND ISOLATION PROCEDURES

Ectothiorhodospira species can be selectively enriched under photoautotrophic conditions with sulfide as the electron donor in saline and alkaline media (3% NaCl and pH 7.6–8.5), even when high proportions of *Chromatiaceae* species occur in the natural sample. Using Pfennig's medium for purple sulfur bacteria with the addition of appropriate concentrations of NaCl and adjusted to an alkaline pH (Trüper, 1970) and alternatively with a medium based on the mineral composition of the soda lakes of the Wadi Natrun (Jannasch, 1957; Imhoff and Trüper, 1977; Imhoff et al., 1979) or modifications thereof, *Ectothiorhodospira* strains have been isolated from marine sources, hypersaline ponds, salt swamps, and salt lakes from many parts of the world. From such sources, isolation can be achieved in agar dilution series with natural samples without prior enrichment. A recipe of the medium specifically designed for species from alkaline lakes containing 5% (and 20%) salts is given below (see Imhoff, 1988a).¹

1. The medium is prepared as a basal solution with (final content per liter medium): KH_2PO_4 , 0.8 g; trace element solution SLA (see below), 1 ml; 1 M sodium carbonate buffer pH 9.0, 200 ml; and NaCl, 30 g. This basal solution is sterilized in an autoclave. Solutions of 20% NH_4Cl (4 ml/l), 1% $CaCl_2 \cdot 2H_2O$ (5 ml/l), 2% $MgCl_2 \cdot 6H_2O$ (5 ml/l), and 5% $Na_2S \cdot 9H_2O$ (5–10 ml/l) are sterilized separately and added after cooling to room temperature. Sodium thiosulfate (1.0 g/l), if used as electron donor, and sodium acetate (1–2 g/l) may be added to the basal solution before sterilization. The final salinity is adjusted by changing the concentrations of sodium chloride, keeping under consideration the constant level of carbonate buffer, which is equivalent to ~2% total salts. The amounts yield a medium with 5% total salts. Part of the sodium chloride (10% of the total final salt content by weight) may be replaced by sodium sulfate. In order to adjust the salinity to a total

The recipe of the standard medium according to Pfennig is given in the description of the genus *Chromatium*.

MAINTENANCE PROCEDURES

Cultures can be maintained at the lab bench in dim light for several months. They can be stored in agar deeps kept in the dark at 4–10°C for up to a year. For long-term preservation, storage in liquid nitrogen is recommended. In this case, cell suspensions of well-grown cultures supplemented with DMSO to a final concentration of 5% are dispensed in plastic ampules, sealed, and frozen at the temperature of liquid nitrogen.

DIFFERENTIATION OF THE GENUS *ECTOTHIORHODOSPIRA* FROM OTHER GENERA

Species of the genus *Ectothiorhodospira* are differentiated from other purple sulfur bacteria of the *Chromatiaceae* family by the deposition of elemental sulfur outside their cells, when grown with sulfide as the photosynthetic electron donor. Differentiation of *Ectothiorhodospira* and *Halorhodospira* species is possible based on both molecular and physiological properties (Table BXII.γ.7). *Ectothiorhodospira* species contain MK-7 and either Q-7 or Q-8 as major components, whereas *Halorhodospira* species lack significant proportions of homologs with 7 isoprenoid units, but instead have MK-8, Q-8, and a short chain MK component as major components (Imhoff, 1984a; Ventura et al., 1993). Both genera form separate groups according to their fatty acid compositions (Thiemann and Imhoff, 1996; Table BXII.γ.8). They differ significantly according to their 16S rDNA sequences, as can be seen by a number of signature sequences and sequence similarities of 87.2–89.9% (Table BXII.γ.7). In contrast to *Ectothiorhodospira* species that form sulfur globules outside their cells, *Thiorhodospira* also forms sulfur globules enclosed by the outer membrane (Bryantseva et al., 1999b).

Metabolic properties such as utilization of carbon, sulfur, and nitrogen sources are very similar among the species of both genera. Whereas several species of the genus *Ectothiorhodospira* are able to grow under chemotrophic conditions in the dark, *Halorhodospira* species are obligately phototrophic. *Ectothiorhodospira* species have growth optima under conditions of 1–8% total salts, whereas *Halorhodospira* species do not grow below 10% total salts.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *ECTOTHIORHODOSPIRA*

Characteristic differential properties of *Ectothiorhodospira* species are summarized in Table BXII.γ.9. The genetic relationship of *Ectothiorhodospira* species based on 16S rDNA sequences is shown

TAXONOMIC COMMENTS

After reevaluation of the taxonomy of *Ectothiorhodospiraceae* on the basis of complete 16S rDNA sequences, the extremely halophilic species were removed from *Ectothiorhodospira* and reassigned to the new genus *Halorhodospira* (Imhoff and Söling, 1996). Analyses of 16S rDNA sequences and fatty acid compositions supported the identification and species assignments of some strains, but demonstrated the misclassification of others (Imhoff and Söling, 1996; Thiemann and Imhoff, 1996). A number of strains have been tentatively assigned to *E. mobilis*. Strains 8112 (DSMZ 237), 8113 (DSMZ 238), and 8815 (DSMZ 240, Trüper, 1970; Mandel et al., 1971) were confirmed as strains of *E. mobilis*. Strain 8115 (DSMZ 239, Trüper, 1970; Mandel et al., 1971) was recognized as belonging to *E. shaposhnikovii*. Strains 51/7 (BN 9903, ATCC 51935) and C (BN 9902, Imhoff et al., 1978) were recognized as belonging to a new species, *E. haloalkaliphila*. Strain BA 1010 (DSMZ 241, Matheron and Baulaigue, 1972) was found to represent the new species *E. marina*, and strain BA 1011 (Matheron and Baulaigue, 1972) was found to be most closely related to strain BA 1010. The recognition of strain DSMZ 239 as belonging to *E. shaposhnikovii*, the close relationship between *E. shaposhnikovii* and *E. vacuolata*, and that between *E. mobilis* and *E. marismortui* are supported by the DNA–DNA hybridization studies of Ivanova et al. (1985) and DNA restriction pattern analysis by Ventura et al. (1993). These results were substantiated by DNA–DNA reassociation studies by Ventura et al. (2000); these authors proposed the transfer of *E. vacuolata* to *E. shaposhnikovii* and *E. marismortui* to *E. mobilis*. Given the present state of knowledge, we do not follow this proposal here because of clearly distinctive phenotypic properties and ribotyping patterns. In addition, strain BN 9903 (ATCC 51935) has been intensively studied and, though it is known from the literature as *E. mobilis* (Stackebrandt et al., 1984; Imhoff et al., 1982a; Imhoff, 1984a; Imhoff and Riedel, 1989; Imhoff and Thiemann, 1991; Imhoff et al., 1991; Zahr et al., 1992; Ventura et al., 1993), recent studies have led to the definition of this strain as the type strain of *E. haloalkaliphila* (Imhoff and Söling, 1996).

in Fig. BXII.γ.19 of the chapter describing the family *Ectothiorhodospiraceae*.

List of species of the genus *Ectothiorhodospira*

1. *Ectothiorhodospira mobilis* Pelsh 1936, 120^{AL} emend. Ventura, Viti, Pastorelli and Giovannetti 2000, 589.
mol. bi. lis. L. adj. *mobilis* mobile.

Cells vibrioid, curved in a short spiral or slightly bent rods, 0.7–1.0 × 2.0–2.6 μm (Fig. BXII.γ.20). Motile by

of 20% salts, either 180 g NaCl or 20 g Na₂SO₄ and 160 g NaCl are added to a final volume of 1 liter medium. The medium is dispensed into tightly closed screw-capped bottles that are completely filled and stored until use. Prior to inoculation, it is necessary to preincubate the bottles, after removal of an appropriate small volume, at the incubation temperature in order to achieve volume expansion.

The trace element solution SLA has the following composition: FeCl₂·4H₂O, 1.8 g; CoCl₂·6H₂O, 250 mg; NiCl₂·6H₂O, 10 mg; CuCl₂·5H₂O, 10 mg; MnCl₂·4H₂O, 70 mg; ZnCl₂, 100 mg; H₃BO₃, 500 mg; Na₂MoO₄·2H₂O, 30 mg; and Na₂SeO₃·5H₂O, 10 mg. These components are dissolved in 1 liter of doubly distilled water. The pH of the solution is adjusted with HCl to 2–3.

means of a polar tuft of flagella. Color of cell suspensions free of polysulfides and S⁰ is red. Absorption spectra of living cells show maxima at 378, 488, 516, 550, 590, 798, and 854 nm, with a shoulder at 892 nm. Photosynthetic pigments are bacteriochlorophyll *a* (esterified with phytol) and carotenoids of the spirilloxanthin series with spirilloxanthin as the major component.

Obligately phototrophic and strictly anaerobic. Grow under anoxic conditions in the light with reduced sulfur compounds or organic carbon sources as electron donors. Photoautotrophic growth occurs with sulfide, thiosulfate, S⁰, sulfite and hydrogen. Acetate, pyruvate, malate, succinate, and fumarate used as organic carbon sources and electron donors. Some strains also use fructose, glucose, lactate, butyrate, and propionate. Nitrogen sources are ammonia, glu-

TABLE BXII.γ.9. Differential characteristics of *Ectothiorhodospira* species^a

Characteristics	<i>Ectothiorhodospira mobilis</i>	<i>Ectothiorhodospira haloalkaliphila</i>	<i>Ectothiorhodospira marina</i>	<i>Ectothiorhodospira marismortui</i>	<i>Ectothiorhodospira shaposhnikovii</i>	<i>Ectothiorhodospira vacuolata</i>
Cell diameter (μm)	0.7–1.0	0.7–1.2	0.8–1.2	0.9–1.3	0.8–0.9	1.5
Motility	+	+	+	+	+	+
Color of cell suspensions	red	red	red	red	red	red
NaCl optimum (%)	2–3	5	2–6	3–8	3	1–6
Salinity range (%)	1–5	2.5–15	0.5–10.0	1–20	0–7	0.5–10.0
pH optimum	7.6–8.0	8.5–10.0	7.5–8.5	7.0–8.0	8.0–8.5	7.5–9.5
Growth medium	mPF	mPF/EM5%	mPF	mPF	mPF/EM5%	mPF/EM5%
Chemolithotrophic growth	–	+	U	–	+	U
Sulfate assimilation	+	+	(+)	–	+	–
Fatty acid cluster	V	VII	VII	V	VI	VI
Major quinones	MK7/Q8	MK7/Q8	MK7/Q8	MK7/Q8	MK7/Q7	MK7/Q7
Mol% G + C of DNA of the species	67.3–68.4	62.2–63.5	62.8	65.0	62.0–64.0	61.4–63.6
Mol% G + C of DNA of the type strain	67.3 (Bd)	63.5 (<i>T_m</i>)	62.8 (Bd)	65.0 (<i>T_m</i>)	62.0 (<i>T_m</i>)	63.6 (<i>T_m</i>)
<i>Substrates used:</i> ^b						
Hydrogen	+	U	U	+	+	+
Sulfide	+	+	+	+	+	+
Thiosulfate	+	+	+	–	+	+
Sulfur	+	+	+	+	+	+
Sulfite	+	U	+	U	+	U
Acetate	+	+	+	+	+	+
Pyruvate	+	+	+	+	+	+
Propionate	±	U	+	(+)	+	+
Butyrate	±	U	–	–	+	–
Lactate	±	U	+	(+)	+	–
Fumarate	+	+	+	+	+	+
Succinate	+	+	+	+	+	+
Malate	+	+	+	+	+	+
Fructose	±	–	–	–	+	(+)
Glucose	±	–	–	–	U	–
Ethanol	–	–	–	–	–	–
Propanol	–	–	–	–	–	–

^aSymbols: +, characteristic positive or substrate utilized by most strains; ±, characteristic positive or substrate utilized by some strains; –, characteristic negative or substrate not utilized by most strains; U, unknown—no data available; (+) weak activity or growth.

^bNone of the species uses formate, methanol, ethanol, propanol, citrate, or benzoate.

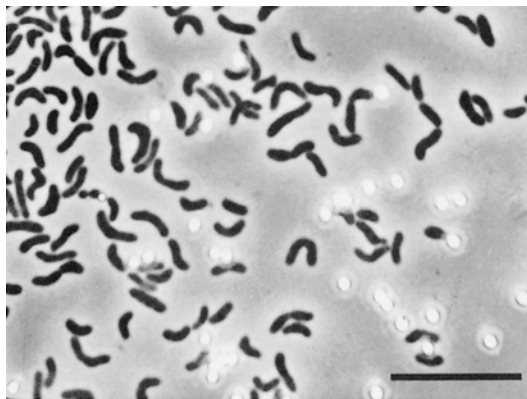


FIGURE BXII.γ.20. *Ectothiorhodospira mobilis* DSMZ 237. Note extracellular sulfur globules. Phase-contrast micrograph. Bar = 10 μm. (Reproduced with permission from H.G. Trüper and J.F. Imhoff. 1977. In Starr, Stolp, Trüper, Balows and Schlegel (Editors), The Prokaryotes. A Handbook on Habitats, Isolation, and Identification of Bacteria. Springer-Verlag, Berlin, p. 277.)

tamine, and some other amino acids. Sulfate is assimilated as sole sulfur source. Growth of some strains is enhanced by vitamin B₁₂.

Sodium chloride required for growth. Optimal devel-

opment is at salt concentrations of 2–3% (salinity range 1–7%), pH 7.6–8.0, and 25–30°C.

Habitat: Salt marshes, marine coastal sediments containing sulfide and exposed to the light.

Major quinone components are MK-7 and Q-8.

The mol% G + C of the DNA is: 67.3–68.4 (Bd); type strain 67.3 (Bd).

Type strain: DSMZ 237.

GenBank accession number (16S rRNA): BN 9911, X93481.

2. *Ectothiorhodospira haloalkaliphila* Imhoff and Süling 1997, 915^{VP} (Effective publication: Imhoff and Süling 1996, 112.)

ha' lo. al. ka. li' phi. la. Gr. n. *hals* salt; Arabic n. *al kali* potash; Gr. adj. *philos* loving; M.L. fem. adj. *haloalkaliphila* loving salt and alkaline conditions.

Cells are vibrios or curved in a short spiral, 0.7–1.2 × 2.0–3.0 μm. Motile by means of a polar tuft of flagella. Color of cell suspensions free of polysulfides and S⁰ is red. Absorption spectra of living cells show maxima at 379, 488, 513, 552, 593, 798, and 851 nm, with a shoulder at 892 nm. Photosynthetic pigments are bacteriochlorophyll *a* (esterified with phytol) and carotenoids of the spirilloxanthin series with spirilloxanthin as the major component.

Grow preferably anaerobically in the light with reduced sulfur compounds or organic carbon sources as electron donors. Photoautotrophic growth occurs with sulfide, thio-

sulfate, and S^0 . Chemoautotrophic and chemoheterotrophic growth is possible under microoxic conditions in the presence of sulfide. Organic carbon sources and electron donors used are acetate, pyruvate, malate, succinate, and fumarate. Nitrogen sources are ammonia and some amino acids. Sulfate is assimilated as the sole sulfur source.

Sodium chloride required for growth. Optimal development is at salt concentrations of 5% (range 2.5–15%), pH 8.5–10, and 26–40°C. Habitat: alkaline salt and soda lakes containing sulfide and exposed to the light.

Major quinone components are MK-7 and Q-8.

The mol% G + C of the DNA is: 63.5 (T_m).

Type strain: ATCC 51935, BN 9903.

GenBank accession number (16S rRNA): X93479.

3. **Ectothiorhodospira marina** Imhoff and Süling 1997, 915^{VP} (Effective publication: Imhoff and Süling 1996, 112.)
ma.ri'na. L. fem. adj. *marina* marine.

Cells curved, sometimes slightly bent rods, 0.8–1.2 × 1.5–4.0 µm. Motile by means of a polar tuft of flagella. Color of cell suspensions that are free of polysulfides and S^0 is red. Absorption spectra of living cells show characteristic absorption bands at 379, 483, 513, 552, 593, 798, and 853 nm, with a shoulder at 893 nm. Photosynthetic pigments are bacteriochlorophyll *a* (esterified with phytol) and carotenoids of the spirilloxanthin series with spirilloxanthin as the major component.

Grow preferably anaerobically in the light under photoautotrophic and photoheterotrophic conditions. Reduced sulfur compounds or organic carbon sources serve as electron donors. Photoautotrophic growth is possible with sulfide, thiosulfate, S^0 , and sulfite. Acetate, pyruvate, lactate, propionate, malate, succinate, and fumarate used as organic carbon sources and electron donors. Oxoglutarate and peptone also support growth. Nitrogen sources are ammonia, dinitrogen, and glutamine. Sulfate is assimilated as sole sulfur source.

Sodium chloride required for growth. Optimal development is at salt concentrations of 2–6% (salinity range 0.5–10‰), pH 7.5–8.5, and 30–40°C. Habitat: Coastal marine sediments containing sulfide and exposed to the light.

Major quinone components are MK-7 and Q-8.

The mol% G + C of the DNA is: 62.8 (Bd).

Type strain: BA 1010, BN 9914, DSMZ 241.

GenBank accession number (16S rRNA): X93476.

4. **Ectothiorhodospira marismortui*** Oren, Kessel and Stackebrandt 1990, 105^{VP} (Effective publication: Oren, Kessel and Stackebrandt 1989, 529.)
ma'ris.mor'tu.i. L. gen. n. *maris* of the sea; L. adj. *mortuus* dead; M.L. gen. n. *marismortui* of the Dead Sea.

Pleomorphic irregular short rods, often curved, 0.9–1.3 × 1.5–3.3 µm. Motile by means of a polar tuft of flagella. Color of liquid cultures is red to purple. *In vivo* absorption spectra show distinct maxima at 486, 515, 550, 591, 796, and 856 nm, with a shoulder at 885 nm. Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids of the spirilloxanthin series with spirilloxanthin as the major component.

Strictly phototrophic and obligately anaerobic. Grow photoautotrophically with sulfide and S^0 as electron donors, or photoheterotrophically with simple organic substrates. Carbon sources used are acetate, succinate, fumarate, malate, and pyruvate; poor substrates are lactate, glycerol, propionate, and yeast extract. Not utilized: methanol, ethanol, *n*-propanol, isopropanol, *n*-butanol, hydrogen, formate, butyrate, isobutyrate, citrate, benzoate, glycine, L-alanine, L-glutamate, glucose, fructose, mannitol, and sorbitol. Ammonia and dinitrogen can serve as nitrogen sources. Assimilatory sulfate reduction absent. No growth factors required, but small concentrations of yeast extract stimulate growth.

Sodium chloride required for growth. Optimum development is at salt concentrations of 3–8% (salinity range 1–20‰), pH 7–8, and 35–45°C.

Habitat: Microbial mats of hypersaline sulfur springs.

Major quinone components are MK-7 and Q-8.

The mol% G + C of the DNA is: 65.0 (T_m).

Type strain: EG-1, BN 9410, DSMZ 4180.

GenBank accession number (16S rRNA): X93482.

5. **Ectothiorhodospira shaposhnikovii** Cherni, Solovieva, Fedorova and Kondratieva 1969, 483^{AL} emend. Ventura, Viti, Pastorelli and Giovannetti 2000, 589.
sha'posh'ni.kov'i.i. M.L. gen. n. *shaposhnikovii* of Shaposhnikov, named for D. I. Shaposhnikov, a Russian microbiologist.

Cells rod-shaped, usually slightly bent vibrios or short spirilla. 0.8–0.9 × 1.5–2.5 µm when grown on propionate as carbon source. Motile by means of a polar tuft of flagella. Color of cell suspensions in the absence of polysulfides and S^0 is red. Absorption spectra of living cells show maxima at 378, 481, 513, 556, 593, 798, and 854 nm, with a shoulder at 892 nm. Photosynthetic pigments are bacteriochlorophyll *a* (esterified with phytol) and carotenoids of the spirilloxanthin series with spirilloxanthin as the major component.

Cells preferably grow under anoxic conditions in the light. Photoautotrophic growth occurs with reduced sulfur compounds or molecular hydrogen as the electron donor. Chemoautotrophic and chemoheterotrophic growth possible under microoxic conditions in the dark. Photosynthetic electron donors are sulfide, S^0 , thiosulfate, sulfite, molecular hydrogen, acetate, propionate, butyrate, lactate, pyruvate, malate, succinate, fumarate, and fructose. Not used: formate, methanol, ethanol, propanol, glycerol, citrate, and benzoate. Ammonia, dinitrogen, nitrate, and some amino acids used as nitrogen sources. Sulfate is assimilated as the sole sulfur source under photoheterotrophic conditions. Growth factors are not required.

Optimal development is at salt concentrations of 3% (salinity range 0–7‰), pH 8.0–9.0, and 30–35°C.

Habitat: Marine sediments containing sulfide and exposed to the light.

Major quinone components are MK-7 and Q-7.

The mol% G + C of the DNA is: 62.3 (Bd), 64.0 (chem. anal.), and 62.0 (T_m).

Type strain: BN 9711, DSMZ 243, KMMGU N1.

GenBank accession number (16S rRNA): M59151.

6. **Ectothiorhodospira vacuolata*** Imhoff, Tindall, Grant and Trüper 1982b, 266^{VP} (Effective publication: Imhoff, Tindall, Grant and Trüper 1981, 240.)
va.cu.o.la'ta. L. fem. adj. *vacuolata* containing vacuoles.

*Editorial Note: Ventura et al. (2000) consider *E. marismortui* a junior heterotypic synonym of *E. mobilis*.

Cells rod-shaped, sometimes slightly bent, $1.5 \times 2\text{--}4\text{ }\mu\text{m}$. Motile by means of a polar tuft of flagella. Motile forms often develop gas vesicles and later become immotile and float to the top because of the gas vesicles. At low sulfide concentrations and low light intensities, motile cells without gas vesicles predominate. Stationary-phase cells generally become immotile and form gas vesicles. Color of cell suspensions is pink to red. Absorption spectra of living cells show maxima at 378, 486, 513, 555, 593, 798, and 855 nm, with a shoulder at 893 nm. Photosynthetic pigments are bacteriochlorophyll *a* (esterified with phytol) and carotenoids of the spirilloxanthin series with spirilloxanthin as the major component.

Grow under anoxic conditions in the light. Photoautotrophic growth occurs with sulfide, thiosulfate, or S^0 . Organic carbon sources and electron donors used are acetate,

propionate, pyruvate, malate, succinate, and fumarate. Nitrogen sources are ammonia and dinitrogen. Cysteine, but not sulfate or methionine, serves as the sulfur source under photoheterotrophic growth conditions. Growth factors are not required.

Sodium chloride required for growth. Optimal development is at salt concentrations of 1–6% (salinity range 0.5–10%), pH 7.5–9.5, and 30–40°C.

Habitat: Salt flats and salt lakes containing sulfide and exposed to the light.

Major quinone components are MK-7 and Q-7.

The mol% G + C of the DNA is: 61.4–63.6 (T_m), type strain 63.6 (T_m).

Type strain: ATCC 43036, BN 9512, DSMZ 2111.

GenBank accession number (16S rRNA): X93478.

Genus II. *Arhodomonas* Adkins, Madigan, Mandelco, Woese and Tanner 1993, 518^{VP}

RALPH S. TANNER AND JOHANNES F. IMHOFF

A.rho.do.mo' nas. Gr. pref. *a* not; Gr. n. *rhodon* the rose, also the color or odor of the rose; Gr. n. *monas* lonely, alone; M.L. n. *Arhodomonas* the *monas* that is not rose colored.

Cells are **rods that occur singly or in pairs, are motile by means of flagella**, and multiply by binary fission. Gram negative. Members of the *Ectothiorhodospiraceae* of the *Gammaproteobacteria*. **Gas vesicles and endospores are not formed.**

Growth occurs under chemoorganotrophic conditions with oxygen or nitrate as the electron acceptor. Phototrophic growth is not possible and photosynthetic pigments are not produced. Catalase and oxidase positive.

Mesophilic and halophilic bacteria that require sodium for growth.

The mol% G + C of the DNA is: 67 (T_m).

Type species: *Arhodomonas aquaeolei* Adkins, Madigan, Mandelco, Woese and Tanner 1993, 518.

FURTHER DESCRIPTIVE INFORMATION

Arhodomonas aquaeolei is one of the few purely chemotrophic bacteria known that is phylogenetically related to the phototrophic purple sulfur bacteria of the *Gammaproteobacteria* and that belongs to the phylogenetic rather coherent group of the *Ectothiorhodospiraceae*. According to its 16S rDNA sequence, it is genetically most similar to *Halorhodospira* species but is sufficiently different so as to be recognized as a separate genus (Fig. BXII.γ.19 of the chapter describing the family *Ectothiorhodospiraceae* Imhoff and Söling, 1996). The characteristic loop at position 420 (UGCG) and the addition of C in the loop covering position 1361 are also present in the sequence of *Arhodomonas aquaeolei* (Adkins et al., 1993). *Arhodomonas* shares both the requirement for high salt concentrations and the restricted spectrum of carbon sources used with *Halorhodospira* and *Ectothiorhodospira* species (Imhoff, 1989). Most carbohydrates do not support growth, but other simple organic compounds are used instead. *Arhodomonas* does not have the ability to perform a phototrophic type of metabolism.

ENRICHMENT AND ISOLATION PROCEDURES

Arhodomonas aquaeolei was found in a subterranean brine produced from a petroleum reservoir, a habitat that has been without sunlight for very long time periods. It was isolated from this brine by direct plating on plate broth count agar (Difco) containing 15% NaCl and subsequent incubation under aerobic conditions at 23°C. Current information is based on a single isolate.

MAINTENANCE PROCEDURES

Cultures can be preserved by conventional freeze-drying.

DIFFERENTIATION OF THE GENUS *ARHODOMONAS* FROM OTHER GENERA

The 16S rDNA sequence similarity shows that *A. aquaeolei* is related to *Ectothiorhodospiraceae* and is most similar to *Halorhodospira* (Fig. BXII.γ.19 of the chapter describing the family *Ectothiorhodospiraceae*, Imhoff and Söling, 1996). It is clearly distinguished from these phototrophic bacteria by its strictly chemotrophic metabolism, its inability to grow phototrophically, and its lack of photosynthetic pigments. It shares the requirement for high salt concentrations with *Halorhodospira* species and in addition has a restricted spectrum of carbon sources, similar to those of its phototrophic relatives (Imhoff, 1989, 1992). Some properties of *A. aquaeolei* that differentiate it from other genera of the *Ectothiorhodospiraceae* are shown in Table BXII.γ.7 of the chapter describing the family *Ectothiorhodospiraceae*.

FURTHER READING

Adkins, J.P., M.T. Madigan, L. Mandelco, C.R. Woese and R.S. Tanner. 1993. *Arhodomonas aquaeolei* gen. nov., sp. nov., an aerobic, halophilic bacterium isolated from a subterranean brine. Int. J. Syst. Bacteriol. 43: 514–520.

Imhoff, J.F. and J. Söling. 1996. The phylogenetic relationship among *Ectothiorhodospiraceae*: a reevaluation of their taxonomy on the basis of 16S rDNA analyses. Arch. Microbiol. 165: 106–113.

*Editorial Note: Ventura et al. (2000) consider *E. vacuolata* a junior heterotypic synonym of *E. shaposhnikovii*.

List of species of the genus Arhodomonas

1. **Arhodomonas aquaeolei** Adkins, Madigan, Mandelco, Woese and Tanner 1993, 518^{VP}
a. quae. o' lei. L. gen. n. *aquae* of the water; L. gen. n. *olei* of the oil; M.L. gen. n. *aquaeolei* from water of oil.

Cells are straight, short rods, $0.8\text{--}1.0 \times 2.0\text{--}2.5 \mu\text{m}$ and motile by means of a single polar flagellum. Metabolism is strictly chemotrophic. Simple organic compounds such as acetate, propionate, lactate, butyrate, valerate, isovalerate, ethanol, glycerol, glutamic acid, glutamine, and xylose are used as substrates for aerobic chemotrophic growth. Glucose, fructose, sucrose, mannose, maltose, lactose, citrate, alanine, aspartic acid, asparagine, histidine, arginine, ser-

ine, and tryptophan are not used. Nitrate, but not nitrite, is reduced. Oxidase, catalase, and urease are present, but arginine, lysine, and ornithine decarboxylases are absent. Hydrolysis of Tween 80 is positive, while casein, DNA, esculin, gelatin, and starch are not hydrolysed. Cells are sensitive to chloramphenicol, penicillin, and tetracycline.

Mesophilic and extremely halophilic bacterium with optimum growth at 15% NaCl (range: 6–20%).

Habitat: oil field brine.

The mol% G + C of the DNA is: 67 (T_m).

Type strain: HA-1, ATCC 49307.

GenBank accession number (16S rRNA): M26631.

Genus III. Halorhodospira Imhoff and Süling 1997, 915^{VP} (Effective publication: Imhoff and Süling 1996, 112.)

JOHANNES F. IMHOFF

Ha' lo. rho' do. spi' ra. Gr. gen. n. *halos* of the salt; Gr. n. *rhodon* the rose; Gr. n. *spira* the spiral; M.L. fem. n. *Halorhodospira* the spiral rose from salt lakes.

Cells spirals or rods, $0.5\text{--}1.2 \mu\text{m}$ in diameter, motile by means of bipolar flagella, multiply by binary fission. **Gram negative and belong to the Gammaproteobacteria. Internal photosynthetic membranes appear as lamellar stacks** that are continuous with the cytoplasmic membrane. **Photosynthetic pigments are bacteriochlorophyll *a* or *b* and carotenoids.**

Grow **photoautotrophically** under anoxic conditions with reduced sulfur compounds as electron donors, or photoheterotrophically with a limited number of simple organic compounds. **Sulfide oxidized to S^0 , which is deposited outside the cells** and may be further oxidized to sulfate.

Growth dependent on highly saline and alkaline conditions. Greater than 10% (w/v) total salt concentration required by all known species, some of which grow in saturated salt solutions. **Glycine betaine, ectoine, and trehalose accumulated as compatible solutes in response to salt and osmotic stress.** Growth factors not required. Storage products are polysaccharides, poly- β -hydroxybutyrate, and polyphosphate.

Halorhodospira species are found in hypersaline and extremely saline environments with slightly to extremely alkaline pH (up to pH 11–12) that contain sulfide and are exposed to light, such as salt flats, salt lakes, and soda lakes.

The mol% G + C of the DNA is: 50.5–69.7.

Type species: **Halorhodospira halophila** (Raymond and Siström 1969) Imhoff and Süling 1997, 915 (Effective publication: Imhoff and Süling 1996, 110) (*Ectothiorhodospira halophila* Raymond and Siström 1969, 125.)

FURTHER DESCRIPTIVE INFORMATION

Cells of *Halorhodospira* species are spirals, but under severe osmotic stress, i.e., at low salt concentrations, they tend to become vibrioid to rod-shaped and increase in volume. The photosynthetic membranes of *Halorhodospira halochloris* have been intensively studied. In negatively stained preparations under the electron microscope, a regularly granulated structure is observed (Imhoff and Trüper, 1977). More detailed work, including image analysis, has demonstrated that the photosynthetic complexes are

arranged in a regular hexagonal array in *Halorhodospira halochloris* and *Halorhodospira abdelmalekii* (as well as in other bacteria containing bacteriochlorophyll *b*) (Engelhardt et al., 1983). Bacteriochlorophyll *b* of *H. halochloris* and *H. abdelmalekii* is esterified with Δ -2,10-phytydienol, not with phytol as is bacteriochlorophyll *a* of *H. halophila* (Steiner et al., 1981; R. Steiner, personal communication). Carotenoids of the normal spirilloxanthin series are present in *H. halophila* with spirilloxanthin as predominant component and negligible amounts of rhodopin (Schmidt and Trüper, 1971). The content of carotenoids in *H. halochloris* and *H. abdelmalekii* is low. Major components were methoxy-hydroxylycopene glucoside (methoxy-rhodopin glucoside), hydroxylycopene glucoside (rhodopin glucoside), and dihydroxylycopene diglucoside (Takaichi et al., 2001). Substantial amounts of dihydroxylycopene diglucoside diester were also found.

Halorhodospira species are among the most halophilic eubacteria known (Imhoff, 1988a, b). With regard to their salt responses and the salt concentrations required for optimum growth, *Halorhodospira* species are extremely halophilic (see Genus *Ectothiorhodospira* in this volume and Imhoff, 1993). In alkaline and highly hypersaline salt and soda lakes, they commonly develop massive blooms that cause red or green coloration of the water or sediment horizons (Jannasch, 1957; Imhoff et al., 1979). To adapt to high concentrations of salts and to cope with the high external osmotic pressure, bacteria and other unicellular microorganisms have to accumulate osmotically active molecules that are compatible with the molecular cell structures and metabolic processes; these are called compatible solutes or osmotica (see Imhoff, 1986, 1993). In contrast to halophilic algae and archaeobacteria that selectively accumulate glycerol and potassium ions, respectively, *Halorhodospira* species accumulate glycine betaine, ectoine, and trehalose. In *H. halochloris*, glycine betaine was initially reported as the main osmotic active cytoplasmic component (Galinski and Trüper, 1982). More recently, a novel component that is active as an osmoprotectant and accumulates in high concentrations in cells of *H. halochloris* and other *Halorhodospira* species has been identified as 1,4,5,6-tetra-

hydro-2-methyl-4-pyrimidinecarboxylic acid (Galinski et al., 1985). The trivial name ectoine has been given to this component because of its first discovery in an *Ectothiorhodospiraceae* (at that time *Ectothiorhodospira*) species. This compound was later found to be widely distributed among the halophilic eubacteria (Severin et al., 1992). Unlike the *Halorhodospira* species, ectoine and trehalose have not been found as osmotica in *Ectothiorhodospira* species (*E. haloalkaliphila* and *E. marismortui*) that characteristically accumulate glycine betaine, sucrose, and an unidentified component (Oren et al., 1991; Imhoff and Riedel, unpublished results).

ENRICHMENT AND ISOLATION PROCEDURES

Halorhodospira species have been isolated from various hypersaline environments, such as salt and soda lakes, from many parts of the world. Alkaline soda lakes show a natural abundance of *Ectothiorhodospira* species, which can be taken as proof of their successful adaptation to these environments. From such sources, isolation can be achieved by agar dilution series using natural samples without prior enrichment. The tolerance of and dependence on high salinity and alkalinity are strong selective conditions for the enrichment of *Halorhodospira* species.

A medium based on the mineral composition of the soda lakes of the Wadi Natrun (Jannasch, 1957; Imhoff and Trüper, 1977; Imhoff et al., 1979) is appropriate for the cultivation of all currently known *Halorhodospira* species. A recipe for this medium containing 20% total salts is given with the description of the genus *Ectothiorhodospira* (see also Imhoff, 1988a).

MAINTENANCE PROCEDURES

Most cultures can be maintained at the lab bench in dim light for several months; they can be stored for up to a year in agar deeps kept in the dark. Species that have bacteriochlorophyll *b* are more sensitive to prolonged storage, particularly with respect to oxygen and light. These should be transferred more frequently than *H. halophila*. For long-term preservation, storage in liquid nitrogen is recommended as the standard maintenance procedure.

DIFFERENTIATION OF THE GENUS *HALORHODOSPIRA* FROM OTHER GENERA

Halorhodospira species represent the most halophilic species known in the phototrophic purple sulfur bacteria, and they are well adapted to form massive blooms in hypersaline environments and in alkaline soda lakes. *Halorhodospira* species are extremely halophilic bacteria and do not grow at total salt concentrations below 10%, whereas *Ectothiorhodospira* species have growth optima at salt concentrations below 10%.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *HALORHODOSPIRA*

Characteristic properties for differentiation of *Halorhodospira* species are summarized in Table BXII.γ.10 of the chapter describing the family *Ectothiorhodospiraceae*. The genetic relationship of *Ha-*

lorhodospira species based on 16S rDNA sequences is shown in Fig. BXII.γ.19 of the chapter describing the family *Ectothiorhodospiraceae*.

Species of the genus *Halorhodospira* are easily differentiated from other purple sulfur bacteria of the *Chromatiaceae* by the deposition of S⁰ outside their cells, when grown with sulfide as the photosynthetic electron donor. Differentiation of *Ectothiorhodospira* and *Halorhodospira* species is possible based on both molecular and physiological properties (Table BXII.γ.7 of the chapter describing the family *Ectothiorhodospiraceae*). Absorption spectra of *H. halophila* have a clear maximum at approximately 890 nm, but do not show absorbance at 830 nm. The long wavelength absorption maximum of the species having bacteriochlorophyll *b* is between 1010 and 1020 nm. Whereas several species of the genus *Ectothiorhodospira* are able to grow under chemotrophic conditions in the dark, all *Halorhodospira* species are obligately phototrophic. In contrast to *Ectothiorhodospira* species, *Halorhodospira* species do not contain significant proportions of quinone homologs with 7 isoprenoid units, but instead have MK-8, Q-8, and a short chain MK component as major components (Imhoff, 1984a; Ventura et al., 1993). Both genera also form separate groups according to their fatty acid compositions (Thiemann and Imhoff, 1996; Table BXII.γ.8 of the chapter describing the family *Ectothiorhodospiraceae*). Their 16S rDNA sequences are significantly different; this is evident from a number of signature sequences as well as sequence similarities of 87.2–89.9% (Table BXII.γ.7 of the chapter describing the family *Ectothiorhodospiraceae*).

TAXONOMIC COMMENTS

After reevaluation of the taxonomy of *Ectothiorhodospiraceae* based on complete 16S rDNA sequence comparison, the extremely halophilic species were removed from *Ectothiorhodospira* and reassigned to the new genus *Halorhodospira* (Imhoff and Süling, 1996).

Two different but closely related groups of *H. halophila* strains (12 strains included in the study) have been recognized based on fatty acid composition (Thiemann and Imhoff, 1996), 16S rDNA sequences (Imhoff and Süling, 1996), and DNA restriction pattern analysis (Ventura et al., 1993). This diversity is considered to be intraspecies variation, most likely related to different salt responses. The mol% G + C content of the DNA of *H. halochloris* is surprisingly low when compared to that of the two other species of this genus. Such a difference of nearly 20% within the same genus is unusual, particularly when given the rather similar sequences of the 16S rRNA gene. However, the mol% G + C values in Table BXII.γ.9 of the chapter describing the family *Ectothiorhodospiraceae* have been confirmed by two independent analyses of a total of three strains (personal communication of H. Hippe, Braunschweig).

lorhodospira species based on 16S rDNA sequences is shown in Fig. BXII.γ.19 of the chapter describing the family *Ectothiorhodospiraceae*.

List of species of the genus *Halorhodospira*

1. ***Halorhodospira halophila*** (Raymond and Siström 1969) Imhoff and Süling 1997, 915^{VP} (Effective publication: Imhoff and Süling 1996, 110) (*Ectothiorhodospira halophila* Raymond and Siström 1969, 125.)
ha.lo'phi.la. Gr. n. *hals* salt; Gr. adj. *philos* loving; M.L. fem. adj. *halophila* salt-loving.

Cells curved in a spiral, 0.6–0.9 × 2–8 μm (Fig.

BXII.γ.21). Color of cell suspensions in the absence of polysulfides and S⁰ is red. Absorption spectra of living cells show maxima at 378, 488, 520, 554, 590, 799, 831–835, and 886–889 nm. Photosynthetic pigments are bacteriochlorophyll *a* (esterified with phytol) and carotenoids of the spirilloxanthin series with spirilloxanthin as the predominant component.

TABLE BXII.γ.10. Differential characteristics of *Halorhodospira* species^a

Characteristic	<i>Halorhodospira halophila</i>	<i>Halorhodospira abdelmalekii</i>	<i>Halorhodospira halochloris</i>
Cell diameter (μm)	0.6–0.9	0.9–1.2	0.5–0.6
Motility	+	+	+
Color of cell suspensions	red	green	green
Bchl	<i>a</i>	<i>b</i>	<i>b</i>
NaCl optimum (%)	11–32	12–18	14–27
pH optimum	8.5–9.0	8.0–9.2	8.1–9.1
Sulfate assimilation	–	–	+
Fatty acid cluster	I + II	III	IV
Major quinones	MK8/Q8 ^b	MK8/Q8 ^b	MK8/Q8 ^b
Mol% G + C of DNA of the species	66.5–69.7	63.3–63.8	50.5–52.9
Mol% G + C of DNA of the type strain	68.4 (Bd)	63.8 (<i>T_m</i>)	52.9 (<i>T_m</i>)
<i>Substrates used:</i> ^c			
Hydrogen	U	U	U
Sulfide	+	+	+
Thiosulfate	+	–	–
Acetate	+	+	+
Pyruvate	+	+	+
Propionate	+	+	+
Lactate	±	–	–
Fumarate	+	+	+
Succinate	+	+	+
Malate	+	+	+

^aSymbols: +, characteristic positive or substrate utilized by some strains; ±, characteristic positive or substrate utilized by some strains; –, characteristic negative or substrate not utilized by most strains; U, unknown—no data available; (+) weak activity or growth.

^bIn addition a short chain isoprenoid quinone is present in large amounts.

^cNo species uses formate, butyrate, glucose, fructose, glycerol, methanol, ethanol, or propanol.

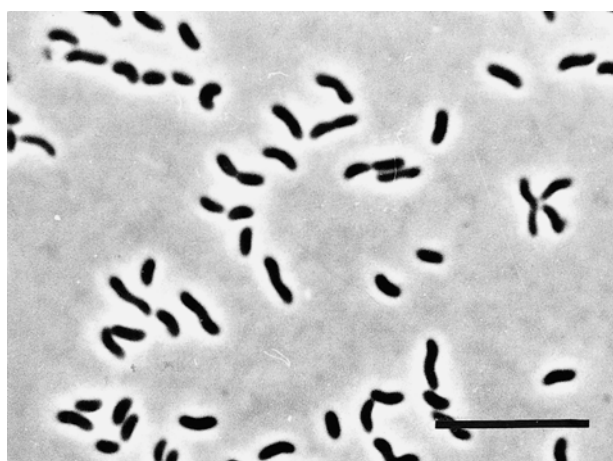


FIGURE BXII.γ.21. *Halorhodospira halophila* BN 9621. Phase-contrast micrograph. Bar = 10 μm. (Reproduced with permission from H.G. Trüper and J.F. Imhoff. 1977. In Starr, Stolp, Trüper, Balows and Schlegel (Editors), The Prokaryotes. A Handbook on Habitats, Isolation, and Identification of Bacteria. Springer-Verlag, Berlin, p. 277.)

Obligately phototrophic and strictly anaerobic. Photoautotrophic growth occurs with sulfide and thiosulfate as electron donors. When inorganic carbonate is present, acetate, propionate, pyruvate, malate, succinate, and fumarate are used as organic carbon sources. Ammonia, glutamate, and glutamine serve as nitrogen sources. Sulfate assimilation absent. Cysteine used as a sulfur source under photoheterotrophic growth conditions. Growth factors not required, but vitamin B₁₂ enhances growth of some strains.

Extremely high concentrations of sodium chloride are required for growth. Optimal development is at 30–47°C, pH 8.5–9.0, and 11–32% salt concentration. Cells lyse at low salt concentrations (below approximately 0.5 M NaCl).

Habitat: mud and waters of highly saline and alkaline environments containing sulfide and exposed to the light, such as salt and soda lakes.

The mol% G + C of the DNA is: 66.5–69.7 (*T_m*), type strain 68.4 (Bd).

Type strain: SL1, BN 9610, DSMZ 244.

GenBank accession number (16S rRNA): M26630.

2. *Halorhodospira abdelmalekii* (Imhoff and Trüper 1982)

Imhoff and Süling 1997, 915^{VP} (Effective publication: Imhoff and Süling 1996, 111) (*Ectothiorhodospira abdelmalekii* Imhoff and Trüper 1982, 266.)

abd'el.ma.lek' i.i. M.L. gen. n. *abdelmalekii* of Abd-El-Malek, named for Yousef Abd-El-Malek, an Egyptian microbiologist.

Cells spirals or rods, 0.9–1.2 × 4–6 μm, or longer. Motile by means of bipolar, sheathed flagella. Color of cell suspensions is pale green, sometimes with a brownish tinge. Photosynthetic pigments are bacteriochlorophyll *b* (esterified with phytadienol) and small amounts of carotenoids.

Grow under strictly anoxic conditions in the presence of sulfide, inorganic carbonate, and an organic carbon source. Acetate, propionate, pyruvate, succinate, fumarate, and malate are used as carbon sources and electron donors. Growth factors are not required. Ammonia is used as nitrogen source.

Extremely high concentrations of sodium chloride are required for growth. Optimal development is at 30–44°C, pH 8.0–9.2, and 12–18% salts.

Habitat: mud and waters of highly saline and alkaline environments containing sulfide and exposed to the light, such as salt and soda lakes.

The mol% G + C of the DNA is: 63.3–63.8 (*T_m*), type strain 63.8 (*T_m*).

Type strain: BN 9840, ATCC 35917, DSMZ 2110.

GenBank accession number (16S rRNA): X93477.

3. **Halorhodospira halochloris** (Imhoff and Trüper 1979) Imhoff and Süling 1997, 915^{VP} (Effective publication: Imhoff and Süling 1996, 111) (*Ectothiorhodospira halochloris* Imhoff and Trüper 1979, 79.)

ha.lo.chlo'ris. Gr. gen. n. *halos* of the salt; Gr. adj. *chloros* green; M.L. adj. *halochloris* green-colored and from salt lakes.

Cells spiral-shaped, 0.5–0.6 μm wide and, depending on the culture conditions, 2.5–8.0 μm long (Fig. BXII.γ.22). Color of cell suspensions is pale green to gooseberry-green, in dense populations with a brownish tinge. Living cells have absorption maxima at 374, 389, 598, 796, 884, and 1018 nm. Photosynthetic pigments are bacteriochlorophyll *b* (esterified with phytadienol) and small amounts of carotenoids (mainly rhodopin glucoside and its methoxy derivative).

Obligately phototrophic and strictly anaerobic. In the presence of sulfide as the photosynthetic electron donor and inorganic carbonate, acetate, propionate, pyruvate, succinate, fumarate, and malate are used. Nitrogen sources are ammonia and glutamine. Growth factors are not required.

Extremely high concentrations of sodium chloride required for growth. Optimal development is at 30–47°C, pH 8.1–9.1, and 14–27% salt concentration, no growth below approximately 10% total salts.

Habitat: mud and waters of highly saline and alkaline environments containing sulfide and exposed to the light, such as salt and soda lakes.

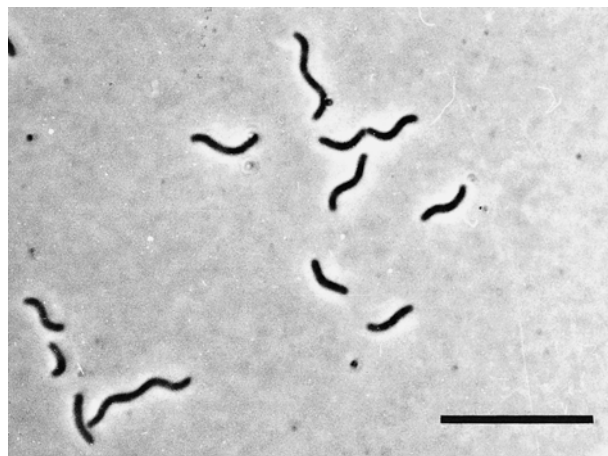


FIGURE BXII.γ.22. *Halorhodospira halochloris* DSMZ 1059. Phase-contrast micrograph. Bar = 10 μm . (Reproduced with permission from H.G. Trüper and J.F. Imhoff. 1977. In Starr, Stolp, Trüper, Balows and Schlegel (Editors), *The Prokaryotes. A Handbook on Habitats, Isolation, and Identification of Bacteria*. Springer-Verlag, Berlin, p. 277.)

The mol% G + C of the DNA is: 50.5–52.9 (T_m), type strain 52.9 (T_m).

Type strain: BN 9850, ATCC 35916, DSMZ 1059.

GenBank accession number (16S rRNA): M59152.

Genus IV. **Nitrococcus** Watson and Waterbury 1971, 224^{AL}

EVA SPIECK AND EBERHARD BOCK

Ni.tro.coc'cus. L. n. *nitrum* nitrate; Gr. n. *coccus* a grain, berry; M.L. masc. n. *Nitrococcus* nitrate sphere.

Spherical cells 1.5 μm or more in diameter, occurring singly or in pairs. Cells reproduce by binary fission. **Intracytoplasmic membranes occur as tubes**, randomly arranged throughout the cytoplasm. Gram negative. **Motile** by 1 or 2 flagella. **Aerobic**. Energy derived from the **oxidation of nitrite to nitrate**. **Obligate lithoautotrophs**. Optimal growth in 70–100% seawater enriched with nitrite and other inorganic salts. No organic growth factors required. Isolated from marine environments.

The mol% G + C of the DNA is: 61.2.

Type species: **Nitrococcus mobilis** Watson and Waterbury 1971, 224^{AL}.

FURTHER DESCRIPTIVE INFORMATION

Nitrococcus mobilis belongs to the family *Ectothiorhodospiraceae* in the class *Gammaproteobacteria* (Woese et al., 1985). It is the only example of a nitrite-oxidizing bacterium that is phylogenetically related to ammonia-oxidizing bacteria (*Nitrosococcus oceani* and "*Nitrosococcus halophilus*") in the same class. All these organisms are from marine habitats. No growth occurs in freshwater media even if NaCl is included. The temperature range for growth is 15–30°C and the pH range is 6.8–8.0. No terrestrial strains are known so far. *Ectothiorhodospira*, *Arhodomonas*, *Halorhodospira*, and *Thiorhodospira* are the closest relatives of *Nitrococcus*.

Additional details and a comparison of the biochemical properties of *Nitrococcus* to those of other nitrite-oxidizing genera can be found in the introductory chapter "Lithoautotrophic Nitrite-Oxidizing Bacteria." Detailed treatments of the ecology of nitrite-

oxidizing bacteria and of the phylogeny of these organisms can be found in the introductory chapter "Nitrifying Bacteria."

Because of the presence of extensive intracytoplasmic membrane systems in both genera, Teske et al. (1994) hypothesized that *Nitrobacter* and *Nitrococcus* derived from an immediate photosynthetic ancestry. Additional features that the two genera have in common include *a*-type cytochromes, flagella, carboxysomes, and storage materials. The tubular membranes of *Nitrococcus* are quite similar to the internal photosynthetic membrane system of *Thiocapsa pfennigii* (*Thiococcus pfennigii*).

A spherical cell of *Nitrococcus* with two flagella is shown in Fig. BXII.γ.23. The tubular intracytoplasmic membranes are visible in Fig. BXII.γ.24. Cells are elongated before division (1.8 $\times$ 3.5 μm). They may occur as small clumps of cells embedded in a slime matrix. The cell wall is similar to that seen in most other Gram-negative bacteria. The tubular membrane system arises by invaginations of the cytoplasmic membrane and has a branching nature. As in *Nitrobacter*, the cytoplasmic and intracytoplasmic membranes are asymmetric, with an electron-dense layer on the inner side; the structure of the membranes becomes apparent when cells are ruptured and negatively stained. The surface of the membrane is covered with particles, which are arranged in rows (Fig. BXII.γ.24 and BXII.γ.25). Watson and Waterbury (1971) described doughnut-shaped particles 6–8 nm in diameter, which occasionally form rectilinear arrays. The particulate intracytoplasmic membranes of *Nitrococcus* were labeled with gold particles using monoclonal antibodies (MAbs) that recognize the β -

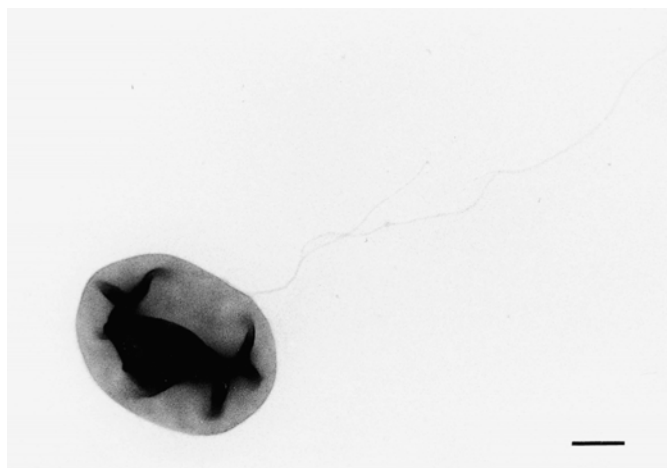


FIGURE BXII.γ.23. Negatively stained cell of *Nitrococcus mobilis* with flagella. Bar = 500 nm.

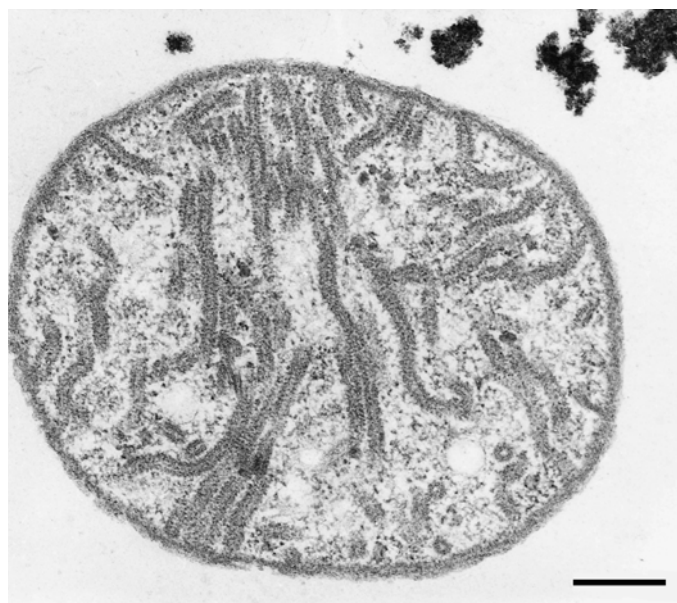
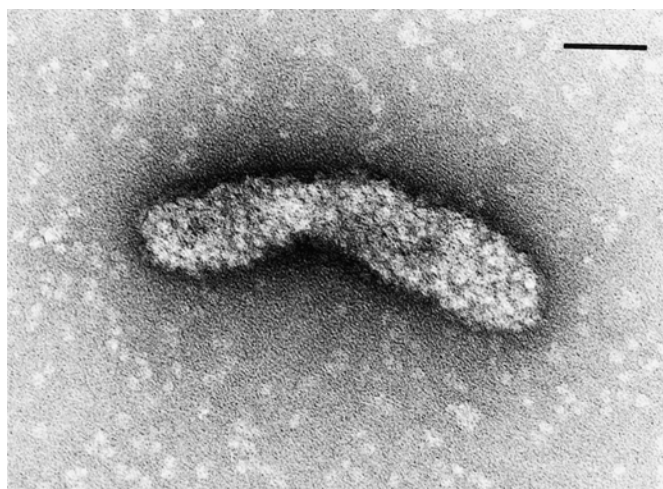


FIGURE BXII.γ.24. Ultrathin section of *Nitrococcus mobilis* with tubular intracytoplasmic membranes which surface is covered with particles. Bar = 250 nm.

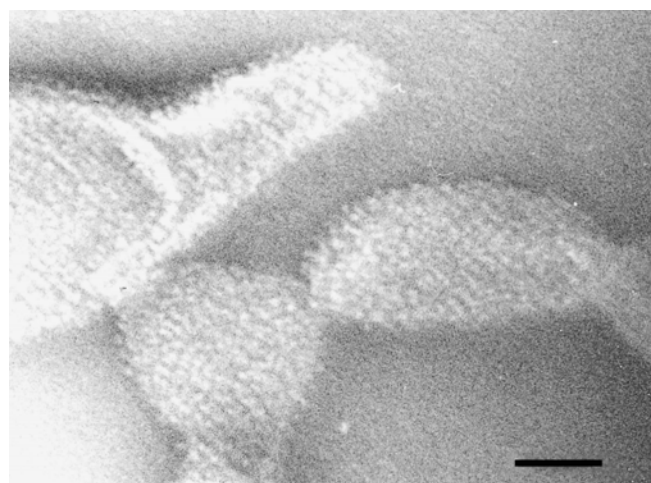
subunit of the nitrite oxidoreductase (β -NOR) of *Nitrobacter* (Fig. BXII.γ.26). The molecular mass of the β -subunit of the nitrite-oxidizing system (β -NOS) of *Nitrococcus* is in the same range as the β -NOR of *Nitrobacter* (65 kDa) (Bartosch et al., 1999). The size of this protein corresponds to that of one of the main proteins (135 and 61 kDa) found in crude extracts of *Nitrococcus* (Krause-Kupsch, personal communication). Cytochromes of the a - and c -types as well as several kinds of inclusion bodies such as carboxysomes and poly- β -hydroxybutyrate granules are present in *Nitrococcus*. Electron-dense bodies believed to be composed of glycogen are observed.

ENRICHMENT AND ISOLATION PROCEDURES

Nitrite oxidizers can be isolated using a mineral medium containing nitrite; the compositions of media for lithotrophic, mixotrophic, and heterotrophic growth are given in Table BXII.γ.11.



A



B

FIGURE BXII.γ.25. Isolated intracytoplasmic membranes of *Nitrococcus mobilis* in negative contrast. The membrane surface is densely packed with particles of about 6 nm in diameter (A) which can be regularly arranged in rows (B). Bars = 50 nm.

Serial dilutions of enrichment cultures must be incubated for one to several months in the dark. Since nitrite oxidizers are sensitive to high partial pressures of oxygen, cell growth on agar surfaces is limited. Nitrite oxidizers can be separated from heterotrophic contaminants by Percoll gradient centrifugation and subsequent serial dilution (Ehrich et al., 1995).

MAINTENANCE PROCEDURES

Nitrifying organisms can survive starvation for more than one year when kept at 17°C in liquid medium. Nevertheless, cells should be transferred to fresh media every four months. Table BXII.γ.11 lists three different growth media for nitrite oxidizers. Freezing in liquid nitrogen is a suitable technique for maintenance of stock cultures suspended in a cryoprotective buffer containing sucrose and histidine.

DIFFERENTIATION OF THE GENUS *NITROCOCCUS* FROM OTHER GENERA

The nitrite oxidizers are a diverse group of long or short rods, cocci, and spirillae. The genera *Nitrobacter* and *Nitrococcus* possess

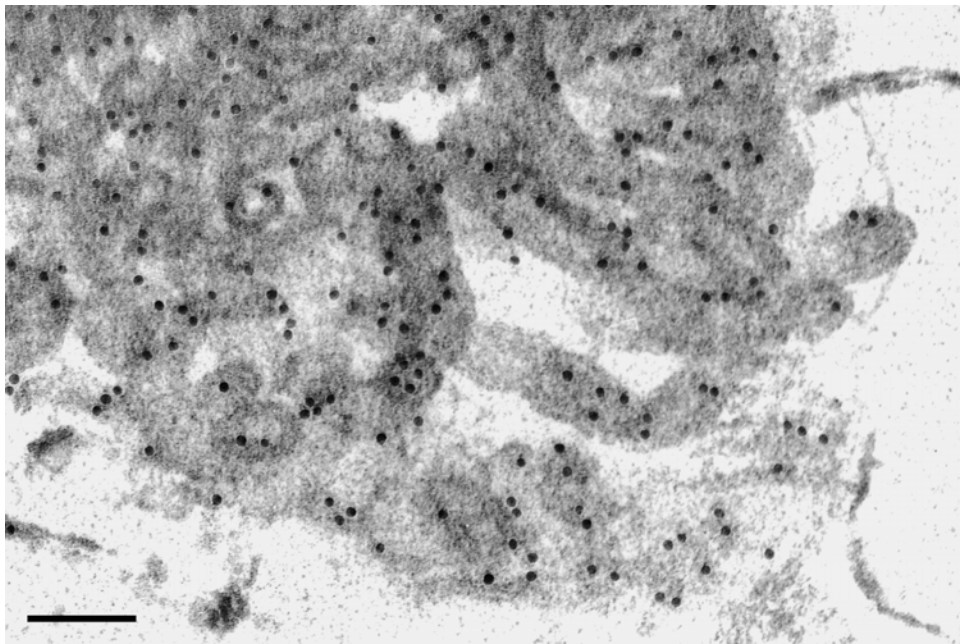


FIGURE BXII.γ.26. Location of the β-NOS in *Nitrococcus mobilis* on the cytoplasmic side of the tubular intracytoplasmic membranes shown by immunogoldlabeling (MAbs 153-1). The cell was partly destroyed by the embedding procedure. Bar = 100 nm.

TABLE BXII.γ.11. Three different media for lithoautotrophic (medium A for terrestrial strains; medium B for marine strains), mixotrophic (medium C), and heterotrophic (medium C without NaNO₂) growth of nitrite oxidizers

Ingredient	Culture medium		
	A ^a	B ^b	C ^{c, d}
Distilled water (ml)	1000	300	1000
Seawater (ml)		700	
NaNO ₂ (mg)	200–2000	69	200–2000
MgSO ₄ ·7H ₂ O (mg)	50	100	50
CaCl ₂ ·2H ₂ O (mg)		6	
CaCO ₃ (mg)	3		3
KH ₂ PO ₄ (mg)	150	1.7	150
FeSO ₄ ·7H ₂ O (mg)	0.15		0.15
Chelated iron (13%, Geigy) (mg)		1	
Na ₂ MoO ₄ ·2H ₂ O (μg)		30	
(NH ₄) ₂ Mo ₇ O ₂₄ ·4H ₂ O (μg)	50		50
MnCl ₂ ·6H ₂ O (μg)		66	
CoCl ₂ ·6H ₂ O (μg)		0.6	
CuSO ₄ ·5H ₂ O (μg)		6	
ZnSO ₄ ·7H ₂ O (μg)		30	
NaCl (mg)	500		500
Sodium pyruvate (mg)			550
Yeast extract (Difco) (mg)			1,500
Peptone (Difco) (mg)			1,500
pH adjusted to ^e	8.6	6	7.4

^aFor terrestrial strains from Bock et al. (1983).
^bFor marine strains modified from Watson and Waterbury (1971).
^cFor terrestrial strains from Bock et al. (1983).
^dFor heterotrophic growth medium C without NaNO₂ is used.
^eAfter sterilization pH should be 7.4–7.8.

a complex arrangement of intracytoplasmic membranes in form of flattened vesicles or tubes. *Nitrococcus* is motile by 1–2 flagella and possesses a tubular membrane system as well as carboxysomes. The taxonomic categorization is based on the work of Sergei and Helene Winogradsky (Winogradsky, 1892). Traditionally, the classification of genera is performed primarily on

cell shape and arrangement of intracytoplasmic membranes. So far, four morphologically distinct genera (*Nitrobacter*, *Nitrococcus*, *Nitrospina*, and *Nitrospira*) have been described; the four genera contain a total of eight species (Watson et al., 1989; Bock and Koops, 1992; Ehrich et al., 1995; Sorokin et al., 1998). Differential traits of the four genera of nitrite-oxidizing bacteria are given in Table BXII.γ.12, further properties in Table BXII.γ.13, and fatty acid profiles in Table BXII.γ.14. Motility and carboxysomes have been observed in *Nitrobacter* and *Nitrococcus* (Tables BXII.γ.12 and BXII.γ.13).

Nitrococcus can be separated from the other genera by means of monoclonal antibodies that recognize the key enzyme (Table BXII.γ.12). Aamand et al. (1996) developed three monoclonal antibodies (MAbs) that recognize the α- and the β-subunit of the NOR of *Nitrobacter*. MAb Hyb 153-2 was specific for the α-NOR of *Nitrobacter*, whereas MAb Hyb 153-1 recognized the β-NOSs of both *Nitrobacter* and *Nitrococcus*. The MAb Hyb 153-3 recognized the β-NOSs of *Nitrobacter*, *Nitrococcus*, *Nitrospina* and *Nitrospira* (Bartosch et al., 1999). Thus, the suite of MAbs permits detection of all known nitrite oxidizers and discrimination of *Nitrobacter* and *Nitrococcus* from *Nitrospina* and *Nitrospira*.

TAXONOMIC COMMENTS

Classification of *Nitrococcus* as a distinct genus of nitrite-oxidizing bacteria is based primarily on its unique morphology and ultrastructure. Sequence analysis of the 16S rRNA confirmed the separation.

FURTHER READING

Watson, S.W. and J.B. Waterbury. 1971. Characteristics of two marine nitrite oxidizing bacteria, *Nitrospina gracilis* nov. gen. nov. sp. and *Nitrococcus mobilis* nov. gen. nov. sp. Arch. Microbiol. 77: 203–230.
Watson, S.W., E. Bock, H. Harms, H.P. Koops and A.B. Hooper. 1989. Nitrifying bacteria. In Staley, Bryant, Pfennig and Holt (Editors), Bergey’s Manual of Systematic Bacteriology, 1st Ed., Vol. 3, The Williams & Wilkins Co., Baltimore. pp. 1808–1833.

TABLE BXII.γ.12. Differentiation of the four genera of nitrite-oxidizing bacteria

Characteristic	<i>Nitrobacter</i>	<i>Nitrococcus</i>	<i>Nitrospina</i>	<i>Nitrospira</i>
Phylogenetic position	<i>Alphaproteobacteria</i>	<i>Gamma</i> proteobacteria	<i>Deltaproteobacteria</i> (preliminary)	Phylum <i>Nitrospirae</i>
Morphology	Pleomorphic short rods	Coccoid cells	Straight rods	Curved rods to spirals
Intracytoplasmic membranes	Polar cap	Tubular	Lacking	Lacking
Size (μm)	0.5–0.9 × 1.0–2.0	1.5–1.8	0.3–0.5 × 1.7–6.6	0.2–0.4 × 0.9–2.2
Motility	+	+	–	–
Reproduction:	Budding or binary fission	Binary fission	Binary fission	Binary fission
Main cytochrome types ^a	<i>a</i> , <i>c</i>	<i>a</i> , <i>c</i>	<i>c</i>	<i>b</i> , <i>c</i>
Location of the nitrite oxidizing system on membranes	Cytoplasmic	Cytoplasmic	Periplasmic	Periplasmic
MAB-labeled subunits (kDa) ^b	130 and 65	65	48	46
Crystalline structure of membrane-bound particles	Rows of particle dimers	Particles in rows	Hexagonal pattern	Hexagonal pattern

^aLithoautotrophic growth.^bMABs, monoclonal antibodies.**TABLE BXII.γ.13.** Properties of the nitrite-oxidizing bacteria

Characteristic	<i>Nitrobacter winogradskyi</i>	<i>Nitrobacter alkalicus</i>	<i>Nitrobacter hamburgensis</i>	<i>Nitrobacter vulgaris</i>	<i>Nitrococcus mobilis</i>	<i>Nitrospina gracilis</i>	<i>Nitrospira marina</i>	<i>Nitrospira moscoviensis</i>
Mol% G + C of the DNA	61.7	62	61.6	59.4	61.2	57.7	50	56.9
Carboxysomes	+	–	+	+	+	–	–	–
<i>Habitat</i> :								
Fresh water	+			+				
Waste water	+			+				
Brackish water				+				
Oceans	+				+	+	+	
Soda lakes		+						
Soil	+		+	+				
Soda soil		+						
Stones	+			+				
Heating system								+

TABLE BXII.γ.14. Primary fatty acids of the described species of nitrite-oxidizing bacteria^{a,b}

Fatty acid	<i>Nitrobacter winogradskyi</i> Engel	<i>Nitrobacter alkalicus</i> AN4	<i>Nitrobacter hamburgensis</i> X14	<i>Nitrobacter vulgaris</i> Z	<i>Nitrococcus mobilis</i> 231	<i>Nitrospina gracilis</i> 3	<i>Nitrospira marina</i> 295	<i>Nitrospira moscoviensis</i> M1
C _{14:1} ω5c						+		
C _{14:0}	+		+		+	+++	+	+
C _{16:1} ω9c							+++	++
C _{16:1} ω7c	+	+	+	+	+++	+++		
C _{16:1} ω5c							+++	+++
C _{16:0} 3-OH						+		+
C _{16:0}	++	++	++	++	+++	++	+++	+++
C _{16:0} 11-CH ₃							+	+++
C _{18:1} ω9c	+		+			+	+	
C _{18:1} ω7	+++	++++	++++	++++	+++	+	+	+
C _{18:0}	+	+	+		+	+	++	+
C _{19:0} ωcyclo7-8	+	+	+	+	+			

^aSymbols: +, <5%; ++, 6–15%; +++, 16–60%; +++, >60%.^bStirred cultures were grown autotrophically at 28°C (*Nitrospira moscoviensis* at 37°C) and collected at the end of exponential growth. Modified from Lipski et al., (2001).*List of species of the genus Nitrococcus***1. *Nitrococcus mobilis*** Watson and Waterbury 1971, 224^{AL}
mo' bi.lis. L. adj. *mobilis* movable, motile.

The morphological, cultural and biochemical characteristics are as described for the genus. The minimum gen-

eration time is 10 h. Optimum temperature range, 25–30°C. Optimum pH range, 7.5–8.0. The type strain was isolated from equatorial Pacific seawater.

The mol% G + C of the DNA is: 61.3 (Bd).

Type strain: ATCC 25380.

Genus V. Thioalkalivibrio Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 578^{VP}

THE EDITORIAL BOARD

Thi.o.al.kal.i.vi' bri.o. Gr. n. *thios* sulfur; N.L. n. *alkali* Arabic *al the*; *qaliy* soda ash; L. v. *vibrio* vibrate; M.L. n. *Thioalkalivibrio* sulfur alkaline vibrio.

Motile Gram-negative curved rods or spirilla (0.4–0.6 × 0.8–3.0 µm). **One polar flagellum. Cell wall rippled.** Possess carboxysomes. **Obligate chemolithoautotrophs. Oxidize sulfide, thiosulfate, sulfur, and tetrathionate.** Carbon assimilation via Calvin cycle. **Halotolerant** up to 1.2–1.5 M Na⁺; require at least 0.3 M Na⁺.

The mol% G + C of the DNA is: 61–65.6.

Type species: **Thioalkalivibrio versutus** Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 579 (*Thioalkalivibrio versutus* (sic) Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 579.)

FURTHER DESCRIPTIVE INFORMATION

These organisms were found in the water and sediments of alkaline lakes in which sulfide is produced in the underlying sediments. Growth in the laboratory was studied in carbonate/bicarbonate buffered medium containing thiosulfate, sulfide, polysulfide, sulfur, and polythionates. All strains produce elemental sulfur from thiosulfate at the beginning of the exponential phase in batch culture; this sulfur vanishes later in growth. Tetrathionate and elemental sulfur were used by most strains. The pH optimum was 9–10. Ammonia, nitrite, and nitrate were used as nitrogen sources. Some strains reduced nitrate to nitrite under reduced O₂ levels. The type strain of *T. denitrificans*, ALJD, is a

facultative anaerobe able to grow with thiosulfate and nitrous oxide as electron donor and acceptor, respectively (Sorokin et al., 2001a).

Analysis of the 16S rDNA sequence of the type strains of *T. versutus*, *T. denitrificans*, and *T. nitratis* showed that the genus *Thioalkalivibrio* belongs to the *Gammaproteobacteria*; two species of the genus *Ectothiorhodospira* (anaerobic purple sulfur bacteria) were their closest relatives (Sorokin et al., 2001a).

ENRICHMENT AND ISOLATION PROCEDURES

These bacteria were obtained by aerobic enrichment and isolation in a carbonate/bicarbonate buffered liquid salts medium containing thiosulfate. For some strains the amount of NaCl in the base medium had to be increased. Strains that reduce nitrate to nitrite can be obtained by adding 20 mM nitrate to the base medium. The single denitrifying strain was obtained using thiosulfate and N₂O in an anaerobic enrichment culture (Sorokin et al., 2001a).

MAINTENANCE PROCEDURES

Thioalkalivibrio spp. can be stored at 4°C for 2 months between transfers in liquid salts medium (pH 10) containing 5 mM MgCl₂ to stabilize the cell walls. They can also be stored in 10% glycerol (v/v) at –80°C (Sorokin et al., 2001a).

List of species of the genus *Thioalkalivibrio*

1. **Thioalkalivibrio versutus** Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 579 (*Thioalkalivibrio versutus* (sic) Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 579.) *ver.su'tus*. L. adj. *versutus* versatile.

Description as for the genus with the following additional characteristic. Best growth occurs under atmospheric levels of O₂.

The mol% G + C of the DNA is: 63–65.6 (*T_m*).

Type strain: AL 2, DSM 13738.

GenBank accession number (16S rRNA): AF126546.

2. **Thioalkalivibrio denitrificans** Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 579^{VP} (*Thioalkalivibrio denitrificans* (sic) Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 579.) *de.ni.tri'fi.cans* M.L. v. *denitrifico* denitrify; M.L. part. adj. *denitrificans* denitrifying.

Description as for the genus with the following additional characteristics. Facultatively anaerobic denitrifier. Best growth is obtained with thiosulfate and nitrous oxide under anaerobic conditions.

The mol% G + C of the DNA is: 62.3–65.0 (*T_m*).

Type strain: ALJD, DSM 13742.

GenBank accession number (16S rRNA): AF126545.

3. **Thioalkalivibrio nitratis** Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 579^{VP} (*Thioalkalivibrio nitratis* (sic) Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 579.) *ni.trat'is*. M.L. adj. *nitratis* nitrate.

Description as for the genus with the following additional characteristic. Nitrate reduced to nitrite under microaerophilic conditions.

The mol% G + C of the DNA is: 61.3–62.1 (*T_m*).

Type strain: ALJ 12, DSM 13741.

GenBank accession number (16S rRNA): AF126547.

Other Organisms

Sorokin et al. (2001c) described two further alkaliphilic chemolithoautotrophs whose 16S rDNA sequences were related to *Thioalkalivibrio* spp.: one type utilized thiocyanate as a nitrogen

source while using thiosulfate as an electron donor; the other utilized thiocyanate as both a source of nitrogen and of electrons.

Genus VI. Thiorhodospira Bryantseva, Gorlenko, Kompantseva, Imhoff, Süling and Mityushina 1999b, 700^{VP}

JOHANNES F. IMHOFF AND VLADIMIR M. GORLENKO

Thi'o.rho'do.spi'ra. Gr. n. *thios* sulfur; Gr. n. *rhodon* the rose; Gr. n. *spira* spiral; M.L. fem. n. *Thiorhodospira* the spiral rose with sulfur.

Cells vibrioid or spiral-shaped, motile by means of a monopolar flagellar tuft, multiplies by binary fission. Gram-negative and belong to the *Gammaproteobacteria*. Internal photosynthetic membranes are parallel lamellae piercing the cytoplasm lengthwise or underlying the cytoplasmic membrane. **Photosynthetic pigments are bacteriochlorophyll *a* and carotenoids.**

Obligately phototrophic and strictly anaerobic. **During photolithoautotrophic growth with sulfide as the electron donor, globules of S⁰ are formed outside the cytoplasm**, in the medium, attached to cells, or in the periplasm. The final oxidation product is sulfate. In the presence of sulfide, organic substances may be photoassimilated. Development is **dependent on sodium salts** in low concentrations and on alkaline conditions.

Habitat: surface of sediments rich in organic matter and microbial mats from brackish soda lakes that contain hydrogen sulfide.

The mol% G + C of the DNA is: 56.0–57.4 (*T_m*).

Type species: Thiorhodospira sibirica Bryantseva, Gorlenko, Kompantseva, Imhoff, Süling and Mityushina 1999b, 700.

ENRICHMENT AND ISOLATION PROCEDURES

Thiorhodospira species can be selectively enriched under photoautotrophic conditions with sulfide as the electron donor and in saline and alkaline media similar to those for *Ectothiorhodospira* and *Halorhodospira* species. From suitable environmental sources, isolation can be achieved using a basal medium¹ in agar dilution series inoculated with natural samples without prior enrichment. Cultures are grown phototrophically in 50-ml screw-capped bottles at 25–30°C and a light intensity of 2000 lux.

MAINTENANCE PROCEDURES

For long-term preservation, storage in liquid nitrogen is recommended as the standard maintenance procedure. For this purpose, cell suspensions of well-grown cultures supplemented with DMSO to a final concentration of 5% are dispensed in plastic ampules, sealed, and frozen at the temperature of liquid nitrogen.

DIFFERENTIATION OF THE GENUS *THIORHODOSPIRA* FROM OTHER GENERA

Thiorhodospira species may be differentiated from other purple sulfur bacteria by the deposition of S⁰ both outside and inside the cells, as seen under the light microscope. Differentiation from *Ectothiorhodospira* and *Halorhodospira* species is possible on the basis of sulfur deposition both outside and inside the cells, the irregular arrangement of the membrane stacks of photosynthetic membranes, the low mol% G + C content, and some characteristic differences in the 16S rDNA sequence (Table BXII.γ.7 of the chapter describing the family *Ectothiorhodospiraceae*). The 16S rDNA sequence of *Thiorhodospira* is ~88–92% similar to those of *Ectothiorhodospira* species. Metabolic properties, such as utilization of carbon, sulfur, and nitrogen sources, are very similar to those of other *Ectothiorhodospiraceae*. *Thiorhodospira* species grow at salt concentrations between 0.5–8%, as do *Ectothiorhodospira* species.

List of species of the genus *Thiorhodospira*

1. ***Thiorhodospira sibirica*** Bryantseva, Gorlenko, Kompantseva, Imhoff, Süling and Mityushina 1999b, 700^{VP}
si.bi'ri.ca. M.L. fem. n. *sibirica* related or belonging to Siberia.

Cells vibrioid or spirals, 3–4 × 7–20 µm, multiply by binary fission with formation of septa. Motile by means of a monopolar tuft of flagella, which is approximately the same length as the cells. Internal photosynthetic membranes are parallel lamellae piercing the cytoplasm lengthwise or underlying the cytoplasmic membrane. Color of cell suspensions is brownish-red to red. Suspensions of intact cells show absorption maxima at 377, 485, 509, 537–550, 591, 711, 799, 830, 858, and 901 nm. Photosynthetic pig-

ments are bacteriochlorophyll *a* and carotenoids of the spirilloxanthin series, with anhydrorhodovibrin, rhodopin, and spirilloxanthin as major components.

Strictly anaerobic and obligately phototrophic. Photolithoautotrophic growth occurs with sulfide and sulfur as electron donors. Thiosulfate is not oxidized. When sulfide and carbonates are present, acetate, pyruvate, malate, succinate, propionate, and fumarate are photoassimilated.

Mesophilic and alkaliphilic brackish-water bacterium with optimum growth at pH 9.0–9.5 (pH-range 7.5–10.5) and 25–30°C. Requires sodium carbonate and/or sodium chloride for growth. Salinity range is 0.5–8% sodium bicarbonate or 0–6% NaCl (in the presence of 0.5% sodium carbonate).

Habitat: Sulfide-containing surface layers of the Lake Malý Kasytů sediment.

The mol% G + C of the DNA is: 56.0–57.4 (*T_m*).

Type strain: A12, ATCC 700588, BN 9312.

GenBank accession number (16S rRNA): AJ006530.

1. The basal medium used for isolation and cultivation of *Thiorhodospira* contains (g/l distilled water): KH₂PO₄, 0.5 g; NH₄Cl, 0.5 g; NaCl, 0.5 g; MgCl₂·6H₂O, 0.2 g; CaCl₂·2H₂O, 0.05 g; NaHCO₃, 2.5 g; Na₂CO₃, 2.5 g; Na-acetate, 0.5 g; Na-malate, 0.5 g; yeast extract, 0.1 g; Na₂S·9H₂O, 0.7 g; and trace element solution SL8, 1 ml. The pH is adjusted to 9.0–9.5 (Bryantseva et al., 1999b).

Family III. **Halothiobacillaceae** *fam. nov.* Kelly and Wood 2003

DONOVAN P. KELLY AND ANN P. WOOD

Hal.o.thi.o.ba.cil'la'ce.æ. M.L. neut. n. *Halothiobacillus* type genus of the family; *-aceae* ending to denote a family; M.L. fem. pl. n. *Halothiobacillaceae* the *Halothiobacillus* family.

Cells are rods, motile by polar flagella, contain ubiquinone Q-8, and are obligately aerobic and chemolithotrophic. **Gram negative, belong to the Gammaproteobacteria, halotolerant and halophilic, obligately chemolithoautotrophic.** Aerobic chemolithoautotrophic growth is dependent on the oxidation of inorganic sulfur compounds or elemental sulfur and the fixation of carbon dioxide by the Calvin-Bassham-Benson reductive pentose phosphate cycle. In nature, members of the *Halothiobacillaceae* occur in seawater and marine mud, fresh water, salt lakes, and hydrothermal vents.

The mol% G + C of the DNA is: 56–58.

Type genus: **Halothiobacillus** Kelly and Wood 2000, 515 emend. Sievert, Heidorn and Kuever 2000, 1235.

TAXONOMIC COMMENTS

The family *Halothiobacillaceae* has been newly created to include the single genus *Halothiobacillus*, and the characteristics of the family are currently those given in the Genus *Halothiobacillus* chapter.

The four currently recognized and validly published species

are physiologically and morphologically similar, and exhibit a relatively wide spread of mol% G + C content of their DNA. Their distinctive physiological properties and biochemical requirements distinguish them all other species. Analysis of nearly complete 16S rDNA sequences from all four species showed them to form a single phylogenetic cluster (Kelly et al., 1998; Kelly and Wood, 2000; Sievert et al., 2000).

Pairwise comparisons of the relative identities of their 16S rDNA sequences using BLASTN and BLAST2 (<http://www.ncbi.nlm.gov>) showed the type species, *H. neapolitanus*, to share 91.0–92.9% identity with the other three; *H. kellyi* showed 93.4% identity to both *H. halophilus* and *H. hydrothermalis*; and *H. halophilus* and *H. hydrothermalis* shared 98.6% identity. For comparison, 16S rDNA from *Allochromatium vinosum* ATCC 17899 (M26629) showed only 86.7% and 88.0% identity to *H. neapolitanus* and *H. halophilus*. Several other strains of *Halothiobacillus* have been identified by 16S rDNA sequencing (strains MS02, RA13, W5 and WJ18), and additional species are likely to be defined and validly published in due course.

Genus I. Halothiobacillus Kelly and Wood 2000, 515^{VP} emend. Sievert, Heidorn and Kuever 2000, 1235

DONOVAN P. KELLY AND ANN P. WOOD

Hal.o.thi.o.ba.cil'lus. Gr. n. *hals* sea, salt; Gr. n. *thios* sulfur; L. n. *bacillus* a small rod; L. masc. n. *Thiobacillus* salt-loving sulfur rodlet.

Obligately chemolithoautotrophic Gram-negative rods, 0.3–0.6 × 1.0–2.5 µm, occurring singly or in pairs. Energy for growth is obtained from the oxidation of reduced inorganic sulfur compounds (thiosulfate, tetrathionate, trithionate, sulfur, and sulfide, but not thiocyanate); unable to oxidize ferrous iron. Tolerant of high concentrations of solutes (e.g., 4 M NaCl; 0.25 M sodium thiosulfate), with some strains showing moderate halophily with a requirement for NaCl (optima 0.4–1.0 M NaCl). Optimum temperature for growth 30–42°C and optimum pH 6.5–8.0. During growth on reduced sulfur compounds, pH is changed from neutrality to pH 2.5–3.0. Contain ubiquinone Q-8. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 56–67.

Type species: **Halothiobacillus neapolitanus** (Parker 1957) Kelly and Wood 2000, 515 (*Thiobacillus neapolitanus* Parker 1957, 86.)

FURTHER DESCRIPTIVE INFORMATION

This genus was created to accommodate the neutrophilic, obligately chemolithoautotrophic, and halophilic or halotolerant species previously described as *Thiobacillus*. Characteristics of the four species of *Halothiobacillus* currently recognized are shown in Table BXII.γ.15. The properties, taxonomy, and differentiation of the genera of sulfur-oxidizing, chemolithoautotrophic, Gram-negative, rod-shaped bacteria are summarized in the tables and the figure in the chapter on the genus *Thiobacillus* which appear in Volume 2 Part C in this *Manual*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *HALOTHIOBACILLUS*

The properties, similarity, and distinctive features of the species of *Halothiobacillus* are fully discussed elsewhere (Kelly et al., 1998; Kelly and Wood, 2000; Sievert et al., 2000).

List of species of the genus *Halothiobacillus*

1. **Halothiobacillus neapolitanus** (Parker 1957) Kelly and Wood 2000, 515^{VP} (*Thiobacillus neapolitanus* Parker 1957, 86.)
ne.a.po.li.ta'nus. L. adj. *neapolitanus* Neapolitan; pertaining to the seawater at Naples from which this species was probably first isolated by Nathansohn in 1902.

Small rods, 0.3–0.5 × 1.0–1.5 µm. Motile by means of a polar flagellum. Colonies grown on thiosulfate agar are small (1–2 mm), circular, convex, glistening, and whitish

yellow due to precipitated sulfur. The center of old colonies becomes pink. In static culture in liquid thiosulfate medium, sulfur and polythionates may accumulate, and the medium becomes uniformly turbid with a sulfur pellicle. Aerated cultures may show transitory accumulation of tetrathionate and trithionate. pH drops to 2.8–3.3. Chemostat cultures do not accumulate intermediates and convert thiosulfate quantitatively to sulfate. This organism oxidizes sulfur, sulfide, thiosulfate, tetrathionate, and trithionate but

TABLE BXII.γ.15. Basic characteristics of species of the genus *Halothiobacillus*

Characteristics	1. <i>H. neapolitanus</i>	2. <i>H. halophilus</i>	3. <i>H. hydrothermalis</i>	4. <i>H. kellyi</i>
Mol% G + C	56	64	67–68	62
Cell size (μm)	0.3–0.5 × 1.0–1.5	0.5 × 1.0	0.5 × 1.0	0.4–0.6 × 1.2–1.5
Motility	+	+	+	+
Carboxysomes	+	nd	nd	nd
Obligately chemolithoautotrophic	+	+	+	+
Halotolerant	+	+	+	+
Optimum NaCl (M)	nd	0.8–1.0	0.43	0.4–0.5
Maximum NaCl (M)	0.86	4	2	2.5
Optimum pH	6.5–6.9	7.0–7.3	7.5–8.0	6.5
pH limits	4.5–8.5	6.0–8.0	6.0–9.0	3.5–8.5
Optimum temperature, °C	28–32	30–32	35–40	37–42
Nitrate reduction to N ₂	–	–	–	–
Growth on/oxidation of:				
Sulfur	+	+	+	+
Thiosulfate	+	+	+	+
Tetrathionate	+	+	+	+
Trithionate	+	+	nd	nd

not thiocyanate, and uses ammonium salts or nitrates as nitrogen source. Obligately chemolithotrophic and autotrophic. Strict aerobe. Optimum temperature for growth: 28–32°C; growth range: 8–39°C. Optimum pH for growth: 6.5–6.9; growth range pH: 4.5–8.5. Isolated from seawater, canal water, and corroding concrete. Frequently, this organism is found in marine mud and in seawater. Presumably widely distributed in freshwater, soil, and marine environments. Member of the *Gamma*proteobacteria.

The mol% G + C of the DNA is: 56 (Bd).

Type strain: ATCC 23641, DSM 15147, NCIB 8539.

GenBank accession number (16S rRNA): AF173169.

2. **Halothiobacillus halophilus** (Wood and Kelly 1991) Kelly and Wood 2000, 515^{VP} (*Thiobacillus halophilus* Wood and Kelly 1991, 280.)

hal.o.phil.us. L. n. *halophil* salt-loving; M.L. adj. *halophilus* loving salt.

Cells are Gram-negative small rods. Obligately halophilic, optimum growth with 1 M NaCl and tolerating up to 4 M NaCl. Obligately chemolithoautotrophic; grows aerobically with thiosulfate, tetrathionate or sulfur; lowers the pH to 5.5–6. Also oxidizes sulfide, trithionate, and hexathionate. On thiosulfate agar, colonies are small (1–3 mm), circular, opaque, smooth, and convex, and become yellow/white with deposited sulfur. Optimum temperature for growth 30–32°C and optimum pH for growth 7.0–7.3. Nitrate reduced to nitrite but anaerobic growth with thiosulfate does not occur. Contains ubiquinone Q-8. Isolated from a salt lake in the wheat belt of Western Australia. Member of the *Gamma*proteobacteria.

The mol% G + C of the DNA is: 64.2 (HPLC).

Type strain: DSM 6132, ATCC 49870.

GenBank accession number (16S rRNA): U58020.

3. **Halothiobacillus hydrothermalis** (Durand, Reysenbach, Prieur and Pace 1993) Kelly and Wood 2000, 515^{VP} (*Thiobacillus hydrothermalis* Durand, Reysenbach, Prieur and Pace 1993, 43.)

hy.dro.ther.mal'is. M.L. *hydrothermalis* pertaining to a hydrothermal vent in the North Fiji basin.

Cells are Gram negative, motile, nonsporeforming, short rods, 0.5 × 1.0 μm. Obligately chemolithotrophic, using oxidation of thiosulfate, tetrathionate, sulfide, or sulfur for growth. Lowers the pH of thiosulfate medium to 4.8, without formation of tetrathionate. Growth occurs in the ranges

pH 6–9, 11–45°C, and 0–2 M NaCl; no growth at pH 5 or 9.5, or at 8°C or 50°C. Optimum pH 7.5–8 and temperature 35–40°C; optimum salinity 0.43 M NaCl, tolerates 2 M NaCl. Contains ubiquinone Q-8. Isolated from samples taken from a depth of 2 km in an active hydrothermal vent in a rift system of the North Fiji Basin. Member of the *Gamma*proteobacteria.

The mol% G + C of the DNA is: 67.1–67.4 (*T_m*, Bd, HPLC).

Type strain: ATCC 51453, DSM 7121.

GenBank accession number (16S rRNA): M90662.

4. **Halothiobacillus kellyi** Sievert, Heidorn and Kuever 2000, 1235^{VP}

kel'ly.i. M.L. gen. n. *kellyi* of Kelly; named after Donovan P. Kelly, a British microbiologist who has made important contributions to research on sulfur-oxidizing bacteria and their physiology.

Cells are Gram negative, motile, and rod shaped (0.4–0.6 × 1.2–2.5 μm). The organism is strictly aerobic and grows autotrophically on thiosulfate, tetrathionate, sulfur, and sulfide, but not thiocyanate. The organism does not grow heterotrophically. When thiosulfate is used as the primary energy source, a transient formation of sulfur occurs. During growth on reduced sulfur compounds, the pH decreases from neutrality to around pH 2.8. Thiosulfate is completely oxidized to sulfate. Autotrophic growth on thiosulfate occurs between pH 3.5 and 8.5 and at a temperature of 35–49°C; optimum growth occurs at pH 6.5 and at 37–42°C. The optimal NaCl concentration for growth is 0.4–0.5 M; growth is possible between NaCl concentrations of 0–2 M. Nitrate is not used as terminal electron acceptor. The average maximum specific growth rate on thiosulfate is 0.45 h⁻¹ for the type strain, Milos-BIII1. On thiosulfate agar, cells produce white to yellowish, smooth, entire colonies (diameter on 1.2% w/v agar is 1–4 mm), in which sulfur is deposited and acid is produced. Ubiquinone Q-8 is present in the respiratory chain. As determined by 16S rRNA gene sequence analysis, *Halothiobacillus kellyi* belongs to the *Gamma*proteobacteria and is most closely related to other members of the *Halothiobacillus* genus. The type strain Milos-BIII1 was isolated from a marine, shallow-water, hydrothermal vent system and is deposited at the DSM.

The mol% G + C of the DNA is: 62 (HPLC).

Type strain: Milos-BIII, DSM 13162.

GenBank accession number (16S rRNA): AF170419.

Order II. **Acidithiobacillales** ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

A.ci.di.thi.o.ba.cil.la'les. M.L. masc. n. *Acidothiobacillus* type genus of the order; *-ales* ending to denote order; M.L. fem. n. *Acidithiobacillales* the *Acidithiobacillus* order.

The order *Acidithiobacillales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the order contains the families *Acidithiobacillaceae* and *Thermithiobacillaceae*.

Autotrophs utilizing reduced sulfur compounds; mesophilic or moderately thermophilic.

Type genus: Acidithiobacillus Kelly and Wood 2000, 513^{VP}.

Family I. **Acidithiobacillaceae** fam. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

A.ci.di.thi.o.ba.cil.la'ce.ae. M.L. masc. n. *Acidothiobacillus* type genus of the family; *-aceae* ending to denote family; M.L. fem. pl. n. *Acidithiobacillaceae* the *Acidithiobacillus* family.

The family *Acidithiobacillaceae* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the family contains the genus *Acidithiobacillus* (type genus).

Description is the same as for the genus *Acidithiobacillus*.

Type genus: Acidithiobacillus Kelly and Wood 2000, 513^{VP}.

Genus I. **Acidithiobacillus** Kelly and Wood 2000, 513^{VP}

DONOVAN P. KELLY AND ANN P. WOOD

A.ci.di.thi.o.ba.cil'lus. L. adj. *acidus* sour; Gr. n. *thios* sulfur; L. n. *bacillus* a small rod; M.L. masc. n. *Thiobacillus* small acid-loving sulfur rod.

Obligately acidophilic, aerobic, Gram-negative rods, 0.4×2.0 μm , motile by one or more flagella. Optimum growth below pH 4.0, using reduced sulfur compounds to support autotrophic growth. Some species oxidize ferrous iron or use natural and synthetic metal sulfides for energy generation; some species oxidize hydrogen. Optimum temperature 30–35°C for mesophilic species and 45°C for moderately thermophilic species. Contain ubiquinone Q-8. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 52–64.

Type species: Acidithiobacillus thiooxidans (Waksman and Joffe 1922) Kelly and Wood 2000, 513 (*Thiobacillus thiooxidans* Waksman and Joffe 1922, 239.)

FURTHER DESCRIPTIVE INFORMATION

The characteristics of the four current recognized species of *Acidithiobacillus* are shown in Table BXII.7.16. The properties, taxonomy, and differentiation of the genera of sulfur-oxidizing, chemolithoautotrophic, Gram-negative, rod-shaped bacteria are summarized in the chapter on the genus *Thiobacillus* which appears in Volume 2 Part C in this *Manual*. Some species of *Acidithiobacillus* grow chemolithoautotrophically on sulfide minerals, and *A. ferrooxidans* also uses the oxidation of ferrous iron [and probably uranium (IV) and copper (I) oxidation] as the source of metabolic energy. One strain of *A. ferrooxidans* (ATCC 21834) has been shown to have a circular chromosome (2.9 Mb) and one small plasmid (8.6 kb) (Irazabal et al., 1997). Other bacteria able to oxidize sulfide minerals or iron (II) that appear superficially similar to *Acidithiobacillus*, but that are taxonomically and biochemically very different, have been described. These include the Gram-negative genus *Leptospirillum* and the Gram-positive genera *Acidimicrobium* and *Sulfobacillus* (Balashova et al., 1974; Golovacheva and Karavaiko, 1979; Harrison and Norris, 1985; Clark and Norris, 1996; Norris et al., 1996).

TAXONOMIC COMMENTS

This genus was created to accommodate the obligately acidophilic *Thiobacillus* species that were shown by 16S rRNA gene sequence analysis to fall into the *Gammaproteobacteria* (McDonald et al., 1997; Kelly and Wood, 2000).

The assignment of *T. albertis* to the new genus as *Acidithiobacillus albertensis* is very tentative. The 16S rRNA gene sequence, required for placement in a family in the *Proteobacteria*, is not yet available. Its relatively high mol% G + C value (61.5) and the possession of a tuft of flagella and a glycocalyx could indicate significant differences from the other three species included here. Unfortunately, the original isolate may have been lost from culture (B.M. Goebel, personal communication).

Since the first description of *Thiobacillus ferrooxidans* (Temple and Colmer, 1951), numerous organisms from mineral leaching environments have been described as strains of *T. ferrooxidans*, because they were extremely acidophilic, obligately chemolithoautotrophic with ferrous iron as an energy substrate, and could degrade pyrite and various other sulfide minerals. It became clear that selective culture for organisms exhibiting this growth phenotype actually resulted in the isolation of morphologically and genomically diverse strains (Harrison, 1982; Kelly and Harrison, 1989; Kondrat'eva and Karavaiko, 1997; Goebel et al., 2000). Eight different DNA homology groups of isolates described as *T. ferrooxidans* were identified by Harrison (Harrison, 1982; Kelly and Harrison, 1989). One of these groups was subsequently identified as *Leptospirillum*, and DNA–DNA hybridization values between the group containing the type strain (ATCC 23270) and the other six groups ranged from 0–19% to 58–73%. Using PCR-based techniques to assess this genomic variability, similarity coefficients between various isolates ranged from nearly 0 to >98% (Novo et al., 1996), and while several culture collection isolates were very closely related to the type strain (ATCC 23270^T), one

TABLE BXII.γ.16. Basic characteristics of species of the genus *Acidithiobacillus*

Characteristic	1. <i>A. thiooxidans</i>	2. <i>A. albertensis</i>	3. <i>A. caldus</i>	4. <i>A. ferrooxidans</i>
Mol% G + C	52	61–62	63–64	58–59
Cell size (μm)	0.5 × 1.0–2.0	0.45 × 1.2–1.3	0.7 × 1.2–1.8	0.5 × 1.0
Motility	+	+	+	+
Single polar flagellum	+	–	+	+
Tuft of polar flagella	–	+	–	–
Glycocalyx	–	+	–	–
Carboxysomes	+	nd	nd	+ ^a
Obligately chemolithoautotrophic	+	+	+	+
Optimum pH	2.0–3.0	3.5–4.0	2.0–2.5	2.5
pH limits	0.5–5.5	2.0–4.5	1.0–3.5	1.3–4.5
Optimum temperature, °C	28–30	28–30	45	30–35
Nitrate reduction	–	–	–	–
<i>Growth on:</i>				
Sulfur	+	+	+	+
Thiosulfate	+	+	+	+
Metal sulfides	+	–	–	+
Ferrous iron	–	–	–	+
Methylated sulfides	–	nd	–	–
Complex media	–	–	– ^b	–

^aUnder carbon dioxide limitation.^bGrows mixotrophically with tetrathionate plus glucose or yeast extract.

strain (ATCC 33020) was not taxonomically related to the others (Selenska-Pobell et al., 1998). It is, therefore, inevitable that some strains assigned to *Acidithiobacillus ferrooxidans* will in due course be reassigned to new species or genera.

Thiobacillus concretivorus (Parker, 1945) was once listed as a

distinct species in *Bergey's Manual* (Parker, 1957) and appeared in the Approved Lists of Bacterial Names (Skerman et al., 1980). It is, however, now recognized as a synonym of *T. thiooxidans* (now *Acidithiobacillus thiooxidans*) (Vishniac, 1974; Kelly and Harrison, 1989).

List of species of the genus *Acidithiobacillus*

1. ***Acidithiobacillus thiooxidans*** (Waksman and Joffe 1922) Kelly and Wood 2000, 513^{VP} (*Thiobacillus thiooxidans* Waksman and Joffe 1922, 239.)

thio.ox'i.dans. Gr. n. *thios* sulfur; M.L. v. *oxido* make acid, oxidize; M.L. part adj. *thiooxidans* oxidizing sulfur.

Short rods, single, paired or in short chains, 0.5 × 1.0–2.0 μm. Motile by means of a polar flagellum (Doetsch et al., 1967). Minute colonies (0.5–1.0 mm) grown on thio-sulfate agar appear transparent or whitish yellow and clear on prolonged incubation; edges appear complete. This organism grows in liquid medium on S⁰, thiosulfate, or tetrathionate. Tetrathionate and sulfur may be produced transiently during growth on thiosulfate. This organism cannot oxidize iron or pyrite but has been shown to grow on sulfur from pyrite in co-culture with “*Leptospirillum ferrooxidans*”, an iron-oxidizing, non-sulfur-oxidizing vibrio. Reduces pH of sulfur media to values of 0.5–0.8.

Obligate chemolithotroph and autotroph; remarkable ability to oxidize S⁰ rapidly. Strictly aerobic. Ammonium sulfate used as nitrogen source. Optimum temperature: 28–30°C; temperature range: 10–37°C. Optimum pH: 2.0–3.0; growth range pH: 0.5–5.5. Isolated from soil, sulfur springs, acid mine drainage waters, and corroding concrete and steel environments. Likely to be widely distributed in acidic habitats with elemental sulfur or soluble reduced sulfur compounds. Member of the *Gamma*proteobacteria.

The mol% G + C of the DNA is: 52 (*T_m*, Bd).

Type strain: ATCC 19377, NCIMB 8343.

GenBank accession number (16S rRNA): Y11596.

2. ***Acidithiobacillus albertensis*** (Bryant, McGroarty, Costerton and Laishley 1988) Kelly and Wood 2000, 514^{VP} (*Thiobacillus*

albertis Bryant, McGroarty, Costerton and Laishley 1988, 221.)

al.ber'ten.sis. M.L. adj. *albertensis* named for Alberta, Canadian province of Alberta.

Rods, 0.45 × 1.2–1.5 μm. Motile by means of a tuft of polar flagella. Condensed glycocalyx present, extending outwards from outer membrane, apparently involved in cell adhesion to surfaces such as sulfur. This organism grows on sulfur, thiosulfate, or tetrathionate; tetrathionate is accumulated transiently during growth on thiosulfate but not on sulfur. Intracellular sulfur granules have been observed in stationary phase cells. Obligate chemolithotroph and autotroph. Aerobic. Ammonium sulfate is used as nitrogen source. Optimum temperature: 28–30°C. Optimum pH: 3.5–4.0; growth range pH: 2.0–4.5. Isolated from extremely acidic soil adjacent to a sulfur stockpile. Probably occurs in other similar environments.

The mol% G + C of the DNA is: 61.5 (UV spectrum).

Type strain: ATCC 35403.

3. ***Acidithiobacillus caldus*** (Hallberg and Lindström 1995) Kelly and Wood 2000, 514^{VP} (*Thiobacillus caldus* Hallberg and Lindström 1995, 619.)

cal'dus. M.L. adj. *caldus* warm, liking warmth.

Cells are short, motile, Gram-negative rods, 0.7–0.8 × 1.2–1.8 μm. Capable of chemolithoautotrophic growth with thiosulfate, tetrathionate, sulfide, sulfur, and molecular hydrogen; no growth with ferrous iron or sulfidic ores. Mixotrophic growth with tetrathionate and glucose or yeast extract. Colonies on tetrathionate agar are small, circular, convex, smooth, and transparent, with precipitated sulfur in their center. Growth occurs between 32 and 52°C and

pH 1–3.5, with optimum growth at 45°C and pH 2–2.5. Contains ubiquinone Q-8 and a menaquinone. No significant DNA hybridization (2–20%) with *T. ferrooxidans*, *T. thiooxidans*, *T. cuprinus*, or *T. thioparus*. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 63.1–63.9 (T_m).

Type strain: KU, ATCC 51756, DSM 8584.

GenBank accession number (16S rRNA): Z29975.

4. **Acidithiobacillus ferrooxidans** (Temple and Colmer 1951) Kelly and Wood 2000, 513^{VP} (*Thiobacillus ferrooxidans* Temple and Colmer 1951, 605.)

ferro.ox'i.dans. L. n. *ferrum* iron; M.L. v. *oxido* oxidize, make acid; M.L. part. adj. *ferrooxidans* iron-oxidizing.

Rods, usually single or in pairs, 0.5 × 1.0 µm. Motile; the type strain is motile by a single polar flagellum. Small colonies (0.5–1 mm) grown on thiosulfate or tetrathionate agar are round, sometimes with irregular margins, and white with sulfur. On solid media (agar or agarose) with ferrous sulfate, microscopic colonies with low iron concentrations are formed, resulting in the appearance of an amber zone in the medium around them. With higher iron

concentrations, round colonies (1 mm) that become red, brown, and hard with deposited ferric salts are produced. Liquid medium with ferrous sulfate at pH 1.6 changes from clear pale green to amber to red-brown with ferric sulfate. At pH 1.9 and above, considerable precipitation and encrustation with basic ferric sulfates (jarosite) take place.

Obligate chemolithotroph and autotroph. Oxidizes and grows on ferrous iron, pyrite, numerous sulfide minerals, sulfur, thiosulfate, or tetrathionate. Strictly aerobic. Ammonium salts and probably nitrate can be used as nitrogen source; reported to be able to fix dinitrogen. Optimum temperature: 30–35°C; growth range temperature: 10–37°C (no growth at 42°C). Optimum pH: about 2.5; growth range pH: 1.3–4.5. Isolated from many locations worldwide where oxidizable iron, sulfide mineral, and sulfur are exposed; particularly prevalent in acid drainage waters of mines for sulfide minerals, mineral leach dumps, and drainage waters from coal mines and coal or spoil heaps. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 58–59 (T_m).

Type strain: ATCC 23270, NCIB 8455.

Family II. Thermithiobacillaceae fam. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Therm.i.thi.o.ba.cil.la'ce.ae. M.L. masc. n. *Thermithiobacillus* type genus of the family; -aceae ending to denote family; M.L. fem. pl. n. *Thermithiobacillaceae* the *Thermithiobacillus* family.

The family *Thermithiobacillaceae* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the family contains the genus *Thermithiobacillus* (type genus).

Description is the same as for the genus *Thermithiobacillus*.

Type genus: **Thermithiobacillus** Kelly and Wood 2000, 515^{VP}.

Genus I. Thermithiobacillus Kelly and Wood 2000, 515^{VP}

ANN P. WOOD AND DONOVAN P. KELLY

Therm.i.thi.o.ba.cil'lus. L. n. *thermae* warm baths; Gr. n. *thios* sulfur; L. n. *bacillus* a small rod; L. masc. n. *Thermithiobacillus* warmth-loving sulfur rodlet.

Gram-negative, nonsporeforming rods, 0.5 × 1–2 µm. Obligate aerobes, moderately thermophilic and obligately chemolithoautotrophic on reduced inorganic sulfur compounds (thiosulfate, polythionates, sulfur, and sulfide, but not thiocyanate); unable to oxidize ferrous iron. Contain ubiquinone Q-8. Other general characteristics are those of *Thiobacillus*. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 66–67.

Type species: **Thermithiobacillus tepidarius** (Wood and Kelly 1985) Kelly and Wood 2000, 515 (*Thiobacillus tepidarius* Wood and Kelly 1985, 436.)

FURTHER DESCRIPTIVE INFORMATION

This genus was created to accommodate the single species of moderate thermophile, *Thiobacillus tepidarius*. While this species showed many physiological similarities to *Thiobacillus neapolitanus* (now *Halothiobacillus neapolitanus*) (Wood and Kelly, 1985, 1986), it was shown by 16S rRNA gene sequence analysis to be unrelated to that genus (Kelly and Wood, 2000). The properties, taxonomy, and differentiation of the genera of sulfur-oxidizing chemolithoautotrophic Gram-negative rod-shaped bacteria are summarized in Tables 1 and 2 and Fig. 1 in the chapter on the genus *Thiobacillus* which appear in Volume 2 Part C in this *Manual*.

List of species of the genus Thermithiobacillus

1. **Thermithiobacillus tepidarius** (Wood and Kelly 1985) Kelly and Wood 2000, 515^{VP} (*Thiobacillus tepidarius* Wood and Kelly 1985, 436.)
tep.i.dar'ius. L. n. *tepidarium* a warm bath fed by natural thermal water; M.L. adj. *tepidarius* warm-bathing (*Thiobacillus*).

Small rods, 0.5 × 1.0–2.0 µm. Motile by a single polar flagellum (5–10 µm in length). Gram negative, nonspore-forming. Colonies grown on thiosulfate agar at 43°C are small (1–2 mm), circular, convex and smooth and become white or yellow with precipitated sulfur. In liquid batch culture on thiosulfate, quantitative production of tetrathionate

takes place before growth continues to effect complete oxidation to sulfate. Sulfur may be precipitated. pH drops to 4.5–5.0. Chemostat cultures do not accumulate intermediates during growth on sulfide, thiosulfate, tetrathionate, or trithionate. This organism oxidizes sulfide, thiosulfate, trithionate, tetrathionate, hexathionate, heptathionate, and sulfite, but not thiocyanate. Obligately chemolithotrophic and autotrophic. Uses ammonium salts as nitrogen source.

Aerobic. Optimum temperature: 43–45°C; range: 20–52°C (no growth at 15 or 55°C). Optimum pH: 6.8–7.5; range: 5.2–8.0. Isolated from the Great Roman Bath at the Temple of Sulis Minerva, Bath, Avon, England; distribution unknown. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 66.6 (UV ratios; Bd).

Type strain: DSM 3134, ATCC 43215.

Order III. *Xanthomonadales* ord. nov.

GERARD S. SADDLER AND JOHN F. BRADBURY

Xan.tho.mo.na.da'les. *Xanthomonas* type genus of the order; *-ales* ending to denote order; M.L. fem. pl. n. *Xanthomonadales* the *Xanthomonas* order.

Gram-negative straight rods lacking prosthecae; hyaline stalks found in one genus (*Nevskia*). Endospores are not produced, no accumulation of poly-β-hydroxybutyrate inclusions. Motile with polar flagellation, flexing/gliding (*Lysobacter*), or nonmotile. Obligate aerobes, having a strictly respiratory metabolism with oxygen as the terminal electron acceptor. Catalase positive. Nitrate not reduced, with the exception of *Stenotrophomonas*. Oxidase

production varies. Fatty acid profiles are complex; branched chain and/or hydroxy fatty acids are found. Ubiquinone Q8 is generally found. Optimum temperature 20–35°C; except *Thermomonas*, 37–50°C.

Type genus: *Xanthomonas* Dowson 1939, 187, emend. Vauterin, Hoste, Kersters and Swings 1995, 483.

Family I. *Xanthomonadaceae* fam. nov.

GERARD S. SADDLER AND JOHN F. BRADBURY

Xan.tho.mo.na.da'ce.ae. M.L. fem. n. *Xanthomonas* type genus of the family; *-aceae* ending to denote a family; M.L. fem. pl. n. *Xanthomonadaceae* the *Xanthomonas* family.

The characteristics are as given for the order. There are twelve genera: *Xanthomonas*, *Frateria*, *Fulvimonas**, *Luteimonas*, *Lysobacter*, *Nevskia*, *Pseudoxanthomonas*, *Rhodanobacter*, *Schineria*, *Stenotrophomonas*, *Thermomonas*, and *Xylella*. Differential characters are

given in Table BXII.γ.17 of the chapter describing the genus *Xanthomonas*.

Type genus: *Xanthomonas* Dowson 1939, 187, emend. Vauterin, Hoste, Kersters and Swings 1995, 483.

Genus I. Xanthomonas Dowson 1939, 187^{AL} emend. Vauterin, Hoste, Kersters and Swings 1995, 483

GERARD S. SADDLER AND JOHN F. BRADBURY

Xan.tho'mo.nas or *Xan.tho.mo'nas*. * Gr. adj. *xanthus* yellow; Gr. fem. n. *monas* unit, monad; M.L. fem. n. *Xanthomonas* yellow monad.

Straight rods, 0.4–0.6 × 0.8–2.0 μm, mostly single or in pairs, occasionally short chains, filaments rarely seen. Gram negative. Do not produce poly-β-hydroxybutyrate inclusions, nor have sheaths, prosthecae, or resting stages. **Motile by a single polar flagellum**. **Obligately aerobic, having a strictly respiratory type of metabolism** with oxygen as the terminal electron acceptor. **No denitrification or nitrate reduction occurs**. **Colonies are usually yellow**, smooth and butyrous, mucoid or viscid. The pigments are highly characteristic brominated aryl polyenes or “xanthomonadins”. A characteristic extracellular acidic heteropolysaccharide called xanthan is produced by most strains giving the

viscous consistency. Growth is inhibited by 6% NaCl, 30% glucose, 0.01% lead acetate, methyl green, or thionin, and by 0.1% (and usually by 0.02%) triphenyl tetrazolium chloride. **Catalase positive; oxidase negative or weak**; urease not produced. H₂S is usually produced, but not indole or acetoin. Acid is not produced in litmus milk or purple milk. Chemoorganotrophic; able to use various carbohydrates and salts of organic acids as sole carbon sources. Small amounts of acid are produced from many carbohydrates, but not from L-rhamnose, adonitol, sorbose, D-sorbitol, meso-inositol, or meso-erythritol. Metabolic activity is shown in Biolog GN microplate tests with D-fructose, D-glucose, D-mannose and methylpyruvate, but not with α-cyclodextrin, adonitol, D-arabitol, meso-erythritol, meso-inositol, xylitol, D-glucosamine, γ-hydroxybutyrate, itaconate, sebacate, L-ornithine, L-pyrroglutamate, D-serine, D,L-carnitine, γ-aminobutyrate, phenylethylamine, putrescine, 2-aminoethanol, or 2,3-butanediol. L-asparagine, L-glutamine, and glycine cannot be used as sole sources of both

*Editorial Note: The genus *Fulvimonas* was described after the cut-off date for inclusion of new taxa in this volume of the *Manual*; hence, there is no separate chapter devoted to it. Characteristics of the genus *Fulvimonas* are given in Table BXII.γ.17 of the chapter describing the Genus *Xanthomonas*.

*Editorial Note: The first pronunciation is according to strict Latin usage while the second is more commonly used.

takes place before growth continues to effect complete oxidation to sulfate. Sulfur may be precipitated. pH drops to 4.5–5.0. Chemostat cultures do not accumulate intermediates during growth on sulfide, thiosulfate, tetrathionate, or trithionate. This organism oxidizes sulfide, thiosulfate, trithionate, tetrathionate, hexathionate, heptathionate, and sulfite, but not thiocyanate. Obligately chemolithotrophic and autotrophic. Uses ammonium salts as nitrogen source.

Aerobic. Optimum temperature: 43–45°C; range: 20–52°C (no growth at 15 or 55°C). Optimum pH: 6.8–7.5; range: 5.2–8.0. Isolated from the Great Roman Bath at the Temple of Sulis Minerva, Bath, Avon, England; distribution unknown. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 66.6 (UV ratios; Bd).

Type strain: DSM 3134, ATCC 43215.

Order III. *Xanthomonadales* ord. nov.

GERARD S. SADDLER AND JOHN F. BRADBURY

Xan.tho.mo.na.da'les. *Xanthomonas* type genus of the order; *-ales* ending to denote order; M.L. fem. pl. n. *Xanthomonadales* the *Xanthomonas* order.

Gram-negative straight rods lacking prosthecae; hyaline stalks found in one genus (*Nevskia*). Endospores are not produced, no accumulation of poly-β-hydroxybutyrate inclusions. Motile with polar flagellation, flexing/gliding (*Lysobacter*), or nonmotile. Obligate aerobes, having a strictly respiratory metabolism with oxygen as the terminal electron acceptor. Catalase positive. Nitrate not reduced, with the exception of *Stenotrophomonas*. Oxidase

production varies. Fatty acid profiles are complex; branched chain and/or hydroxy fatty acids are found. Ubiquinone Q8 is generally found. Optimum temperature 20–35°C; except *Thermomonas*, 37–50°C.

Type genus: *Xanthomonas* Dowson 1939, 187, emend. Vauterin, Hoste, Kersters and Swings 1995, 483.

Family I. *Xanthomonadaceae* fam. nov.

GERARD S. SADDLER AND JOHN F. BRADBURY

Xan.tho.mo.na.da'ce.ae. M.L. fem. n. *Xanthomonas* type genus of the family; *-aceae* ending to denote a family; M.L. fem. pl. n. *Xanthomonadaceae* the *Xanthomonas* family.

The characteristics are as given for the order. There are twelve genera: *Xanthomonas*, *Frateria*, *Fulvimonas**, *Luteimonas*, *Lysobacter*, *Nevskia*, *Pseudoxanthomonas*, *Rhodanobacter*, *Schineria*, *Stenotrophomonas*, *Thermomonas*, and *Xylella*. Differential characters are

given in Table BXII.γ.17 of the chapter describing the genus *Xanthomonas*.

Type genus: *Xanthomonas* Dowson 1939, 187, emend. Vauterin, Hoste, Kersters and Swings 1995, 483.

Genus I. Xanthomonas Dowson 1939, 187^{AL} emend. Vauterin, Hoste, Kersters and Swings 1995, 483

GERARD S. SADDLER AND JOHN F. BRADBURY

Xan.tho'mo.nas or *Xan.tho.mo'nas*. * Gr. adj. *xanthus* yellow; Gr. fem. n. *monas* unit, monad; M.L. fem. n. *Xanthomonas* yellow monad.

Straight rods, 0.4–0.6 × 0.8–2.0 μm, mostly single or in pairs, occasionally short chains, filaments rarely seen. Gram negative. Do not produce poly-β-hydroxybutyrate inclusions, nor have sheaths, prosthecae, or resting stages. **Motile by a single polar flagellum**. **Obligately aerobic, having a strictly respiratory type of metabolism** with oxygen as the terminal electron acceptor. **No denitrification or nitrate reduction occurs**. **Colonies are usually yellow**, smooth and butyrous, mucoid or viscid. The pigments are highly characteristic brominated aryl polyenes or “xanthomonadins”. A characteristic extracellular acidic heteropolysaccharide called xanthan is produced by most strains giving the

viscous consistency. Growth is inhibited by 6% NaCl, 30% glucose, 0.01% lead acetate, methyl green, or thionin, and by 0.1% (and usually by 0.02%) triphenyl tetrazolium chloride. **Catalase positive; oxidase negative or weak**; urease not produced. H₂S is usually produced, but not indole or acetoin. Acid is not produced in litmus milk or purple milk. Chemoorganotrophic; able to use various carbohydrates and salts of organic acids as sole carbon sources. Small amounts of acid are produced from many carbohydrates, but not from L-rhamnose, adonitol, sorbose, D-sorbitol, meso-inositol, or meso-erythritol. Metabolic activity is shown in Biolog GN microplate tests with D-fructose, D-glucose, D-mannose and methylpyruvate, but not with α-cyclodextrin, adonitol, D-arabitol, meso-erythritol, meso-inositol, xylitol, D-glucosamine, γ-hydroxybutyrate, itaconate, sebacate, L-ornithine, L-pyrroglutamate, D-serine, D,L-carnitine, γ-aminobutyrate, phenylethylamine, putrescine, 2-aminoethanol, or 2,3-butanediol. L-asparagine, L-glutamine, and glycine cannot be used as sole sources of both

*Editorial Note: The genus *Fulvimonas* was described after the cut-off date for inclusion of new taxa in this volume of the *Manual*; hence, there is no separate chapter devoted to it. Characteristics of the genus *Fulvimonas* are given in Table BXII.γ.17 of the chapter describing the Genus *Xanthomonas*.

*Editorial Note: The first pronunciation is according to strict Latin usage while the second is more commonly used.

TABLE BXII-γ.17. Differential characteristics of the genera in the family *Xanthomonadaceae*^a

Characteristic	<i>Xanthomonas</i>	<i>Fruteuria</i>	<i>Fulvimonas</i>	<i>Luteimonas</i>	<i>Lysobacter</i>	<i>Neustia</i>	<i>Pseudoxanthomonas</i>	<i>Rhodanobacter</i>	<i>Schineria</i>	<i>Stenotrophomonas</i>	<i>Thermomonas</i>	<i>Xylella</i>
Motility	+	+	+	-	+	+	+	-	-	+	V	-
Flagellation	(exceptions) Single, polar	Single, polar	Single, polar	None	+	Single, polar		None	None	Multiple, polar	Single, polar	None
Cell shape	Rods	Rods	Rods		Rods	Rods		Rods	Rods	Rods	Rods	Rods
Cell size (mm)	0.4-0.6 × 1.0-2.9	0.5-0.7 × 0.7-3.5	0.5 × 2		0.2-0.5 × 1.0-15 (sometimes 70)				2-3 × 0.8- 0.9	0.5 × 1.5	1.0-12.5 × 0.5-0.75	0.25-0.35 × 0.9-3.5
Temperature optimum	28				20-25	28		30	28-37	V+	37-50	26-28
Oxidase	-	-	+		+			+	V		+	-
Catalase	+	+	+		+			+	+	+	+	+
Nitrate reduction	-	-			-			-	-	+	-	-
Aesculin hydrolysis												
<i>Susceptibility to:</i>												
Ampicillin		+		-								
Penicillin G		-		-	V-							
Erythromycin	+	-		-								
Kanamycin		-		-								
Neomycin		-		V-								
Streptomycin		+		-	V-							
Quinones	Q8	Q8		Q8		Q8			Q8	Q8	Q8	
Polyamines	Spermine									Spermidine and	Spermidine	
Mol% G + C of the DNA	63.3-69.7	62-64	71.5-71.9		65.4-70.1	67-69		63	42	66.1-67.8	67.1-68.7	51-52.4
Ecology	Plant pathogens	Plant associated	Soil; plastic degrader	Biofilter	Soil, freshwater; lysing microbes	Freshwater surfaces	Biofilter	Soil; lindane degrader	Fly larvae	Clinical materials; widespread	Kaolin slurry	Plant xylem

[illegible]

carbon and nitrogen. Among the nine fatty acids that predominate in whole cell preparations are 9-methyl decanoic acid ($C_{11:0 \text{ iso}}$), 3-hydroxy-9-methyl decanoic acid ($C_{11:0 \text{ iso } 3\text{OH}}$), and 3-hydroxy-11-methyl dodecanoic acid ($C_{13:0 \text{ iso } 3\text{OH}}$), which are highly characteristic of this genus. The ubiquinone that is present has eight isoprene units. Spermidine is the main polyamine; spermine is usually detectable, but not 2-hydroxyputrescine or 1,3-diaminopropane. Species so far described are plant pathogens or are plant associated.

The mol% G + C of the DNA is: 63.3–69.7 (T_m).

Type species: Xanthomonas campestris (Pammel 1895) Dowson 1939, 190, emend. Vauterin Hoste, Kersters and Swings 1995, 484 (*"Bacillus campestris"* Pammel 1895, 130.)

FURTHER DESCRIPTIVE INFORMATION

Cell morphology Cells examined directly from plant tissues, or from old cultures, may show irregularities in size and shape, but those from actively growing cultures are reasonably uniform rods with rounded ends. The vast majority of cells are within the range $0.4\text{--}0.6 \times 0.8\text{--}2.0 \mu\text{m}$ and predominantly motile by a single polar flagellum. Capsules are visible in many isolates using simple staining techniques though results are variable and depend on handling and conditions prior to staining. The capsular polysaccharides are loosely associated with the cells and the production of large amounts of extracellular polysaccharide results in the slimy or mucoid colonies seen on carbohydrate-rich media.

Fine structure The structure of the cell envelope in *Xanthomonas* appears to be similar to that of other Gram-negative cells, as reviewed by Costerton et al. (1974). Schnaitman (1970) found that the cell wall morphology of a *X. campestris* isolate, as viewed with the electron microscope, was similar to that described for *Escherichia coli* by De Petris (1967). The cell wall is relatively electron transparent, with nuclear material, ribosomes, and granular-like inclusions visible within the cell (Horino, 1973). In cells actively growing in broth cultures, ribosomes are observed throughout the cytoplasm with DNA strands occupying the central part of the cell. Polyribosomes and osmophilic vacuoles were found to be present only in actively growing cells (Al-Mousawi and Richardson, 1990). Studies on *X. hyacinthi*, which causes yellow disease in *Hyacinthus*, have shown the presence of polar fimbriae of ($5 \text{ nm} \times 6 \mu\text{m}$), thought to be involved in the early stages of infection as they were found to attach selectively to stomatal cells (van Doorn et al., 1994). The single polar flagellum may be demonstrated by flagella staining or by electron microscopy of cells from the syneresis water of young cultures. The wavelength of normal flagella averages $1.79 \mu\text{m}$, but occasionally variants with wavelengths of $2\text{--}3 \mu\text{m}$ occur. Very rarely cells with two polar flagella occur (Leifson, 1960).

Cell wall composition As with all Gram-negative bacteria, the cell wall of xanthomonads contains three distinct layers: cytoplasmic or inner membrane, peptidoglycan layer, and outer membrane. The inner and outer membranes are made up of proteins and phospholipids, in addition, lipopolysaccharides are found in the outer membrane. Lipopolysaccharides consist of a heteropolysaccharide portion to which "lipid A" is covalently bonded. The backbone of lipid A in strains from various genera, including *Xanthomonas*, has been shown to contain β -1',6-linked glucosamine disaccharides carrying two phosphate groups, one glycosidically linked and one with ester linkage (Hase and Rietschel, 1976). Hydroxy fatty acids are joined to the LPS backbone through ester and amide linkages (Rietschel et al., 1975). Early

studies of the polysaccharide portion of the LPS of *Xanthomonas* were conducted by Volk (1966, 1968a, b), who obtained D-galacturonic acid-1-phosphate, glucose, mannose, and 2-keto-3-deoxyoctonate. The latter, predictably, predominates in the outer membrane of the cell wall (dos Santos and Dianese, 1985). Volk (1968b) found different ratios of uronic acid:mannose, roughly 1.2:1 and 0.66:1. Most also contained xylose or fucose, but not both. Low amounts of rhamnose were present in two isolates. In addition, 4,7-anhydro- and 4,8-anhydro-3-deoxyoctulosonic acids (Volk et al., 1972) and 3-acetamido-3,6-dideoxy-D-galactose (Hickman and Ashwell, 1966) have been isolated. The high rhamnose content, low content of 2-keto-3-deoxyoctonate, presence of D-galacturonic acid, and absence of heptoses are also found in *Stenotrophomonas maltophilia* (Wilkinson, 1968b; Neal and Wilkinson, 1979).

In early studies on cell envelope proteins, Schnaitman (1970) found that the electrophoretic pattern from *Xanthomonas* was broadly similar to that from *E. coli*, but the major protein constituent had a slightly lower mobility. More detailed analysis demonstrated that strains of *X. campestris* pathovar *campestris* could be differentiated from other xanthomonads and a range of Gram-negative bacteria on the basis of membrane protein patterns (Dianese and Schaad, 1982; Minsavage and Schaad, 1983), and that the antigenic determinants of pathovar-specific peptides were resistant to a range of treatments, with the exception of protease Type 1 digestion (Thavechai and Schaad, 1986). Analysis of the cell envelopes of cassava pathogens, *X. axonopodis* pathovar *manihotis* and *X. cassavae*, revealed two major components relating to the inner and outer membranes of the cell wall (dos Santos and Dianese, 1985). The lighter fraction showed high nicotinamide adenine dinucleotide oxidase and succinate dehydrogenase activities, and larger numbers of peptides than are usually found in the inner membrane of other bacteria. The heavier component showed low enzyme activity and was characteristic of an outer membrane pattern. Heat-modifiable proteins were found in the cell envelopes of both pathovars. Membrane-bound proteins associated with the pigments (xanthomonadins) in *Xanthomonas* have been found in several other genera including a number of Gram-positive bacteria (Lawani et al., 1990).

Studies of the fatty acid component of xanthomonad LPS gave preliminary data on the uniqueness of the fatty acid profile (Rietschel et al., 1975). The taxonomic significance of fatty acid profiles within the genus *Xanthomonas* was further established in a preliminary study of 14 *X. campestris* pathovars pathogenic for hosts in the family *Gramineae* (Stead, 1989). Profiles were found to be complex, containing up to 40 acids, most of which were *iso*- and *anteiso*-branched acids. Grouping of profile types generally correlated with pathovar, although pathovar *translucens*, pathovar *hordei*, pathovar *cerealis*, pathovar *secalis*, and pathovar *undulosa* could not be differentiated. In a comprehensive study of 975 *Xanthomonas* strains 65 different fatty acids were detected and nine were predominantly found in all members of the genus (Yang et al., 1993d): 9-methyl decanoic acid ($C_{11:0 \text{ iso}}$), 3-hydroxy-9-methyl decanoic acid ($C_{11:0 \text{ iso } 3\text{OH}}$), 3-hydroxydodecanoic acid ($C_{12:0 \text{ } 3\text{OH}}$), 3-hydroxy-11-methyl dodecanoic acid ($C_{13:0 \text{ iso } 3\text{OH}}$), 13-methyl tetradecanoic acid ($C_{15:0 \text{ iso}}$), *cis*-9-hexadecenoic acid ($C_{16:1 \text{ } \omega 7c}$), hexadecanoic acid ($C_{16:0}$), 15-methyl hexadecenoic acid isomer F ($C_{17:1 \text{ iso } F}$), and 15-methyl hexadecanoic acid ($C_{17:0 \text{ iso}}$), and three of which, $C_{11:0 \text{ iso}}$, $C_{11:0 \text{ iso } 3\text{OH}}$, and $C_{13:0 \text{ iso } 3\text{OH}}$, are useful in differentiating xanthomonads from other bacteria (Stead, 1992; Yang et al., 1993d). Fatty acids are a valuable tool

in systematic studies and serve as a distinguishing characteristic in assigning new species to the genus (Hu et al., 1997), describing hitherto unrealized diversity within the species *X. fragariae* (Roberts et al., 1998), *X. albilineans* (Yang et al., 1993c), and *X. vesicatoria* (Stall et al., 1994), and differentiating between pathogenic and nonpathogenic strains isolated from the same host (Sahin and Miller, 1996; Vauterin et al., 1996a).

Pigmentation Yellow pigments are present in all species of *Xanthomonas*, but nonpigmented strains sometimes occur. *X. axonopodis* pathovar *manihotis* and some strains of pathovar *ricini* occur naturally as nonpigmented organisms. Occasionally nonpigmented mutants arise in culture (Bryan, 1932; Durgapal, 1977) and studies have shown that the absence of xanthomonadins, and other characters such as EPS production have a limiting effect on epiphytic survival and host infection (Jenkins and Starr, 1982a; Poplawsky and Chun, 1998). Although the presence of yellow pigment is an important characteristic for identification, its absence does not exclude an organism from the genus if other characteristics are in agreement. Starr and Stephens (1964), examining a number of isolates belonging to *X. campestris*, found that all contained one or more pigments that showed absorption maxima at 418 (a shoulder), 437, and 463 nm when dissolved in petroleum ether. The pigments were released from disrupted cells at the same rate as a component of the cytoplasmic membrane, suggesting their attachment to the cell envelope (Stephens and Starr, 1963). Later they were shown to be brominated aryl polyenes and were named xanthomonadins. From two to five different xanthomonadins are present in the various species examined (Andrewes et al., 1973; Starr et al., 1977). Xanthomonadin I, the main component of the pigment of "*Xanthomonas juglandis*", now *X. arboricola* pathovar *juglandis*, was found to be 17 (4-bromo-3-methoxyphenyl) 17-bromo-heptadeca-2,4,6,8,10,12,14,16-octaenoic-acid (Andrewes et al., 1976). The detection of xanthomonadins is highly significant and their presence (Irey and Stall, 1982; Jenkins and Starr, 1982b; Gitaitis et al., 1987; Angeles-Ramos et al., 1991), or that of the gene cluster coding for their synthesis (Poplawsky et al., 1993), can be used to assign unknown strains to the genus *Xanthomonas*.

Quinones Isoprenoid quinones occur in the cytoplasmic membranes of most procaryotes, where they have an important role as electron carriers in electron transport. Members of the *Alphaproteobacteria*, *Betaproteobacteria*, and *Gammaproteobacteria* contain only ubiquinones, with the latter possessing Q7, Q8, Q9, and Q12 (Suzuki et al., 1993a). The only ubiquinone detected in members of the genus *Xanthomonas*, thus far, is ubiquinone Q8 (Ikemoto et al., 1980).

Colonial or cultural characteristics and life cycles Colonies of all species of *Xanthomonas* are normally smooth, round, entire, and butyrous, at least when young, but may show surface markings such as striations and become lobed when older. The production of mucoid, butyrous colonies is an essential feature for discriminating certain pathogenic xanthomonads from saprophytes (Samson et al., 1989). Mutant colonies of different appearance occasionally occur. These may be less mucoid and tend toward rough (Corey and Starr, 1957a) or may be crenated (Whitfield et al., 1981).

When they are first visible, colonies are transparent and buff to very pale yellow, but the color soon deepens and then varies according to the species and the medium used. Pathovars of *X. campestris* and *X. axonopodis* darken towards a buttercup yellow or light cadmium, especially on nutrient agar supplemented with

5% glucose or sucrose. *X. fragariae* remains small and pale on nutrient agar, but gives abundant, spreading, orange-yellow growth on addition of 1% glucose. *X. albilineans* grows on an agar medium containing 2% sucrose and 1% peptone, as buff-yellow or honey-colored colonies.

Growth rates vary widely in the genus. The faster-growing strains produce visible colonies from single cells in 24–36 h at 25°C. Slower-growing strains may take 2–3 days for single colonies to become visible. The time taken to produce colonies about 1 mm in diameter at this temperature varies from 2 to 3 days, to a week or 10 days. The slowest-growing species take about a week to appear, and often longer on isolation plates.

Differences in colonial morphology have been used for identification. CKTM medium¹ (Sijam et al., 1991) can be used to differentiate pepper and tomato strains of *X. vesicatoria* (Sijam et al., 1992). After 2–3 d growth on this medium, all strains produce circular, raised colonies that are yellow and are surrounded by a clear ring. Minute crystals can be seen within the clear ring within 4 d. Strains isolated from tomato then develop opaque haloes around the colonies during a further 2–6 d, unlike pepper strains. Of 69 strains studied, 60 were correctly identified to host by colony characteristics on this medium. Similarly, it is possible to differentiate colonies of *X. arboricola* pathovar *pruni* from other yellow-pigmented organisms by viewing colony morphology on nutrient agar through a lined template, subilluminated by a fluorescent light box (Gitaitis et al., 1988). The clarity and refractive quality of colonies of the pathogen contrasted with distorted and opaque patterns obtained from other organisms.

There is some evidence that colony morphology may be linked to pathogenicity and virulence characteristics. Early studies on *X. phaseoli* found that the size and number of lesions on the host was reduced when using rough colonial variants instead of the mucoid wild type (Corey and Starr, 1957a). Similarly, *X. oryzae* pathovar *oryzae* (*X. oryzae*) exhibited a range of colonial variants. Nwigwe (1973) described two such variants: yellow, with wax-yellow, entire, glistening colonies of 2.5–3.5 mm diameter, and white, with spherical, entire, butyrous colonies of 3.5–5.5 mm diameter. The nonpigmented larger colonies grew faster and were more apparent in cultures known to be less virulent. Subsequent reports have given details of further colonial variants, again linking these variations to differences in pathogenicity and virulence (Choi et al., 1981; Dath and Devadath, 1982; Tsuchiya et al., 1982). Similar findings linking colony morphology and virulence have been presented for *X. oryzae* pathovar *oryzicola* (Rao and Devadath, 1977), *X. axonopodis* pathovar *vasculorum* (Peros, 1988), and *X. axonopodis* pathovar *phaseoli* (Jindal and Patel, 1984). Plasmid loss has been implicated in this phenomenon (Chakrabarty et al., 1995; Ulaganathan and Mahadevan, 1991), as have the actions of transposable elements (Dai et al., 1991).

Nutrition and growth conditions *Xanthomonas* species are chemoorganotrophic. In purely synthetic media containing minerals, ammonium nitrogen, and a suitable carbon source such

1. CKTM medium consists of (g/900 ml, unless otherwise stated): Soy peptone, 2.0; Bacto tryptone, 2.0; dextrose, 1.0; L-glutamine 6.0; L-histidine, 1.0; (NH₄)₂HPO₄, 0.8; KH₂PO₄, 1.0; MgSO₄·7H₂O, 0.4; anhydrous CaCl₂, 0.25; Bacto Agar (Difco), 12.0; Tween 80, 10 ml; Pourite (Analytical Products, Belmont CA), 1 drop. Subsequent to autoclaving, the medium is cooled to 45°C and the following are added; cycloheximide, 0.5 ml (1.0 g in 5 ml methanol); bacitracin, 2.0 ml (1.0 g in 20 ml distilled H₂O); neomycin sulfate, 0.5 ml (0.4 g in 20 ml distilled H₂O); cephalixin (65 mg); 5-fluorouracil (12 mg); and tobramycin (0.4 mg); all dissolved in 100 ml distilled H₂O.

as glucose, amino acid supplements are required by most strains of all species. Usually glutamate or methionine, but occasionally both, are needed. There is evidence that methionine may play a role in pathogenesis, as studies on methylthiopropionic acid (MTPA), a phytotoxin produced by *X. axonopodis* pathovar *manihotis*, have demonstrated that methionine is a precursor of MTPA (Ewbank and Maraite, 1990). *X. axonopodis* pathovar *axonopodis* and *X. fragariae* will grow when vitamin-free casein hydrolysate is added, but their exact amino acid requirements are unknown (Kennedy and King, 1962), and *X. arboricola* pathovar *pruni* also requires nicotinic acid (Starr, 1946). Experiments have shown that in this pathovar nicotinic acid can be replaced by tryptophan, which is then converted to nicotinic acid (Wilson and Henderson, 1963).

Ammonium salts, e.g., ammonium phosphate, can serve as nitrogen sources, but nitrates are often less acceptable. Patel and Kulkarni (1949) reported that *X. axonopodis* pathovar *malvacearum* uses potassium nitrate, but not sodium nitrate. Kotasthane et al. (1965) examined 16 pathovars of *X. campestris* and found that all would accept DL-alanine, L-glutamate, and L-proline as sole carbon sources. Glycine, L-leucine, DL-methionine, DL-serine, and DL-norleucine were not used. As sole nitrogen source, DL-alanine, L-glutamate, L-proline, DL-methionine, DL-threonine, DL-aspartate, L-asparagine, L-hydroxyproline, and L-histidine were accepted by all; DL-serine, DL-norleucine, and L-tyrosine by none; while results with glycine, DL-valine, L-tryptophan, L-leucine, DL-isoleucine, L-arginine, DL-lysine, and L-cystine varied with the pathovar. Asparagine can serve as a nitrogen source if glucose is supplied, but it cannot serve as a simultaneous source of both carbon and nitrogen (Starr and Weiss, 1943; Starr, 1946). This is used as a diagnostic test for *Xanthomonas* (Dye, 1962), since yellow *Enterobacteriaceae* and many *Pseudomonas* species will grow with asparagine as sole source of both elements (da Silva Romeiro and Moura, 1998; Schaad and Stall, 1988; Pernezny et al., 1995). Glutamate or alanine can serve in this way for *Xanthomonas* (Lewis, 1930).

Studies from the optimization of xanthan gum production have shown that production is influenced by the type and initial concentrations of the carbon and nitrogen sources (de Vuyst and Vermeire, 1994). The optimal sole carbon source was found to be either 4% glucose, 4–5% sucrose (optionally as 10% molasses) or 2.8% Sirodex A (a glucose syrup, of which the glucose content was 95–96%); other components were 2.0% corn steep liquor as a combined nitrogen-phosphate source. Similar studies found that a medium containing 8% sucrose and 4% peptone was optimal, and it was noted that the bacterium preferentially consumed glucose and fructose formed during sterilization of the sucrose-containing medium (Bruggeman et al., 1998).

The optimum growth temperature is usually in the range 25–30°C, but lower for *X. populi* at about 20°C. The minimum is above 4°C and the maximum varies from 27.5 to 39°C.

Respiratory chain Xanthomonads are obligately aerobic and use oxygen as the terminal electron acceptor. They are oxidase negative by the Kovacs test, indicating that cytochrome *c* is not present (Jones and Poole, 1985). Hochster and Nozzolillo (1960) postulated that the respiratory chain in *X. axonopodis* pathovar *phaseoli* is comprised of cytochrome a_1 , a_2 , b_1 , and flavoprotein. The spectra found at liquid air temperature showed a slight shoulder at 549 nm, suggesting a very small cytochrome *c* content, and with cell-free extracts, they showed some reduced nicotinamide adenine dinucleotide (NADH)-cytochrome *c* reductase activity. Similar results were found by Rye et al. (1988), who

reported cytochromes *b*, *c*, aa_3 , *o* type, and possibly a_1 , present, with cytochrome *c* only present in reduced amounts. The difference spectra of Sands et al. (1967) for *Pseudomonas syringae*, and of Stanier et al. (1966) for *S. maltophilia*, are very similar, but do not show any noticeable peak or shoulder at or near 549 nm. Both of these latter species are oxidase negative and thought to contain no cytochrome *c*.

Carbohydrate metabolism Studies of eight of the currently accepted species indicate that glucose catabolism in *Xanthomonas* is similar to that in *Pseudomonas*. Glucose is oxidized readily, but not gluconate. No gluconate or 2-keto-gluconate is produced from glucose (Lockwood et al., 1941). The Entner–Doudoroff pathway is present (Katznelson, 1955, 1958; Whitfield et al., 1982) and is the predominant pathway of glucose catabolism, with only 8–16% of the glucose following the pentose phosphate pathway. The hexose cycle is not significantly used (Zagallo and Wang, 1967). The tricarboxylic acid and glyoxylate cycles are present in *X. axonopodis* pathovar *phaseoli* (Madsen and Hochster, 1959), and the presence of the TCA cycle confirmed in *X. campestris* (Whitfield et al., 1982). Two enzymes of the tricarboxylic acid cycle that show interesting variations of taxonomic significance are citrate synthase and succinate thiokinase (Weitzman, 1980). *X. hyacinthi* has a citrate synthase that is typical of Gram-negative aerobes: it is a large molecule (molecular weight is ~250,000), is inhibited by NADH, and is reactivated by adenosine monophosphate. The succinate thiokinase is also typical of Gram-negative bacteria, being large in size (molecular weight is 140,000–150,000).

With respect to the breakdown of oligosaccharides, Hayward (1977) assessed the glycosidase activity of various bacteria, mostly plant pathogens; of 39 strains of *Xanthomonas* from nine of the currently accepted species, all showed β -glucosidase activity, as did *Stenotrophomonas maltophilia*. α -Glucosidase activity was weak in *X. albilineans* and absent in both of the two strains of *X. hortorum* pathovar *vitians* and *X. vasicola* pathovar *holcicola*. It was also absent from the four strains of *X. arboricola* pathovar *pruni*, but present in *X. arboricola* pathovar *juglandis*. Dekker and Candy (1979) showed that β -mannanases and carboxymethylcellulases produced by *X. campestris* were present both as membrane-bound constituents and extracellularly. Fructose is phosphorylated by a specific phosphoenolpyruvate-dependent phosphotransferase system in *X. campestris* pathovar *campestris* (de Crecy-Lagard et al., 1991a, b). Transposon mutagenesis studies revealed that mutants incapable of metabolizing fructose also failed to utilize mannose, sucrose, and mannitol. Pectolytic activity has been shown for xanthomonads, but varies among strains. Small variations in methodology can also cause variation in results. Dye (1960) examined 142 isolates of various pathovars of *X. campestris* and 3 isolates of *X. albilineans*. Some of the *X. campestris* strains had as much activity as *Erwinia carotovora*, but many showed little or no activity. Two of the *X. albilineans* strains showed only weak polygalacturonase activity, while one also showed some pectin methylesterase activity. Using pectate gels at various pH levels, Hildebrand (1971) found that the pitting produced by two pathovars of *X. campestris* was strong and unaffected by pH, whereas with *X. arboricola* pathovar *juglandis* and *X. axonopodis* pathovar *phaseoli* and pathovar *vitians* there was greatest activity at pH 5. Two isolates of *X. fragariae* and one of *X. axonopodis* pathovar *dieffenbachiae* showed slight activity. Lange and Knösel (1970) examined eight strains belonging to three species and detected strong pectin methylesterase, pectic transeliminase, and some polygalacturonase activity in *X. vesicatoria* (one isolate); *X. axonopodis* path-

ovar *begoniae* and *X. hortorum* pathovar *pelargonii* showed slight or no activity. These authors also demonstrated significant C_xcellulase and proteolytic activity in all strains except one of pathovar *begoniae*.

Metabolism of aromatic compounds Degradation of protocatechuate, phenylalanine, and synephrine by six pathovars from four species of *Xanthomonas* and four pathovars of uncertain affiliation was found to occur by meta cleavage (William and Mahadevan, 1980). These compounds could serve as carbon sources for growth, but benzoate and *o*-hydroxybenzoate could not be used. More recently, studies on the degradation of xenobiotic compounds such as 1-chloro-2,4-dinitrobenzene and the herbicide alachlor, have demonstrated the presence of glutathione-S-transferase, a key enzyme in the detoxification of xenobiotics, in *X. campestris* (Zablotowicz et al., 1995).

In the synthesis of aromatic amino acids by *Xanthomonas*, the first step is the combination of the multifunctional metabolites D-erythrose-4-phosphate and phosphoenolpyruvate to form 3-deoxy-D-arabinoheptulosonate-7-phosphate (DAHP), the first metabolite that is specific to the pathway. The step is catalyzed by the enzyme DAHP synthetase. The subsequent pathway is multibranched and the activity of DAHP synthetase may be governed by various regulatory systems. Thirty-two genera were examined by Jensen et al. (1967) and found to have one of six different control patterns. Control in *X. campestris* and *X. hyacinthi* was unusual. It was identified as sequential feedback inhibition by chorismate. Arogenate also produced some inhibition, but prephenate, phenylalanine, and tyrosine had little or no effect. There was also some inhibition by tryptophan in *X. hyacinthi* but little or none in *X. campestris*. Whitaker et al. (1981a) examined, among other species, five isolates of *X. campestris* (pathovars not specified), one of *X. albilineans*, one of *X. axonopodis*, two of *S. maltophilia*, one of "*Pseudomonas gardneri*", and the type strain of *Pseudomonas geniculata*, and found that all had DAHP synthetases that showed sequential feedback inhibition by chorismate and lack of stimulation by divalent cations. These properties have not been found in any other groups of bacteria. It is interesting to note that Dye (1966b) found an isolate of "*P. gardneri*" to be a synonym of *X. vesicatoria*, and De Ley (1978) considered it *Xanthomonas* based on DNA-rRNA hybridization. Whitaker et al. (1981a) point out that the isolate of *P. geniculata* was originally deposited in the ATCC as a strain of *P. maltophilia*.

The enzymes involved in the biosynthesis of tyrosine and phenylalanine from prephenate also have taxonomic importance. The dehydrogenases of the two pathways to tyrosine were studied by Byng et al. (1980). They found that although both dehydrogenase activities (i.e., prephenate dehydrogenase and arogenate dehydrogenase) were shown by nearly all species of *Pseudomonas* and *Xanthomonas* studied, the enzymes varied in the cofactor required and in their inhibition by tyrosine (feedback control). Isolates of *X. axonopodis* (pathovar *axonopodis*, pathovar *malvacearum* and pathovar *phaseoli*), *X. hyacinthi*, *X. hortorum* pathovar *pelargonii*, *X. albilineans*, and *S. maltophilia* all showed an NAD-dependent prephenate dehydrogenase that was inhibited by tyrosine, and all showed an NAD-dependent arogenate dehydrogenase, except *X. axonopodis* pathovar *axonopodis*, which showed no activity.

The dehydratases of the two pathways to phenylalanine (i.e., prephenate dehydratase and arogenate dehydratase) were studied by Whitaker et al. (1981b). They found that not only was one or the other of these enzymes absent in some groups, but that the degree of activation of prephenate dehydratase by tyrosine

varied considerably between groups. The combination of these characteristics gave a fine taxonomic tool for distinguishing not only the five rRNA-DNA hybridization groups, but also two subgroups of group II and three subgroups of group III, along the same lines as found with DNA hybridization. Group V organisms contained arogenate dehydratase and prephenate dehydratase, which was not activated by tyrosine. They could not be distinguished from group Ib using these enzymes. Assigned to group V were *S. maltophilia*, *X. campestris* (three isolates), *X. albilineans*, *X. axonopodis*, "*P. gardneri*", and *P. geniculata*.

A further metabolic pathway involving aromatic amino acids is relatively well known, at least in one pathovar of *X. arboricola*, pathovar *pruni*, which requires nicotinic acid for growth, but can accept tryptophan or 3-hydroxyanthranilic acid as a substitute. This it does by synthesis of nicotinic acid from these substances by a pathway that leads to synthesis of nicotinamide adenine dinucleotide (NAD) from tryptophan (Wilson and Henderson, 1963). This pathway has been found in animals, the fungus *Neurospora crassa*, yeasts, and some *Streptomyces* species, but is absent from various bacteria that have been examined (Yanofsky, 1954; Lingens et al., 1966). The first three enzymes involved, tryptophan pyrrolase, kynurenine formamidase, and kynureninase, are coordinately induced by L-tryptophan (Brown and Wagner, 1970). Wagner and Brown (1970) presented evidence that the pathway is regulated by the first step from tryptophan. The activity of the allosteric enzyme involved, tryptophan pyrrolase, is under feedback control (inhibition) by the end products of the pathway, NADH and NADH phosphate.

Van Eys (1960) has shown that nicotinic acid may also be replaced by NAD for growth of *X. arboricola* pathovar *pruni*. The final step in degradation of NAD to nicotinic acid is catalyzed by a pyridine ribosidase that shows unusual preference for the pyridine-riboside linkage over the purine-riboside linkage.

Catabolism of nucleoside triphosphates Huang et al. (1975) found that ATPase is bound to the cell membrane of *X. oryzae* pathovar *oryzae* by a mechanism that involves magnesium ions, which also stabilize this enzyme. Nucleoside tri- and diphosphates are hydrolyzed, but not monophosphates. The pathway for degradation of ATP has been elucidated by Hochster and Madsen (1959). Working with *X. oryzae* pathovar *oryzae*, Yang et al. (1975) obtained evidence for the extracellular breakdown of deoxycytidine triphosphate to the diphosphate, monophosphate, and finally to deoxycytidine. The material was taken up by the cell and incorporated into DNA, probably by stepping up again through mono-, di-, and triphosphates.

Genetics Early studies demonstrated genetic transformation in *X. axonopodis* pathovar *phaseoli* using DNA from strains differing in colony type or in streptomycin resistance (Corey and Starr, 1957b, c). Subsequently, exchange of genetic material has been achieved by conjugal plasmids (Daniels et al., 1984) transformation by electroporation (Wang and Tseng, 1992; Ferreira et al., 1995) and transducing phage (Weiss et al., 1994). Studies using a transposon (Tn5-SC), constructed to quantify genetic deletions or amplifications, demonstrated that the genomic stability of *X. campestris* pathovar *campestris* was as great as that of other Gram-negative bacteria (Martinez-Salazar et al., 1993). Similarly, homologous recombination between plasmid sequences was found to occur at a similar level to that reported for other Gram-negative bacteria.

Much study has been directed towards pathogenicity, xanthomonadin, and xanthan gum production (see Daniels and Leach,

1993). Transposon-induced mutagenesis has been employed widely to characterize components within the pathogenicity function. Hu et al. (1992) found that nonpathogenic mutants of *X. campestris* pathovar *campestris* accumulated extracellular polygalacturonate lyase, α -amylase, and endoglucanase in the periplasm, but were deficient in a signal peptidase involved in protein secretion. Similarly, Tn5-induced mutants of *X. campestris* pathovar *campestris* deficient in protease and extracellular polysaccharide production were found to be nonpathogenic to turnip seedlings (Rosato et al., 1994). In contrast, no differences in the production of protease, polygalacturonase, extracellular polysaccharides or growth rate were found between an avirulent, transposon-induced, mutant of *X. vesicatoria* and the wild type (Yun et al., 1995). A 5.4-kb DNA fragment from the type strain of *X. campestris* pathovar *campestris* was found to alter the phenotype of *X. campestris* pathovar *armoraciae* 417 when inoculated onto cabbage (Chen et al., 1994). This transfer conferred the ability to cause blight on the recipient, which generally causes a non-systemic leaf spot of crucifers. It was suggested that the genes conferring the ability to induce blight symptoms may be unique to *X. campestris* pathovar *campestris* and can affect the apparent pathovar status of the recipient strain.

The *hrp* gene cluster determines functions necessary for pathogenicity on host plants.

Cross-complementation showed that *X. axonopodis* pathovar *vitians* and *X. campestris* pathovar *campestris* *hrp* sequences are functionally interchangeable, and there is evidence to suggest that there is some functional homology between the *hrp* gene cluster of *Ralstonia solanacearum* and a range of xanthomonads (Arlat et al., 1991). Other complementation studies have revealed that the *hrp* region is organized into at least six different groups (Bonas et al., 1991) and in some pathovars homology was high enough to permit marker gene exchange or complementation of nonpathogenic mutants. Further, *X. campestris* pathovar *campestris* and *X. oryzae* pathovar *oryzae* were found to contain a 1428-base pair *hrpX* gene, involved in the regulation of *hrp* genes (Oku et al., 1998). Sixteen distinct xanthomonad pathovars and 12 strains of *X. oryzae* pathovar *oryzae* were shown to contain homologs of *hrpX*, which were not apparent in heterologous bacteria such as *Agrobacterium*, *Erwinia*, and *Pseudomonas* spp.

Expression of *hrp* genes was found to be repressed when the organisms were grown on rich media, and in minimal medium the level of expression depended on the carbon source supplied to the cells (Arlat et al., 1991). Similar results were obtained by Schulte and Bonas (1992b) who found that expression of *hrp* genes and the hypersensitive response were suppressed in complex media, but induced in plants. Using a minimal medium, both sucrose and methionine were required for induction of *hrp* genes at expression levels similar to those found by induction in plants. Further studies showed that plant filtrates were able to induce *hrp* genes (Schulte and Bonas, 1992a); the inducing molecule(s) were heat stable and hydrophilic.

Seven transcriptional units (*pigA* through *pigG*) involved in xanthomonadin production have been identified (Poplawsky and Chun, 1997). Inactivation of the *pigB* unit resulted in reduced xanthomonadin and extracellular polysaccharide slime production. This phenomenon was reversible when *pigB* mutant strains were grown with other xanthomonadin transcriptional mutants, the wild type, or other xanthomonads. This finding suggests that *pigB* is involved in a regulatory function, which can be restored by a nontransforming diffusible factor (DF). The DF is thought

to be novel and different from other signal molecules, such as homoserine lactone derivatives found in *Vibrio*, *Agrobacterium*, *Erwinia*, *Pseudomonas* and *Burkholderia* spp. (Chun et al., 1997). This finding was similar to that obtained from a recent survey of 106 isolates from seven genera of Gram-negative plant-associated bacteria (Chung et al., 1998a), in which relatively few xanthomonads produced any detectable homoserine lactones.

Nonmucoid mutants deficient in xanthan gum production have been restored by conjugal mating (Thorne et al., 1987). Groups of clones that contained overlapping homologous DNA were found to complement specific mutations, suggesting clustering of the genetic loci involved in xanthan synthesis. Further work has shown that the xanthan gum gene cluster is made up of 12 genes, mainly expressed as an operon under the control of a promoter situated upstream from the first gene, *gumB* (Katzen et al., 1996).

The chromosome map of *X. campestris* pathovar *campestris* has recently been elucidated (Tseng et al., 1999) and the positions of eight loci involved in EPS (xanthan gum) production, nine auxotrophic markers, the gene cluster for xanthomonadin production, and two 16S rRNA operons have been determined.

Molecular biology A range of fingerprinting methods have been applied to xanthomonads, usually at the species/pathovar/strain level. Hartung and Civerolo (1987) digested genomic DNA from a collection of *X. axonopodis* pathovar *citri* isolates with the restriction endonuclease *EcoRI*, the fragments were separated by PAGE, and the resulting genomic DNA fingerprints compared. Fingerprints varied among separate disease outbreaks and within single outbreaks. Subsequent studies using a conventional RFLP approach, probing *EcoRI* and *PvuII* polymorphisms with seven cosmid clones selected from a library of *X. axonopodis* pathovar *citri* XC62, demonstrated that *X. axonopodis* pathovar *citri* has a clonal population structure, consistent with previous findings (Hartung and Civerolo, 1989). A relatively heterogeneous group of related strains was isolated only from Florida citrus nurseries, and these were circumscribed into two groups in a subsequent study (Hartung and Civerolo, 1991). RFLP analysis has formed the basis of characterization studies for "*X. campestris* pathovar *citrumelo*" (Gottwald et al., 1991), *X. axonopodis* pathovar *vignicola* (Verdier et al., 1998a), "*X. campestris* pathovar *zeae*" (Qhobela et al., 1990), *X. axonopodis* pathovar *manihotis* (Verdier et al., 1998b), *X. oryzae* pathovar *oryzae* (Leach et al., 1992), *X. campestris* pathovar *mangiferaeindicae* (Gagnevin et al., 1997), and *X. vesicatoria* (Chung et al., 1996). Variations on this basic method have been used to detect polymorphisms in *hrp* genes (Leite et al., 1994a, b) and insertion sequences (Berthier et al., 1994; Adhikari et al., 1995; Chen et al., 1999a), for strain characterization or in conjunction with pulsed-field gel electrophoresis (Cooksey and Graham, 1989; Davis et al., 1997; Roberts et al., 1998).

DNA sequences corresponding to conserved motifs of repetitive bacterial elements (REP, ERIC, and BOX) are widely distributed in phytopathogenic *Xanthomonas* and *Pseudomonas* strains (Louws et al., 1994) and have been used to characterize *X. hortorum* pathovar *pelargonii* (Sulzinski et al., 1995), *X. vesicatoria* (Louws et al., 1995), and *X. fragariae* (Opgenorth et al., 1996). The AFLP technique (Vos et al., 1995), based on the selective PCR amplification of restriction fragments from a total digest of genomic DNA, has been used to study *X. axonopodis* pathovars and *X. vasicola* (Janssen, et al., 1996; Restrepo et al., 1999). From the former analysis, pathovars of *X. axonopodis*, which mainly cause disease in citrus and grasses, were recovered

in two groups separate from *X. vasicola*, which correlated well with studies using other techniques.

Plasmids Early studies by Lai et al. (1977a) demonstrated transmission of plasmids RP4 and RK2 from *Escherichia coli* to *X. vesicatoria*, conferring resistance to various antibiotics and production of penicillinase. The transconjugants were able to transmit the plasmids to other strains of the same pathovar, as well as to other pathovars, and also to other bacterial phytopathogens belonging to a number of Gram-negative genera. Stability of the plasmids in the transconjugants varied. A high proportion were retained in bacteria in leaf lesions on plants, in detached leaves kept moist at 5°C, and in dried leaves, but were rapidly lost from cells grown on agar (Lai et al., 1977b). More recently, copper resistance was found to be plasmid borne and self-transmissible in strains of *X. vesicatoria* (Stall et al., 1986). Copper resistance appears to be linked to avirulence, as avirulent, copper-resistant transconjugants were detected subsequent to mating copper-resistant strains with virulent, copper-sensitive strains. Further studies into the emergence of copper resistance have confirmed the wide transmissibility of plasmids coding for this attribute (Bender et al., 1990; Park and Cho, 1996). Copper resistance was transferred to sensitive strains of the same pathovar at a rate of 4.3×10^{-3} to 1.0×10^{-5} , and could be transferred to other pathovars of *X. campestris*. Within *X. vesicatoria*, there is also evidence that race determinants are coded on plasmids, as race shifts have been observed (Swanson et al., 1988; Kousik and Ritchie, 1996). Race shifting was thought to result from the loss of a plasmid carrying avirulence genes or the inactivation of the avirulence gene by an insertion element. In an extensive study of 522 strains of *X. vesicatoria*, the number and size of plasmids was found to vary widely, the latter ranging from 3 to 300 kb (Canteros et al., 1995). Seventy-one different profiles were observed, with a maximum of five plasmid classes present in any one profile. No plasmid or plasmid profile was found to be characteristic for the species (pathovar) and exchange of plasmids in natural populations was thought to be widespread.

Although plasmids are not found universally in xanthomonads (Lazo and Gabriel, 1987), they have been found in *X. axonopodis* pathovar *cyamopsidis*, pathovar *dieffenbachiae*, and pathovar *glycines*, and *X. hortorum* pathovar *hederae* (Lazo and Gabriel, 1987), *X. arboricola* pathovar *pruni* and *X. axonopodis* pathovar *vitians* (Kado and Liu, 1981; Lazo and Gabriel, 1987), *X. campestris* pathovar *sesami* (Sheela et al., 1994), *X. axonopodis* pathovar *manihotis* (Lin et al., 1979), *X. axonopodis* pathovar *malvacearum* (Lazo and Gabriel, 1987; de Feyter and Gabriel, 1991), *X. oryzae* pathovar *oryzae* (Choi et al., 1989b; Amuthan and Mahadevan, 1994), *X. axonopodis* pathovar *vignicola* (Lazo and Gabriel, 1987; Ulaganathan and Mahadevan, 1988), *X. axonopodis* pathovar *vasculorum* (Ezavin et al., 1992), and *X. axonopodis* pathovar *citri* (Civiero, 1985; Lazo and Gabriel, 1987). In the case of the latter, the number of plasmids varies from 1 to 5, sizes ranging from 7 to 100 kb (Pruvost et al., 1992). In general, strains associated with the same form of citrus bacterial canker disease shared the same plasmid types and these exhibited similar restriction patterns. This uniformity, also present in other pathovars within the genus, has enabled the exploitation of plasmids as high-resolution markers in population studies of *X. vesicatoria* (Hwang et al., 1995), *X. axonopodis* pathovar *manihotis* (Restrepo and Verdier, 1997), and *X. axonopodis* pathovar *citri* itself (Hartung et al., 1996). In the case of the latter, a nested PCR approach relied on the use of labeled primers to allow detection of the amplification product in a microtiter plate. All strains of *X. axonopodis*

pathovar *citri* tested were detected with this system, as were some strains of *X. axonopodis* pathovar *aurantifolii* but not other xanthomonads such as *X. axonopodis* *citrumelo*, which causes citrus bacterial spot disease.

Antigenic structure Polyclonal antibodies as taxonomic tools have been limited in their impact in the study of *Xanthomonas*. Work has mostly resulted in multiple groupings of pathovars and many cross-reactions. Thus, Elrod and Braun (1947a), using whole cells as antigens for agglutination, found that 36 nomenclature species fell into five groups. Cross-reactions were reduced if the exopolysaccharides were removed by thorough washing, or if the cells were grown on medium low in carbohydrate. Further work on one group (Elrod and Braun, 1947b), using cross-absorbed antisera, separated two "serologically good species," "*Xanthomonas juglandis*" (four isolates) and "*Xanthomonas carotae*" (three isolates). With *X. vesicatoria*, results have varied. Lovrekovich and Klement (1965) found two serovars among 34 strains and reported correlation with phagovars and host of origin. Charudattan et al. (1973) also found two serovars in 72 strains and Schaad (1976), using ribosome preparations from 25 strains, found two serovars and a single isolated strain. Both used double gel-diffusion and neither found any correlation with host origin or pathogenicity. Yano et al. (1979) found nine serovars in 31 isolates belonging to 14 pathovars of *X. campestris*. They used whole cells washed in formalized saline for injection, preparations of extracellular polysaccharide as antigens, and indirect hemagglutination. Cross-reactions were evident, as serovars did not correspond to pathovars. Similar results were obtained by Anderson and Nameth (1990) using polyclonal antisera raised against *X. hortorum* pathovar *pelargonii*. The antisera, when used to develop an immunogold silver stain dot-immunobinding assay and ELISA, were found to be highly reactive for *X. hortorum* pathovar *pelargonii*, but five of seven other pathovars tested also gave moderate reactions. When used in conjunction with other methods, such as immunofluorescence microscopy (Schaad, 1978; Franken, 1992) or for immunocapture prior to use of a PCR assay (Lopes and Damann, 1996; Hartung et al., 1997), polyclonals clearly have value.

The use of monoclonal antibodies (MAbs) has greatly increased specificity. Thus Alvarez et al. (1985) were able to distinguish *Xanthomonas* spp. from a large number of other bacteria using one carefully selected MAb. Using three others they could separate *X. campestris* pathovar *campestris* from various other pathovars, and using a further five MAbs they separated pathovar *campestris* into six groups, finding interesting epidemiological correlations. Extension of these methods allowed identification of *X. albilineans* (Alvarez et al., 1996), *X. axonopodis* pathovar *citri* (Benedict et al., 1985; Alvarez et al., 1991), *X. axonopodis* pathovar *citrumelo* (Alvarez et al., 1991; Gottwald et al., 1991), *X. axonopodis* pathovar *dieffenbachiae* (Lipp et al., 1992), *X. vesicatoria* (Bouzar et al., 1994), *X. oryzae* pathovar *oryzae* and pathovar *oryzicola* (Benedict et al., 1989), and *X. hortorum* pathovar *pelargonii* and *X. axonopodis* *begoniae* (Benedict et al., 1990). Franken et al. (1992), using polyclonal and monoclonal antibodies made with whole cells and with flagella preparations of *X. campestris* pathovar *campestris*, found that four MAbs reacted with lipopolysaccharides, whereas two reacted with proteins of 39 and 29 kDa. These proteins were apparently present only in *Xanthomonas* and may be the same proteins found in the outer membrane extracts (Min-savage and Schaad, 1983; Laakso et al., 1990). There was an interesting correlation with ability to hydrolyze starch, but again

no correlation with pathogenicity. Out of 16 other *Xanthomonas* isolates included in the tests, only two isolates of *X. campestris* pathovar *campestris* cross-reacted with *X. vesicatoria*.

Antibiotic and drug sensitivity Isolates of *X. oryzae* pathovar *oryzae* were found to be sensitive to streptomycin, ampicillin, novobiocin, erythromycin, oleandomycin, vancomycin, rifampicin, and oxymycin (all at 5 µg/ml). The MIC of penicillin G was 40 µg/ml and pimarcin did not inhibit growth up to 100 µg/ml (Choi et al., 1988). In a study of *X. campestris* pathovar *mangiferaeindicae*, strains were also found to be sensitive to streptomycin, in addition to kanamycin, gentamycin, oxytetracycline, and erythromycin (Pruvost et al., 1998). Other antibiotics active against some xanthomonads include ascamycin (Sudo et al., 1996), xanthobacidin (Huang and Chang, 1975), and zhongshenmycin (Zhang et al., 1996).

Pathogenicity and ecology Species and pathovars of *Xanthomonas* have been reported to cause disease in at least 124 monocotyledons and 268 dicotyledons, but no gymnosperms, ferns, or lower plants (Leyns et al., 1984). The host ranges and geographical distributions of these pathovars vary widely (Bradbury, 1986). The symptoms produced also vary widely and are helpful, sometimes essential, for identification of the pathogen involved. Most species and pathovars produce leaf spots, at least initially. These may spread very little, or be limited by the leaf veins, causing angular leaf spots in dicotyledons. They may spread to include the veins and then may develop systemically. Systemic infections produce wilts, death of shoots, or cankers on twigs and branches, or combinations of these symptoms, and if severe the whole plant may be killed. The extent of symptom development and type of symptoms expressed may also depend on the environmental conditions and the cultivar of the host plant (see Rudolph, 1993).

The mechanisms involved in symptom production are incompletely known. Metabolic products such as toxins and enzymes have been shown to be involved. In the case of the latter, protease-deficient mutants showed considerable loss of virulence in pathogenicity tests (Dow et al., 1990) and endoglucanase is thought to play a role, albeit minor, in the early stages of disease development (Gough et al., 1988). Certain enzymes involved in pectin and polygalacturonate degradation appear to have a limited role in pathogenesis (Dow et al., 1989; Beaulieu et al., 1991), though Boher et al. (1995) found that during the infection process the middle lamellae, primary, and secondary plant cell walls were degraded by cellulolytic and pectolytic activity. In addition, their results suggest that bacterial exocellular polysaccharides were involved in plant cell surface degradation (Boher et al., 1997). Similar findings have been made in other studies (Kingsley et al., 1993; Dow et al., 1995) and the involvement of extracellular polysaccharides has long been considered (Morris et al., 1977). El Banoby and Rudolph (1979) found that specific fractions of these polysaccharides from several xanthomonads produced persistent water-soaked spots when introduced into leaves of the host plants from which the bacteria came. Similar results were found with EPS prepared from culture filtrates of *X. vesicatoria*, in which watersoaking and, in some cases, foliar discoloration or necrosis, could be induced by EPS alone (Walkes and O'Garro, 1996). Ramirez et al. (1988) found a correlation between virulence and parameters such as the final viscosity of the culture, the viscifying capacity of the polymer, and the amount of acetyl substituents in the gum. Infrared spectral analysis revealed that intramolecular interaction of gum constituents could play a significant

role in virulence. Dow et al. (1995) found that a pathogenicity locus within *X. campestris* pathovar *campestris* comprises two genes governing the biosynthesis of lipopolysaccharide within the bacterial cell. They suggested that the LPS protects the bacteria against antibacterial substances produced by the plant early in the infection process. Subsequent studies on the early phases of infection detected the LPS produced by *X. axonopodis* pathovar *manihotis* on the outer surface of the bacterial envelope and in areas of the plant middle lamellae in the vicinity of the pathogen (Boher et al., 1997). Lipopolysaccharides are also thought to be involved in the induction of the hypersensitive response and resistance in susceptible hosts (Newman et al., 1995, 1997; Romero and Kimura, 1997); they may also have value in epidemiological studies, as differences in the LPS of *X. albilineans* isolated in different locations has been reported (Pillay et al., 1995).

Host specificity is high in pathovars of *Xanthomonas*. In the lists of pathovars below, only about half a dozen have been found to have hosts in more than one plant family. On the other hand, Schnathorst (1966) found that *X. axonopodis* pathovar *malvacearum*, when inoculated onto *Phaseolus* sp., produced early stages of infection very similar to those of pathovar *phaseoli*. The two pathovars were also serologically related, according to earlier experiments. His experiments helped to disprove earlier ideas that host specificity could be changed by artificial passage through nonhost plants.

A critical point in the life cycle of xanthomonads, as with other plant pathogens, is transmission to a new host, particularly if a period of survival in the absence of the host is necessary (see Stall et al., 1993). Such survival may be achieved in many ways, such as with seed, plant residues, perennial hosts, epiphytically, saprophytically in soil, and in insects (Schuster and Coyne, 1974). Many xanthomonads solve this problem by transmission with the seed of their host. These organisms are listed by Richardson (1990). The bacteria may be carried in detritus with the seed, on or in the seed coat, or deeper inside the tissues of the seed itself. Epiphytic survival is becoming well known, e.g., *X. axonopodis* pathovar *manihotis* and *X. arboricola* pathovar *pruni* are both known to spend the interseasonal time epiphytically (Persley, 1978; Young, 1978). In agriculture, many pathogens are most effectively carried by the activities of man, either in infected planting material, which may be symptomless (Hayward, 1974), or on tools, wheeled vehicles, and even grazing animals, as for *X. axonopodis* (Castaño et al., 1964). Survival in soil saprophytically is unusual, but has been suggested by the work of Goto et al. (1978) for *X. axonopodis* pathovar *citri*. Xanthomonads have occasionally been detected in run-off water and ditches around fields of infected plants. Results suggest that survival in this situation would be short (Steadman et al., 1975).

All organisms currently named in the genus *Xanthomonas* are plant pathogens, but isolates that are apparently non-pathogenic are also found (Vauterin et al. 1996a). Some have been identified as known pathogens growing epiphytically, e.g., *X. axonopodis* pathovar *phaseoli* isolated from symptomless weeds (Angeles-Ramos et al., 1991). Most of those examined by SDS-PAGE of proteins and fatty acid analysis can be identified with the currently accepted *Xanthomonas* spp. But identification to the pathovar level in the absence of a pathogenicity test is more difficult (Vauterin et al., 1996a). In any case, the naming of a nonpathogenic organism, with no known connection with disease, as a pathovar is unacceptable. The vast majority of *Xanthomonas* spp. found so far have been associated with plants, but recently organisms be-

longing to this genus or closely related to it have been shown to form a significant proportion of the bacteria present in microbial communities of hydrothermal vents. These organisms were also found to be related to several *Thiobacillus* spp. (Moyer et al., 1995).

ENRICHMENT AND ISOLATION PROCEDURES

With plant-pathogenic bacteria, the plant itself is probably the ideal "enrichment medium" and isolation is best made from plant material using a dilution procedure, unless contraindicated, onto a nonselective medium. Nutrient agar is usually satisfactory for isolation because it is not sufficiently rich to encourage growth of saprophytes. Supplementing Difco nutrient agar with yeast extract (5 g/l) has been recommended for general isolation work (Dye, 1980; Schaad and Stall, 1988). YSP² medium has been recommended for the isolation of *X. albilineans* (Dye, 1962) or SP³ medium supplemented with cycloheximide and penicillin G (Hayward, 1979). Dye's GYCA medium⁴ is useful for pigment production, and it is good for *X. cassavae*, which grows poorly and is short lived on media containing peptone. GYCA is also good for keeping stock cultures, as is YDC⁵ (Schaad and Stall, 1988).

Semi-selective media for particular xanthomonads have also been described: CKTM for *X. vesicatoria* (Sijam et al., 1991). The selectivity depends on cycloheximide, bacitracin, neomycin, cephalixin, 5-fluorouracil, tobramycin, and Tween 80. Strains of *X. vesicatoria* can be distinguished from strains of other xanthomonads by the formation of a clear ring around their colonies. *X. arboricola* pathovar *pruni* has been recovered from soil and plant material using XPSM⁶ medium (Civerolo et al., 1982), which also suppresses the growth of saprophytic bacteria and fungi generally found in soil samples. The use of nutrient agar supplemented with sodium deoxycholate (200 mg/l), has been used for the selective isolation of *X. vesicatoria* (Bashan and Asouline, 1983), as has a medium containing carboxymethyl cellulose and gelatin (Gitaitis et al., 1991). For a more detailed account, see Kado and Heskett (1970) and Schaad and Stall (1988).

A number of plating assays for detecting *X. campestris* pathovar *campestris* in crucifer seeds have been evaluated (Franken, et al., 1991). Centrifugation prior to plating had no appreciable effect, but the addition of Tween 20, benomyl, and chlorothalonil to the saline (0.85% NaCl) centrifugation solution generally resulted in lower recovery of the target organism. It was found that shaking at room temperature for 2.5 h and soaking at 4–6°C for

1.5 h increased the number of colony forming units recovered, but did not result in more infected seed lots being detected. No significant differences were found between the plating media tested: FS (Yuen et al., 1987), NSCA, and NSCAA (Randhawa and Schaad, 1984). In a similar study, CS20ABN medium was developed by incorporation of bacitracin, neomycin, and cycloheximide into CS20A (Chang et al., 1991). Comparison with CS20A, NSCA, NSCAA, and FS showed that all media recovered high numbers of saprophytic bacteria, that were too numerous to count and overgrew *X. campestris* pathovar *campestris* colonies, with the exception of CS20ABN, where 59–100% of recovered colonies were identified as *X. campestris campestris*.

MAINTENANCE PROCEDURES

Stock cultures may be maintained on GYCA for periods up to six months, with regular subculturing at approximately monthly intervals. Hokawat and Rudolph (1991) found that there was no appreciable decrease in virulence in cultures stored for two years on YDC agar slants. Less frequent subculturing is possible with most isolates at lower-than-room-temperature, e.g., in a refrigerator at ~7–10°C, but care should be taken to ensure that the isolate can tolerate the temperature. *X. axonopodis* pathovar *citri*, cultured on agar plates inverted and stored at 4°C in darkness, could be maintained for at least 26 weeks on a semi-enriched minimal medium (Wu et al., 1990), with no loss of virulence or alteration in colonial morphology. Cultures on richer media did not survive storage.

For longer periods of storage, most isolates can be freeze dried and will keep for years (Lelliott, 1965; Hokawat and Rudolph, 1991). Freeze-dried cultures of *X. oryzae* suspended in 10% sucrose and 1% gelatin retained viability and virulence in a suspending medium containing 2% dextrin, 0.5% ascorbic acid, 0.5% ammonium chloride, 0.5% thiourea, and 0.85% NaCl (Hwang and Cho, 1986). Liquid cultures, applied and then dried onto Whatman 3M paper, have also been used effectively to store xanthomonads (Kidby et al., 1977). *X. albilineans*, which does not keep well in the lyophilized condition, can be kept on an agar slope under sterile mineral oil for periods up to approximately two years.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

The significance of polyamines as a tool in differentiating xanthomonads from other members of the *Gammaproteobacteria* has been established (Auling et al., 1991; Hamana and Matsuzaki, 1993), and in an extensive study of polyamine profiles of 140 *Xanthomonas* strains, spermidine was found to predominate in all members of the genus (Yang et al., 1993b). In contrast, *Stenotrophomonas maltophilia*—at that time encompassed within the genus *Xanthomonas*—contained spermidine and large amounts of cadaverine.

Exopolysaccharides are generally heteropolysaccharides that have repeat structures with carbohydrate and noncarbohydrate substituents (Leigh and Coplin, 1992). The EPS produced by *Xanthomonas* is similar in structure to class 1 capsular polysaccharides and is known commercially as xanthan gum. Xanthans are high-molecular-weight, acidic, anionic heteropolysaccharides composed mainly of D-glucose, D-mannose, and D-glucuronic acid in a ratio of 2:2:1, with small amounts of pyruvic and acetic acids that vary with strains and conditions (Cadmus et al., 1978; Whitfield et al., 1981; Sutherland, 1993). Because of their physical properties and lack of toxicity to man and animals, xanthans

2. YSP medium consists of (g/l): yeast extract, 5.0; sucrose, 20.0; peptone, 10.0; and agar 15.0.

3. SP medium consists of (g/l): peptone, 5.0; K₂HPO₄, 0.5; MgSO₄·7H₂O, 0.25; sucrose, 10.0 or 20.0; agar, 15.0. To control growth of saprophytes, cycloheximide (100 mg/l) and penicillin G (200 IU/ml) can be added.

4. GYCA medium consists of (g/l): yeast extract, 5.0; glucose, 5.0, CaCO₃, 40.0, and agar, 15.0. The carbonate is mixed well into the medium just before solidification.

5. YDC medium consists of (g/l): yeast extract, 5.0; dextrose, 20.0, CaCO₃, 20.0, and agar, 15.0. The carbonate is mixed well into the medium just before solidification.

6. XPSM medium consists of (g/l except where indicated): alginic acid, 2.0; 8-azaguanine, 0.2; nicotinic acid, 2 mg; cysteine, 3 mg; KH₂PO₄, 0.8; K₂HPO₄, 0.8; MgSO₄, 0.1; Bacto agar, 15.0. Subsequent to autoclaving, the medium is cooled to 45°C and chlorothalonil (Diamond Shamrock, Cleveland, OH) and kasugamicin are added aseptically to a final concentration of 80 and 16 mg/ml, respectively.

have a wide range of uses in industry. They are gelling agents, emulsifiers, stabilizing agents, and plasticizers, particularly in the food industry (Jeannes, 1974; Sutherland and Ellwood, 1979). For the genetic control of xanthan synthesis, see Köplin et al. (1992).

DIFFERENTIATION OF THE GENUS *XANTHOMONAS* FROM OTHER GENERA

Characteristics useful for differentiating *Xanthomonas* from other genera are listed in Table BXII.7.17.

TAXONOMIC COMMENTS

Xanthomonas is a well-defined, homogeneous genus. This is particularly true now that the closely related yet distinctive species, *Xanthomonas maltophilia*, as described by Swings et al. (1983), is accommodated in its own genus, *Stenotrophomonas* (Palleroni and Bradbury, 1993). In addition, *Xanthomonas ampelina* (Panagopoulos, 1969), as described in the previous edition, has been relocated to *Xylophilus* (Willems et al., 1987) in the family *Pseudomonadaceae*, from which *Xanthomonas* was removed. The exact position of *Xylella* may need further investigation, as its very different G + C ratio suggests a genus rather distantly related to other members of the family.

Phylogenetic treatment Our current understanding of the phylogenetic position of the genus *Xanthomonas* owes much to early nucleic acid hybridization studies (Palleroni et al., 1973; De Ley, 1978; De Vos and De Ley, 1983). In this work, the first indications of the multigeneric nature of the genus *Pseudomonas* were presented, as well as strong evidence that the genus *Xanthomonas*, traditionally associated with pseudomonads, was well defined and distinct (De Ley, 1992). The pioneering work of Palleroni and co-workers (1973) assigned the genus *Xanthomonas* to rRNA hybridization group V, distinct from the majority of fluorescent and nonfluorescent taxa of the genus *Pseudomonas*. Congruent results were presented in subsequent studies, and in these broader analyses encompassing a wider range of Gram-negative genera, both the genus *Xanthomonas* and *Pseudomonas sensu stricto* were placed in rRNA super-family II (De Ley, 1978). In addition, results from these studies demonstrated that *Stenotrophomonas maltophilia* (at that time considered a member of the genus *Pseudomonas*) was related to the genus *Xanthomonas* and was subsequently renamed as *Xanthomonas maltophilia* (Swings et al., 1983; see also Van Zyl and Steyn, 1992; Palleroni and Bradbury, 1993).

Subsequently, cataloging of 16S rRNA sequences placed the genus *Xanthomonas* in the *Gammaproteobacteria* (purple bacteria; Stackebrandt et al., 1988), along with *Pseudomonas sensu stricto* (Woese et al., 1985), the subdivision being broadly synonymous with rRNA super families I and II (De Ley, 1978). In a more detailed overview of the phylogeny of the genus *Pseudomonas* and traditionally associated genera and species, the genera *Xanthomonas*, *Stenotrophomonas*, and *Xylella* were recovered together based on 16S rDNA sequence information (Kerstens et al., 1996). This grouping broadly equated with rRNA hybridization group V (Palleroni et al., 1973). In this analysis, however, some doubt was expressed as to the true phylogenetic position of the group, as it was not possible to unequivocally place it within either the *Gammaproteobacteria* or *Betaproteobacteria*. Subsequently, the position of all three genera was confirmed within the *Gammaproteobacteria* and the monophyletic lineage of the genus *Xanthomonas*

demonstrated (Moore et al., 1997). In this analysis, a core set of species was found to be closely related; *X. axonopodis* pathovar *axonopodis*, *X. campestris* pathovar *campestris*, *X. fragariae*, *X. populi* pathovar *populi*, and *X. oryzae* pathovar *oryzae* all clustered together, with *X. albilineans* more distant. A high degree of homogeneity was evident within the genus, and indeed *S. maltophilia* was found to share 95–97% sequence similarity with the xanthomonads studied here. In a subsequent, more detailed study by Hauben et al. (1997), similar findings were presented. The mean similarity value derived from the comparison of 16S rDNA sequence data from 46 xanthomonads, including 20 type strains, was found to be 98.2%. The genus was recovered in three clusters, one of which contained *X. albilineans*, *X. hyacinthi*, *X. theicola*, and *X. translucens*, while the second contained *X. sacchari*, and the remaining cluster the “core” species of the genus. In addition, *S. maltophilia*, although closely related to *Xanthomonas*, was clearly distinct. This finding, and similar results reported by Moore et al. (1997), was in keeping with an earlier study of restriction maps of the 16S rRNA gene, which demonstrated that the genera *Stenotrophomonas* and *Xanthomonas* were closely related, sharing 69 restriction sites, though a number of sites specific to *Stenotrophomonas* gave indications of the distinctiveness of the genera (Nesme et al., 1995).

Pathovars Within the genus *Xanthomonas*, it has long been known that *X. campestris*, with its many pathovars, was heterogeneous. Some pathovars are very similar; others show clear distinctions in phenotype and genotype, especially when sensitive methods are used. DNA–DNA hybridization studies of 54 pathovars then considered to belong to *X. campestris* indicated six groups and four ungrouped pathovars (Hildebrand et al., 1990; Vauterin et al., 1990; Palleroni et al., 1993). Extending these studies with more strains confirmed the six groups and added 14 more, to make 20 genomic species in the genus as a whole (Vauterin et al., 1995). Four of these equate to the existing species *X. albilineans*, *X. fragariae*, *X. oryzae*, and *X. populi*. The remaining 16 are derived from the heterogeneous *X. campestris*. Rather surprisingly, Vauterin et al. (1995) found that the phenotypically distinct species *X. axonopodis*, as studied by classical methods, belongs to the largest DNA hybridization group, which therefore takes its name. In the case of the latter, the emended species was found to encompass 37 pathovars, together with a further 11 included based on a previous study by Palleroni et al. (1993). In a similar way, *X. campestris* pathovar *plantaginis* has been added to the six pathovars in the emended description of *X. campestris*, and pathovar *carotae* and pathovar *taraxaci* added to the three already in *X. hortorum*. The inclusion of these pathovars on the basis of previously collected results is open to debate, as Vauterin et al. employed spectrophotometric determination of renaturation rates to measure DNA hybridization values (De Ley et al., 1970), whilst Palleroni et al. used an S1 nuclease radiolabeling procedure (Hildebrand et al., 1990). In addition, in the study of Palleroni et al., some of the individual reassociation values reported were very low. The reassociation values reported for *X. axonopodis* pathovar *betlicola* did not exceed 43% to any other member of the species *X. axonopodis*, as defined by Vauterin et al. (1995), and average reassociation values for each of the other pathovars studied were seldom in excess of 50% to other members of the species. These findings, coupled with ambiguous results derived from fatty acid analysis (Yang et al., 1993d) and SDS-PAGE analysis of whole cell proteins (Vauterin et al., 1991), further call into question the assumptions made about these pre-

viously studied pathovars in the current treatment of the genus. Clearly, further study is required to confirm the species affiliations of the pathovars studied by Palleroni et al. (1993); for the purposes of this review, we have highlighted those concerned.

Subsequent to the recent treatment of the genus by Vauterin et al. (1995), the number of recognized pathovars within *X. axonopodis* was reduced, as some of them were considered invalid in terms of the Standards for Naming Pathovars (Dye et al., 1980) viz: *X. axonopodis* pathovar *aurantifolii* and *X. axonopodis* pathovar *citrumelo* (See Young et al., 1991b), and *X. axonopodis* pathovar *cassavae*, *X. axonopodis* pathovar *phaseoli* biovar *fuscans* and *X. axonopodis* pathovar *vesicatoria* (Young et al., 1996b). Similarly, *X. arboricola* pathovar *poinsetticola*, *X. hortorum* pathovar *vitians*, *X. translucens* pathovar *hordei*, and *X. vasicola* pathovar *vasculorum*, all described by Vauterin et al. (1995), were also considered defective in terms of the Standards for Naming Pathovars and not included in the list of Young et al. (1996b). In the case of the latter, by invalidating *X. vasicola* pathovar *vasculorum*, the need to maintain *X. vasicola* pathovar *holcicola* is removed and the pathovar is reduced to a synonym of *X. vasicola*. The remaining pathovars are listed as *Pathovars Incertae Sedis*, together with all other pathovars not studied by Vauterin et al. (1995) under their names as pathovars of *X. campestris sensu lato*.

In conclusion, the complexity of the genus *Xanthomonas* is recognized and few would argue with Vauterin and Swings (1997) that the genus represents a continuum of genotypic and phenotypic diversity, albeit encompassing a number of distinctive taxa. The heterogeneity within *X. campestris sensu lato* was self-

evident and much valuable information has been produced over the years to support this. This said, it is our opinion that the classification produced by Vauterin et al. (1995) and described below is unconvincing, especially for some of the more heterogeneous taxa, such as *X. axonopodis*. The power and value of DNA hybridization data for bacterial taxonomy is well established, but when the groupings derived from this method are not conclusively backed up by other techniques such as 16S rRNA gene sequencing (Hauben et al., 1997), SDS-PAGE analysis of whole-cell proteins (Vauterin et al., 1991), fatty acids (Yang et al., 1993d; Vauterin et al., 1996b), and biochemical tests (Vauterin et al., 1995), which is particularly true for *X. axonopodis*, a classification driven by these data alone is weakened, as the method cannot be used in any practical sense to perform identifications. Diagnosticians with poor resources will have problems operating within the current system, even when the polyphasic methods concur, as classical phenotypic data are of marginal relevance. Polyphasic methods will only work for the four existing species of *Xanthomonas*, viz: *X. albilineans*, *X. fragariae*, *X. oryzae*, and *X. populi* (see Table BXII.γ.18), as the 16 recently proposed species were derived from *X. campestris sensu lato* and classical phenotypic tests were not included in their description (Vauterin et al., 1995). It should be recognized that many diagnosticians are currently identifying putative xanthomonads to the genus level, noting the diseased host and identifying straight to the pathovar level, paying little heed to the species affiliation. Clearly, this situation is unacceptable in the long term, and much work remains to be done to develop an alternative strategy for characterization within the genus *Xanthomonas*.

TABLE BXII.γ.18. Characteristics of some *Xanthomonas* species that can be determined by classical methods^{a,b}

Characteristic	<i>X. albilineans</i>	<i>X. fragariae</i>	<i>X. oryzae</i>	<i>X. populi</i>
Mucoid growth on nutrient agar + 5% glucose	—	+	+	+
Hydrolysis of gelatin	D	+	— ^c	—
H ₂ S from peptone	—	—	+	—
Maximum growth temperature, °C	37	33	32	27.5
Maximum salt tolerance (% w/v)	0.5	0.5–1.0	0.5–2.0	0.4–0.6
Acid production from: ^d				
Arabinose	—	—	d ^e	—
Glucose, sucrose	+	+	+	+
Mannose	+	+	+	+
Galactose	d	—	+	+
Trehalose	—	—	+	+
Cellobiose	—	—	+	—
Fructose	—	+	+	+
Lactose	—	—	—	—
Maltose	—	—	—	—
Xylose	+	—	—	—
Utilization of:				
Acetate, citrate, malate			+	
Propionate		—	+	
Succinate		+	+	
DL-Tartrate	—	—		
Benzoate	—	—		
Growth on Oxoid NA:				
None	+			
Good			+	
Poor to very poor		+		+

^afor symbols see standard definitions.

^bData from Dye (1962, 1966b); Ridé and Ridé (1978, 1992; Swings et al. (1990); Vera Cruz et al. (1984).

^cReaction is positive for pathovar *oryzicola*.

^dIn the medium of Hayward (1964).

^eIn the medium of Starr (1946).

FURTHER READING

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- Swings, J.G. and E.L. Civerolo (Editors). 1993. *Xanthomonas*, Chapman & Hall, London.
- Vauterin, L. and J. Swings. 1997. Are classification and phytopathological diversity compatible in *Xanthomonas*? J. Ind. Microbiol. Biotechnol. 19: 77–82.

List of species of the genus *Xanthomonas*

1. ***Xanthomonas campestris*** (Pammel 1895) Dowson 1939, 190^{AL} emend. Vauterin, Hoste, Kersters and Swings 1995, 484 ("Bacillus campestris" Pammel 1895, 130.) *cam.pe's'tris*. L. gen. n. *campestris* of a level field; also the specific epithet of *Brassica campestris*, a host plant.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The original isolates of this species caused a vascular disease of *Brassica* spp. Currently, the species includes a number of pathovars that cause diseases predominantly within the family Cruciferae. In general, pathovars are not distinguishable by phenotypic characterization and identification is reliant on knowledge of their hosts.

Seventeen strains isolated from host plants in the family Cruciferae fell into DNA hybridization group 15 with the type strain of *X. campestris* (Vauterin et al., 1995). The average DNA binding value within the group was 87% ± 7%. Analysis of strains by fatty acid methyl esters recovered the majority in Cluster 2 (Yang et al., 1993d), which also contains strains from *X. arboricola*, *X. axonopodis*, and *X. vesicatoria* (Table BXII.γ.19). Subsequent analysis of fatty acid data demonstrated a high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in cluster 1 (Vauterin et al., 1991), though only strains from pathovar *campestris* were examined in this study. Metabolic activity on carbon compounds, as shown by Biolog GN microplates, clearly distinguishes 14 of the other 19 species, but not the remaining five viz *X. hortorum*, *X. arboricola*, *X. cassavae*, *X. axonopodis*, and *X. vasicola* (see Table BXII.γ.20). Laboratory identification of the species without complex apparatus and processes would seem to be uncertain.

In the following list of the seven currently acceptable pathovars of *X. campestris* and their pathotype strains (Vauterin et al., 1995; Young et al., 1996b), names of host plants and their respective families are given. It should be noted that *X. campestris* pathovar *plantaginis* was not studied by Vauterin et al. (1995) and its inclusion in the species was based on the work of Palleroni et al. (1993). The validity of this action could be questioned, as DNA from this pathovar was not hybridized against the full range of pathovars covered in the study of Vauterin et al. and operational differences between the analyses may have occurred.

The mol% G + C of the DNA is: 65.8–66.6; 68.0–68.3 (*T_m*) for *Xanthomonas campestris* pathovar *incanae*.

Type strain: ATCC 33913, CFPB 2350, DSM 3586, ICMP 13, LMG 568, NCPPB 528.

GenBank accession number (16S rRNA): X95917.

- a. ***Xanthomonas campestris* pathovar *campestris*** (Pammel 1895) Dowson 1939.

Hosts: *Brassica* spp., *Capsella bursa-pastoris*, *Lepidium sativum*, *Matthiola* spp., *Raphanus sativus*, *Amorica rusticana*, (fam. Cruciferae). The one report on *Boerhaavia*

erecta (fam. Nyctaginaceae) seems doubtful and has not been confirmed.

Type strain: ATCC 33913, CFPB 2350; ICMP 13; LMG 568; NCPPB 528.

- b. ***Xanthomonas campestris* pathovar *aberrans*** (Knösel 1961) Dye 1978.

Host: *Brassica oleracea* var. *botrytis*, (fam. Cruciferae).

Type strain: ICMP 4805, LMG 9037, NCPPB 2986.

- c. ***Xanthomonas campestris* pathovar *armoraciae*** (McCulloch 1929) Dye 1978.

Hosts: *Armoracia rusticana*. By inoculation *Brassica oleracea* var. *botrytis*, *B. oleracea* var. *capitata* (fam. Cruciferae), *Phaseolus vulgaris* (fam. Leguminosae).

Type strain: ICMP 7, LMG 535, NCPPB 347.

- d. ***Xanthomonas campestris* pathovar *barbareae*** (Burkholder 1941) Dye 1978.

Host: *Barbarea vulgaris* (fam. Cruciferae).

Type strain: ATCC 13460, ICMP 438, LMG 547, NCPPB 983.

- e. ***Xanthomonas campestris* pathovar *incanae*** (Kendrick and Baker 1942) Dye 1978.

Host: *Matthiola incanae* (fam. Cruciferae).

Type strain: ATCC 13462, CFPB 2527, ICMP 574, LMG 7490, NCPPB 937.

- f. ***Xanthomonas campestris* pathovar *plantaginis*** (Thornberry and Anderson 1937) Dye 1978.

Hosts: *Plantago* spp. (fam. Plantaginaceae).

Type strain: ATCC 23382, ICMP 1028, LMG 848, NCPPB 1061.

- g. ***Xanthomonas campestris* pathovar *raphani*** (White 1930) Dye 1978.

Hosts: *Brassica* spp., *Raphanus sativus*, (fam. Cruciferae), *Capsicum annuum*, *Lycopersicon esculentum*, *Nicotiana tabacum* (fam. Solanaceae).

Type strain: ATCC 49079, ICMP 1404, LMG 860, NCPPB 1946.

2. ***Xanthomonas albilineans*** (Ashby 1929) Dowson 1943, 11^{AL} emend. van den Mooter and Swings 1990, 367 ("Bacterium albilineans" Ashby 1929, 135.)

al.bi.lin'e.ans. L. adj. *albus* white; L. part. adj. *lineans* striping; M.L. adj. *albilineans* white striping.

The characteristics are as described for the genus and as listed in Tables BXII.γ.18, BXII.γ.19, and BXII.γ.20. Causes leaf scald disease of sugar cane (*Saccharum officinarum*); other natural hosts include *Brachiaria piligera*, *Imperata cylindrica* var. *major*, *Paspalum conjugatum* (fam. Gramineae). Colonies are yellowish buff on Dye's YSP medium or similar sucrose-peptone media. Growth usually fails or is very poor on nutrient agar. Glutamate and methionine are required for growth. Esculin is hydrolyzed, but not starch.

TABLE BXII-γ.19. Percentage concentration of fatty acid methyl esters useful for species differentiation within the genus *Xanthomonas*^{a,b}

Fatty Acid	1. X.	2. X.	3. X.	4. X.	5. X.	6. X.	7. X.	8. X.	9. X.	10. X.	11. X.	12. X.	13. X.	14. X.	15. X.	16. X.	17. X.	18. X.	19. X.	20. X.
<i>campestris</i>																				
<i>albilineans</i>																				
<i>arboricola</i>																				
<i>axonopodis</i>																				
<i>bronni</i>																				
<i>cassavae</i>																				
<i>codiae</i>																				
<i>cucurbitae</i>																				
<i>hyacinthi</i>																				
<i>melonis</i>																				
<i>oryzae</i>																				
<i>populi</i>																				
<i>sacchari</i>																				
<i>thericola</i>																				
<i>vesicatoria</i>																				
<i>Saturated acids:</i>																				
C _{16:0}	3.6	13.4	4	4.1	4.9	1.4	6.3	6.7	3.7	2	3.8	4.3	18.5	3.4	7.4	4.2	8.8	4.3	15.8	3.3
<i>Unsaturated acids:</i>																				
C _{15:1} o8c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.9	0	0	0	0	0
C _{16:1} o9c	0.9	0	0.9	0.7	0.7	0.5	1.5	0.1	0	0.1	0	1.6	2.3	1	16.9	1.1	0.2	1	0.7	0.8
C _{16:1} o7c	12.7	13.2	15	16.2	8.9	10.3	15.1	16	11.7	18.6	12.1	10.4	24.9	13.8	0	15	12.9	21.1	25	15.6
<i>Branched saturated acids:</i>																				
C _{14:0} iso	0.7	0	0.8	0.5	0	0	0.6	1.2	0.2	0.7	3.7	0.2	0	0.7	0	14	9.8	2.6	0	0.9
C _{15:0} iso	26.5	2.4	31.1	28.2	21.3	31.8	26	31.5	36.8	33.2	11.1	28.5	6.3	31.8	23.4	17.3	12.9	29.4	15.5	24.6
C _{15:0} anteiso	13.9	0.1	13.7	9.3	2.6	9.5	7.4	10.2	10.5	15.7	26.8	6.6	0.7	9.4	1.6	5.6	15.9	5	3	15.5
C _{16:0} iso	3.2	4.7	2	2.6	1.7	1.1	3.6	3.2	0.9	1.7	6.6	1.9	0.2	1	0	18.8	9.9	3.6	1.3	4.1
C _{17:0} iso	6.8	19.3	4.5	6.7	10.9	9.5	7.4	3.9	3.7	2.7	2.1	10.2	14.6	5.3	2.5	2.7	2.4	4.8	7.3	6.8
C _{17:0} anteiso	0.8	2.9	0.3	0.4	0.1	1	0.5	0.2	0.2	0.1	1.1	0.2	0.3	0.5	0	0.2	1.2	0	1.1	1.1
C _{19:0} iso	0	1.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>Branched unsaturated acids:</i>																				
C _{15:1} iso G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.6	0	0	0	0	0
C _{17:1} iso o9c	8.5	23.3	3.8	6.5	25.7	17.5	7.5	6.1	3.5	3.9	1.7	10.4	9.1	6.2	0	2.2	2.4	7.4	11.2	8.3
<i>Others:</i>																				
C _{16:0} alc	0	0.3	0	0	0	0	0	0	0	0	1.6	0	0	0	0	0	0	0	0	0
C _{17:0} cyclo	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

^aThe most distinctive components are highlighted in bold.^bAdapted from Vauterin et al. (1996b).

TABLE BXII-γ.20. Some diagnostic tests for species of *Xanthomonas* with Biolog GN[®] plates^{a,b}

Carbon source	1. X. <i>campestris</i> (19)	2. X. <i>albilineans</i> (5)	3. X. <i>carboricola</i> (12)	4. X. <i>axonopodis</i> (96)	5. X. <i>bonni</i> (2)	6. X. <i>casaruae</i> (5)	7. X. <i>codiae</i> (2)	8. X. <i>cucurbitae</i> (5)	9. X. <i>fragariae</i> (2)	10. X. <i>hortorum</i> (29)	11. X. <i>lyacinthi</i> (5)	12. X. <i>melonis</i> (2)	13. X. <i>oryzae</i> (10)	14. X. <i>pisi</i> (2)	15. X. <i>populi</i> (2)	16. X. <i>sacchari</i> (2)	17. X. <i>theicola</i> (3)	18. X. <i>translucens</i> (32)	19. X. <i>vasicola</i> (9)	20. X. <i>vesicatoria</i> (7)
Dextrin	+	+	+	+	+	d	+	+	+	d	+	+	+	+	+	+	+	+	d ^c	+
N-Acetyl-D- galactosamine	-	-	-	d	-	d	-	-	-	-	-	-	-	-	-	-	-	-	-	d
Cellobiose	+	d	+	+	+	d	+	+	-	+	+	+	+	+	+	+	+	d	d ^c	d
L-Fucose	d ^c	d	+	d	d	d	+	d	d	d	+	+	d	+	+	+	+	d	d	+
D-Galactose	+	+	+	+	+	d	+	d	d	d	+	+	d	+	+	+	+	d	d	-
Gentiobiose	+	-	d	+	+	+	+	-	+	+	+	d	d	+	+	+	+	d	d ^c	+
α-D-Lactose	-	d	d	d	d	-	+	-	-	+	+	+	+	+	+	+	+	-	-	-
Maltose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	d ^c	+
Melibiose	+	-	+	d	d	+	+	d	-	+	-	+	-	+	+	+	-	-	-	-
β-Methyl-D- glucoside	-	-	-	-	d	-	+	-	-	-	D	d	-	+	d	+	-	-	-	-
D-Psicose	+	+	+	+	+	d	+	+	+	+	+	+	+	+	+	+	+	+	+	+
L-Rhamnose	-	-	-	-	d	-	-	-	-	-	-	-	-	-	-	-	-	-	-	d
Acetic acid	D	-	+	d	d	d	+	-	-	d	-	+	-	-	d	+	d	-	d	-
cis-Aconitic acid	D	-	d	-	-	d	+	-	-	d	-	+	-	-	d	+	-	-	d	-
D-Glacturonic acid	-	-	-	-	-	-	-	-	-	-	-	-	-	+	-	d	-	-	-	-
D-Glucuronic acid	-	-	-	-	+	-	-	-	-	-	-	-	-	+	-	d	-	-	-	-
β-Hydroxybutyrate	-	-	d	d	d	-	d	-	-	-	+	d	-	+	-	+	+	d	d	-
p-Hydroxyphenyl- acetate	-	-	-	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-	-
α-Ketobutyric acid	D	-	d	d	d	d	+	-	-	d	-	+	-	+	-	+	-	-	d	d
α-Ketovaleric acid	-	-	-	-	-	d	-	-	-	-	-	d	-	-	-	+	+	-	-	-
D-L-Lactic acid	D	+	+	d	+	d	-	+	-	+	-	d	d	+	d	+	+	d	d	-
Malonic acid	d ^c	-	d	d	-	-	+	-	-	d	-	+	-	+	d	-	-	-	d	d
D-Saccharic acid	d ^c	-	-	d	-	d	+	d	d	-	-	+	-	+	-	d	-	-	d	d
Bromosuccinic acid	+	-	+	+	+	+	+	+	+	+	+	+	d	+	+	+	-	d	d ^c	+
Alaninamide	D	D	d	d ^c	+	+	+	d	-	+	-	+	d	+	+	+	+	+	d ^c	+
D-Alanine	D	-	d	+	-	+	+	-	-	d	-	+	d	+	d	+	+	d	d ^c	d
L-Alanine	D	-	d	+	d	+	+	d	-	-	-	+	d	+	d	+	+	d	d ^c	+
L-Alanylglycine	D	D	d	+	+	+	+	d	-	+	-	+	d	+	d	+	+	+	d ^c	d
L-Glutamic acid	+	D	+	+	+	+	+	+	+	+	+	+	d	+	-	+	+	+	+	+
Hydroxy-L- proline	D	-	d	d	-	d	+	-	-	d	D	+	-	d	-	+	+	d	d ^c	d
L-Leucine	-	-	d	d	-	-	d	-	-	-	-	+	-	d	-	+	-	-	d	d
L-Serine	D	-	d	d	+	+	+	+	-	+	D	+	-	+	d	+	d	d	d	+
Glycerol	D	-	+	d	d	-	+	d	-	d	+	d	-	+	d	+	d	d	d	d
Glucose-1- phosphate	D	-	d	d	d	d	-	-	-	d	+	-	-	+	-	-	d	d	-	-
Glucose-6- phosphate	D	-	-	d	d	d	-	-	-	d	+	-	-	+	-	-	d	d	-	-

^aFor symbols see standard definitions. Numbers in parentheses indicate the number of strains tested.^bAdapted from Vauterin et al. (1995).^c89% positive.

H₂S is not produced from peptone. The maximum sodium chloride tolerance is 0.5%. Maximum growth temperature is 37°C. Acid is produced within 21 days in Dye's medium C from glucose, sucrose, mannose, and xylose, but not from arabinose, fructose, trehalose, cellobiose, rhamnose, adonitol, mannitol, sorbitol, dulcitol, *meso*-inositol, salicin, inulin, or α -methylglucoside. Two strains isolated from the host plant *Saccharum officinarum* showed an average DNA binding value of 97% (Vauterin et al., 1995). The fatty acid methyl ester profile (Table BXII.7.19) recovers the majority of strains studied in Cluster 31 of Yang et al. (1993d). Subsequent analysis of fatty acid data demonstrated a high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins was unable to recover the strains in a single cluster, though the three strains included in this study showed greater similarity to each other than to any other xanthomonad examined (Vauterin et al. 1991). Metabolic activity on carbon compounds as shown by Biolog GN microplates is shown in Table BXII.7.20.

Certain strains isolated from *Saccharum officinarum*, exclusively from Guadeloupe, were originally included in this species but are now encompassed within *X. sacchari*, as they are clearly distinct on the basis of DNA hybridization studies, fatty acids, and protein profiling.

The mol% G + C of the DNA is: 63.1–64.5 (Bd, T_m).

Type strain: ATCC 33915, CFBP 2523, ICMP 196, LMG 494, NCPPB 2969.

GenBank accession number (16S rRNA): X95918.

3. **Xanthomonas arboricola** Vauterin, Hoste, Kersters and Swings 1995, 484^{VP}
ar.bo.ri' co.la. L. n. *arbor* a tree; L. suff. *-cola* dweller; M.L. adj. n. *arboricola* tree dweller.

The characteristics are as described for the genus and as listed in Tables BXII.7.19 and BXII.7.20. The isolates cause disease mainly in temperate trees, including hazelnut, walnut, and *Prunus* spp. In general, pathovars are not distinguishable by phenotypic characterization and identification is reliant on knowledge of their hosts. The species was recovered as DNA hybridization group 4 by Vauterin et al. (1995). The average DNA binding value within the group was 79% $\pm$ 15%. Three of the five pathovars were closely related, with hybridization values of 89%, and were distinguished from other strains by their quinate metabolism (Lee et al., 1992). *X. arboricola* pathovar *populi* and pathovar *poinsettiiicola* were related to other group 4 pathovars at average DNA hybridization values of 57% and 68%, respectively, and ranged from 48 to 76%, indicating some lack of homogeneity in the species. Fatty acid methyl ester composition (Table BXII.7.19) placed the strains in Cluster 2 of Yang et al. (1993d), a position shared by three pathovars of *X. axonopodis*, several pathovars of *X. campestris*, and *X. vesicatoria*. Subsequent analysis of fatty acid data demonstrated a high degree of homogeneity within the species (Vauterin et al., 1996b). Most strains were recovered in Cluster 7c by protein SDS-PAGE analysis (Vauterin et al., 1991), though some strains of *X. arboricola* pathovar *juglandis* were recovered in Cluster 2 and *X. arboricola* pathovar *poinsettiiicola* LMG 5403 was unclustered in this study. Metabolic activity on various carbon sources, determined using GN microplates, distinguishes this species from 15 of the 20 in the genus (see Table BXII.7.20), but distinction from *X. hor-*

torum, *X. cassavae*, *X. axonopodis*, *X. vasicola*, and *X. campestris* is not clear. Laboratory identification of the species without complex apparatus and processes would seem to be uncertain.

In the following list of currently acceptable pathovars of *X. arboricola* and their pathotype strains (Vauterin et al., 1995; Young et al., 1996b), the synonyms, names of host plants, and their respective families are given. It should be noted that *X. campestris* pathovar *celebensis* was not studied by Vauterin et al. (1995) and its inclusion in this species was based on the work of Palleroni et al. (1993). The validity of this action could be questioned, as DNA from this pathovar was not hybridized against the full range of pathovars covered in the study of Vauterin et al. and operational differences between the analyses may have occurred.

The mol% G + C of the DNA is: 66.0–67.0 (T_m).

Type strain: ATCC 49083, ICMP 35, LMG 747, NCPPB 411.

GenBank accession number (16S rRNA): Y10757.

Additional Remarks: This is also the pathotype strain of *X. arboricola* pathovar *juglandis* (there being no *X. arboricola* pathovar *arboricola*).

- a. **Xanthomonas arboricola** pathovar *celebensis* (Gäumann 1923) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *celebensis* (Gäumann 1923) Dye 1978.)
 Hosts: *Musa* spp. (fam. Musaceae).
 Type strain: ATCC 19045, ICMP 1488, LMG 677, NCPPB 1832.
- b. **Xanthomonas arboricola** pathovar *corylina* (Miller, Bollen, Simmons, Gross and Barss 1940) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *corylina* (Miller, Bollen, Simmons, Gross and Barss 1940) Dye 1978.)
 Hosts: *Corylus* spp. (fam. Betulaceae).
 Type strain: ATCC 19313, CFBP 1159, ICMP 5726, LMG 689, NCPPB 935.
- c. **Xanthomonas arboricola** pathovar *juglandis* (Pierce 1901) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *juglandis* (Pierce 1901) Dye 1978.)
 Hosts: *Juglans* spp. (fam. Juglandaceae).
 Type strain: ATCC 49083, CFBP 2528, ICMP 35, LMG 747, NCPPB 411.
- d. **Xanthomonas arboricola** pathovar *populi* (ex de Kam 1984) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *populi* (ex de Kam 1984) Young, Bradbury, Davis, Dickey, Ercolani, Hayward and Vidaver 1991a.)
 Hosts: *Populus* spp. (fam. Salicaceae).
 Type strain: ICMP 8923, LMG 12141.
- e. **Xanthomonas arboricola** pathovar *pruni* (Smith 1903) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *pruni* (Smith 1903) Dye 1978.)
 Hosts: *Prunus* spp., *Sorbus japonica* (fam. Rosaceae).
 Type strain: ATCC 19316, CFBP 2535, ICMP 51, LMG 852, NCPPB 416.
4. **Xanthomonas axonopodis** Starr and Garcés 1950, 81^{AL} emend. Vauterin, Hoste, Kersters and Swings 1995, 484 (*Xanthomonas axonoperis* (sic) Starr and Garcés 1950, 81.)
a.xo.no'po.dis. M.L. n. *Axonopus* generic name of a grass; M.L. gen. n. *axonopodis* of *Axonopus*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. As originally constituted, this species was slower growing and less metabolically active than *X. campestris*, from which it was clearly differentiated phenotypically. The emended description now includes isolates from a wide variety of host plants and *X. axonopodis* is the most heterogeneous species currently recognized in the genus. In general, pathovars are not distinguishable by phenotypic characterization and identification is reliant on knowledge of their hosts. The species was recovered as DNA hybridization group 9 by Vauterin et al. (1995). The average DNA binding value within the group was $77\% \pm 15\%$. The type strain of the species shows a mean DNA binding within the group of only 66%. *X. axonopodis* and *X. campestris* pathovar *vasculorum* type A were closely related and were assigned to Group 9 with an overall level of DNA hybridization of 58% with other strains of the group. The individual values ranged from 47 to 71%, indicating a lack of homogeneity in the group. Fatty acid methyl ester composition (Table BXII.γ.19) recovered the bulk of isolates in Cluster 1, but isolates were also found in Clusters 2, 4, 5, 15, 18, 21, 28, and 29, together with several pathovars belonging to different species, as currently described (Yang et al., 1993d). Subsequent analysis of fatty acid data demonstrated a low homogeneity within the species (Vauterin et al., 1996b). Most isolates were recovered in Cluster 3 by protein SDS-PAGE analysis (Vauterin et al., 1991), alongside isolates of *X. cucurbitae*. Other isolates were recovered in clusters 5, 6, 8, 9, and a sizeable number were unclustered by this method. Metabolic activity on various carbon sources, determined using Biolog GN microplates, distinguishes this species from 10 others, but not from the remaining nine, viz *X. hortorum*, *X. arboricola*, *X. cassavae*, *X. oryzae*, *X. vasicola*, *X. melonis*, *X. vesicatoria*, *X. campestris*, and *X. translucens* (Table BXII.γ.20).

In the following list of currently acceptable pathovars of *X. axonopodis* and their pathotype strains (Vauterin et al., 1995; Young et al., 1996b), the synonyms, names of host plants, and their respective families are given. It should be noted that *X. campestris* pathovar *bellicola*, *X. campestris* pathovar *biophyti*, *X. campestris* pathovar *fascicularis*, *X. campestris* pathovar *khayae*, *X. campestris* pathovar *maculifoliogardeniae*, *X. campestris* pathovar *martyniicola*, *X. campestris* pathovar *mehusii*, *X. campestris* pathovar *nakataecorchori*, *X. campestris* pathovar *pedalii*, *X. campestris* pathovar *physalidicola*, and *X. campestris* pathovar *punicae* were not studied by Vauterin et al. (1995) and their inclusion in this species was based on the work of Palleroni et al. (1993). The validity of this action could be questioned, as DNA from these pathovars were not hybridized against the full range of pathovars covered in the study of Vauterin et al. and operational differences between the analyses may have occurred.

The mol% G + C of the DNA is: 62.6–65 (Bd, T_m).

Type strain: ATCC 19312, ICMP 50, LMG 538, NCPPB 457.

GenBank accession number (16S rRNA): X95919.

- a. *Xanthomonas axonopodis* pathovar *axonopodis* Starr and Garcés 1950.

Hosts: *Axonopus scoparius*, *A. micay*, *A. compressus*, and *A. affinis*. By inoculation, *Digitaria decumbens*, *Hypharrhenia rufa*, *Panicum* sp., and *Saccharum officinarum* (fam. Gramineae).

Type strain: ATCC 19312, ICMP 50, LMG 538, NCPPB 457.

- b. *Xanthomonas axonopodis* pathovar *alfalfae* (Riker, Jones and Davis 1935) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *alfalfae* (Riker, Jones and Davis 1935) Dye 1978.)

Hosts: *Medicago sativa*, *Melilotus indica*, *Pisum sativum*, *Phaseolus vulgaris*, *Trigonella foenum-graecum* (fam. Leguminosae).

Type strain: ICMP 5718, LMG 497, NCPPB 2062.

- c. *Xanthomonas axonopodis* pathovar *bauhiniae* (Padhya, Patel and Kotasthane 1965a) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *bauhiniae* (Padhya, Patel and Kotasthane 1965a) Dye 1978.)

Host: *Bauhinia racemosa* (fam. Leguminosae).

Type strain: ICMP 5720, LMG 548, NCPPB 1335.

- d. *Xanthomonas axonopodis* pathovar *begoniae* (Takimoto 1934) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *begoniae* (Takimoto 1934) Dye 1978.)

Host: *Begonia* spp. (fam. Begoniaceae).

Type strain: ATCC 49082, CFBP 2524, ICMP 194, LMG 7303, NCPPB 1926.

- e. *Xanthomonas axonopodis* pathovar *betlicola* (Patel, Kulkarni and Dhande 1951b) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *betlicola* (Patel, Kulkarni and Dhande 1951b) Dye 1978.)

Hosts: *Piper betle*, *P. hookeri*, *P. longum* (fam. Piperaceae).

Type strain: ATCC 11677, ICMP 312, LMG 555, NCPPB 2972.

- f. *Xanthomonas axonopodis* pathovar *biophyti* (Patel, Chauhan, Kotasthane and Desai 1969) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *biophyti* (Patel, Chauhan, Kotasthane and Desai 1969) Dye 1978.)

Host: *Biophytum sensitivum* (fam. Oxalidaceae).

Type strain: ICMP 2780, LMG 556, NCPPB 2228.

- g. *Xanthomonas axonopodis* pathovar *cajani* (Kulkarni, Patel and Abhyankar 1950) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *cajani* (Kulkarni, Patel and Abhyankar 1950) Dye 1978.)

Host: *Cajanus cajan* (fam. Leguminosae).

Type strain: ATCC 11639, ICMP 444, LMG 558, NCPPB 573.

- h. *Xanthomonas axonopodis* pathovar *cassiae* (Kulkarni, Patel and Dhande 1951) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *cassiae* (Kulkarni, Patel and Dhande 1951) Dye 1978.)

Hosts: *Cassia tora*, *C. occidentalis*, *Cicer arietinum*, *Pisum sativum* (fam. Leguminosae).

Type strain: ATCC 11638, ICMP 358, LMG 675, NCPPB 2973.

- i. *Xanthomonas axonopodis* pathovar *citri* (Hasse 1915) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas citri* (ex Hasse 1915) Gabriel, Kingsley, Hunter and Gottwald 1989; *Xanthomonas campestris* pathovar *citri* (Hasse 1915) Dye 1978.)

Hosts: *Aegle marmelos*, *Atalantia* spp. *Balsamocitrus paniculata*, *Casimiroa edulis*, *Chaetospermum glutinosa*, *Citropsis*

- schweinfurthii*, *Citrus* spp. and hybrids, *Clausena lansium*, *Eremocitrus glauca*, *Evodia* spp., *Feronia* spp., *Feroniella* spp., *Fortunelia* spp., *Hesperethusa crenulata*, *Limonia* spp., *Melicope triphylla*, *Microcitrus* spp., *Murraya eratica*, *Paramignya longipedunculata*, *Poncirus trifoliata* and hybrids, *Severina buxifolia*, *Toddalia asiatica*, *Zanthoxylum* spp. (fam. Rutaceae).
- Type strain: CFBP 2525, ICMP 24, LMG 682, NCPPB 409.
- j. ***Xanthomonas axonopodis* pathovar *clitoriae*** (Pandit and Kulkarni 1979) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *clitoriae* (Pandit and Kulkarni 1979) Dye, Bradbury, Goto, Hayward, Lelliott and Schroth 1980.)
- Host: *Clitoria biflora* (fam. Leguminosae).
- Type strain: ICMP 6574, LMG 9045, NCPPB 3092.
- k. ***Xanthomonas axonopodis* pathovar *coracanae*** (Desai, Thirumalachar and Patel 1965) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *coracanae* (Desai, Thirumalachar and Patel 1965) Dye 1978.)
- Host: *Eleusine coracana* (fam. Gramineae).
- Type strain: ICMP 5724, LMG 686; NCPPB 1786.
- l. ***Xanthomonas axonopodis* pathovar *cyamopsidis*** (Patel, Dhande and Kulkarni 1953) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *cyamopsidis* (Patel, Dhande and Kulkarni 1953) Dye 1978.)
- Host: *Cyamopsis tetragonoloba* (fam. Leguminosae).
- Type strain: ICMP 616, LMG 691, NCPPB 637.
- m. ***Xanthomonas axonopodis* pathovar *desmodii*** (Patel 1949) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *desmodii* (Patel 1949) Dye 1978.)
- Host: *Desmodium diffusum* (fam. Leguminosae).
- Type strain: ATCC 11640, ICMP 315, LMG 692, NCPPB 481.
- n. ***Xanthomonas axonopodis* pathovar *desmodiigangetici*** (Patel and Moniz 1948) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *desmodiigangetici* (Patel and Moniz 1948) Dye 1978.)
- Host: *Desmodium gangeticum* (fam. Leguminosae).
- Type strain: ATCC 11671, ICMP 577, LMG 693, NCPPB 577.
- o. ***Xanthomonas axonopodis* pathovar *desmodiilaxiflori*** (Pant and Kulkarni 1976a) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *desmodiilaxiflori* (Pant and Kulkarni 1976a.)
- Hosts: *Desmodium laxiflorum*, *Tamarindus indica* (fam. Leguminosae).
- Type strain: ICMP 6502, LMG 9046, NCPPB 3086.
- p. ***Xanthomonas axonopodis* pathovar *desmodiitundifolii*** (Desai and Shah 1960) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *desmodiitundifolii* (Desai and Shah 1960) Dye 1978.)
- Host: *Desmodium rotundifolium* (fam. Leguminosae).
- Type strain: ICMP 168, LMG 694, NCPPB 885).
- q. ***Xanthomonas axonopodis* pathovar *dieffenbachiae*** (McCulloch and Pirone 1939) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *dieffenbachiae* (McCulloch and Pirone 1939) Dye 1978.)
- Hosts: *Aglaonema robellinii*, *Anthurium andraeanum*, *Dieffenbachia* spp. (fam. Araceae), *Dracaena fragrans* (fam. Liliaceae).
- Type strain: ICMP 5727, LMG 695, NCPPB 1833.
- r. ***Xanthomonas axonopodis* pathovar *erythrinae*** (Patel, Kulkarni and Dhande 1952a) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *erythrinae* (Patel, Kulkarni and Dhande 1952a) Dye 1978.)
- Host: *Erythrina indica* (fam. Leguminosae).
- Type strain: ATCC 11679, ICMP 446, LMG 698, NCPPB 578.
- s. ***Xanthomonas axonopodis* pathovar *fascicularis*** (Patel and Kotasthane 1969b) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *fascicularis* (Patel and Kotasthane 1969b) Dye 1978.)
- Host: *Corchorus fascicularis* (fam. Tiliaceae).
- Type strain: ICMP 5731, LMG 9047, NCPPB 2230.
- t. ***Xanthomonas axonopodis* pathovar *glycines*** (Nakano 1919) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *glycines* (Nakano 1919) Dye 1978.)
- Hosts: *Brunnichia cirrhosa* (fam. Polygonaceae), *Dolichos uniflorus*, *Glycine* spp. *Phaseolus lunatus*, *P. vulgaris* (fam. Leguminosae).
- Type strain: ATCC 43911, CFBP 2526, ICMP 5732, LMG 712, NCPPB 554.
- u. ***Xanthomonas axonopodis* pathovar *khayae*** (Sabet 1959) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *khayae* (Sabet 1959) Dye 1978.)
- Hosts: *Khaya senegalensis*, *K. grandifoliola* (fam. Meliaceae).
- Type strain: ICMP 671, LMG 753, NCPPB 536.
- v. ***Xanthomonas axonopodis* pathovar *lespedezae*** (Ayers, Lefebvre and Johnson 1939) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *lespedezae* (Ayers, Lefebvre and Johnson 1939) Dye 1978.)
- Host: *Lespedeza* spp. (fam. Leguminosae).
- Type strain: ATCC 13463, ICMP 439, LMG 757, NCPPB 993.
- w. ***Xanthomonas axonopodis* pathovar *maculifoliigardeniae*** (Ark and Barrett 1946) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *maculifoliigardeniae* (Ark and Barrett 1946) Dye 1978.)
- Hosts: *Gardenia* spp., *Ixora coccinea* (fam. Rubiaceae).
- Type strain: CFBP 1155, ICMP 318, LMG 758, NCPPB 971.
- x. ***Xanthomonas axonopodis* pathovar *malvacearum*** (Smith 1901) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *malvacearum* (Smith 1901) Dye 1978.)
- Hosts: *Gossypium* spp., *Thespesia lambas* (fam. Malvaceae), *Ceiba pentandra* (fam. Bombacaceae).
- Type strain: CFBP 2530, ICMP 5739, LMG 761, NCPPB 633.
- y. ***Xanthomonas axonopodis* pathovar *manihotis*** (Bondar 1915) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *manihotis* (Bondar 1915) Dye 1978.)
- Host: *Manihot* spp. (fam. Euphorbiaceae).
- Type strain: ATCC 49073, CFBP 2603, ICMP 5741, LMG 773, NCPPB 1834.

- z. ***Xanthomonas axonopodis* pathovar *martyniicola*** (Moniz and Patel 1958) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *martyniicola* (Moniz and Patel 1958) Dye 1978.)
Host: *Martynia diandra* (fam. Martyniaceae).
Type strain: ICMP 82, LMG 9049, NCPPB 1148.
- aa. ***Xanthomonas axonopodis* pathovar *melhusii*** (Patel, Kulkarni and Dhande 1952a) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *melhusii* (Patel, Kulkarni and Dhande 1952a) Dye 1978.)
Host: *Tectona grandis* (fam. Verbenaceae).
Type strain: ATCC 11644, ICMP 619, LMG 9050, NCPPB 994.
- ab. ***Xanthomonas axonopodis* pathovar *nakataecorchori*** (Padhya and Patel 1963b) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *nakataecorchori* (Padhya and Patel 1963b) Dye 1978.)
Host: *Corchorus acutangulus* (fam. Tiliaceae).
Type strain: ICMP 5742, LMG 786, NCPPB 1337.
- ac. ***Xanthomonas axonopodis* pathovar *patelii*** (Desai and Shah 1959) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *patelii* (Desai and Shah 1959) Dye 1978.)
Host: *Crotalaria juncea* (fam. Leguminosae).
Type strain: ICMP 167, LMG 811, NCPPB 840.
- ad. ***Xanthomonas axonopodis* pathovar *pedalii*** (Patel and Jindal 1972) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *pedalii* (Patel and Jindal 1972) Dye 1978.)
Host: *Pedaliium murex* (fam. Pedaliaceae).
Type strain: ICMP 3030, LMG 812, NCPPB 2368.
- ae. ***Xanthomonas axonopodis* pathovar *phaseoli*** (Smith 1897) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas phaseoli* (ex Smith 1897) Gabriel, Kingsley, Hunter and Gottwald 1989; *Xanthomonas campestris* pathovar *phaseoli* (Smith 1897) Dye 1978.)
Hosts: *Lablab purpureus* (syn. *Dolichos lablab*), *Lupinus polyphyllus*, *Phaseolus lunatus*, *Phaseolus vulgaris* (fam. Leguminosae).
Type strain: ATCC 9563, CFBP 2534, ICMP 5834, LMG 7455, NCPPB 3035.
- af. ***Xanthomonas axonopodis* pathovar *phyllanthi*** (Sabet, Ishag and Khalil 1969) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *phyllanthi* (Sabet, Ishag and Khalil 1969) Dye 1978.)
Host: *Phyllanthus niruri* (fam. Euphorbiaceae).
Type strain: ICMP 5745, LMG 844, NCPPB 2066.
- ag. ***Xanthomonas axonopodis* pathovar *physalidicola*** (Goto and Okabe 1958) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *physalidicola* (Goto and Okabe 1958) Dye 1978.)
Host: *Physalis alkekengi* var. *francheti* (fam. Solanaceae).
Type strain: ATCC 49077, ICMP 586, LMG 845, NCPPB 761.
- ah. ***Xanthomonas axonopodis* pathovar *poinsettiicola*** (Patel, Bhatt and Kulkarni 1951a) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *poinsettiicola* (Patel, Bhatt and Kulkarni 1951a) Dye 1978.)
Hosts: *Euphorbia pulcherrima*, *E. milii*, *Manihot esculenta* (fam. Euphorbiaceae).
Type strain: ATCC 11643, ICMP 5779, LMG 849, NCPPB 581.
- ai. ***Xanthomonas axonopodis* pathovar *punicae*** (Hingorani and Singh 1959) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *punicae* (Hingorani and Singh 1959) Dye 1978.)
Host: *Punica granatum* (fam. Punicaceae).
Type strain: ICMP 360, LMG 859, NCPPB 466.
- aj. ***Xanthomonas axonopodis* pathovar *rhynchosiae*** (Sabet, Ishag and Khalil 1969) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *rhynchosiae* (Sabet, Ishag and Khalil 1969) Dye 1978.)
Hosts: *Lupinus termis*, *Mucuna pruriens* (syn. *Stizolobium alterrimum*), *Rhynchosia memnonia* (fam. Leguminosae).
Type strain: ICMP 5748, LMG 8021, NCPPB 1827.
- ak. ***Xanthomonas axonopodis* pathovar *ricini*** (Yoshii and Takimoto 1928) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *ricini* (Yoshii and Takimoto 1928) Dye 1978.)
Host: *Ricinus communis* (fam. Euphorbiaceae).
Type strain: ATCC 19317, ICMP 5747, LMG 861, NCPPB 1063.
- al. ***Xanthomonas axonopodis* pathovar *sesbaniae*** (Patel, Kulkarni and Dhande 1952b) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *sesbaniae* (Patel, Kulkarni and Dhande 1952b) Dye 1978.)
Host: *Sesbania aegyptiaca* (fam. Leguminosae).
Type strain: ATCC 11675, (ICMP 367, LMG 867, NCPPB 582.
- am. ***Xanthomonas axonopodis* pathovar *tamarindi*** (Patel, Bhatt and Kulkarni 1951a) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *tamarindi* (Patel, Bhatt and Kulkarni 1951a) Dye 1978.)
Hosts: *Caesalpinia sepiaria*, *Tamarindus indica* (fam. Leguminosae).
Type strain: ATCC 11673, ICMP 572, LMG 955, NCPPB 584.
- an. ***Xanthomonas axonopodis* pathovar *vasculorum*** (Cobb 1893) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *vasculorum* (Cobb 1893) Dye 1978.)
Hosts: *Bambusa vulgaris*, *Brachiaria mutica*, *Coix lacryma-jobi*, *Panicum maximum*, *Pennisetum purpureum*, *Saccharum officinarum*, *Sorghum* spp., *Thysanolaena maxima*, *Zea mays* (fam. Gramineae), *Cocos nucifera*, *Dictyosperma alba* (fam. Palmae).
Type strain: ATCC 35938, CFBP 2602, ICMP 5757, LMG 901, NCPPB 796.
- ao. ***Xanthomonas axonopodis* pathovar *vignaeradiatae*** (Sabet, Ishag and Khalil 1969) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *vignaeradiatae* (Sabet, Ishag and Khalil 1969) Dye 1978.)
Hosts: *Lablab purpureus* (syn. *Dolichos lablab*), *Vigna radiata* (fam. Leguminosae).
Type strain: ICMP 5759, LMG 936, NCPPB 2058.
- ap. ***Xanthomonas axonopodis* pathovar *vignicola*** (Burkholder 1944) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *vignicola* (Burkholder 1944) Dye 1978.)

Hosts: *Phaseolus vulgaris*, *Vigna pubigera*, *V. unguiculata* (fam. Leguminosae).

Type strain: ATCC 11648, ICMP 333, LMG 8752, NCPPB 1838.

- aq. ***Xanthomonas axonopodis* pathovar *vitians*** (Brown 1918) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *vitians* (Brown 1918) Dye 1978.)

Host: *Lactuca* spp. (fam. Compositae).

Type strain: ATCC 19320, CFBP 2538, ICMP 336, LMG 937, NCPPB 976.

Additional Remarks: This strain may originally have been inappropriately chosen as the pathotype strain (Vauterin et al., 1995).

5. ***Xanthomonas bromi*** Vauterin, Hoste, Kersters and Swings 1995, 485^{VP}

bro' mi. L. masc. gen. M.L. n. *Bromus* generic name for the brome grass; M.L. gen. n. *bromi* of *Bromus*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates were all recovered from *Bromus* spp. (fam. Gramineae) and cause a wilting disease. Three strains studied were recovered in DNA hybridization group 7 (Vauterin et al., 1995). The average DNA binding value within the group was 88% ± 6%. The fatty acid methyl ester profile (Table BXII.γ.19) belongs to cluster 25 of Yang et al. (1993d), which only contains isolates from *Bromus* spp. Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species (Vauterin et al., 1996b). All *Bromus* isolates were recovered in Cluster 7 by protein SDS-PAGE analysis (Vauterin et al., 1992), distinct from other xanthomonads isolated from cereals and grasses. Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes the two strains studied from all other species within the genus (Table BXII.γ.20).

The mol% G + C of the DNA is: 65.6 (T_m).

Type strain: CFBP 1976, ICMP 12545, LMG 947.

GenBank accession number (16S rRNA): Y10764.

6. ***Xanthomonas cassavae*** (ex Wiehe and Dowson 1953) Vauterin, Hoste, Kersters and Swings 1995, 485^{VP}
cas.sa'vae. M.L. fem. gen. n. *Cassava* the English form of the flour made from *Manihot esculenta*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates cause leaf spotting and necrosis of *Manihot* spp. (fam. Euphorbiaceae). Of the five strains received as this pathogen and studied in DNA hybridization experiments (Vauterin et al., 1995), four, including the pathotype strain, showed an average DNA binding value of 94% ± 2%. The remaining strain showed binding at 87% with group 9 and was included in *X. axonopodis* as pathovar *cassavae*. The fatty acid methyl ester profile (Table BXII.γ.19) of the majority of strains studied belongs to Cluster 5 of Yang et al. (1993d). Subsequent analysis of fatty acid data demonstrated a high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in cluster 5 (Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from 14 of the other 19 species, but not the remaining five viz *X. hortorum*, *X. arboricola*, *X. axonopodis*, *X. vesicatoria*, or *X. campestris* (Table BXII.γ.20).

The mol% G + C of the DNA is: 64.2–66.1 (T_m).

Type strain: ICMP 204, LMG 673, NCPPB 101.

GenBank accession number (16S rRNA): Y10762.

7. ***Xanthomonas codiae*** Vauterin, Hoste, Kersters and Swings 1995, 485^{VP}

co.di.ae'i. M.L. neut. n. *codiaeum* generic name of croton; M.L. gen. n. *codiae* of *Codiaeum*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates cause a wilt of *Codiaeum variegatum* (fam. Euphorbiaceae). Two of four strains of *X. campestris* pathovar *poinsetticola* studied were recovered in DNA hybridization group 6 in the study of Vauterin et al. (1995), with an average DNA binding value of 99%. They were considered sufficiently distinct from the pathotype and from all other groups to be separated as a species. These strains were recovered in Cluster 2 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al. 1993d), alongside strains of *X. arboricola*, *X. axonopodis*, *X. campestris*, and *X. vesicatoria*. Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species, though it should be noted that only two strains were studied (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 3c (Vauterin et al. 1991), alongside *X. axonopodis* pathovar *alfalfae* and *X. axonopodis* pathovar *ricini*. Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others (Table BXII.γ.20).

The mol% G + C of the DNA is: 66.3 (T_m).

Type strain: ICMP 9513, LMG 8678.

GenBank accession number (16S rRNA): Y10765.

8. ***Xanthomonas cucurbitae*** (ex Bryan 1926) Vauterin, Hoste, Kersters and Swings 1995, 485^{VP}

cu.cur'bit.ae. M.L. fem. n. *Cucurbita* generic name of squash; M.L. gen. n. *cucurbitae* of *Cucurbita*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates cause necrotic spots on leaves, and occasionally stems, petioles, and fruits of *Citrullus vulgaris*, *Cucumis sativus*, and *Cucurbita* spp. (fam. Cucurbitaceae). Two strains of *X. campestris* pathovar *cucurbitae* were recovered in DNA hybridization group 8 in the study of Vauterin et al. (1995), with an average DNA binding value of 88%. They were considered sufficiently distinct from all other groups to be separated as a species. These strains were recovered in Cluster 1 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al., 1993d), alongside many strains of *X. axonopodis*. Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species, though it should be noted that only two strains were studied (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 10 (Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others, except *X. translucens* from which there are no clear differences (Table BXII.γ.20). The negative reaction with gentiobiose is particularly useful for diagnosis.

The mol% G + C of the DNA is: 66.1–66.8 (T_m).

Type strain: ICMP 2299, LMG 690, NCPPB 2597.

GenBank accession number (16S rRNA): Y10760.

9. **Xanthomonas fragariae** Kennedy and King 1962, 875^{AL} emend. van den Mooter and Swings 1990, 367.
fra.gar'i.ae. M.L. n. *Fragaria* generic name of strawberry;
 M.L. gen. n. *fragariae* of *Fragaria*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.18, BXII.γ.19, and BXII.γ.20. The isolates cause leaf spot disease of *Fragaria* spp. (fam. Rosaceae). Two strains of *X. fragariae* studied were recovered in DNA hybridization group 1 in the study of Vauterin et al. (1995), with an average DNA binding value of 91%. These strains were recovered in Cluster 12 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al., 1993d). Subsequent analysis of fatty acid data demonstrated a high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 18 (Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others (Table BXII.γ.20).

The mol% G + C of the DNA is: 62.6–63.3 (Bd, T_m).

Type strain: ATCC 33239, CFBP 2157, ICMP 5715, LMG 708, NCPPB 1469.

GenBank accession number (16S rRNA): X95920.

10. **Xanthomonas hortorum** Vauterin, Hoste, Kersters and Swings 1995, 485^{VP}
horto'rum. L. gen. n. *hortorum* of gardens.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. In general, pathovars are not distinguishable by phenotypic characterization and identification is reliant on knowledge of their hosts. Six strains isolated from necrotic spots on a variety of plants, mainly on leaves, were recovered in DNA hybridization group 2 in the study of Vauterin et al. (1995), with an average DNA binding value of $92\% \pm 10\%$. These strains were recovered in Cluster 10 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al., 1993d), though *X. campestris* pathovar *vitis* type B (LMG 938), now included in this species, was found to be loosely associated with Cluster 1. Subsequent analysis of fatty acid data demonstrated a high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 7e (*X. hortorum* pathovar *hederae*) and 15 (*X. hortorum* pathovar *pelargonii*; Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from 14 of the other 19 species, but not the remaining five *viz* *X. arboricola*, *X. cassavae*, *X. axonopodis*, *X. vesicatoria*, and *X. campestris* (Table BXII.γ.20). Laboratory identification of the species without complex apparatus and processes would seem to be uncertain.

In the following list of currently acceptable pathovars of *X. hortorum* and their pathotype strains (Vauterin et al., 1995; Young et al., 1996b), the synonyms, names of host plants, and their respective families are given. It should be noted that *X. campestris* pathovar *carotae* and *X. campestris* pathovar *taraxaci* were not studied by Vauterin et al. (1995) and their inclusion in this species was based on the work of Palleroni et al. (1993). The validity of this action could be questioned, as DNA from this pathovar was not hybridized against the full range of pathovars covered in the study of Vauterin et al. and operational differences between the analyses may have occurred.

The mol% G + C of the DNA is: 63.6–65.1 (T_m).

Type strain: ICMP 453, LMG 733, NCPPB 939.

GenBank accession number (16S rRNA): Y10759.

Additional Remarks: This is also the pathotype strain of *X. hortorum* pathovar *hederae* (there being no *X. hortorum* pathovar *hortorum*).

- a. **Xanthomonas hortorum** pathovar *carotae* (Kendrick 1934) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *carotae* (Kendrick 1934) Dye 1978.)

Host: *Daucus carota* (fam. Umbelliferae).

Type strain: ICMP 5723, LMG 8646, NCPPB 1422.

Additional Remarks: This strain has been reported to be unsuitable as a pathotype strain (Young et al. 1991a).

- b. **Xanthomonas hortorum** pathovar *hederae* (Arnaud 1920) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *hederae* (Arnaud 1920) Dye 1978.)

Host: *Hedera helix* (fam. Araliaceae).

Type strain: ICMP 453, LMG 733, NCPPB 939.

- c. **Xanthomonas hortorum** pathovar *pelargonii* (Brown 1923) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *pelargonii* (Brown 1923) Dye 1978.)

Host: *Geranium* spp., *Pelargonium* spp. (fam. Geraniaceae).

Type strain: CFBP 2533, ICMP 4321, LMG 7314, NCPPB 2985.

- d. **Xanthomonas hortorum** pathovar *taraxaci* (Niederhauser 1943) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *taraxaci* (Niederhauser 1943) Dye 1978.)

Host: *Taraxacum bicorne* (fam. Compositae).

Type strain: ATCC 19318, CFBP 410, ICMP 579, LMG 870, NCPPB 940.

11. **Xanthomonas hyacinthi** (ex Wakker 1883) Vauterin, Hoste, Kersters and Swings 1995, 486^{VP}
hy.a.cin'thi. Gr. *hyakinthos* used as M.L. masc. n. *Hyacinthus* generic name; M.L. gen. n. *hyacinthi* of *Hyacinth*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates cause yellow leaf stripes and wilt in *Hyacinthus orientalis* (fam. Liliaceae). Three strains of *X. campestris* pathovar *hyacinthi* studied were recovered in DNA hybridization group 17 of Vauterin et al. (1995), with an average DNA binding value of $98\% \pm 2\%$. They were considered sufficiently distinct from all other groups to be separated as a species. These strains were recovered in Cluster 19 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al., 1993d). Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 4 (Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others (Table BXII.γ.20).

The mol% G + C of the DNA is: 69.2–69.3 (T_m).

Type strain: ATCC 19314, ICMP 189, LMG 739, NCPPB 599.

GenBank accession number (16S rRNA): Y10754.

12. **Xanthomonas melonis** Vauterin, Hoste, Kersters and Swings 1995, 486^{VP}
me.lo'nis. L. fem. gen. n. *melonis* of melon.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates cause a soft rot of fruit of *Cucumis melo* (fam. Cucurbitaceae). Two strains of *X. campestris* pathovar *melonis* studied were recovered in DNA hybridization group 13 of Vauterin et al. (1995), with an average DNA binding value of 88%. They were considered sufficiently distinct from all other groups to be separated as a species. These strains were recovered in Cluster 4 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al., 1993d), alongside strains of *X. axonopodis* pathovar *citri*, *X. campestris* pathovar *lantanae*, and *X. campestris* pathovar *viticola*, the latter two pathovars were not examined by Vauterin et al. (1995) and their exact taxonomic position is at present uncertain. Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 19 (Vauterin et al., 1991), alongside two strains of *X. campestris* pathovar *poinsetticola*; the latter were not examined by Vauterin et al. (1995) and their exact taxonomic position is at present uncertain. Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others, except *X. axonopodis* (see Table BXII.γ.20).

The mol% G + C of the DNA is: 66.1 (T_m).

Type strain: ICMP 8682, LMG 8670, NCPPB 3434.

GenBank accession number (16S rRNA): Y10756.

13. **Xanthomonas oryzae** (ex Ishiyama 1922) Swings, van den Mooter, Vauterin, Hoste, Gillis, Mew and Kersters 1990, 309^{VP} emend. van den Mooter and Swings 1990, 367.
ory'zae. M.L. gen. n. *Oryza* the generic name for rice; M.L. gen. n. *oryzae* of *Oryza*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.18, BXII.γ.19, and BXII.γ.20. The isolates cause blight and leaf streak of *Oryza* spp. (fam. Gramineae). Six strains of *X. campestris* pathovar *oryzae* and *X. campestris* pathovar *oryzicola* studied were recovered in DNA hybridization group 10 of Vauterin et al. (1995), with an average DNA binding value of $91\% \pm 7\%$. These strains were recovered in Cluster 22 (*X. oryzae* pathovar *oryzicola*) and 24 (*X. oryzae* pathovar *oryzae*) by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al., 1993d). Subsequent analysis of fatty acid data demonstrated a high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 14 (*X. oryzae* pathovar *oryzae*) and 15 (*X. oryzae* pathovar *oryzicola*) (Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others (see Table BXII.γ.20).

In the following list of currently acceptable pathovars of *X. oryzae* and their pathotype strains (Vauterin et al., 1995; Young et al., 1996b), the synonyms, names of host plants, and their respective families are given.

The mol% G + C of the DNA is: 64.6–65 (T_m).

Type strain: YK9, ATCC 35933, CFBP 2532, ICMP 3125, LMG 5047, NCPPB 3002.

GenBank accession number (16S rRNA): X95921.

- a. **Xanthomonas oryzae** pathovar **oryzae** (Ishiyama 1922) Swings, Van den Mooter, Vauterin, Hoste, Gillis, Mew and Kersters 1990 (*Xanthomonas campestris* pathovar *oryzae* (Ishiyama 1922) Dye 1978.)

Hosts: *Isachne globosa*, *Leersia* spp., *Leptochloa* spp., *Oryza* spp., *O. sativa*, *Phalaris arundinaceae*, *Phragmites communis*, *Zizania aquatica* (fam. Gramineae).

Type strain: ATCC 35933, CFBP 2532, ICMP 3125, LMG 5047, NCPPB 3002.

- b. **Xanthomonas oryzae** pathovar **oryzicola** (Fang, Ren, Chen, Chu, Faan and Wu 1957) Swings, Van den Mooter, Vauterin, Hoste, Gillis, Mew and Kersters 1990 (*Xanthomonas campestris* pathovar *oryzicola* (Fang, Ren, Chen, Chu, Faan and Wu 1957) Dye 1978.)

Hosts: *Oryza sativa*, *Oryza* spp., *Leersia hexandra* (weak) (fam. Gramineae).

Type strain: ATCC 49072, CFBP 2286, ICMP 5743, LMG 797, NCPPB 1585.

14. **Xanthomonas pisi** (ex Goto and Okabe 1958) Vauterin, Hoste, Kersters and Swings 1995, 486^{VP}
pi'si. M.L. gen. n. *Pisum* genus of peas; M.L. gen. n. *pisi* of *Pisum*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates cause necrotic spots and rotting of *Pisum sativum* (fam. Leguminosae), and are strongly pectolytic, causing soft rot in vegetable tissue. The one strain of *X. campestris* pathovar *pisi* studied was recovered in DNA hybridization group 12 of Vauterin et al. (1995) and considered sufficiently distinct from all other groups to be separated as a species. This strain was distantly related to Cluster 1 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al., 1993d), the latter encompassing many strains of *X. axonopodis*. As only one strain has been studied to date, no assessment of the homogeneity of this species can be made. SDS-PAGE analysis of total proteins could not cluster this strain with any other xanthomonad examined in this study (Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others (see Table BXII.γ.20). This is by far the most reactive species of *Xanthomonas*, being metabolically active on 70 out of the 95 carbon compounds on microplates (Vauterin et al., 1995). Especially useful diagnostically are the activities shown on *N*-acetyl-D-galactosamine, formate, D-galacturonate, *p*-hydroxyphenylacetate, glucuronamide, and thymidine, which are positive for *X. pisi* and negative for all other species studied.

The mol% G + C of the DNA is: 64.6 (T_m).

Type strain: ATCC 35939, ICMP 570, LMG 847, NCPPB 762.

GenBank accession number (16S rRNA): Y10758.

15. **Xanthomonas populi** (ex Ridé 1958) van den Mooter and Swings 1990, 367^{VP}
po'pu.li. M.L. masc. n. *Populus* generic name of the poplar tree; M.L. n. *populi* of *Populus*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.18, BXII.γ.19, and BXII.γ.20. In Dye's Medium C, acid was produced within 21 days from glucose, sucrose, mannose, galactose, trehalose, and fruc-

tose (Rid   and Rid  , 1978), but not from arabinose and cellobiose, which agrees with the results obtained from GN plates. Sodium chloride tolerance is low at 0.4–0.6%, and maximum growth temperature is the lowest in all xanthomonads at 27.5  C. Few of the cells in culture have a flagellum, but it can be detected by immunofluorescence. The isolates cause a weeping canker of poplar (*Populus* spp.) and all species in the sections *Aigeros*, *Deltoidea*, *Tacamahaca*, *Leuce* (subsections *Albidae* and *Trepidae*) and their hybrids are susceptible (fam. Salicaceae). Three strains of *X. populi* studied were recovered in DNA hybridization group 3 of Vauterin et al. (1995), with an average DNA binding value of 96% $\pm$ 1%. These strains were recovered in Cluster 30 by fatty acid methyl ester analysis (Table BXII.  .19; Yang et al., 1993d). Subsequent analysis of fatty acid data demonstrated a low internal homogeneity within the species, which may be due to poor or non-reproducible growth (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 13 (Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others. One of the least reactive species, it is unequivocally positive for only 10 compounds; these are glycogen, sucrose, D-trehalose, D-fructose,   -D-glucose, and maltose, as well as the four shown in Table BXII.  .20.

The mol% G + C of the DNA is: 62–65 (T_m).

Type strain: ATCC 51165, ICMP 5816, LMG 5743, NCPPB 2959.

GenBank accession number (16S rRNA): X95922.

16. **Xanthomonas sacchari** Vauterin, Hoste, Kersters and Swings 1995, 486^{VP}
sac' cha.ri. L. gen. n. *sacchari* from sugar.

The characteristics are as described for the genus and as listed in Tables BXII.  .19 and BXII.  .20. The isolates, all of which have been isolated from Guadeloupe, cause a disease of sugarcane (*Saccharum officinarum*; fam. Gramineae) very like leaf scald. Two strains studied were recovered in DNA hybridization group 19 of Vauterin et al. (1995), with an average DNA binding value of 98%. They were considered sufficiently distinct from *X. albilineans* and all other groups to be separated as a species. All six strains were recovered in Cluster 27 by fatty acid methyl ester analysis (Table BXII.  .19; Yang et al., 1993d). Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species (Vauterin et al., 1996b). All Guadeloupe isolates were recovered in Cluster 6 by protein SDS-PAGE analysis (Vauterin et al., 1992), distinct from other xanthomonads from cereals and grasses, except for *X. campestris* pathovar *vasculorum* LMG 8285; the latter was not examined by Vauterin et al. (1995) and its exact taxonomic position is at present uncertain. Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others (Table BXII.  .20). Other clear results not shown in the table include metabolic activity on quinate and L-phenylalanine, which also distinguish this species from most others.

The mol% G + C of the DNA is: 68.5 (T_m).

Type strain: LMG 471.

GenBank accession number (16S rRNA): Y10766.

17. **Xanthomonas theicola** Vauterin, Hoste, Kersters and Swings 1995, 486^{VP}
the.i' co.la. L. fem. gen. n. *Thea* the generic name for tea; L.v. *colere* to inhabit; M.L. adj. *theicola* living in tea.

The characteristics are as described for the genus and as listed in Tables BXII.  .19 and BXII.  .20. The isolates cause bacterial canker of tea (*Camellia sinensis*; fam. Theaceae). A strain of *X. campestris* pathovar *theicola* was recovered in DNA hybridization group 18 in the study of Vauterin et al. (1995), and considered sufficiently distinct from all other groups to be separated as a species. Three strains of the species were recovered in Cluster 20 by fatty acid methyl ester analysis (Table BXII.  .19; Yang et al., 1993d). Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species (Vauterin et al., 1996b), though only three strains were studied. SDS-PAGE analysis of total proteins placed strains in Cluster 16 (Vauterin et al., 1991). Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from all others (see Table BXII.  .20). *X. theicola* differs for three or more compounds from 15 species, differs clearly for two compounds from *X. axonopodis*, and only for   -ketovaleate from *X. vasicola* and *X. translucens*.

The mol% G + C of the DNA is: 69.2 (T_m).

Type strain: ICMP 6774, LMG 8684.

GenBank accession number (16S rRNA): Y10763.

18. **Xanthomonas translucens** (ex Jones, Johnson and Reddy 1917) Vauterin, Hoste, Kersters and Swings 1995, 487^{VP}
trans.lu' cens. L. v. *transluceo* to be translucent; L. part. adj. *translucens* being translucent.

The characteristics are as described for the genus and as listed in Tables BXII.  .19 and BXII.  .20. The isolates cause disease in various members of the family Gramineae. Twenty-three strains were recovered in DNA hybridization group 16 in the study of Vauterin et al. (1995), with an average DNA binding value of 78% $\pm$ 11%. These strains were recovered in three closely related clusters, Clusters 7, 8, and 9, by fatty acid methyl ester analysis (Table BXII.  .19; Yang et al. 1993d). Subsequent analysis of fatty acid data demonstrated a relatively low level of internal homogeneity within the species (Vauterin et al., 1996b). All isolates were recovered in Cluster 1 by protein SDS-PAGE analysis of total proteins (Vauterin et al., 1992), distinct from other xanthomonads isolated from cereals and grasses. Metabolic activity on carbon compounds, determined using Biolog GN microplates, enables clear distinction from 14 of the species, although seven of these used only one C-compound for clear differentiation (see Table BXII.  .20). The remaining five species, not clearly differentiated by any of the 95 compounds, were *X. cucurbitae*, *X. axonopodis*, *X. oryzae*, *X. vasicola*, and *X. vesicatoria*.

In the following list of currently acceptable pathovars of *X. translucens* and their pathotype strains (Vauterin et al., 1995; Young et al., 1996b), the synonyms, names of host plants, and their respective families are given.

The mol% G + C of the DNA is: 69.1–69.7 (T_m).

Type strain: ATCC 19319, ICMP 5752, LMG 876, NCPPB 973.

GenBank accession number (16S rRNA): X99299.

- a. ***Xanthomonas translucens* pathovar *translucens*** (Jones, Johnson and Reddy 1917) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *hordei* (Hagborg 1942) Dye 1978) *Xanthomonas campestris* pathovar *translucens* (Jones, Johnson and Reddy 1917) Dye 1978.)
Hosts: *Hordeum* spp. and other cereals (fam. Gramineae).
Type strain: ATCC 19319, CFBP 2054, ICMP 5752, LMG 876, NCPPB 973.
 - b. ***Xanthomonas translucens* pathovar *arrhenatheri*** (Egli and Schmidt 1982) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *arrhenatheri* Egli and Schmidt 1982.)
Host: *Arrhenatherum elatius* (fam. Poaceae).
Type strain: ATCC 33803, CFBP 2056, ICMP 7727, LMG 727, NCPPB 3229.
 - c. ***Xanthomonas translucens* pathovar *cerealis*** (Hagborg 1942) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *cerealis* (Hagborg 1942) Dye 1978.)
Hosts: *Agropyron* spp., *Avena* spp., *Bromus* spp., *Hordeum* spp., *Secale cereale*, *Triticum* spp. (fam. Gramineae).
Type strain: CFBP 2541, ICMP 1409, LMG 679, NCPPB 1944.
 - d. ***Xanthomonas translucens* pathovar *graminis*** (Egli, Goto and Schmidt 1975) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *graminis* (Egli, Goto and Schmidt 1975) Dye 1978.)
Hosts: *Alopecurus pratensis*, *Dactylis glomerata*, *Festuca* spp., *Lolium multiflorum*, *L. perenne*, *Phleum pratense* (fam. Gramineae).
Type strain: ATCC 29091, CFBP 2053, ICMP 5733, LMG 726, NCPPB 2700.
 - e. ***Xanthomonas translucens* pathovar *phlei*** (Egli and Schmidt 1982) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *phlei* Egli and Schmidt 1982.)
Host: *Phleum pratense* (fam. Gramineae).
Type strain: ATCC 33805, CFBP 2062, ICMP 7725, LMG 730, NCPPB 3231.
 - f. ***Xanthomonas translucens* pathovar *phleipratensis*** (Wallin and Reddy 1945) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *phleipratensis* (Wallin and Reddy 1945) Dye 1978.)
Host: *Phleum pratense* (fam. Gramineae).
Type strain: CFBP 2540, ICMP 5744, LMG 843, NCPPB 1837.
 - g. ***Xanthomonas translucens* pathovar *poae*** (Egli and Schmidt 1982) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *poae* Egli and Schmidt 1982.)
Host: *Poa trivialis* (fam. Gramineae).
Type strain: ATCC 33804, CFBP 2057, ICMP 7726, LMG 728, NCPPB 3230.
 - h. ***Xanthomonas translucens* pathovar *secalis*** (Reddy, Godkin and Johnson 1924) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *secalis* (Reddy, Godkin and Johnson 1924) Dye 1978.)
Hosts: *Secale cereale*, *Hordeum* and *Triticum* spp. may become artificially infected (fam. Gramineae).
Type strain: CFBP 2539, ICMP 5749, LMG 883, NCPPB 2822.
 - i. ***Xanthomonas translucens* pathovar *undulosa*** (Smith, Jones and Reddy 1919) Vauterin, Hoste, Kersters and Swings 1995 (*Xanthomonas campestris* pathovar *undulosa* (Smith, Jones and Reddy 1919) Dye 1978.)
Hosts: *Triticum* spp. (fam. Gramineae), *Secale cereale*, *Hordeum* spp. by inoculation.
Type strain: ATCC 35935, CFBP 2055, ICMP 5755, LMG 892, NCPPB 2821.
19. ***Xanthomonas vasicola*** Vauterin, Hoste, Kersters and Swings 1995, 487^{VP}
vas.i' co.la. L. neut. n. *vas* a vessel; M.L. a duct or vessel as in a vascular bundle; *cola* dwelling in; M.L. adj. (indecl.) *vasicola* living in vascular bundles.
- The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates cause leaf streak of *Sorghum* spp. (fam. Gramineae), which may cover large areas and produce much bacterial exudate. Strains have also been found to cause a gummosis of sugar cane (*Saccharum officinarum*; fam. Gramineae). The three strains of *X. campestris* pathovar *holcicola* studied and three of the seven strains originally assigned to *X. campestris* pathovar *vasculorum* were recovered in Group 11, with an average DNA binding value of 90% ± 4%. As the remaining four strains of *X. campestris* pathovar *vasculorum*, including the pathotype strain, were quite different, Group 11 was considered a separate species and named *X. vasicola* (Vauterin et al., 1995). These strains were recovered in Cluster 21 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al., 1993d). Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species (Vauterin et al., 1996b). All isolates were recovered in Cluster 8 by protein SDS-PAGE analysis (Vauterin et al., 1992), distinct from other xanthomonads isolated from cereals and grasses. Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from 13 other *Xanthomonas* species, but no clear results distinguish it from the remaining six species viz *X. fragariae*, *X. arboricola*, *X. axonopodis*, *X. oryzae*, *X. vesicatoria*, and *X. translucens* (see Table BXII.γ.20).
- Vauterin et al. (1995) proposed two pathovars, pathovar *holcicola* attacking *Sorghum* spp. and pathovar *vasculorum* attacking *Saccharum* sp.; however as no direct comparisons of their pathogenicity have been conducted, their status is uncertain.
- The mol% G + C of the DNA is: 64.2 (T_m).
Type strain: ICMP 3103, LMG 736, NCPPB 2417.
GenBank accession number (16S rRNA): Y10755.
20. ***Xanthomonas vesicatoria*** (ex Doidge 1920) Vauterin, Hoste, Kersters and Swings 1995, 487^{VP}
ve.si.ca.to' ri.a. L. *vesica* a blister; M.L. adj. *vesicatorius* causing a blister.
- The characteristics are as described for the genus and as listed in Tables BXII.γ.19 and BXII.γ.20. The isolates cause necrotic lesions on leaves, stems, fruit and other parts of *Capsicum* spp. and *Lycopersicon* spp. (fam. Solanaceae). The two strains of *X. campestris* pathovar *vesicatoria* type B studied were recovered in DNA hybridization group 14 of Vauterin et al. (1995), with an average DNA binding value of 99%, and were quite different from three pathovar *ve-*

sicatoria type A strains that fell into *X. axonopodis*. They were considered sufficiently distinct from all other groups to be separated as a species. These strains were recovered in Cluster 2 by fatty acid methyl ester analysis (Table BXII.γ.19; Yang et al. 1993d), alongside strains of *X. arboricola*, *X. axonopodis*, and *X. campestris*. Subsequent analysis of fatty acid data demonstrated a very high degree of internal homogeneity within the species (Vauterin et al., 1996b). SDS-PAGE analysis of total proteins placed strains in Cluster 7 (Vauterin et al., 1991), alongside strains of *X. arboricola* and *X. hortorum*. Metabolic activity on carbon compounds, determined using Biolog GN microplates, clearly distinguishes this species from 13 of the other *Xanthomonas* spp., but not from *X. hortorum*, *X. cassavae*, *X. axonopodis*, *X. vasicola*, or *X. translucens* (Table BXII.γ.20). The only clear methods to distinguish this species from *X. axonopodis* seem to be SDS-PAGE of cell proteins and DNA hybridization. More suitable phenotypic tests are needed. Because of the re-arrangement of the original taxon into two species, the host ranges of both need re-examination.

The mol% G + C of the DNA is: 65.6 (T_m).

Type strain: ICMP 63, LMG 911, NCPPB 422.

GenBank accession number (16S rRNA): Y10761.

*Species Incertae Sedis**

1. ***Xanthomonas campestris* pathovar *alangii*** (Padhya and Patel 1962) Dye 1978.
Host: *Alangium lamarckii* (fam. Alangiaceae).
Deposited strain: ICMP 5717, LMG 470; NCPPB 1336.
2. ***Xanthomonas campestris* pathovar *amaranthicola*** (Patel, Wankar and Kulkarni 1952c) Dye 1978.
Host: *Amaranthus* spp. (fam. Amaranthaceae).
Deposited strain: ATCC 11645, ICMP 441, LMG 498, NCPPB 570.
3. ***Xanthomonas campestris* pathovar *amorphophalli*** (Jindal, Patel and Singh 1972) Dye 1978.
Host: *Amorphophallus campanulatus* (fam. Araceae).
Deposited strain: ICMP 3033, LMG 499, NCPPB 2371.
4. ***Xanthomonas campestris* pathovar *aracearum*** (Berniac 1974) Dye 1978.
Host: *Xanthosoma sagittifolium* (fam. Araceae).
Deposited strain: ICMP 5381, LMG 532, NCPPB 2832.
5. ***Xanthomonas campestris* pathovar *arecae*** (Rao and Mohan 1970) Dye 1978.
Host: *Areca catechu* (fam. Palmaceae).
Deposited strain: ICMP 5719, LMG 533, NCPPB 2649.
6. ***Xanthomonas campestris* pathovar *argemones*** (Srinivasan, Patel and Thirumalachar 1961b) Dye 1978.
Host: *Argemone mexicana* (fam. Papaveraceae).
Deposited strain: ICMP 1617, LMG 534, NCPPB 1593.
7. ***Xanthomonas campestris* pathovar *arracaciae*** (Pereira, Paradella and Zagatto 1971) Dye 1978.
Host: *Arracacia xanthorrhiza* (fam. Umbelliferae).
Deposited strain: ICMP 3158, LMG 536, NCPPB 2436.
8. ***Xanthomonas campestris* pathovar *asclepiadis*** Flynn and Vidaver 1995.
Host: *Asclepias* spp. (fam. Asclepiadaceae).
Deposited strain: ICMP 10007.
9. ***Xanthomonas campestris* pathovar *azadirachtae*** (Desai, Gandhi, Patel and Kotasthane 1966) Dye 1978.
Host: *Azadirachta indica* (fam. Meliaceae).
Deposited strain: ICMP 3102, LMG 543, NCPPB 2388.
10. ***Xanthomonas campestris* pathovar *badrii*** (Patel, Kulkarni and Dhande 1950) Dye 1978.
Hosts: *Xanthium strumarium* (fam. Compositae), *Pisum sativum* (fam. Leguminosae).
Deposited strain: ATCC 11672, ICMP 571, LMG 546, NCPPB 571.
11. ***Xanthomonas campestris* pathovar *betae*** Robbs, Kimura and Ribeiro 1981.
Host: *Beta vulgaris* (fam. Chenopodiaceae).
Deposited strain: ICMP 8917, LMG 9040, NCPPB 2592.
12. ***Xanthomonas campestris* pathovar *bilvae*** Chakravarti, Sarma, Jain and Prasad 1984.
Host: *Aegle marmelos* (fam. Rutaceae).
Deposited strain: ICMP 8918, NCPPB 3213.
13. ***Xanthomonas campestris* pathovar *blepharidis*** (Srinivasan and Patel 1956) Dye 1978.
Hosts: *Blepharis boerhaavifolia*, *B. molluginifolia* (fam. Acanthaceae).
Deposited strain: ATCC 17995, ICMP 5722, LMG 557, NCPPB 1757.
14. ***Xanthomonas campestris* pathovar *boerhaaviae*** (Mathur, Swarup and Sinha 1964) Bradbury 1986.
Host: *Boerhaavia repens* (fam. Nyctaginaceae).
Deposited strain: ICMP 9423, LMG 9041, NCPPB 1612.
15. ***Xanthomonas campestris* pathovar *brunneivaginae*** (Luo, Liao and Chen 1988) Young, Saddler, Takikawa, De Boer, Vauterin, Gardan, Gvozdyak and Stead 1996b.
Host: *Oryza sativa* (fam. Gramineae).
Deposited strain: ICMP 9991.
16. ***Xanthomonas campestris* pathovar *cannabis*** Severin 1978.
Host: *Cannabis sativa* (fam. Moraceae).
Deposited strain: ICMP 6570, LMG 9042, NCPPB 2877.
17. ***Xanthomonas campestris* pathovar *cannae*** Easwaramurthy, Kaviyaran and Gnanamanickam 1984.
Host: *Canna generalis* (fam. Cannaceae).
Deposited strain: ICMP 8306, LMG 9043.
18. ***Xanthomonas campestris* pathovar *carissae*** (Moniz, Sabley and More 1964) Dye 1978.
Hosts: *Carissa congesta*, *C. carandas*, *Thevetia nerifolia* (fam. Apocynaceae), *Cestrum nocturnum* (fam. Solanaceae).
Deposited strain: ICMP 3034, LMG 669, NCPPB 2373.
19. ***Xanthomonas campestris* pathovar *centellae*** Basnyat and Kulkarni 1979.
Host: *Centella asiatica* (fam. Umbelliferae).
Deposited strain: ICMP 6746, LMG 9044, NCPPB 3245.
20. ***Xanthomonas campestris* pathovar *clerodendri*** (Patel, Kulkarni and Dhande 1952b) Dye 1978.
Host: *Clerodendron phlomoides* (fam. Verbenaceae).

*Editorial Note: The following pathovars of *X. campestris* were not examined by Vauterin et al. (1995), and their exact taxonomic position is uncertain.

- Deposited strain*: ATCC 11676, ICMP 445, LMG 684, NCPPB 575.
21. ***Xanthomonas campestris* pathovar *convolvuli*** (Nagarkoti, Banerjee and Swarup 1973) Dye 1978.
Host: *Convolvulus arvensis* (fam. Convolvulaceae).
Deposited strain: ICMP 5380, LMG 685, NCPPB 2498.
 22. ***Xanthomonas campestris* pathovar *coriandri*** (Srinivasan, Patel and Thirumalachar 1961a) Dye 1978.
Hosts: *Coriandrum sativum*, *Foeniculum vulgare* (fam. Umbelliferae).
Deposited strain: ATCC 17996, ICMP 5725, LMG 687, NCPPB 1758.
 23. ***Xanthomonas campestris* pathovar *daturae*** (Jain, Dange and Siradhana 1975) Bradbury 1986.
Host: *Datura metel* (fam. Solanaceae).
Deposited strain: ICMP 12546, NCPPB 2932.
 24. ***Xanthomonas campestris* pathovar *durantae*** (Srinivasan and Patel 1957) Dye 1978.
Host: *Duranta repens* (fam. Verbenaceae).
Deposited strain: ICMP 5728, LMG 696, NCPPB 1456.
 25. ***Xanthomonas campestris* pathovar *esculenti*** (Rangaswami and Easwaran 1962) Dye 1978.
Host: *Hibiscus esculentus* (fam. Malvaceae).
Deposited strain: ICMP 5729, LMG 699, NCPPB 2190.
 26. ***Xanthomonas campestris* pathovar *eucalypti*** (Truman 1974) Dye 1978.
Hosts: *Eucalyptus citriodora*, *E. maculata* (fam. Myrtaceae).
Deposited strain: ICMP 5382, LMG 700, NCPPB 2337.
 27. ***Xanthomonas campestris* pathovar *euphorbiae*** (Sabet, Ishag and Khalil 1969) Dye 1978.
Host: *Euphorbia acalyphoides* (fam. Euphorbiaceae).
Deposited strain: ICMP 5730, LMG 863, NCPPB 1828.
 28. ***Xanthomonas campestris* pathovar *fici*** (Cavara 1905) Dye 1978.
Host: *Ficus religiosa* (fam. Moraceae).
Deposited strain: ICMP 3036, LMG 701, NCPPB 2372.
 29. ***Xanthomonas campestris* pathovar *guizotiae*** (Yirgou 1964) Dye 1978.
Host: *Guizotia abyssinica* (fam. Compositae).
Deposited strain: ICMP 5734, LMG 731, NCPPB 1932.
 30. ***Xanthomonas campestris* pathovar *gummisudans*** (McCulloch 1924) Dye 1978.
Host: *Gladiolus* sp. (fam. Iridaceae).
Deposited strain: ICMP 5780, LMG 732, NCPPB 2182.
 31. ***Xanthomonas campestris* pathovar *heliotropii*** (Sabet, Ishag and Khalil 1969) Dye 1978.
Hosts: *Heliotropium aegypticum*, *H. sudanicum* (fam. Boraginaceae).
Deposited strain: ICMP 5778, LMG 735, NCPPB 2057.
 32. ***Xanthomonas campestris* pathovar *ionidii*** (Padhya and Patel 1963a) Dye 1978.
Host: *Ionidium heterophyllum* (fam. Violaceae).
Deposited strain: ICMP 5736, LMG 744, NCPPB 1334.
 33. ***Xanthomonas campestris* pathovar *lantanae*** (Srinivasan and Patel 1957) Dye 1978.
Host: *Lantana camara* var. *aculeata* (fam. Verbenaceae).
Deposited strain: ICMP 5737, LMG 754, NCPPB 1455.
 34. ***Xanthomonas campestris* pathovar *laureliae*** (Dye 1963a) Dye 1978.
Host: *Laurelia novae-zelandiae* (fam. Monimiaceae).
Deposited strain: ICMP 84, LMG 755, NCPPB 1155.
 35. ***Xanthomonas campestris* pathovar *lawsoniae*** (Patel, Bhatt and Kulkarni 1951a) Dye 1978.
Host: *Lawsonia alba* (fam. Lythraceae).
Deposited strain: ATCC 11674, ICMP 319, LMG 756, NCPPB 579.
 36. ***Xanthomonas campestris* pathovar *leeana*** (Patel and Kothanathan 1969a) Dye 1978.
Host: *Leea edgeworthii* (fam. Vitaceae).
Deposited strain: ICMP 5738, LMG 9048, NCPPB 2229.
 37. ***Xanthomonas campestris* pathovar *leersiae*** (ex Fang, Ren, Chen, Chu, Faan and Wu 1957) Young, Bradbury, Davis, Dickey, Ercolani, Hayward and Vidaver 1991a.
Host: *Leersia hexandra* (fam. Gramineae).
Deposited strain: ICMP 8788.
 38. ***Xanthomonas campestris* pathovar *malloti*** Goto 1993.
Host: *Mallotus japonicus* (fam. Euphorbiaceae).
Deposited strain: ATCC 51262, ICMP 11536.
 39. ***Xanthomonas campestris* pathovar *mangiferaeindicae*** (Patel, Moniz and Kulkarni 1948) Robbs, Ribeiro and Kimura 1974.
Hosts: *Anacardium occidentale*, *Mangifera indica*, *Spondias mangifera* (fam. Anacardiaceae).
Deposited strain: ATCC 11637, CFBP 1716, ICMP 5740, LMG 941, NCPPB 490.
 40. ***Xanthomonas campestris* pathovar *merremiae*** (Pant and Kulkarni 1976b) Dye, Bradbury, Goto, Hayward, Lelliott and Schroth 1980.
Host: *Merremia gangetica* (fam. Convolvulaceae).
Deposited strain: ICMP 6747, LMG 9051, NCPPB 3114.
 41. ***Xanthomonas campestris* pathovar *mirabilis*** (Durgapal and Trivedi 1976) Young, Bradbury, Davis, Dickey, Ercolani, Hayward and Vidaver 1991a.
Host: *Mirabilis jalapa* (fam. Nyctaginaceae).
Deposited strain: ICMP 8949.
 42. ***Xanthomonas campestris* pathovar *musacearum*** (Yirgou and Bradbury 1968) Dye 1978.
Hosts: *Ensete ventricosum*, *Musa* spp. (fam. Musaceae).
Deposited strain: ATCC 49084, ICMP 2870, LMG 785, NCPPB 2005.
 43. ***Xanthomonas campestris* pathovar *nigromaculans*** (Taki-moto 1927) Dye 1978.
Host: *Arctium lappa* (fam. Compositae).
Deposited strain: ATCC 23390, ICMP 80, LMG 787, NCPPB 1935.
 44. ***Xanthomonas campestris* pathovar *obscurae*** Chand and Singh 1994.
Host: *Ipomoea obscura* (fam. Convolvulaceae).
Deposited strain: ICMP 12547, NCPPB 3759 (incorrectly reported as NCPPB 3359 in Chand and Singh 1994).
 45. ***Xanthomonas campestris* pathovar *olitorii*** (Sabet 1957) Dye 1978.
Host: *Corchorus olitorius* (fam. Tiliaceae).
Deposited strain: ICMP 359, LMG 9052, NCPPB 464.

46. *Xanthomonas campestris* *pathovar papavericola* (Bryan and McWhorter 1930) Dye 1978.
Hosts: *Argemone mexicana*, *Eschscholzia californica*, *Meconopsis baileyi*, *Papaver* spp. (fam. Papaveraceae).
Deposited strain: ATCC 14179, ICMP 220, LMG 809, NCPPB 2970.
47. *Xanthomonas campestris* *pathovar parthenii* Chand, Singh, Singh and Singh 1995.
Host: *Parthenium hysteriphorus* (fam. Compositae).
Deposited strain: ICMP 12476, NCPPB 3888.
48. *Xanthomonas campestris* *pathovar passiflorae* (Pereira 1969) Dye 1978.
Host: *Passiflora edulis* (fam. Passifloraceae).
Deposited strain: ICMP 3151, LMG 810, NCPPB 2346.
49. *Xanthomonas campestris* *pathovar paullinae* Robbs, Medeiros and Kimura 1982.
Host: *Paullinia cupana* var. *sorbilis* (fam. Sapindaceae).
Deposited strain: ICMP 8919, LMG 9053, NCPPB 3079.
50. *Xanthomonas campestris* *pathovar pennamericanum* Qhobela and Claflin 1988.
Hosts: *Panicum milaceum*, *Pennisetum americanum* (fam. Gramineae).
Deposited strain: ATCC 49152, ICMP 9627.
51. *Xanthomonas campestris* *pathovar phormiicola* (Takimoto 1933) Dye 1978.
Host: *Phormium tenax* (fam. Agavaceae).
Deposited strain: ICMP 4294, LMG 702, NCPPB 2983.
52. *Xanthomonas campestris* *pathovar physalidis* (Srinivasan, Patel and Thirumalachar 1962) Dye 1978.
Hosts: *Physalis minima*, *P. peruviana* (fam. Solanaceae).
Deposited strain: ATCC 17994, ICMP 5746, LMG 846, NCPPB 1756.
53. *Xanthomonas campestris* *pathovar sesami* (Sabet and Dowson 1960) Dye 1978.
Host: *Sesamum orientale* (fam. Pedaliaceae).
Deposited strain: ICMP 621, LMG 865, NCPPB 631.
54. *Xanthomonas campestris* *pathovar spermacoces* (Srinivasan and Patel 1956) Dye 1978.
Host: *Spermacoce hispida* (fam. Rubiaceae).
Deposited strain: ATCC 17998, ICMP 5751, LMG 868, NCPPB 1760.
55. *Xanthomonas campestris* *pathovar syngonii* Dickey and Zumoff 1987.
Host: *Syngonium podophyllum* (fam. Araceae).
Deposited strain: ICMP 9154, LMG 9055, NCPPB 3586.
56. *Xanthomonas campestris* *pathovar tardicrescens* (McCulloch 1937) Dye 1978.
Hosts: *Belamcanda* sp., *Iris* spp. (fam. Iridaceae).
Deposited strain: ICMP 4295, LMG 9056, NCPPB 2984.
57. *Xanthomonas campestris* *pathovar thespesiae* Patil and Kulkarni 1981.
Host: *Thespesia populnea* (fam. Malvaceae).
Deposited strain: ICMP 7466, LMG 9057.
58. *Xanthomonas campestris* *pathovar thirumalacharii* (Padhya and Patel 1964) Dye 1978.
Host: *Triumfetta pilosa* (fam. Tiliaceae).
Deposited strain: ATCC 23577, ICMP 5852, LMG 872, NCPPB 1452.
59. *Xanthomonas campestris* *pathovar tribuli* (Srinivasan and Patel 1956) Dye 1978.
Host: *Tribulus terrestris* (fam. Zygophyllaceae).
Deposited strain: ICMP 5753, LMG 873, NCPPB 1454.
60. *Xanthomonas campestris* *pathovar trichodesmae* (Patel, Kulkarni and Dhande 1952a) Dye 1978.
Host: *Trichodesma zeylanicum* (fam. Boraginaceae).
Deposited strain: ATCC 11678, ICMP 5754, LMG 874, NCPPB 585.
61. *Xanthomonas campestris* *pathovar uppalii* (Patel 1948) Dye 1978.
Hosts: *Ipomoea muricata* (fam. Convolvulaceae), *Tropaeolum majus* (fam. Tropaeolaceae).
Deposited strain: ATCC 11641, ICMP 5756, LMG 893, NCPPB 586.
62. *Xanthomonas campestris* *pathovar vernoniae* (Patel, Desai and Patel 1968) Dye 1978.
Host: *Vernonia cinerea* (fam. Compositae).
Deposited strain: ICMP 5758, LMG 9058, NCPPB 1787.
63. *Xanthomonas campestris* *pathovar viegasii* Robbs, Neto, Malavolta and Kimura 1989.
Host: *Pachystachys lutea* (fam. Acanthaceae).
Deposited strain: ICMP 9261.
64. *Xanthomonas campestris* *pathovar viticola* (Nayudu 1972) Dye 1978.
Hosts: *Azadirachta indica* (fam. Meliaceae), *Phyllanthus maderaspatensis* (fam. Euphorbiaceae), *Vitis vinifera* (fam. Vitaceae).
Deposited strain: ICMP 3867, LMG 965, NCPPB 2475.
65. *Xanthomonas campestris* *pathovar vitiscarnosae* (Moniz and Patel 1958) Dye 1978.
Host: *Vitis carnosae* (fam. Vitaceae).
Deposited strain: ICMP 90, LMG 939, NCPPB 1149.
66. *Xanthomonas campestris* *pathovar vitistrifoliae* (Padhya, Patel and Kotasthane 1965b) Dye 1978.
Host: *Vitis trifolia* (fam. Vitaceae).
Deposited strain: ICMP 5761, LMG 940, NCPPB 1451.
67. *Xanthomonas campestris* *pathovar vitiswoodrowii* (Patel and Kulkarni 1951a) Dye 1978.
Host: *Vitis woodrowii* (fam. Vitaceae).
Deposited strain: ATCC 11636, ICMP 3965, LMG 954, NCPPB 1014.
68. *Xanthomonas campestris* *pathovar zantedeschiae* (Joubert and Truter 1972) Dye 1978.
Host: *Zantedeschia aethiopica* (fam. Araceae).
Deposited strain: ICMP 2372, LMG 9059, NCPPB 2978.
69. *Xanthomonas campestris* *pathovar zingibericola* (Ren and Fang 1981) Bradbury 1986.
Host: *Zingiber mioga* (fam. Zingiberaceae).
Deposited strain: ICMP 8787, LMG 9060.
70. *Xanthomonas campestris* *pathovar zinniae* (Hopkins and Dowson 1949) Dye 1978.
Host: *Zinnia elegans* (fam. Compositae).
Deposited strain: ICMP 5762, LMG 8692, NCPPB 2439.

Genus II. Frateuria Swings, Gillis, Kersters, De Vos, Gosselé and De Ley 1980, 555^{VP}

JEAN SWINGS AND MARTIN SIEVERS

Fra.teur i.a. M.L. fem. n. *Frateuria* named after the late Joseph Frateur (1903–1974), eminent Belgian microbiologist.

Regular straight rods, $0.5\text{--}0.7 \times 0.7\text{--}3.5\text{ }\mu\text{m}$, occurring singly or in pairs. Gram negative. **Generally motile by polar flagella or nonmotile. Obligately aerobic.** Optimum temperature for growth $25\text{--}30^\circ\text{C}$. Colonies on mannitol–yeast extract–peptone (MYP) agar¹ are yellow to orange. On glucose–yeast extract–CaCO₃ (GYC) agar,² most strains produce a typical brown water-soluble pigment. Oxidase negative. Grows at pH 3.6. No nitrate reduction. Starch and gelatin are not hydrolyzed. H₂S is produced. Chemoorganotrophic. Acid is produced from ethanol and a number of carbon sources. On D-glucose and D-xylose, the pH drops below 4.0. From D-glucose, 2-keto-, and 2,5-diketogluconic acids are formed, but not 5-ketogluconic acid. No requirements for growth factors. Isolated from *Lilium auratum* and from the fruit of *Rubus parvifolius* (raspberry) in Japan.

The mol% G + C of the DNA is: 62–64 (*T_m*).

Type species: Frateuria aurantia Swings, Gillis, Kersters, De Vos, Gosselé and De Ley 1980, 555.

FURTHER DESCRIPTIVE INFORMATION

On GYC agar, cells of *Frateuria* appear mostly as rods, rarely as ovoids, never in chains. Some strains form filaments. *Frateuria* grows luxuriantly on this medium and forms typical dark coffee-brown colonies, which measure 2–5 mm in diameter with a dark center, and are glistening or rough, flat, or raised, and circular with a regular edge. The pigment also diffuses into the agar. Colonies on MYP agar are yellow to orange, glistening or rough, regular or irregular, highly convex, measuring 1–3 mm in diameter. All strains can be grown in a defined Hoyer medium³ with ammonium sulfate as the sole nitrogen source and mannitol

as the carbon source. Luxuriant growth is obtained when mixtures of amino acids are supplied instead of NH₄⁺. Some strains grow in yeast extract broth or in peptone broth, but for routine cultivation of *Frateuria*, D-mannitol and D-glucose are incorporated in the media as carbon sources. In a mannitol–salts–vitamins medium⁴, several single amino acids are used by *Frateuria* as sole nitrogen sources. L-Alanine, L-aspartic acid, L-glutamic acid, and L-proline are not utilized as carbon and nitrogen sources by *Frateuria*.

The following physiological tests are positive for all the strains: catalase, production of H₂S, ketogenesis on glycerol and on D-mannitol, and oxidation of DL-lactate to CO₂ and water. The following tests are universally negative: oxidase, reduction of nitrates, indole formation, gelatin hydrolysis, and oxidation of acetate.

Injected into apples or pears, *Frateuria* causes browning and rotting symptoms (Vanden Abeele et al., 1980).

Frateuria is able to grow and to degrade aniline at pH 4.0–7.0 (Aoki et al., 1984).

For growth on natural gas, co-cultures of *Methylococcus capsulatus* with *Frateuria aurantia* have been proposed. *Frateuria aurantia* consumes ethanol and acetate, both exometabolites of the underoxidation of the ethane (present in natural gas) by *Methylococcus capsulatus*. Both exometabolites inhibit the growth of *M. capsulatus* (Romanovskaya et al., 1992).

Frateuria strains produce the polyene antibiotics enacyloxin IIa, IIIa and IVa (Watanabe et al., 1992a, b, c).

1. Mannitol–yeast extract–peptone (MYP) agar has the following composition (g/l): mannitol, 25.0; yeast extract, 5.0; peptone, 3.0; and agar, 25.0.

2. Glucose–yeast extract–CaCO₃ (GYC) agar consists of (g/l): D-glucose, 50.0; yeast extract, 10.0; CaCO₃, 30.0; and agar, 25.0.

3. The Hoyer medium consists of (g/l): mannitol, 30.0; (NH₄)₂SO₄, 1.0; K₂HPO₄, 1.0; K₂HPO₄, 0.1; KH₂PO₄, 0.9; MgSO₄·7H₂O, 0.25; and FeCl₃, 0.005.

4. The medium for testing utilization of L-amino acids as sole nitrogen sources consists of D-mannitol, 30.0 g; D-biotin, 0.001 g; Ca-D-pantothenate, 0.001 g; thiamine, 0.001 g; pyridoxal hydrochloride, 0.0015 g; niacin, 0.0015 g; riboflavin, 0.0015 g; *p*-aminobenzoic acid, 0.001 g; vitamin B₁₂, 0.001 g; folic acid, 0.001 g; salt solution A, 5 ml; salt solution B, 5 ml; L-amino acid, 1 g; in 1 liter of 0.2 M Tris-maleate buffer, pH 5.4. Salt solution A contains (g/l distilled water): KH₂PO₄, 10.0; and K₂HPO₄, 10.0. Salt solution B contains (g/l distilled water): MgSO₄·7H₂O, 4.0; NaCl, 0.2; FeSO₄·7H₂O, 2.0; MnSO₄·H₂O, 0.15.

TABLE BXII.γ.21. Characteristics differentiating the genera *Frateuria*, *Gluconobacter*, and *Acetobacter*^a

Characteristics	<i>Frateuria</i>	<i>Gluconobacter</i>	<i>Acetobacter</i>
<i>Flagellar arrangement in motile strains:</i>			
Polar	+	+	–
Peritrichous	–	–	+
Overoxidation of ethanol	–	–	+
Oxidation of DL-lactate to CO ₂ and H ₂ O	+	–	+
Oxidation of acetate to CO ₂ and H ₂ O	–	–	+
Formation of brown water-soluble pigments on GYC agar	+ ^b	–	–
Formation of H ₂ S	+	–	–
<i>Type of ubiquinone formed:</i>			
Q ₈	+	–	–
Q ₉	–	–	D
Q ₁₀	–	+	D
Growth on Frateur's Hoyer mannitol medium	+	–	–
<i>Carbon sources for growth:</i>			
D-Mannose, L-arabinose, D-xylose, L-xylose	+	–	–
Mol% G + C of DNA	62–64	57–64	51–65

^aFor symbols see standard definitions.

^bSome strains are negative.

TABLE BXII.γ.22. Other characteristics of *Frateruria aurantia*^a

Characteristic	Result or Reaction
Growth in 0.5% yeast extract broth or 0.5% peptone broth	d
<i>Growth in Frateur's Hoyer medium with:</i>	
Ethanol	—
Glucose	d
Mannitol	+
Growth in SM ^b containing 0.5–2% NaCl	d
Growth in SM at pH 3.6–8.1	+
<i>Temperature range, growth in SM at:</i>	
4°C	—
30–37°C	+
<i>Ethanol tolerance, growth in SM containing:</i>	
1% Ethanol	+
2–5% Ethanol	d
10% Ethanol	—
<i>Glucose tolerance, growth in 0.5% yeast extract containing:</i>	
20% Glucose	+
25–30% Glucose	d
35% Glucose	—
<i>Tolerance to metals and dyes, growth in presence of:</i>	
Malachite green, 0.001%; crystal violet, 0.0001%; brilliant green, 0.001%	+
Cd(CH ₃ COO) ₂ ·2H ₂ O, 0.01%; CoSO ₄ ·7H ₂ O, 0.01%	—
HgCl ₂ , 0.001%	+
CH ₃ COOTl	d
Vitamin requirements: <i>p</i> -aminobenzoic acid, thiamine, niacin, pantothenate	—
Final pH in D-glucose, D-xylose, or D-galactose media	4,5
<i>Acid produced from:</i> ^c	
Ethanol, D-ribose, D-mannose, <i>n</i> -propanol, <i>i</i> -inositol	+
<i>m</i> -Erythritol, D-arabinose ^d , D-fructose, sucrose, <i>n</i> -butanol	d
Glycerol ^d , L-rhamnose ^d , L-sorbose, D-mannitol, sorbitol, D-cellobiose ^d , D-lactose ^d , maltose, raffinose, dextrin, starch, <i>n</i> -amyl alcohol	—
<i>Carbon sources for growth:</i> ^c	
Glycerol, <i>m</i> -erythritol, D-ribose, D-fructose, D-glucose, D-mannose, D-mannitol, D-sorbitol, L-arabinose, D-lyxose, L-lyxose, adonitol, <i>i</i> -inositol, acetate, glycerate, lactate	+
Ethanol, D-arabinose, D-xylose, L-rhamnose, D-galactose, L-sorbose, D-cellobiose, D-lactose, maltose, sucrose, raffinose, starch, methanol, ethanediol, L-arabitol, <i>m</i> -xylitol, oxalate, malonate, tartrate, malate, gluconate, dextrin	d
<i>n</i> -Propanol, dulcitol, formate, citrate	—
<i>Amino acids as sole nitrogen source:</i>	
Arginine, asparagine, glutamine, glutamate, isoleucine, leucine, phenylalanine	+
Alanine, aspartate, histidine, methionine, proline, serine, valine	d
Cysteine, glycine, lysine, threonine, tryptophan	—
<i>Amino acids as sole carbon and nitrogen sources:</i> ^f	
Alanine, aspartate, glutamate, proline	—
<i>Antimicrobial agents (amount per disk):</i>	
Bacitracin, 10 U; chloramphenicol, 30 µg; colistin sulfate, 10 µg; erythromycin, 10 µg; fusidic acid, 10 µg; gentamicin, 10 µg; lincomycin, 2 µg; methicillin, 10 µg; neomycin, 30 µg; nitrofurantoin, 200 µg; penicillin G, 10 U; polymyxin B, 300 U; sulfafurazole, 100 µg	R
Kanamycin, 30 µg; novobiocin, 30 µg; streptomycin, 10 µg; tetracycline, 30 µg	S
Ampicillin, 10 µg; cephaloridine, 25 µg; nalidixic acid, 30 µg	d
Catalase	+
Oxidase	—
Nitrate reduction, indole production, gelatin hydrolysis, change in litmus milk	—
<i>Ferric chloride reaction on media containing:</i>	
D-Glucose	d
D-Fructose	+
D-Galactose	—

^aFor symbols see standard definitions; R, resistant; S, susceptible.^bSM medium contains (g/l): D-glucose, 50.0; yeast extract, 5.0.^cAcid formation was tested in the following medium (g/l): yeast extract, 5.0; bromocresol purple, 0.02; carbon source, 10.0. A final pH below 5.9 was considered as positive for acid formation.^dAfter retesting, some corrections of our previous results (Swings et al., 1980) were necessary.^eGrowth was tested on the following medium (g/l): carbon source, 3.0; yeast extract, 0.5; vitamin-free Casamino acids, 3.0; agar (Oxoid No. 1), 25.0.^fFrater's Hoyer medium was employed, with D-mannitol omitted.

All known authentic *Frateruria* strains have been isolated in Japan.

ENRICHMENT AND ISOLATION PROCEDURES

Yamada et al. (1976) used the following medium for the enrichment of *Frateruria* from raspberries (per liter of 10% potato ex-

tract): D-glucose, 10.0 g; ethanol, 5.0 ml; yeast extract, 5.0 g; peptone, 3.0 g; and acetic acid, 0.3 ml. The pH was 4.5 and the incubation temperature 30°C. Isolation of *Frateruria* is possible on a CaCO₃-containing medium, on which the colonies dissolve the CaCO₃ and thus are surrounded by a clear zone.

MAINTENANCE PROCEDURES

Originally *Frateuria* was grown and maintained on a medium containing (g/l of 20% potato extract): D-glucose, 5.0; glycerol, 15.0; yeast extract, 30.0; peptone, 20.0; and CaCO₃, 10.0. *Frateuria* can be maintained for five months at 4°C on slants of this medium contained in screw-capped vials. *Frateuria* survives lyophilization for many years.

DIFFERENTIATION OF THE GENUS *FRATEURIA* FROM OTHER GENERA

Frateuria resembles the genera of acetic acid bacteria in its ability to oxidize ethanol to acetic acid and to grow at pH 3.6. It resembles mostly *Gluconobacter* because of its polar flagellation; however, it oxidizes lactate, requires no vitamins, and does not produce 5-ketogluconic acid. *Acetobacter* differs from *Frateuria* by its peritrichous flagellation and its ability to oxidize acetate. *Frateuria* makes the ubiquinone Q8, whereas *Gluconobacter* possesses ubiquinone Q10 and *Acetobacter* possesses Q9 or Q10.

Frateuria can be easily distinguished from its phylogenetically nearest neighbors within the *Gammaproteobacteria* (viz, *Lysobacter*, *Stenotrophomonas*, *Xanthomonas*) using FAME analysis, in which it characteristically possesses C_{12:0} 2OH and C_{17:0} cyclo fatty acids, which are absent in *Lysobacter*, *Stenotrophomonas*, and *Xanthomonas*. The latter three genera typically display C_{11:0} iso, C_{11:0} iso 3OH, and C_{15:0} anteiso fatty acids, which are absent in *Frateuria*. The presence of more than 50% C_{15:0} iso fatty acid is characteristic for *Frateuria*.

TAXONOMIC COMMENTS

When either phenotypic features or protein gel electrophoregrams of *Frateuria* strains are compared, a tight cluster is obtained, suggesting a high genetic resemblance among them. In the past, "*Acetobacter aurantius*" strains were considered as polarly flagellated acetic acid bacteria that were intermediate between *Acetobacter* and *Gluconobacter* (Asai, 1968). Yamada et al. (1976) already suspected that these strains form a separate new genus, mainly because of their typical ubiquinone Q8. The reclassification of these strains was performed by Swings et al. (1980).

DNA-rRNA hybridizations had already shown that *Frateuria* was not related to *Acetobacter* or to *Gluconobacter*, and did not belong in rRNA superfamily IV (*sensu* De Ley, 1978), of which the family *Acetobacteraceae* constitutes a separate branch (De Vos, 1980; Gillis and De Ley, 1980; Swings et al., 1980). 16S rDNA gene sequencing confirmed that *Frateuria* belongs in the *Gammaproteobacteria* (the former rRNA superfamily II *sensu* De Ley 1978). *Frateuria* is closely related to *Stenotrophomonas*, *Xanthomonas*, *Xylella*, and *Lysobacter*. 16S rRNA gene sequence similarity values of *F. aurantia* with the type strains of *Stenotrophomonas maltophilia*, *Xanthomonas campestris*, *Xylella fastidiosa*, and *Lysobacter enzymogenes* are 89.6%; 89.6%; 88.8%, and 88.8%, respectively. *Frateuria*, *Stenotrophomonas*, and *Xanthomonas* were shown to contain the polyamine spermidine (Hamana and Matsuzaki, 1993; Yang et al., 1993b).

FURTHER READING

Swings, J., M. Gillis, K. Kersters, P. De Vos, F. Gosselé and J. De Ley. 1980. *Frateuria*, a new genus for "*Acetobacter aurantius*". Int. J. Syst. Bacteriol. 30: 547-556.

List of species of the genus *Frateuria*

1. *Frateuria aurantia* Swings, Gillis, Kersters, De Vos, Gosselé and De Ley 1980, 555^{VP}.

au.ran'tia. L. v. *aurare* to overlay with gold; M.L. adj. *aurantius* gold colored, refers to the gold-yellow color of the strains on MYP agar.

The description of the species is as for the genus. See

Tables BXII.γ.21 and BXII.γ.22 for additional characteristics.

Isolated from *Lilium auratum* and from the fruit of *Rubus parvifolius* in Japan.

The mol% G + C of the DNA is: 62-64 (*T_m*).

Type strain: IFO 3245, ATCC 33424, DSM 6220.

GenBank accession number (16S rRNA): AJ010481.

Genus III. *Luteimonas* Finkmann, Altendorf, Stackebrandt and Lipski 2000, 280^{VP}

ANDRÉ LIPSKI AND ERKO S. STACKEBRANDT

Lu.te.i.mo' nas. M.L. adj. *luteus* yellow; Gr. n. *monas* a unit; M.L. fem. n. *Luteimonas* a yellow unit.

Rod-shaped cells 0.4-0.6 × 0.8-1.8 μm. Gram negative. **Aerobic**, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor. Nitrite is reduced to nitrous oxide, but nitrate is not reduced. Colonies are yellow. Heterotrophic. Ubiquinone with eight isoprene units. **Iso-/anteiso-type fatty acid profile** with C_{15:0} iso and C_{17:1} iso 9c predominating. Isolated from several experimental biofilters supplied with the waste gas of an animal rendering plant.

The mol% G + C of the DNA is: 67.

Type species: *Luteimonas mephitis* Finkmann, Altendorf, Stackebrandt and Lipski 2000, 280.

ENRICHMENT AND ISOLATION PROCEDURES

All strains of the genus were isolated on thiosulfate agar from cultures for the enrichment of methyl-sulfide-degrading bacteria (Bendinger and Reichert, personal communication). However,

the capability for sulfide degradation was not stable for the isolates of this genus and was lost during subcultivation (Reichert, personal communication).

MAINTENANCE PROCEDURES

Short-term maintenance on agar plates for several weeks at room temperature is acceptable. For long-term storage, preservation at below -70°C with 5% dimethylsulfoxide is recommended. Freezing at higher preservation temperatures results in rapid loss of viability.

DIFFERENTIATION OF THE GENUS *LUTEIMONAS* FROM OTHER GENERA

The genus can be separated from the genera *Xanthomonas* and *Stenotrophomonas* by the absence of the fatty acid 3-hydroxy-isotridecanoic acid, and from the species *Xylella fastidiosa* by the

presence of branched-chain fatty acids. No gliding motility, which is a characteristic property of the genus *Lysobacter* (Christensen and Cook, 1978), was observed for the genus *Luteimonas*. Denitrification activity is absent in the genera *Xanthomonas*, *Stenotrophomonas*, *Xylella*, and *Lysobacter* (one *Lysobacter antibioticus* strain is positive), but present in the genus *Luteimonas*.

TAXONOMIC COMMENTS

The genus is a deep-branching member of the *Xanthomonas* group of the *Gamma*proteobacteria. *Luteimonas mephitis* is remotely and approximately equidistantly related to members of *Xanthomonas*, *Pseudoxanthomonas*, *Xylella*, *Lysobacter*, and *Stenotrophomonas* (93–95% 16S rDNA similarity); members of *Schineria*, *Rhodanobacter*, *Frateruia*, and *Nevskia* are even more distantly related.

Using members of *Stenotrophomonas* as a root, analyses by RDP

dendrogram (Maidak et al., 1997; De Soete, 1983), maximum-likelihood, and neighbor-joining (Felsenstein, 1993) place *Luteimonas mephitis* between *Lysobacter antibioticus* and the species *Rhodanobacter lindanclasticus*, *Frateruia aurantia*, and *Nevskia ramosa*, which share a common ancestry with *L. mephitis*. *N. ramosa* is characterized by a significantly longer branch length than the other members of this phylogenetic cluster. Using the 16S rDNA sequence of *N. ramosa* as a root, *Frateruia aurantia* and *Rhodanobacter lindanclasticus* are separated from *Luteimonas mephitis*, *Lysobacter antibioticus*, and members of *Xylella*, *Xanthomonas*, *Stenotrophomonas*, and *Pseudoxanthomonas*. Hence, the branching order of genera constituting the deeply branching lineages of the *Gamma*proteobacteria changes with the selection and number of sequences included in the analysis, and at present cannot be considered stable.

List of species of the genus *Luteimonas*

1. ***Luteimonas mephitis*** Finkmann, Altendorf, Stackebrandt and Lipski 2000, 280^{VP}
me.phi' tis. L. gen. fem. n. *mephitis* of harmful odor.

The characteristics are the same as those given for the genus and as listed in Table BXII.γ.23. The predominant compounds of the fatty acid profile are C_{15:0 iso}, C_{17:1 iso ω9c}, and C_{17:0 iso}. The major hydroxy fatty acid is C_{11:0 iso 3OH}. The fatty acids C_{17:0 cyclo} and C_{13:0 iso 3OH} are absent. At the beginning of the stationary phase in liquid media, cells tended to clump, showing “trembling”-like behavior. Stain-

ing of flagella (Heimbrook et al., 1989) was negative. As cells from this growth phase also showed no evidence of flagella by electron microscopic observation, the type strain of *L. mephitis* is considered nonmotile. The strain showed no growth under anaerobic conditions on Caso agar (Becton Dickinson and Company, Cockeysville, MD, USA). No pathogenic activity toward humans, animals, or plants is known.

The mol% G + C of the DNA is: 67 (HPLC).

Type strain: B1953/27.1, DSM 12574.

GenBank accession number (16S rRNA): AJ012228.

TABLE BXII.γ.23. Characteristics of *Luteimonas mephitis*^a

Characteristic	Reaction
Nitrate reduction to nitrite or N ₂	—
Nitrite reduction to N ₂ O	+
Tween 80 hydrolysis	+
Urease, lecithinase, arginine dihydrolase	—
Citrate alkalization	d
Malonate alkalization	—
Growth at 37°C	—
Acid from:	
Glucose	+
Rhamnose, sucrose, xylose, fructose, lactose, maltose	—
Hydrolysis of:	
pNP-N-acetyl-β-glucosaminide, pNP-β-galactopyranoside, pNP-β-glucuronide, pNP-β-xyloside, bis-pNP-phosphate, pNP-phosphocholine	—
pNP-β-glucopyranoside, pNP-α-D-maltoside, thymidine-5'-monophosphate-pNP-ester	d
L-alanine-pNA, L-arginine-pNA, glycine-pNA, L-leucine-pNA, L-lysine-pNA, L-proline-pNA, L-valine-pNA	—
γ-L-glutamate-pNA	d
Growth in the presence of:	
NaCl (3%), ZnCl ₂ (0.01%)	—
Cadmium acetate (0.001%), lead acetate (0.01%)	d
Resistant to:	
Erythromycin (10 μg/disk), streptomycin (10 μg/disk), nalidixic acid (30 μg/disk), ampicillin (10 μg/disk), penicillin G (10 U)	+
Kanamycin (30 μg/disk)	—
Gentamicin (10 μg/disk), fucidin (10 μg/disk), tetracycline (30 μg/disk), novobiocin (30 μg/disk), neomycin (30 μg/disk)	d

^aSymbols: see standard definitions; pNP, *para*-nitrophenyl; pNA, *para*-nitroanilide. Data from Lipski et al., 1992; Finkmann et al. 2000.

Genus IV. *Lysobacter* Christensen and Cook 1978, 372^{AL}

PENELOPE CHRISTENSEN

Lys.o.bac'ter. Gr. adj. *lysis* loosing; M.L. n. *bacter* equivalent of Gr. neut. n. *bactrum* a rod; M.L. masc. n. *Lysobacter* the loosing rod; intended to mean the lysing rod.

Thin rods, 0.2–0.5 × 1.0–15 (sometimes up to 70) μm, which are Gram negative, nonflagellated, **gliding**, **flexing**, and aerobic. **Colonies are highly mucoid, cream-colored, pink, or yellow-brown**; many strains also produce a brown, water-soluble **pigment**. **Nonfruiting; no microcysts** produced. Growth in broth culture is silky. Most strains are resistant to actinomycin D. This organism **degrades chitin** and often other polysaccharides, but it does not degrade filter-paper cellulose, and it infrequently degrades agar. Strongly proteolytic, characteristically **lysing a variety of microorganisms** (both Gram-negative and Gram-positive bacteria, including actinomycetes, blue-green and green algae, yeasts, and filamentous fungi), as well as nematodes. Habitat: soil and fresh-water.

The mol% G + C of the DNA is: 65.4–70.1.

Type species: Lysobacter enzymogenes Christensen and Cook 1978, 374.

FURTHER DESCRIPTIVE INFORMATION

Thin, flexible, gliding, Gram-negative rods, 0.2–0.5 × 1.0–15 (sometimes up to 70) μm (Figs. BXII.γ.27, BXII.γ.28, and BXII.γ.29). Colonies are slimy or mucoid, with the rubbery type of *L. gummosus* being particularly difficult to handle, and may be white, cream-colored, yellow, pink, or brown; many strains also produce a brown, water-soluble pigment, especially in older cultures. The nature of *Lysobacter* pigments is not known, but they are not carotenoids or flexirubins (Reichenbach et al., 1981). Colonies of *L. brunescens* spread in the typical “*Cytophaga*” fashion, but those of *L. antibioticus* and *L. enzymogenes* are more mucoid, and it is believed that this quality conceals the typical spreading. All strains of the latter two species have been observed to produce a thin, advancing fringe of cells at colony edges on one or more media. However, *L. gummosus* has never been observed to spread, most likely due to both its shorter cells and the large amount of gum produced on every medium on which it has been grown.

Liquid cultures show a characteristic silkiness when gently tapped, a property correlated with the presence of longer cells. Silkiness is not often seen with *L. gummosus* cultures, which the heavy gum seems to preclude lengthy cells. Liquid cultures of all four species are somewhat viscous, and *L. gummosus* is so much so that a 2-d-old tube culture may be inverted without spillage.

Aerobic. Mesophilic. The optimum temperature for growth of these organisms is relatively high (20–40°C) as compared to most other soil and water organisms. pH range for growth: 5–10. Growth of most strains is reduced by 1% NaCl.

Chemoorganotrophic. Metabolism is usually respiratory; molecular oxygen is used as the terminal electron acceptor. NO₃⁻, NH₄⁺, glutamate, and asparaginate are used as sole nitrogen sources, but urea is used by only a few strains. In general, the addition of yeast extract to a salts–glucose–nitrate agar or to a chitin-only medium stimulates growth. Chitin, but not filter-paper cellulose or agar, is hydrolyzed; some species degrade alginate, pectate, starch, and/or carboxymethyl cellulose (CMC). One of the extracellular enzymes of *L. enzymogenes* strain ATCC 27796, previously known as “myxobacter AL-1”, was the first enzyme with both β-1,4-glucanase and chitosanase activities to be described (Hedges and Wolfe, 1974). This enzyme attacks amor-

phous forms of cellulose such as CMC but does not attack crystalline cellulose. It was the first enzyme with chitosanase activity to be purified to homogeneity and could be useful for the characterization of fungal cell walls and as a tool for the classification of fungi. It has been shown to be capable of attacking some fungal cell walls to release glucosamine. The enzyme has a molecular weight of 28,900–31,000 Da and contains large amounts of the basic amino acids lysine and arginine among its 230 residues.

Strongly proteolytic. The enzymes of two strains of *L. enzymogenes* have been the subjects of intensive study. One strain, ATCC 27796, known formerly as “myxobacter AL-1,” excretes a bacteriolytic enzyme, designated “AL-1 protease I”, with a molecular weight of 14,300 and containing 136 amino acids. In contrast to many exocellular enzymes, this protein contains a disulfide bond, 1 mol hexose per mol protein, and a relatively high content of aromatic amino acids; the E₂₈₀ of a 1% solution in a 1 cm cuvette is 15.8 (Jackson and Wolfe, 1968). This enzyme has been found to cleave the pentaglycine bridge in the cell wall of *Staphylococcus* and other bacteria and to remove peptide moieties from the peptidoglycan (Katz and Strominger, 1967; Tipper et al., 1967). ATCC 27796 also produces another extracellular protease (“AL-1 protease II”), which crystallizes as fine needles and has a molecular weight of 17,000 Da, with 157 amino acid residues (Wingard et al., 1972). This enzyme does not possess cell-wall lytic activity, and it has a unique specificity for lysine residues, exhibiting peptide bond cleavage on the amino side of lysine.

It was not realized at the time that the AL-1 protease reported by Jackson and Matsueda (1970) was from the same species of bacterium as the α-lytic protease described in the paper immediately following it in that publication (Whitaker, 1970). The “myxobacterium *Sorangium*”/“myxobacter 495” (now ATCC 29487) has since been classified as *L. enzymogenes*, and it was with this strain that Whitaker’s group did the initial enzymological characterization of α- and β-lytic proteases (Whitaker, 1965, 1967; Whitaker et al., 1965a, b). The proteases of “495” (now ATCC 29487) have provided much fruitful work for biochemists interested in the three-dimensional structure of microbial proteases as compared with mammalian examples. Although the classification of the zinc-containing α-lytic protease is uncertain, β-lytic protease has been shown to be a serine protease with the active site amino acid sequence Asp–Ser–Gly (Whitaker et al., 1966; Whitaker and Roy, 1967). This is the same sequence as that of the mammalian pancreatic serine protease, and this enzyme was the first microbial serine protease isolated that is not a member of the Thr–Ser–Met family. The subsequent elucidation of the complete polypeptide sequence has shown 198 amino acids in a single polypeptide chain, which contains three intrachain disulfide bridges and has an overall molecular weight of 19,869 (Olson et al., 1970). In a 2.8-Å resolution crystallographic study of α-lytic protease, the three-dimensional structure of the enzyme has been determined, facilitating a detailed description of the active site. Comparisons with other microbial serine proteases (viz *Streptomyces griseus* proteases A and B) support the possibility that they arose from a common ancestral gene (Brayer et al., 1979). Sub-

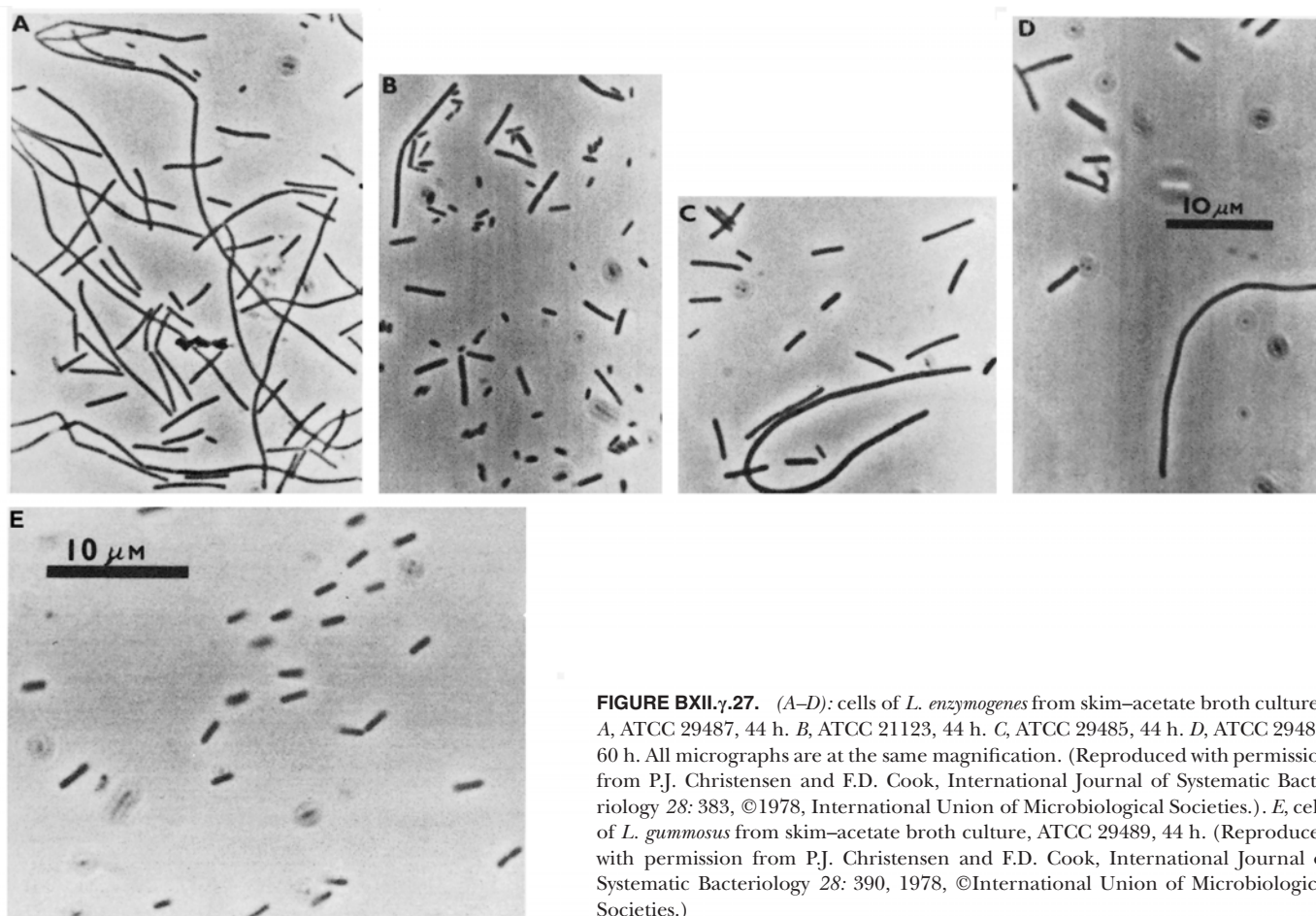


FIGURE BXII.γ.27. (A–D): cells of *L. enzymogenes* from skim–acetate broth cultures. A, ATCC 29487, 44 h. B, ATCC 21123, 44 h. C, ATCC 29485, 44 h. D, ATCC 29488, 60 h. All micrographs are at the same magnification. (Reproduced with permission from P.J. Christensen and F.D. Cook, *International Journal of Systematic Bacteriology* 28: 383, ©1978, International Union of Microbiological Societies.). E, cells of *L. gummosus* from skim–acetate broth culture, ATCC 29489, 44 h. (Reproduced with permission from P.J. Christensen and F.D. Cook, *International Journal of Systematic Bacteriology* 28: 390, 1978, ©International Union of Microbiological Societies.)

sequent work has included further study of the active site and comparison with mammalian elastase (Bauer et al., 1981), investigation of the molecular structure at 1.8-Å resolution (M.N.G. James, personal communication), and D. Agard's isolation of the α -lytic protease gene, sequencing, and point-specific mutations (M.N.G. James, personal communication).

The nucleases of *L. enzymogenes* strain ATCC 29487 (formerly "myxobacter 495") have been characterized by von Tigerstrom (1980, 1981, 1983), with the two major ones being extracellular RNase and DNase. The RNase consists of one polypeptide chain with a molecular weight of 46,000–47,000 Da that is most active at pH 8.0–8.5 and in the presence of Mg^{2+} . The nonspecific nuclease has a molecular weight of 22,000–28,000 Da, is most active at pH 8.0, and requires Mg^{2+} or Mn^{2+} .

Catalase, oxidase, and phosphatase are produced; work is in progress on the phosphatases of *L. enzymogenes* (R.G. von Tigerstrom, personal communication).^{*} The indole, methyl-red, and Voges–Proskauer tests are negative.

Lyses Gram-negative and Gram-positive bacteria, including actinomycetes, yeasts and filamentous fungi, blue-green and green algae, and nematodes. Lysobacters do not attack Gram-negative

bacteria as vigorously as they do Gram-positive species; this may be a self-protective feature.

All strains known at present have been isolated from soil or freshwater habitats; no studies have been aimed at finding them elsewhere, but this would probably be a fruitful area for investigation. Their polysaccharolytic and proteolytic abilities suggest a role in the degradation of biological structural and storage materials. The significance of their lytic abilities toward other organisms has yet to be fully assessed. It is quite possible that marine members of the genus will be found or that these organisms may have medical significance.^{*} *Lysobacter* spp. have probably been confused with "eubacteria" in the past when encountered on regular plating media; nutritionally dilute media are needed in order for lysobacters and other gliding bacteria to be recognized, since they do not exhibit spreading on plate count agar (PCA).

One species, *L. antibioticus*, elaborates a potent, wide-spectrum antibiotic named "myxin," which is a 1-hydroxy-6-methoxyphenazine (Cook et al., 1971; Behki and Lesley, 1972). It is manufactured by Hoffman and Roche, but there are two important drawbacks to its widespread medical usage: it has too wide a spectrum,

^{*}Editorial Note: Since the publication of this chapter in the 1st edition of *Bergey's Manual of Systematic Bacteriology*, several papers on this topic have been published. See von Tigerstrom (1984); von Tigerstrom and Stelmashuk (1985, 1986, 1987); Au et al. (1991).

^{*}Editorial Note: Since the publication of this chapter in the 1st edition of *Bergey's Manual of Systematic Bacteriology*, several papers on this topic have been published. See Bonner et al. (1988), O'Sullivan et al. (1988), Harada et al. (1988), and Tymiak et al. (1989).

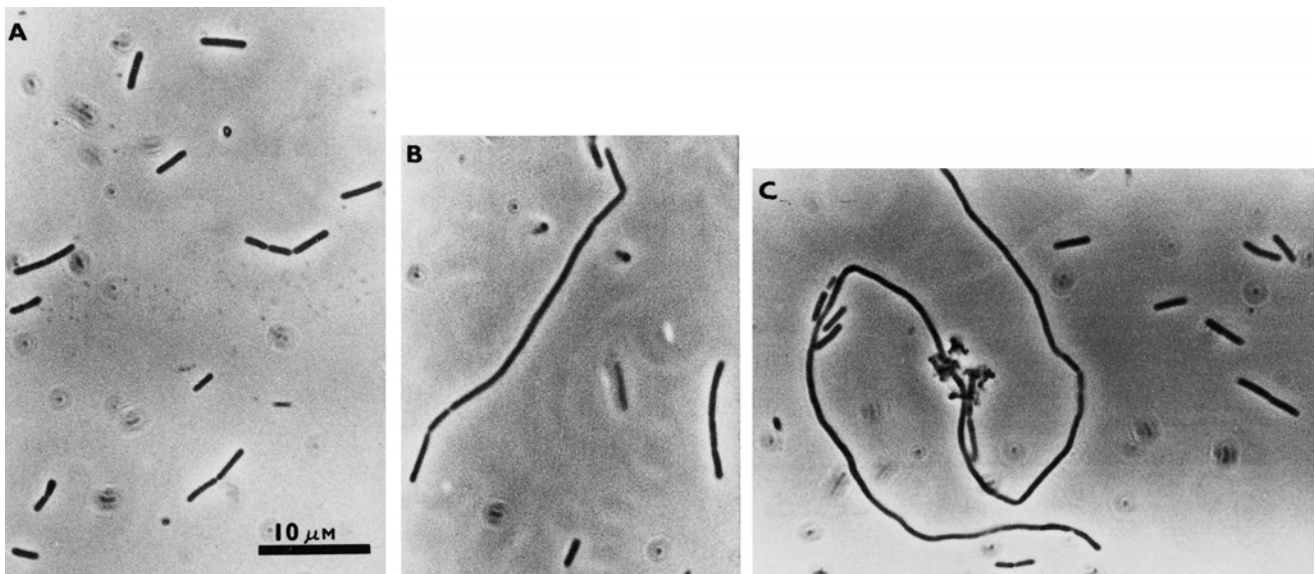


FIGURE BXII.γ.28. Cells of *L. antibioticus* from skim-acetate broth cultures. A, ATCC 29479, 66 h. B, UASM 4593, 32 h. C, UASM 4578, 72 h. All micrographs are at the same magnification. (Reproduced with permission from P.J. Christensen and F.D. Cook, International Journal of Systematic Bacteriology 28: 389, 1978, ©International Union of Microbiological Societies)

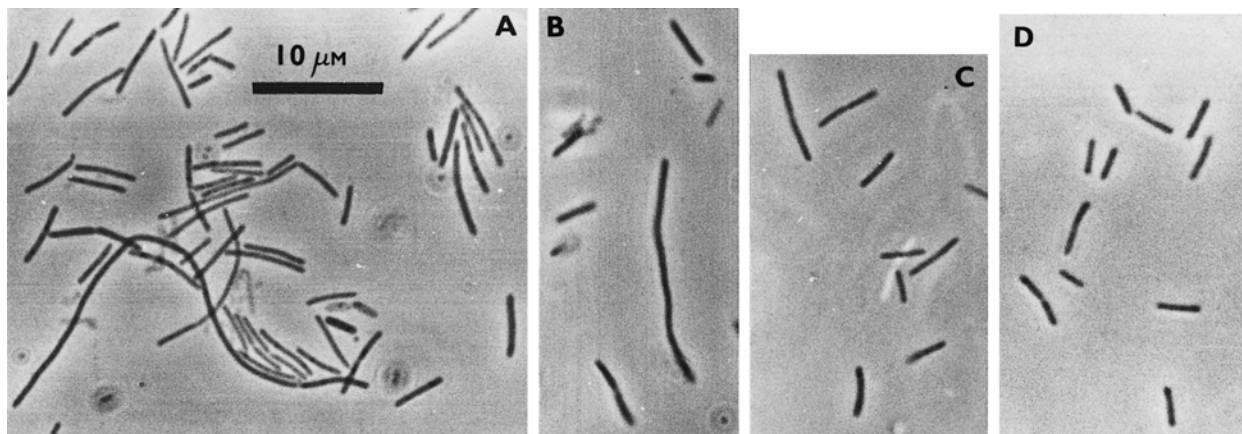


FIGURE BXII.γ.29. Cells of *L. brunescens* from skim-acetate broth cultures. A, ATCC 29483, 60 h. B, ATCC 29482, 60 h. C, UASM 4541, 44 h. D, UASM CB 5, 30 h. All micrographs are at the same magnification. (Reproduced with permission from P.J. Christensen and F.D. Cook, International Journal of Systematic Bacteriology 28: 391, 1978, ©International Union of Microbiological Societies.)

especially for internal use, and it is explosive when dry. Its use is restricted to topical application for resistant fungal skin problems.

ENRICHMENT AND ISOLATION PROCEDURES

Lysobacters have the unusual ability to hydrolyze chitin, and their isolation is facilitated by providing chitin as a suspension, in ground mushrooms, or in autoclaved yeast cells. The simplest method is to enrich soil samples with one of the above sources of chitin, incubate it for at least 1 month, then plate it out onto yeast cell agar (Smit and Clark, 1971), which consists of 0.5% baker's yeast in a 1.5% agar medium. The number of colonies capable of lysing autoclaved yeast cells increases dramatically with

such enrichment procedures (Christensen and Cook, 1978). Water samples can be plated directly onto yeast cell agar.

MAINTENANCE PROCEDURES

Lysobacters grow profusely on PCA (plate count agar, from, for example, Difco), SAA (skim-acetate agar: 0.5% skim milk, 0.05% yeast extract, and 0.02% sodium acetate in 1.5% agar) or broth, or CCA (Cook's *Cytophaga* agar: 0.2% tryptone in 1.0% agar) (Christensen and Cook, 1972). The more dilute the medium, the greater the expression of the spreading morphology, thus CCA is recommended for this purpose and because on this medium, organisms do not appear to lose their polysaccharolytic and proteolytic potentials. On PCA, colonies of lysobacters are

often indistinguishable from nongliding bacteria. Ordinary plating techniques are sufficient with incubation at room temperature for the species so far known; however, difficulty can occur with *L. gummosus* because of its intensely rubbery colonies, and, for this species, it is recommended that young cultures (<24 h) be used when possible. Lyophilization is successful, and cultures are readily obtainable from these preparations, which are recommended for long-term storage. Caution is advised on two aspects, however: (a) colony morphology may change slightly after freeze-drying, and (b) strains of *L. enzymogenes* do not exhibit as strong a microbial lytic action and do not maintain as high a level of certain proteolytic enzymes, after they have been maintained in the laboratory on skim milk media for some time, as

do cells that have not been freeze-dried, and this may also apply to lyophilized cultures.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

In order to differentiate lysobacters from the myxobacteria, it is necessary to demonstrate the absence of fruiting bodies. At least two methods should be attempted, such as inoculation onto sterile wild herbivore pellets in water agar (Smit and Clark, 1971) and induction of myxospores following addition of 0.5 M glycerol during exponential growth (Dworkin and Gibson, 1964).

Suitable methods for demonstration of polysaccharide degradation, antimicrobial lytic action, and susceptibility to actinomycin D have been published by Christensen (1977).

DIFFERENTIATION OF THE SPECIES OF THE GENUS *LYSOBACTER*

Characteristics useful for distinguishing among the species of *Lysobacter* are given in Tables BXII.γ.24 and BXII.γ.25.

TABLE BXII.γ.24. Differential characteristics of the species of the genus *Lysobacter*^{a,b,c}

Characteristics	1. <i>L. enzymogenes</i>	2. <i>L. antibioticus</i>	3. <i>L. brunescens</i>	4. <i>L. gummosus</i>
Broth culture viscous	+	+	—	+
Colonies:				(heavy)
Type of growth	Sloppy, mucoid	Sloppy, mucoid	Spreading, thin	Pulvinate, gummy
Surface smooth	+ or —	+	+ or —	+
Opacity	Tp, Tl or Op	Tl or Op	Tp or Tl	Tl or Op
Color	Dark C to deep Y-C	Pi to Br-Pi or O-Br	Y to Ch	Pale Y-G to Y-G
Brown, water-soluble pigment produced	None to moderate	Weak to heavy	Weak to heavy	—
Myxin crystals in old cultures	—	Usually +	—	—
Urea as nitrogen source	+ or —	—	—	+
Acid from:				
Cellobiose	+	+	—	+
Sucrose	+	—	—	+
Lactose	+	d	—	+
Lipase-Tween 20	+	d	—	+
Hydrolysis of:				
Alginate	+ or —	d	—	—
CMC	+	+	—	+
Pectate	+	— ^d	+	+
SYS or NBS	—	— ^e	+	—
Potato starch	+ or —	— ^f	+	—
Sheep erythrocytes	α, β, or γ	β or α	γ or —	α
H ₂ S produced	d	—	+	—
Citrate as sole carbon source	+	+	—	+
Growth on:				
MacConkey's	+ (colorless or —)	d (colorless)	—	—
EMB	+ (Pi)	+ (Pi)	—	+ (Pi)
Completely inhibited by 0.1% SLS	+ or —	+	+	—
Susceptible to:				
Chloramphenicol (30 μg)	d	—	+	—
Penicillin G (10 U)	—	—	d	—
Actinomycin D	d	—	+	d
Habitat	Soil	Soil	Freshwater	Soil

^aData are from Christensen and Cook (1978), to which the reader is referred for details of tests.

^bSymbols: +, 90% or more of strains are positive; —, 90% or more of strains are negative; and d, 11–89% of strains are positive.

^cAbbreviations: Br, brown(ish); C, cream-colored; Ch, chocolate; EMB, eosin methylene blue; G, gray; NBS, nutrient broth–starch; O, orange; Op, opaque; Pi, Pink(ish); SLS, sodium lauryl sulfate; SYS, salts–yeast extract–starch; Tl, translucent; Tp, transparent; and Y, yellow(ish).

^dGrowth but not liquefaction.

^eOnly one strain recorded on SYS.

^fOnly one strain recorded.

TABLE BXII.γ.25. Other characteristics of the species of the genus *Lysobacter*^{a,b,c}

Characteristics	1. <i>L. enzymogenes</i>	3. <i>L. antibioticus</i>	4. <i>L. brunescens</i>	2. <i>L. gummosus</i>
Cell dimensions (μm)	0.3–0.5 × 4–50	0.4 × 4–40	0.2–0.5 × 7–70	0.4 × 2
Flexing ^d	+	+	+	–
Gliding ^d	+	+	+	+ (short jerks)
Silkeness in broth culture ^d	+	+	+	+
<i>Fruiting bodies induced by:</i>				
Glycerol	–	–	–	–
Pellets	–	–	–	–
Partial inhibition by 1% NaCl	d	+	+	–
% NaCl causing complete inhibition	≤3	3	≤2	3
Preferred atmosphere	Air or 10% O ₂	Air or 10% O ₂	Air or 10% O ₂	Air
Growth range temperature (°C)	5–40	2–40	4–50	10–40
Growth optimum temperature (°C)	25–35	25–33	30–40	20
Initial pH for growth	5 to >10	5 to >10	5 to >10	6 to >10
<i>Nitrogen sources:</i>				
Nitrate	+	+	+	+
Ammonia	+	+	+	+
Glutamate	+	+	+	+
Asparaginate	+	+	d	+
<i>OF test (glucose):</i>				
Oxidation	+	+	+	+
Fermentation	–	d	d	–
<i>Acid from:</i>				
Glucose	+	+	+	+
Glycerol	d	–	–	–
Mannitol	d	–	–	–
Lipase–Tween 40	+	+	d	+
Lipase–Tween 60	+	+	+	+
Lipase–Tween 80	+	+	d	+
<i>Hydrolysis of:</i>				
Agar (gelase)	+ or –	– ^e	+	–
Agar tubes	–	–	–	–
Cellulose (filter paper)	–	–	–	–
Chitin	+	+	+	+
Gelatin liquefaction	+	+	+	+
Complete peptonization of milk (days)	1–3	1	1	1
Casein—growth and NH ₃ produced	+, +	+, +	+, +	+, +
Penassay—growth	+	+ ^e	+	+
Casitone—growth and NH ₃ produced	+, +	+, +	+, +	+, +
Casamino acids—salts—growth and NH ₃ produced	+, +	+, – ^e	+, –	+, –
Grows well on 0.2% tryptone	+	+	+	+
Catalase	+	+	+	+
Oxidase	+	+	+	nd
Phosphatase	+	+	+	+
Indole	–	–	–	–
Methyl red	–	–	–	–
Voges–Proskauer	–	–	–	–
Reduction of NO ₃ [–] to NO ₂ [–]	–	d	–	–
Reduction of NO ₂ [–] to gas	–	–	–	–
Growth reduced by 0.01% SLS	+ or –	–	d	–
<i>Susceptible to:</i>				
10 μg streptomycin	–	d	d	–
300 U polymyxin B (Kirby–Bauer)	d	–	d	d
300 U polymyxin B (Christensen)	+	+	+	+
<i>Antimicrobial lytic action:</i>				
Gram-negative bacteria:				
<i>Escherichia coli</i>	d	d	d	–
<i>Pseudomonas aeruginosa</i>	d	–	d	–
<i>Serratia marcescens</i>	d	d	d	–
Gram-positive bacteria:				
<i>Arthrobacter</i> spp.	+	d	d	+
<i>Bacillus subtilis</i>	+	d	d	+
Actinomycetes:				
UASM 4432	+	–	–	+
UASM 4441	+	d	d	+
Fungi:				
<i>Rhizopus</i> spp.	+	d	d	d
<i>Penicillium notatum</i>	+	d	d	+
<i>Sclerotinia sclerotiorum</i>	d	+	d	+
Baker's yeast	+	d	+	+
Alga:				
<i>Chlorella</i> spp.	d	+	+	+

^aData are from Christensen and Cook (1978), to which the reader is referred for details of tests.^bSymbols: +, 90% or more of strains are positive; –, 90% or more of strains are negative; and d, 11–89% of strains are positive.^cAbbreviations: OF, oxidation/fermentation; and SLS, sodium lauryl sulfate.^dFlexing, gliding, and silkeness are more readily observed with longer cells.^eOnly one strain recorded on nutrient broth.

List of species of the genus Lysobacter

1. **Lysobacter enzymogenes** Christensen and Cook 1978, 374^{AL}. *en.zy.mo'ge.nes*. Gr. n. *zyme* leaven; M.L. n. *enzymum* enzyme; Gr. v. *gennaio* to produce; M.L. adj. *enzymogenes* enzyme-producing.

Two distinct colony forms are known: a dirty-white, mucoid colony and a yellowish, nonmucoid one. The mucoid colony produces nonmucoid mutants, but the yellowish, nonmucoid form does not produce revertants to the dirty-white, mucoid form. The colony descriptions given here and in Table BXII.γ.25 cover the whole range of colony forms observed. The forms are identical in other properties.

On CCA, 5-d-old colonies are dark cream-colored, with Munsell notation 2.5Y 6–8/4–8 and 5–10YR 6–8/4–8, and circular to irregular. Colonies usually have a smooth surface, but occasionally have a rough surface. Edges may be entire, undulate, lobate or erose; elevation is typically effuse with a raised or convex center, but occasionally flat. Colonies are transparent or translucent; no brown, water-soluble pigment is produced.

On SAA, 5-d-old colonies are dark cream-colored, with Munsell notation 2.5–7.5Y 6.5–8.5/4–6 and 7.5–10YR 5.5–7/5–7, and circular to irregular, with a smooth or rough surface. Edges may be entire, undulate or erose; elevation is typically effuse with a raised, convex or umbonate center, but occasionally flat. Colonies are transparent, translucent or opaque; no brown, water-soluble pigment is produced.

On PCA, 5-d-old colonies are deep yellow, cream-colored, or creamy brown, with Munsell notation 1.5–5Y 3–7/4–8, and more or less circular, usually with a smooth surface. Edges may be entire, undulate or erose; elevation is usually convex, but may be effuse with a convex or raised center or may be umbonate. Colonies are translucent or opaque; some brown, water-soluble pigment may be produced.

Older cultures of the majority of strains produce copious, dark-brown, water-soluble pigment on most media.

The mol% G + C of the DNA is: 65.4–70.1 (T_m).

Type strain: ATCC 29487, UASM 495.

Additional Remarks: Other strains: ATCC 27796 (UASM AL-1), ATCC 29488 (UASM 13B), and ATCC 29485/6 (colony types of UASM 18L).

2. **Lysobacter antibioticus** Christensen and Cook 1978, 387^{AL}. *an.ti.bi.o'ti.cus*. Gr. pref. *anti* against; Gr. n. *bios* life; M.L. adj. *antibioticus* against life, antibiotic.

On CCA, 5-d-old colonies are pinkish brown, with Munsell notation 1–10YR 2.5–7/4–8, and circular to irregular, with a smooth surface. Edges are usually erose, but may be entire, undulate, or filamentous. Elevation is usually flat or raised, but sometimes effuse with a convex center. Translucent or opaque; some or much brown, water-soluble pigment is produced.

On SAA, 5-d-old colonies are pinkish brown, with Munsell notation 2.5Y 7/5 and 1.5–7.5YR 3–6/4–8, and more or less circular, with a smooth surface. Edges are usually erose, but may be entire or filamentous. Elevation is usually raised, but may be effuse, flat or convex. Translucent or opaque; some or much brown, water-soluble pigment is produced or may be absent.

On PCA, 5-d-old colonies are orange-brown, with Mun-

sell notation 1–2.5Y 4–6/3–4 and 5–10YR 2–5/2–6, and more or less circular, with a smooth surface. Edges are usually entire, but may be erose or undulate. Elevation is convex or raised. Colonies are usually opaque; weak to heavy, brown, water-soluble pigment is produced.

Older cultures on most media produce copious brown, water-soluble pigment, and deep-red crystals of the antibiotic myxin may be observed within the highly mucoid colonies.

The mol% G + C of the DNA is: 66.2–69.2 (T_m).

Type strain: ATCC 29479, UASM 3C.

GenBank accession number (16S rRNA): AB019582.

Additional Remarks: Other strains: ATCC 29480 (UASM L17) and ATCC 29481 (UASM 4045).

3. **Lysobacter brunescens** Christensen and Cook 1978, 390^{AL}. *bru.nes'cens*. L. v. *brunescio* to become dark brown; L. part. adj. *brunescens* becoming dark brown.

On CCA, 5-d-old colonies are brownish yellow, with Munsell notation 2.5–5Y 3–4/4–8 and 10YR 4–5/6–8, and irregular, with a rough surface. Edges are usually filamentous, but may be lobate, undulate, or erose. Elevation is effuse with an umbonate or convex center. Colonies are transparent; amounts of brown, water-soluble pigment vary from moderate to absent.

On SAA, 5-d-old colonies are yellow-brown, with Munsell notation 2.5Y 4.5/6 and 7.5–10YR 4–6/6–8. Usually irregular, but occasionally circular. Colonies usually have a rough surface. Edges are typically erose or filamentous, but may be entire, undulate, or lobate. Elevation is effuse or raised, with convex or umbonate centers. Transparent or translucent; weak to moderate amounts of brown, water-soluble pigment are produced.

On PCA, 5-d-old colonies are deep yellow-brown, with Munsell notation 1–2.5Y 3–4/4–6 and 10YR 3–4/4–6. Circular to irregular, with a smooth or rough surface. Edges are usually entire or undulate, but sometimes erose or filamentous. Elevation is either effuse with convex or umbonate centers, raised, or convex. Translucent; production of brown, water-soluble pigment is moderate to heavy.

Older cultures on all media produce copious, dark-brown, water-soluble pigment.

The mol% G + C of the DNA is: 67.6–67.8 (T_m).

Type strain: ATCC 29482, UASM D.

Additional Remarks: Other strains: ATCC 29483 (UASM 2) and ATCC 29484 (UASM 6).

4. **Lysobacter gummosus** Christensen and Cook 1978, 388^{AL}. *gum.mo'sus*. L. adj. *gummosus* slime (gum)-producing.

On CCA, 5-day-old colonies are pale yellowish-gray, with Munsell notation 5Y 8/2. Circular, with a smooth surface. Entire edge. Elevation pulvinate. Translucent; no water-soluble pigment produced. Gummy.

On SAA, 5-d-old colonies are pale yellowish-gray, with Munsell notation 5Y 8/2. Circular, with a smooth surface. Entire edge. Elevation pulvinate. Opaque; no water-soluble pigment produced. Very gummy.

On PCA, 5-d-old colonies are yellow-gray, with Munsell notation 5Y 7/4. Circular, with a smooth surface. Entire

edge. Elevation pulvinate. Opaque; no water-soluble pigment produced. Very gummy.

Older cultures are intensely gummy and do not produce a water-soluble pigment. In very old cultures, the viscosity

of the colony may decrease dramatically, although viable cells can still be recovered from this thin slime.

The mol% G + C of the DNA is: 65.7 (T_m).

Type strain: ATCC 29489, UASM 402.

Genus V. Nevskia Famintzin 1892, 484^{AL}

HANS-DIETRICH BABENZIEN AND HERIBERT CYPIONKA

Nev'ski.a. M.L. fem. n. *Nevskia* from the Neva, a river in St. Petersburg.

Cells elongated, rod shaped, usually slightly bent with **acellular hyaline stalks** (Fig. BXII.γ.30). The **stalks branch dichotomously** because of binary fission of mature cells and the preference for laterally excreted slime material (Fig. BXII.γ.31). Undisturbed

and slowly growing cells form **flat, rosette-like to bush-like microcolonies** (Fig. BXII.γ.32), up to 80 μm in diameter, **on the surface of freshwater**. Faster growth results in the formation of a thin surface pellicle without distinct stalks (Fig. BXII.γ.33A) and with

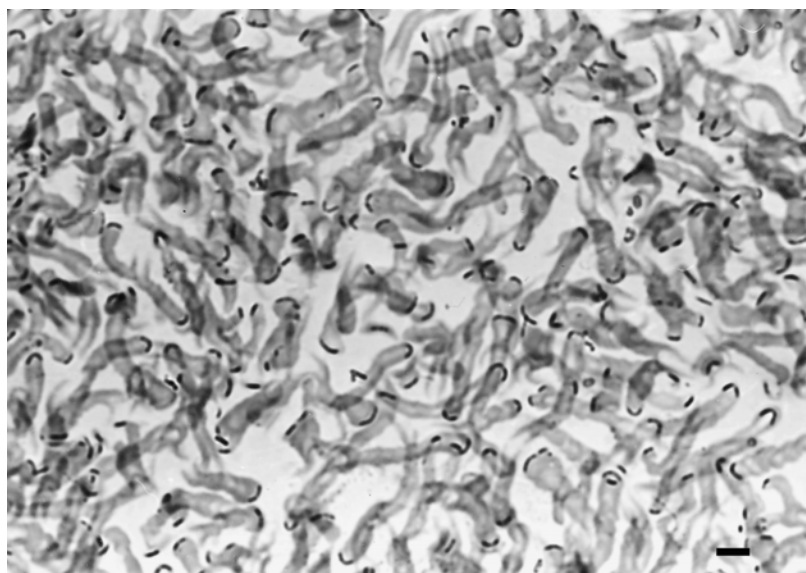


FIGURE BXII.γ.30. Enrichment culture of *N. ramosa*, thin bent rods at the tip of slime stalks. Formalin-toluidine blue. Bar = 10 μm.

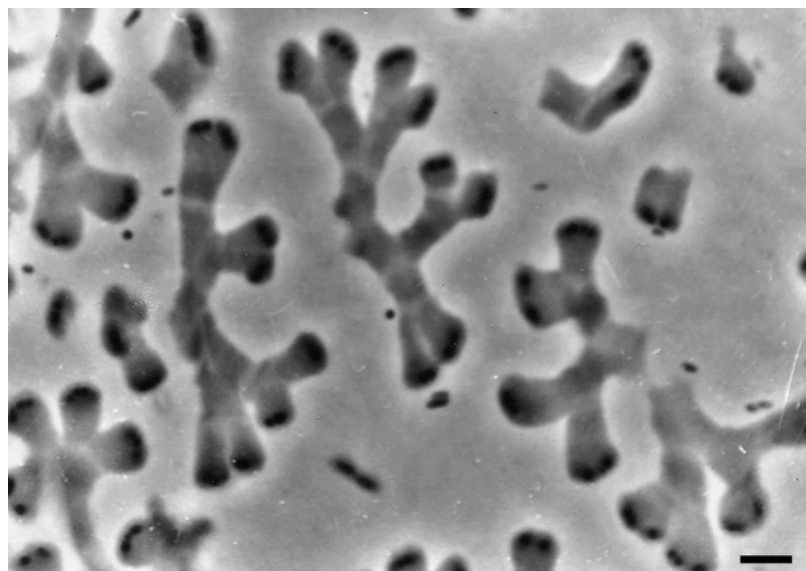


FIGURE BXII.γ.31. *Nevskia ramosa*, strain Soe 1 (DSM 11499). Dichotomously branched slime stalks with *Nevskia ramosa* cells at the tip. Phase-contrast micrograph. Bar = 10 μm.